FERN HALPER, PhD

DATA MAKES THE WORLD GO 'ROUND

THE DATA, TECH, AND TRUST BEHIND AI SUCCESS

DATA
MAKES THE WORLD
GO 'ROUND

FERN HALPER, PhD

DATA MAKES THE WORLD GO 'ROUND

THE DATA, TECH, AND TRUST BEHIND AI SUCCESS

WILEY

Published by John Wiley & Sons, Inc., Hoboken, New Jersey.

Library of Congress Cataloging-in-Publication Data has been applied for:

Paperback ISBN: 9781394390632
ePDF ISBN: 9781394390656
ePub ISBN: 9781394390649
oBook ISBN: 9781394390663

Cover Design: Wiley
Cover Image: © Максим Ивасюк/Getty Images

Set in 10.5/13pt Palatino LT Std by Lumina Datamatics

To my grandson.
AI will shape the world you inherit;
may it be guided by wisdom, integrity, and possibility.
May it be a world that lifts you up and carries you
toward all the places you choose to go.

Contents

Acknowledgments

I would like to thank the many editors, production specialists, and publishing staff at Wiley who contributed their expertise to this project. These include Jajneswar Chhotaray, Sheryl Nelson, and Christine O'Connor. A particular note of gratitude goes to John Sleeva, who served as my primary editor and was a consistent source of direction and support. I would also like to thank James Minatel, who first brought this project into the Wiley portfolio, and Annie Melnick, whose support helped set the foundation for this book.

I spoke with many people while writing this book, and I am deeply grateful for the time these technology leaders and experts devoted to sharing their experience and insight. Their contributions, through case studies, interviews, and expert briefings, added depth and practical relevance to the chapters that follow. My sincere thanks go to Cal Al-Dhubaib, Ronald Brachman, Donna Burbank, Helen Brown-Liburd, PhD, Donald Farmer, Amber Heffner-Cosby, Joanne Heith, PsyD, Ibrahim Itani, Deanne Larson, PhD, Stephen Lee, Evan Levy, Naveed Memon, Ram Kumar Nimmakayala, John O'Brien, Ray Ready, Prashanth Southekal, PhD, Sam Wong, Daniel Ziv, and others who spoke on the condition of anonymity.

My thanks as well to my sister, Adrienne Halper, who reviewed early chapters and offered thoughtful feedback that strengthened the clarity of the writing.

About the Author

Fern Halper, PhD, is a nationally recognized expert on data and artificial intelligence whose work helps business and technology leaders understand what it really takes to succeed with data and AI in practice. She brings a rare combination of hands-on practitioner experience, academic teaching, and industry thought leadership. Over more than 25 years, Dr. Halper has been published hundreds of times on data mining, analytics, machine learning, and artificial intelligence, and she is coauthor of several Wiley-published titles in cloud computing, big data, and analytics.

Halper is widely recognized as a leading voice in the data and AI community. She is a frequent speaker and educator at national conferences and events, where she helps executives and practitioners cut through hype to build AI literacy, develop governance and risk understanding, and design practical roadmaps for enterprise AI adoption.

Halper holds a BA from Colgate University and a PhD from Texas A&M University. She has been a lead analyst at Bell Laboratories, a partner at Hurwitz & Associates, and has taught at both Colgate University and Bentley University. She is currently the vice president of research at TDWI and founder of AI Foundations Group.

Introduction

For the past 30 years, I have had the privilege of working in the data and analytics space. I have been at the center of the field's evolution—from relational databases and data mining to the transformative power of artificial intelligence (AI) that we see today.

I began my data career at AT&T Bell Laboratories, working in a Center of Excellence that brought together experts from diverse fields—economists, computer scientists, operations researchers, statisticians, geophysicists (like me), and even psychologists. Our mission was clear: to find innovative ways to solve business problems using data. One example was pioneering the continuous audit of online systems, a concept that is now foundational in modern auditing and compliance. I also had the opportunity to work with Bell Labs research teams to bring early machine learning models into practical business use. At the time, we could predict customer churn with decent accuracy, but deploying those models in production was a different story. Operational constraints made it nearly impossible.

Since then, I have spent more than two decades as a practitioner, industry analyst, thought leader, and researcher. I've had the opportunity to help organizations navigate and capitalize on the most transformative waves, from the rise of e-commerce and the dot-com bubble to the cloud computing revolution, big data, and now AI. I've been an industry analyst. I've coauthored books on cloud computing and big data. For the past 12 years I've worked with a wonderful team at TDWI, a data and AI education and research firm, as the vice president of research.

A Generational Leap in AI

We are at a unique moment in technological history. The advent of generative AI represents a generational leap. It is a transformation as important as the emergence of the internet or mobile computing. Until recently, organizations followed a relatively predictable analytics maturity curve: they started with reports, moved to dashboards, then self-service analytics, and eventually developed predictive models. As organizations became more advanced, they put machine learning models into production for use cases such as churn prediction, fraud detection, recommendation engines, and predictive maintenance.

Not surprisingly, as organizations see tangible value from AI, demand continues to grow. I've observed a clear pattern: the more mature an organization is in its analytics capabilities, the more measurable value it derives from AI. This creates a virtuous circle; as companies benefit from AI, they become more committed to advancing their capabilities further.

But now, generative and agentic AI are trying to change the game. Organizations that were once methodically advancing through the analytics maturity model are now feeling intense pressure to implement AI-driven solutions immediately. Yet, many companies are not yet ready to utilize some of these advances to the fullest. While they think it may be easy to deploy a chatbot for instance, they don't realize that their data foundation needs to be in place if they hope to make use of customer data. They haven't necessarily thought about what resources would be needed to build AI applications at scale. They haven't thought about how to govern their data, their analytics, or their AI. What I see in my research is that many companies are still working to put the data foundation in place (although they are making progress), democratize business intelligence (BI) to make it easier for business users to access and analyze data, and try to govern what they have. And now they are being asked to implement AI!

Much of this hype, of course, is being driven by generative AI. And there are at least two sides to the generative AI coin. There is the consumerization of AI because of generative AI. That means people in companies making use of generative AI platforms (such as ChatGPT, Perplexity, Claude, or Microsoft Copilot) to make their work more efficient. Then there is the flip side, the side I think about as a data and analytics professional; that is building applications that use this technology, against company data, for competitive advantage. Of course, in between is the movement to infuse AI into applications across the data and analytics life cycle. It is a complex time.

The Purpose of This Book

There have been many books written about succeeding with AI—what companies are doing, what their leaders say work, but they are at the leadership level. This book goes down a level, into the technology, into governance, into the organizational structure. It will give you a window into what goes on behind the scenes of successful AI.

The reality is that many companies aren't ready. While organizations are making progress they aren't there yet. And the point is that a lot of leaders are trying to move forward, but they will only get so far, because they really don't get what is involved. So, I want to help them get it. This book examines the organizational components, data foundations, analytics foundations, skills, and governance needed to succeed with AI. It is geared toward directors and vice president of data and analytics as well as CEOs, CTOs, and other C-suite members who want to understand what is involved.

Many executives claim to support AI and analytics but lack a deep enough understanding of what it takes to build a truly successful program. It is not just about investing in cutting-edge technology; success depends on the right strategy, the right resources, the right tooling, and, most importantly, a cultural shift within the organization.

This book is designed to provide practical insights and real-world best practices drawn from my experiences in the field, my years of research, as well as the dozens of interviews I've done for this book. It is not a technical deep dive. Instead, it is a strategic guide that highlights the organizational and technological factors that lead to success with data, analytics, and AI. It is meant to discuss the landscape in a way that data and business professionals can understand, so they can understand what will be needed to succeed. It discusses the latest trends in data and AI, from data products to the data fabric, from data governance to responsible AI, and from building models to operationalizing them in production.

Throughout this book, you will find case studies and interviews with industry leaders, those who have successfully navigated the complexities of AI adoption. Their stories will provide valuable lessons and actionable strategies that you won't find in other books.

We are at a pivotal moment. AI is reshaping industries, redefining competitive advantage, and revolutionizing the way businesses operate. This book will help you navigate these changes with clarity, confidence, and a strategic mindset.

How This Book Is Structured

This book is divided into distinct sections that weave together to tell the story of what it takes to succeed with data and AI.

Part I—The AI Imperative: Part I sets the stage for the rest of the book. It looks at what is happening in enterprises today, the promise of AI, how some organizations are not yet ready but being pressured to implement it, common mistakes, and the five key dimensions for AI readiness and maturity.

Part II—Building the Data Foundation for AI: Part II examines the data foundation needed for AI. Many organizations try to jump ahead of this, especially with generative AI use cases; however, when they realize they need company data for their applications, they aren't prepared. The section discusses the new kinds of data organizations need to utilize (such as unstructured text data), the problem with data silos, and how organizations are trying to unify their data and create a modern architecture, and what is involved with this process. It also looks at other data management issues such as findability, useability, and data quality and trends in this area. It ends with newer trends in data management such as data products.

Part III—Democratizing Analytics: Part III focuses on how implementing BI has often been a precursor to moving forward on the analytics journey. There are important lessons that companies can learn from implementing BI. Additionally, democratizing BI and making it available to more users serves numerous purposes; it frees up data analysts and other analytical roles to do more advanced work; it also starts to build the culture around analytics and gets people data literate, which will be important as they move into more complex analytics such as AI. This section describes why it is important and best practices for rolling it out into the company and getting buy-in. With the advent of generative BI, this becomes even more important.

Part IV—The AI Build-out: Ecosystem, Models, Operations, and the Shift to Agents: Part IV examines best practices for building models and aligning them with business strategy. It also will look at the skills to build traditional and newer AI applications, a high-level overview of what is involved, as well as what's new with generative and agentic AI. In this section, the notion of using generative AI with company data is introduced, along with the challenges that entails. Here, I'll also talk about operationalizing AI applications and discuss what is involved from a technology and organizational perspective. This is about putting these models into production, the new roles, and processes this involves.

Part V—Governing Data and AI: As we will see in this book, governing data is separate from governing AI. Organizations have been focused on governing their data (and their traditional structured data at that). However, AI involves new data types that will also need to be governed. Likewise, AI models themselves will need to be governed. This includes new generative and agentic AI models as well. This section discusses what is involved in terms of roles, processes, and tools while enabling the business to innovate.

Part VI—Putting It All Together: The final chapter provides a data and AI readiness checklist, a tool that leaders can use to examine where they are, identify gaps, and prioritize next steps.

Some Important Points

I'm sure that I won't surprise any of you reading this book if I say that this space is moving quickly. It is. What I've written reflects the state of the industry at the end of 2025. That is the first point. The second point is that vendors and products mentioned in this book appear solely as examples. I am not endorsing any specific solutions. The descriptions of capabilities reflect the state of the market as of late 2025 and are accurate to the best of my knowledge at that time. Because products evolve rapidly, readers should independently verify current features and suitability before making any technology decisions. No guarantee, warranty, or responsibility for errors, omissions, or subsequent changes is implied.

For additional context on my long-running perspective on the data and analytics space, I have written about these topics for more than 20 years at my blog, *Data Makes the World Go 'Round* (`www.datamakesworld.com`), which inspired the title of this book.

Part

I

The AI Imperative

CHAPTER

1

The Business Case for Analytics and Artificial Intelligence

In this chapter, we will examine the following:

- Modern AI—evolution and competitive advantage
- The democratization of AI and competitive advantage
- Real-world applications of AI and generative AI
- Analytics and AI investment trends
- Justifying investments with concrete outcomes

Five years from now, the leaders who succeeded in this moment may say they made the right call on artificial intelligence (AI). The others may talk about the “hype cycle” and wonder where their advantage went. We are at a generational inflection point. AI is being embedded in daily workflows and reshaping how we compete. From predictive analytics to generative and agentic AI, organizations are discovering new ways to cut costs, boost productivity, and open new markets.

But chasing hype is not a strategy. Too many leadership teams are trying to pilot chatbots, co-pilots, and generative tools without the solid data foundations, governance, or operational models to make them stick. The result can be proof-of-concepts that don’t scale and security and compliance risks that they didn’t anticipate. If you want AI to create measurable value, you need to start with a clear business case, one that ties investment to strategy, outcomes, and the capabilities required to deliver them.

This chapter examines the business case for AI, exploring its history, how it is evolving, and how it can create value.

Modern AI—Evolution and Competitive Advantage

Everyone is talking about AI. It is on the cover of magazines, in the news, and the subject of conferences and events. Legislative bodies, government entities, and others are trying to regulate it. It is a major focus for tech companies, which are embedding it in their software and creating new innovations using AI. I've spoken with many organizations that say they are under executive and board-level pressure to "do something" with AI.

Yet, AI is not new. The idea that machines could act intelligently has been around since the time of the ancient Greeks. In the 1950s, when John McCarthy coined the term at a summer research workshop at Dartmouth College, he described it as "making a machine behave in ways that would be called intelligent if a human were so behaving."[1]

Generally, people recognize several major waves of AI. The first wave focused on logic, rules, and symbolic reasoning. For instance, the ELIZA system, created by Joseph Weizenbaum at MIT in the mid-1960s, gave the impression of a conversation.[2] Known now as the first-ever chatbot, it was a rule-based natural language processing system that could simulate conversation by identifying keywords in user input and responding with predefined scripts.

The second wave began in the 1990s. I think of this period as the rise of the idea of data mining—the process of discovering patterns, trends, and useful insights from large datasets using statistical, machine learning, and analytical techniques. In other words, it marked a shift beyond traditional statistics toward the use of computer algorithms such as decision trees and basic neural networks.

This was when I first started to use AI. I remember experimenting with the C4.5 machine learning algorithm[3] at Bell Labs during that time. This decision tree algorithm (what is now considered a simple machine learning algorithm) was popular because it could classify data into distinct categories with a high degree of accuracy. It also influenced the development of modern ensemble methods such as random forests and gradient boosted trees (we'll talk more about these techniques in Chapter 15), which build on the same core principles. I was impressed by the ability of C4.5 to generate rules that could, for example,

---

[1] McCarthy, J., Minsky, M.L., Rochester, N., and Shannon, C.E. (2006). A proposal for the Dartmouth Summer Research Project on Artificial Intelligence, August 31, 1955. *AI Magazine*, 27(4), p. 12. Available at: `doi.org/10.1609/aimag.v27i4.1904` [Accessed 7 June 2025].

[2] Weizenbaum, J. (1966). ELIZA—a computer program for the study of natural language communication between man and machine. *Communications of the ACM*, 9(1), pp. 36–45.

[3] Quinlan, J.R. (1993). *C4.5: Programs for machine learning*. California: Morgan Kaufman Publishers.

estimate the probability that a customer would churn. The importance of these kinds of algorithms was that they could learn patterns without rules.

Use cases for data mining included fraud detection, customer segmentation, market basket analysis, churn prediction, credit scoring, and demand forecasting. Machine learning algorithms then became more complex, supported by the compute power available to run them. Deep neural networks emerged (neural networks with multiple layers), giving rise to the "modern" AI of today.

Much of the hype around the current wave is a result of the introduction of generative AI—AI systems that can automatically create new content, such as text, images, code, or audio, by learning patterns from existing data and generating outputs. (See the following sidebar for the difference between traditional AI and generative AI.) These systems have emerged as a result of recent advances in algorithms such as the popular GPT-5. Many view these systems as revolutionary—on par with the introduction of the internet, e-commerce, and smartphones. And it is quite remarkable what they can do.

That's a brief history of the evolution of AI to put it into context. But what about modern AI and how it can make a company more competitive? To effectively compete, a company needs a differentiated or distinct product offering that generates high customer satisfaction and retention. It needs to achieve operational efficiencies. It must have the ability to innovate and the ability to adapt to a changing landscape. Of course, it needs the financial and talent resources to accomplish this. I would argue that the ability to make evidence-based decisions and using data are also key to competitive advantage.

Using AI algorithms and building AI products can provide competitive advantage. For instance, if I can predict the probability that a customer is going to churn, I can reach out to them to try to prevent this. If I can understand the sentiment of my customers and what is causing negative sentiment, I can start to improve my products. If I can segment customers into distinct groups, I can better understand and market to them. If I can predict when my machinery needs maintenance, I can avoid excessive downtime and save money.

These examples use AI algorithms. If I can embed these AI algorithms into applications, such as recommendation engines (such as those used on Amazon), I have a better chance of increasing my company's revenue. These recommendation engines analyze a wide range of data, including customers' past purchases, items they've viewed, items in their cart, ratings, and even the behavior of similar customers. More personalized recommendations lead to potentially more purchases and greater revenue.

Numerous industry studies confirm that organizations leveraging AI are more likely to achieve measurable business impact than those that do not. For example, Accenture's *The Art of AI Maturity* found that top performers in AI maturity (what they called AI Achievers) achieved 50% higher revenue growth than peers.[4] McKinsey's *State of AI* survey reported that more than a quarter of respondents attributed at least 5% of earnings before interest and taxes (EBIT) directly to AI adoption.[5] Deloitte found that those further along in AI were significantly more likely to report revenue-generating results, such as "entering new markets/expanding services to new constituents, creating new products/programs or services, or enabling new business/service models."[6]

TRADITIONAL AI VS. GENERATIVE AI—A SIMPLE EXPLANATION

AI is an umbrella term that includes a wide range of methodologies and technologies. It draws on advances in disciplines such as mathematics, computer science, computational linguistics, cognitive sciences, and robotics. Popular AI technologies include machine learning (ML), natural language processing (NLP), and neural networks that form the foundation for the intelligent capabilities seen in modern AI systems.

Traditionally, organizations have applied AI to solve a variety of business problems, such as predicting customer churn, detecting fraud, recommending products, anticipating equipment maintenance, and forecasting disease progression. Many of these use cases rely on supervised learning, one of the most accessible and widely used approaches. In supervised learning, a model is trained on a labeled dataset—data where the outcome (or target of interest) is already known—to recognize patterns and predict future results.

For instance, a telecom company might train a model to predict customer churn using features (e.g., attributes) such as customer tenure, average monthly spend, and number of service plans. The model uses these features to learn how different customer behaviors relate to the likelihood of churn. While these patterns may be difficult for a human to detect in large datasets, machine learning algorithms such as decision trees can surface probabilistic rules—for example, identifying that customers with more than three plans and more than three years of tenure are at higher risk of leaving.

Developing traditional AI models involves a sequence of steps: collecting and preparing the data, training and tuning the model (using known outcomes—e.g., the historical data), validating and testing it for accuracy, and deploying it to production.

(continues)

[4] The art of AI maturity: Advancing from practice to performance 2021. (n.d.). Accenture. Available at: `https://www.accenture.com/content/dam/system-files/acom/custom-code/ai-maturity/Accenture-Art-of-AI-Maturity-Report-Global-Revised.pdf#zoom=40` [Accessed 7 August 2025].

[5] The State of AI in 2021. (2021). McKinsey. Available at: `https://www.mckinsey.com/capabilities/quantumblack/our-insights/global-survey-the-state-of-ai-in-2021` [Accessed 7 August 2025].

[6] Deloitte. (2022). *Deloitte's state of AI in the enterprise,* 5th edition report. Available at: `https://www.deloitte.com/content/dam/assets-zone3/us/en/docs/services/consulting/2024/us-ai-institute-state-of-ai-fifth-edition.pdf` [Accessed 7 August 2025].

TRADITIONAL AI VS. GENERATIVE AI—A SIMPLE EXPLANATION (CONTINUED)

Once in production, the model must be monitored over time to ensure that it continues performing well, as all models tend to degrade if left unchecked.

In generative AI, rather than predicting outcomes, systems are designed to create new outputs—such as text, images, music, or code—based on what they've learned from large training datasets. These systems are powered by transformer models, including large language models (LLMs) trained on a corpus that is (in many cases) the internet. The result is the ability to generate content that mimics or extends the patterns seen in their training data. Probably most of you reading this book have already used some sort of LLM such as a GPT to create an email or do some research. Instead of retrieving facts or calculating outcomes directly, these systems operate by predicting the next word in a sequence based on prior examples in their training data. As discussed later in the book, this can lead to issues and requires governance of both the data and the model.

The Democratization of AI and Competitive Advantage

I believe we've reached an inflection point—and an important one. With the advent of generative AI, AI is becoming consumerized and democratized in the workplace; that is, it is becoming more accessible to everyone. Think about all the ways you use AI in your daily life (e.g., virtual assistants such as Apple's Siri or health tracking apps like Fitbit or Apple Watch). Now, many people in business are using AI as part of their daily work. It's in their spreadsheets, emails, and meeting notes. It's part of the everyday workflow, whether you're in marketing, operations, or finance. Developers are using it to generate code; marketers are using it to generate campaigns. AI is no longer just the realm of data scientists and other quants. This is a major change.

This democratization is powerful—but also tricky because it leads to a key question: *How do you go beyond the surface and really use AI to compete?*

Part of the answer lies with your own data. Generative AI can do amazing things out of the box, but the real power is when you use these models with your proprietary data. Your data can help turn generic AI into a strategic asset. It can help you build smarter tools, uncover unique insights, and differentiate in ways that others can't replicate.

Think about it. There is no doubt that organizations using analytics and AI to gain insights from their data have a competitive advantage. In my research at TDWI, I routinely look at the characteristics of those organizations that state that they have been successful in their analytics programs against those who haven't. They are more likely to be implementing advanced analytics and particularly

AI. Additionally, they make sure to implement certain programs, such as data literacy and data governance that help them to build skills, trust insights, and compete more effectively.[7]

For example, if an organization can determine which customers are likely to churn, it can reach out to those customers with an offer that may prevent them from leaving. Ideally, the organization already has insights about the customer, including segmentation and behavioral understanding. Then, it can target these high-risk customers and use generative AI with AI-generated personalized campaigns.

Similarly, if a manufacturing company uses a predictive maintenance model (built using traditional machine learning) and identifies a system at risk of failing, it can dispatch a technician to fix the issue. That technician, while on site, might use an assistant system (developed using generative AI) that draws from manuals, diagrams, historical maintenance logs, and error code documentation to answer questions as needed and replace the part at risk.

note **TDWI (`www.tdwi.org`) is a leading source of education and research for data and analytics professionals working across AI, business intelligence (BI), and data management. For more than 30 years, TDWI has empowered organizations with the knowledge and skills to build future-ready data and AI strategies and solutions through research-backed learning, its expert faculty, and a vibrant global community. The company provides in-person conferences, virtual events, on-demand courses, research reports, and memberships, designed to meet the evolving needs of today's data- and AI-enabled enterprises.**

Real-world Applications of AI and Generative AI

There are numerous examples of how organizations are using AI together with their company data to compete. Netflix's recommendation engine uses AI to analyze customers' viewing histories and compare them with patterns from users with similar tastes to suggest movies and shows they're likely to enjoy. This creates a personalized experience. Amazon uses AI to optimize its supply chain and delivery routes. It also uses it to forecast demand based on historical information and employs techniques such as deep learning to determine whether merchandise is damaged.

The UPS ORION (On-road Integrated Optimization and Navigation) and its newer navigation tool called UPSNav system use AI to optimize delivery routes

[7] TDWI. (2023). Achieving Success with Modern Analytics. *TDWI Best Practices Report*, p. 12. Available at: `tdwi.org/research/2023/06/adv-all-best-practices-report-achieving-success-with-modern-analytics.aspx` [Accessed 7 June 2025].

and provide directions in precise detail.[8] Sports teams use AI to market to their fans, and retail companies use it to push offers to loyalty members.

There are also numerous examples of how companies are using generative AI with their company data to compete. For instance, chatbots are a very popular use case. Daily Harvest, a meal delivery service, uses AI in chatbots in its customer service channels and lets customers easily make changes to their plans, such as skipping their next order.[9] Virtual assistant chatbots are also popular. Portfolio managers and analysts at JPMorgan use a system called Smart Monitor, which the firm estimates has helped reduce time researching a topic. It pulls in data from earnings calls, filings, and other sources, generating tailored alerts and analysis for their asset managers and wealth management professionals.[10]

I see many companies using generative AI on their call center notes or incident reports to summarize them and classify problems into categories. This provides valuable insights as they work toward improving customer service. Several companies I've spoken to are developing tools for their sales teams to build dossiers on potential target clients before going into meetings with them. This data may come from their customer relationship management (CRM) systems as well as external sources. All the salesperson needs to do is specify the client, and they can obtain useful intelligence about them. Some systems even generate scripts for salespeople to use, complete with the tone they should adopt for specific clients.

Agentic AI moves the needle even further forward. Although still in the early days, the idea here is to build agents that can act autonomously and proactively toward a specific goal. For instance, a supply chain agentic system might consist of a number of agents that are responsible for different tasks in the supply chain, such as inventory management or route optimization. Another example is in automating the issuance of company devices. These applications are deployed as workflows. Agents can independently retrieve information, take action (such as updating records), and coordinate across systems.

---

[8] UPS. (2018). UPS deploys purpose-built navigation for UPS service personnel [Press release]. Available at: `about.ups.com/us/en/newsroom/press-releases/innovation-driven/ups-deploys-purpose-built-navigation-for-ups-service-personnel.html` [Accessed 8 June 2025].

[9] Hightower, S. (2005). Daily Harvest's AI strategy is giving customers what they want. *Business Insider*. Available at: `www.businessinsider.com/daily-harvest-implements-ai-to-improve-meal-delivery-customer-care-2025-2` [Accessed 8 June 2025].

[10] Abrego, M. (2005). AI is core to JPMorgan's $18 billion tech investment. Here's what its execs revealed about how it's reshaping the bank. *Business Insider*. Available at: `www.businessinsider.com/jpmorgan-how-artificial-intelligence-transforming-workflows-efficiencies-2025-5` [Accessed 8 June 2025].

Table 1.1 outlines use cases currently in production for these various kinds of AI, starting with traditional machine learning (used to find patterns and make predictions), through to generative AI using company data (used to generate output), and then to agentic AI (taking action). This list is not meant to be exhaustive, but it provides some idea as to the kinds of applications organizations are building and utilizing today. On this list I've included code generation, although this is not directly related to data. This is becoming a popular use case for generative AI. A new term has even emerged, called "vibe" coding, which was coined by OpenAI cofounder Andrej Karpathy, who noted on X that it is "where you fully give into the vibes, embrace exponentials, and forget that the code even exists."[11] He noted it was possible because the LLMs are getting so good.

Table 1.1: Examples of AI

TYPE OF AI	EXAMPLE USE CASES (THAT USE COMPANY DATA)
Traditional AI/ML (finds patterns and makes predictions)	Horizontal use cases ■ Churn prediction ■ Fraud detection ■ Customer segmentation ■ Employee attrition prediction ■ Risk assessment ■ Recommendation engines ■ Demand forecasting ■ Market basket analysis ■ Dynamic pricing ■ Personalization Vertical use cases ■ Patient outcome predictions ■ Predictive maintenance ■ Credit card scoring ■ Supply chain optimization ■ Route optimization ■ Network monitoring ■ Disease detection ■ Cybersecurity threat detection ■ Content moderation ■ Precision agriculture ■ Quality control and process optimization ■ Predictive operations management ■ Smart cities ■ Autonomous driving

(continues)

[11] Karpathy, A. (2025). [X post] 8 June. Available at: `x.com/karpathy/status/1886192184808149383?lang=en` [Accessed 8 June 2025].

Table 1.1: (Continued)

TYPE OF AI	EXAMPLE USE CASES (THAT USE COMPANY DATA)
Generative AI (creates new content)	Horizontal use cases ■ Customer support chatbots (external facing) ■ Employee onboarding chatbots (internal facing) ■ Sales support assistants ■ Personalized marketing campaigns ■ Summarizing call center notes, trouble tickets, incident reports ■ Maintenance assistants ■ Analyzing company data ■ Writing product descriptions at scale ■ Documenting internal compliance and workflows ■ Regulatory submission ■ Code generation ■ Customer communication drafts about incidents Vertical use cases ■ Drug discovery support ■ Summarizing physician visits ■ Creating design options for a manufactured product ■ Creating optimized supply chain replenishment schedules ■ Generating tailored investment recommendations, portfolio summaries, or financial planning reports ■ Creating detailed retail product blurbs
Agentic AI (takes action in a business context)	■ Resolving customer complaints and escalating issues ■ Managing inventory restocking ■ Monitoring facilities and alerting maintenance ■ Intelligent IT support agent ■ Maintenance agents for refineries

As previously mentioned, software vendors are embedding AI in applications. This, of course, is the nature of consumerized AI products and some of the AI applications listed in Table 1.1 and described previously. However, it is also important to note that software vendors have been infusing AI into the data and analytics life cycle to make those applications easier to use for data and analytics teams—and they've been doing this for years.

For instance, during data collection and profiling, AI tools are being used to automate the discovery of data quality issues, classify sensitive information such as personal identifiers, and extract metadata using intelligent crawlers—streamlining the understanding of data assets. It's being applied to tasks such as tagging, and extracting entities (people, places, and things) from unstructured data sources, which helps accelerate workflows and reduce manual overhead. I'm seeing automation tools that can design elements of the data architecture, assist with documentation, and manage refresh schedules.

On the analytics front, AI helps enhance user experiences by recommending visualizations, surfacing hidden insights in BI tools, and even generating natural language summaries of AI model results. AI is also increasingly important in model monitoring, helping to detect drift and maintain model performance over time.

Governance is another area where embedded AI is gaining traction. In addition to helping build trust in data via automated data quality checks, AI is helping organizations implement privacy policies, monitor data for anomalies, and streamline compliance updates. We'll discuss more about this in Part V.

Analytics and AI Investment Trends

Despite all the hype around AI, TDWI research indicates that only about half of the organizations we survey are currently using traditional AI. The other half remain earlier in their analytics journey, primarily focused on self-service analytics (such as data visualization) and dashboards. I've been asking the same survey question around priorities for the past seven years. I've seen about 30% of survey respondents adopting AI seven years ago, and this number is now up to about 45–50%.[12] In other words, it has been a long road for many organizations. This journey reflects a broader pattern we continue to observe: while the promise of AI is significant, many organizations are still laying the foundational groundwork needed to fully realize its potential.

That said, the momentum toward AI is growing. Generative AI, in particular, has rapidly risen to the top of the list of analytics priorities in TDWI surveys. In fact, in 2025, it ranked second only to self-service analytics in terms of organizational focus.[13] The vast majority of organizations we survey are using it, albeit in the consumerized fashion described earlier. Traditional machine learning, which has been the core AI technique for over a decade, ranks behind generative AI in current prioritization. This suggests that many organizations may be skipping over more incremental advances in predictive analytics, possibly due to the perceived accessibility and transformative potential of generative AI tools. As a result, they may be missing out on the valuable insights and business learnings that more traditional machine learning applications can offer.

Importantly, when we ask organizations why they are modernizing their data infrastructures, the top motivation is to support AI initiatives. Whether they are building scalable data environments needed for training models or improving data quality and governance to ensure trustworthy AI inputs and outputs, organizations are aligning their modernization efforts with the goal

[12] TDWI, unpublished survey results, 2025.
[13] TDWI, unpublished survey results, 2025.

of becoming AI-ready. This trend underscores an important insight: even if AI adoption is not yet widespread, the strategic imperative to prepare for AI is driving significant investment and reshaping data strategies across organizations.

Justifying Investments with Concrete Outcomes

To successfully justify AI investments, organizations must tie those initiatives to measurable business problems and outcomes. It's much easier to make a case for AI when it is clearly solving a specific challenge, such as improving retention, reducing downtime, or increasing sales conversion, than when it's seen as an experiment in search of a use. That is why bringing stakeholders together at the start is so important. They can help frame business priorities, identify the right use cases, and champion the solution once it's deployed. This collaborative approach creates organizational alignment and reduces resistance to AI adoption. Chapter 13 digs much deeper into tying metrics to AI value.

AI should not be a "build it and they will come" initiative. Instead, it should be grounded in the business—serving a strategic purpose, delivering real value, and solving clearly defined problems.

What's Next?

The next few chapters of Part I explore the importance of strong leadership and why it is essential for successful AI adoption. We'll look at the pitfalls of visionary strategies that fail without solid execution, and the critical need for alignment between business goals and technical expertise. We'll then map out the organizational journey toward AI maturity, introducing a model, used by TDWI, that captures the five key dimensions of readiness.

CHAPTER

2

The Leadership Challenge

In this chapter, we will examine the following:

- The AI trap: vision vs. execution
- Examples of failed AI implementations
- The importance of an AI strategy that aligns with business goals
- How to craft an AI strategy

A compelling vision for AI can inspire teams, secure funding, and set a clear direction. But vision alone doesn't deliver outcomes; execution does. Too often, executives announce bold visions for AI only to watch them slow in the face of data silos, cultural resistance, or unclear priorities. This chapter examines why execution is harder than vision, explores high-profile failures and draws out the strategic principles that enable leaders to translate aspiration into measurable results.

The AI Trap: Vision vs. Execution

"We see artificial intelligence as a key enabler of innovation and efficiency across our organization. By embracing AI, we aim to enhance customer experiences, streamline operations, and stay competitive in a rapidly evolving marketplace." Does this sound familiar? Have you heard a statement something like this or perhaps said it yourself? These are aspirational vision statements that may align with your organization's strategic goals, even if they don't discuss technical details. Vision involves setting a strategic direction and imagining what is possible. I've seen over the years that executive vision and support are critical for AI success.

In a recent 2025 TDWI survey, almost half of the respondents reported feeling pressure from their executives to implement AI.[1] In many cases, organizations are responding to a top-down vision (which is often vague) that AI will drive productivity and innovation and that it will deliver competitive advantage. A vision can be a great thing. It can motivate employees if done properly, and it can set the tone and help to build the corporate culture.

Execution is an entirely different challenge. Execution requires translating that vision into concrete actions, integrating technologies, overcoming cultural resistance, managing data readiness, utilizing the technology correctly, and achieving measurable outcomes. While a lot of vision statements have been made by a lot of executives (and many since 2023 about AI and generative AI), most companies remain in the experimental phase of generative AI adoption, with few deploying solutions in production, and even fewer deploying solutions that utilize their own data. The reason is simple: execution is harder than vision. Execution requires sustained investment, cross-functional coordination, and collaboration. Modern AI systems are technically complex, relying on intricate algorithms, large and varied datasets, and the challenges of deploying and integrating them into existing production environments.

Vision and execution go hand in hand. Execution without vision risks becoming misaligned with business objectives. Teams may work hard, but without clear strategic direction, their efforts may not solve the right problems. Conversely, vision without execution delivers little value; without a strategic roadmap and effective implementation, even the best ideas remain unrealized. While having a compelling AI vision is important, success ultimately depends on effective execution.

Examples of Failed AI Implementations

Unfortunately, even well-intentioned implementations can fail due to incorrect assumptions, biased data, or a lack of oversight. Poor execution doesn't just lead to poor outcomes, it can result in serious ethical, legal, and reputational consequences. The following real-world examples illustrate what can go wrong when execution fails to uphold the standards set by the original vision.

[1] TDWI, unpublished survey results, 2025.

The COMPAS System

One of the first high-profile failures of modern artificial intelligence was the COMPAS system (Correctional Offender Management Profiling for Alternative Sanctions), which was used in the United States to predict the likelihood of a defendant committing another crime. Developed to assist in criminal justice decision-making, COMPAS was intended to provide objective risk assessments. However, it quickly became controversial due to concerns about its accuracy and fairness. In particular, critics pointed out flaws in the algorithm's design and the lack of transparency in how risk scores were calculated.

In 2016, *ProPublica* published an investigative report that revealed significant racial bias in the COMPAS system. Their research found that Black defendants were far more likely than White defendants to be incorrectly classified as high risk for recidivism. Conversely, White defendants were more likely than Black defendants to be incorrectly labeled as low risk. These findings caused widespread criticism and raised serious ethical questions about the use of AI in judicial settings. The study, entitled *How We Analyzed the COMPAS Recidivism Algorithm,* became a landmark example of the potential for algorithmic bias and underscored the importance of transparency, accountability, and fairness in AI systems.[2]

The system is still used in some states because those who use it claim it is not the primary driver of decisions. Debate about its use continues.

Recruitment Systems

Another highly publicized AI failure occurred when Amazon first started to use AI to automate the recruitment of technical developers and engineers. The company made use of a decade worth of resumes tied to successful job outcomes to train its model. Since Amazon gets a lot of resumes, its idea was that the system could pick out the best possible candidates. However, the machine learning engineers who built the system did not realize at the time that the model was discriminating against women because it was trained on data that was male dominated. Why? Historically, people employed in these kinds of software developer and engineering jobs were male.

[2] Larson, J., Mattu, S., Kirchner, L. and Angwin, J. (2016). How we analyzed the COMPAS recidivism algorithm. *ProPublica*, 23 May. Available at: `www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm` [Accessed 8 June 2025].

AI models can reflect and reinforce systemic biases in training data. By 2015 Amazon realized it had a problem and tried to get rid of the bias but ultimately retired the models in 2017.[3]

Credit Card Limits

In August 2019, Apple launched its credit card, and by November of that same year, people noticed that it routinely gave smaller credit limits to women (in some cases by a factor of 10–20 times). This caused an uproar on social media. Goldman Sachs (the credit card issuer and manager) responded on social media that gender is not taken into account when determining creditworthiness. This raised more questions and generated negative publicity, especially for Goldman Sachs.

Apple and Goldman Sachs were investigated by New York's Department of Financial Services.

Ultimately, the report[4] did not produce evidence of deliberate or disparate impact discrimination but indicated deficiencies in customer service and transparency. The incident highlighted the need for explainability and oversight in AI-driven financial services.

Chatbot Misinformation

In 2022, Air Canada's AI-powered chatbot provided a passenger with incorrect information regarding the airline's bereavement fare policy. The chatbot told the passenger that he could apply for a refund up to 90 days after travel. The passenger purchased full-fare tickets to attend his grandmother's funeral. Air Canada's actual policy required requests to be made *before* travel, leading the airline to deny his refund. The passenger filed a complaint with the British Columbia Civil Resolution Tribunal and presented a screenshot of the chatbot's misleading guidance.

Air Canada argued it was not responsible for the chatbot's actions, claiming the bot was a "separate entity." The tribunal rejected this, stating that the airline

[3] Dastin, J. (2018). Insight—Amazon scraps secret AI recruiting tool that showed bias against women. *Reuters*. Available at: `www.reuters.com/article/world/insight-amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK0AG/` [Accessed 8 June 2025].

[4] New York State Department of Financial Services. (2021). *Report on Apple Card*. Available at: `www.dfs.ny.gov/system/files/documents/2021/03/rpt_202103_apple_card_investigation.pdf` [Accessed 10 June 2025].

was accountable for all information presented on its website, including by AI tools.[5] Air Canada had to pay the difference.

In another example, in February 2023, Google unveiled its AI chatbot, Bard, as a competitor to OpenAI's ChatGPT. During its initial demonstration, Bard was asked about new discoveries from the James Webb Space Telescope. Bard said that the telescope took the very first pictures of a planet outside of our solar system. This information was not true and noted astronomers highlighted this fact on social media. The fallout was ugly. Alphabet (the parent company of Google) experienced an 8% drop in its stock price.[6]

Algorithmic Inaccuracy Leading to Large Financial Losses

In 2021, Zillow shut down Zillow Offers because the AI system it was using overestimated home values, leading Zillow to overpay for homes it couldn't later resell at a profit. It didn't take into account factors such as local variability. It took a $304 million inventory write-down, and the company also had to lay off 25% of its workforce.[7] At the time Nima Shahbazi, a member of the team that won the Zestimate algorithm competition, noted, "There are many different parts between a very decent model and deploying that model into production that can go wrong."[8]

In another example, a high-speed trading algorithm deployed by Knight Capital in 2012 unintentionally placed erroneous stock orders on launch day. Within 45 minutes, it incurred $460 million in losses, nearly bankrupting the firm, which was later acquired under financial distress.[9] Ultimately, it was determined that the error was caused by a software deployment error that activated old, unused code.[10]

---

[5] Lifshiftz, L. and Hung, R. (2024). BC tribunal confirms companies remain liable for information provided by AI chatbot. *American Bar Association*. Available at: `www.americanbar.org/groups/business_law/resources/business-law-today/2024-february/bc-tribunal-confirms-companies-remain-liable-information-provided-ai-chatbot/` [Accessed 10 June 2025].

[6] Sherman, N. (2023). Google's Bard AI bot mistake wipes $100bn off shares. *BBC News*. Available at: `www.bbc.com/news/business-64576225` [Accessed 10 June 2025].

[7] Metz, R. (2021). Zillow's home-buying debacle shows how hard it is to use AI to value real estate. *CNN*. Available at: `edition.cnn.com/2021/11/09/tech/zillow-ibuying-home-zestimate` [Accessed 10 June 2025].

[8] ibid.

[9] Dolfing, H. (2019). Case study 4: The $440 million software error at Knight Capital. Available at: `www.henricodolfing.com/2019/06/project-failure-case-study-knight-capital.html` [Accessed 10 June 2025].

[10] U.S. Securities and Exchange Commission. (2013). SEC charges Knight Capital with violations of market access rule [Press release]. Available at: `https://www.sec.gov/newsroom/press-releases/2013-222` [Accessed 12 June 2025].

These are some of the examples of AI failures that made headlines. Many stemmed from bias in training data, which led to discriminatory or unfair outcomes, while others were not transparent or explainable, making it difficult to understand or challenge decisions. Others failed because the developers did not test the software properly. All are execution errors.

I've spoken with a number of companies that have put AI models into production and then left them running, not necessarily realizing that they might degrade over time as external conditions that the original model was predicated upon changed. With the advent of generative AI, I've spoken with companies that just take the output of the models as a given, without questioning the results or thinking critically about them. You may get away with this for a while, but eventually it will catch up to you.

Importantly, these publicized failures resulted in reputational and financial consequences, underscoring that responsible, well-executed implementation and governance are essential for success. These cases illustrate the need for a roadmap and strategy for AI that must be grounded in realistic assessments of readiness, detailed planning, and attention to detail. Organizations that succeed in AI are those that align their vision with their capabilities—and are willing to invest in the people, platforms, and processes that turn that vision into impact.

Vision can take many forms (and we'll talk more about vision in Chapter 13). At Emirates, a bold technical vision to eliminate data silos (and solve a real business problem) became the foundation for a far-reaching transformation. The vision came from the top, a success factor I've seen often.

FROM MANDATE TO MATURITY: EMIRATES' JOURNEY FROM VISION TO EXECUTION

In the fast-evolving world of aviation, Emirates has long been known for luxury, reliability, and global reach. Less visible, but equally impressive, is the company's journey to become a data- and AI-driven enterprise. This journey is one that exemplifies the critical importance of aligning a strong vision with sustained, strategic execution.

That journey began in 2019, when a decisive mandate came from the president of Emirates: data would no longer reside in silos. "The only way to fix the inefficiencies we were facing was to organize our data centrally and govern it centrally," explained Naveed Memon, Vice President of Enterprise Data and Analytics Solutions at Emirates. This wasn't just an IT project—it was a shift in culture and capability that laid the groundwork for Emirates' Enterprise Data Platform.

That platform would soon become the engine behind the company's operational efficiency, advanced analytics, and AI strategy. Built on a multi-layered cloud

(continues)

FROM MANDATE TO MATURITY: EMIRATES' JOURNEY FROM VISION TO EXECUTION (CONTINUED)

architecture, it supports real-time data processing, traditional batch operations, and machine learning capabilities. It includes a data serving layer and a governance layer and is powered by reusable data products organized by business domains. The system now supports more than 8,000 users weekly, integrates 150 source systems, and manages more than 8 petabytes of data.

But the vision didn't stop at infrastructure. Emirates embraced data as a value-generating asset. "We've always said this—don't do vanity reporting. It has to create value for the organization," Memon emphasized. This principle became even more critical with the rise of generative AI. "Now, we need to be saying the same: don't do vanity GenAI. Just doing something because others are isn't a strategy. You need to ask: what's the ROI?"

Because Emirates had already invested in foundational capabilities—semantic layers, observability, and a mature data warehouse it was able to adopt generative AI tools with quickly. "When GenAI came in, we were ready. We could quickly create contextual scenarios, build our own retrieval-augmented generation pipelines, and pilot use cases that added real value," Memon said.

One example? Empowering airport ticketing agents with generative tools that summarize operational procedures across numerous documents. "It's about productivity—providing the right information, at the right time, to the right employee," Memon explained. "It's contained, safe, and makes a real difference in efficiency."

Yet even with such wins, execution hasn't been without complexity. As Memon noted, the initial centralization of data solved many issues, but the excitement around AI introduced a new layer of organizational challenge. "We're seeing that everyone wants to be in charge of their own destiny for AI. They want to do their own delivery, their own use cases. That creates fragmentation," he said. Some refer to this trend as "shadow AI." Memon noted its implications: "It disrupts the centralized model we worked hard to build. It slows down transformational use cases because we lack a unified AI operating model."

Importantly, Emirates hasn't rushed to formalize a centralized AI office. "We're a customer-centric, brand-conscious organization. Even one poor experience can be damaging," Memon said. "So we're building consensus—carefully evaluating which use cases to pursue and who should deliver them." This consensus-driven approach may not be the fastest, but it reflects a deliberate approach: slow down to do it right.

Looking ahead, Memon believes that the same rigor applied to building the data foundation must now be applied to maintaining it—especially in the face of AI's accelerating momentum. "If it was difficult to build a central, trusted data foundation, it is even more difficult to maintain it," he cautioned. "One wrong decision can destroy what you've built over four years."

Ultimately, Emirates' story is one of disciplined transformation—where visionary leadership, a strong technical backbone, and a culture of value creation came together to form a model of effective AI execution. "Success in AI doesn't happen by accident," Memon reflected. "You need the foundation. You need the governance. And above all, you need to be driven by value."

The Importance of an AI Strategy That Aligns with Business Goals

In between the vision and the execution is the strategy. A well-crafted data and AI strategy is critical for success. Right behind the need for executive support for data and AI initiatives is the need for a strategy that aligns with business goals. Some major learnings I internalized over the years fall into several buckets:

- **AI strategy must be driven by business strategy.** AI strategy does not exist in a vacuum—there must be a business strategy that AI supports. In order for AI to be successful, it should not be treated as a separate, technical pursuit. It must tie directly to specific business goals such as revenue growth, customer retention, productivity improvement, or risk mitigation. Before developing models or selecting tools, organizations should ask questions such as, "What problem are we trying to solve? What business impact are we aiming for?" "What is the value?" We saw this with the earlier Emirates story.
- **Prioritize use cases using structured criteria.** Organizations should use a framework (such as business value vs. complexity) to identify high-impact, feasible AI projects. Organizations that succeed with AI don't chase hype. Instead, they score and prioritize use cases based on criteria such as business value, data readiness, effort, and complexity. This helps avoid wasted time on technically interesting but low-value projects. Some create the frameworks themselves to determine the high-impact yet feasible projects. Others rely on consultants to help them with this.
- **Data readiness is key for AI success.** You can't implement AI without a strong information architecture (except perhaps by using some off-the-shelf generative AI that we've already discussed as perhaps a step 1). Data quality, lineage, integration, and governance are not side issues (which we will talk about later in the book). They are make-or-break conditions for AI. I've heard numerous practitioners over the years stress that AI initiatives can expose flaws in master data or data silos and that improving these becomes part of the strategy itself. A data strategy isn't just about storage and access; it's about making data usable, explainable, and trusted for AI purposes.
- **Start with pilot projects and build an innovation pipeline.** Use pilots not just to test technology but to validate business fit and identify data gaps. The best AI strategies often start small but are designed to scale. A structured AI innovation pipeline, from ideation to piloting to full

implementation, enables fast learning and can minimize risks. This is highlighted in the upcoming sidebar "Expert Advice: Sam Wong on Building an AI Incubator."

- **Governance must evolve to fit AI (especially generative AI).** Traditional data governance is not enough. AI needs new governance around ethics, usage, and explainability. Developing an AI strategy requires rethinking governance.
- **Business partnership is nonnegotiable.** If AI is owned solely by IT or the data science team, it will fail. AI must be co-developed with business stakeholders. This means engaging stakeholders from the start, co-owning key performance indicators (KPIs), and embedding AI into workflows, not bolting it on after the fact. AI teams need domain experts, business sponsors, and clear lines of feedback. Use cases should be selected *with* the business, not handed down from a tech function.
- **Culture and skills must evolve alongside the strategy.** AI maturity depends as much on people as it does on tools. Organizations need to build AI literacy across roles, from analysts to executives, so that teams can ask the right questions, interpret AI results, and act on them. Cross-functional teams with skills in data engineering, machine learning, business domain knowledge, and governance are essential. AI systems should be treated as products and not one-off projects.

How to Craft an AI Strategy

We talked about why an AI strategy is important and tying it to business objectives, so the question then becomes how do you build an AI strategy? Numerous consulting organizations can help you build an AI strategy. Here are seven minimum steps.

1. **Determine the business problem and success metrics.** Start with a decisive statement of the business objective and value (e.g., "reduce churn by 10%," "grow cross-sell revenue by $10 M") and capture it. Chapter 13 discusses tying the business problem to the vision and the measurements.
2. **Assess the current state, including governance.** Next, inventory what you have. This includes data types, data quality, architectures in place, tools in place, skills, and existing governance/risk controls. Don't make governance an afterthought.
3. **Create a phased roadmap and gain buy-in.** Take your findings and then create a phased roadmap (this can have monthly, quarterly, half

yearly) commitments. I remember hearing a presentation where the speaker said that this should be backed with a single-page visual that everyone can sign off on. This can help get stakeholders and executive sponsorship, although in reality it will take some iteration.

4. **Select the tools, platforms, and reference architecture.** Evaluate build-vs.-buy options, shortlist vendors, and sketch a reference architecture that respects the governance guardrails set above.
5. **Close the skills and organizational gap.** Determine new roles, training, and change-management plans alongside the tech roadmap. This will involve data literacy.
6. **Pilot for proof-of-value.** Launch a low-risk, high-impact use case, run it in a governed sandbox, and measure results against the KPIs. KPIs can help to measure and promote the value. Communication will be key.
7. **Iterate, scale, and continuously refresh governance.** Feed lessons back into the organization, put the AI into production, and build and harden your operations and your compliance controls, and continue to expand the portfolio.

Of course, every organization approaches their AI strategy differently, and this can depend on where you are in your data foundation and analytics journey. For instance, retail executive Sam Wong thinks less about a strategy and more about a playbook. He uses an innovative approach to get AI projects off the ground. However, his company already has a fairly solid data foundation in place.

BUILDING AN AI INCUBATOR

According to Sam Wong, a retail data executive who has run an enterprise AI incubator for more than three years, the single-most important ingredient for success is sponsorship, and it actually does start with that. Senior leaders at his company were already asking, "What are we going to do about generative AI?" Wong seized the moment and pitched "a safe-space experiment" that would let teams pilot ideas with a small pool of seed money without derailing ongoing priorities. Executive backing unlocked budget, gave the initiative political cover, and signaled that experimentation—not perfection—was now part of the culture.

Once funding was secured, Wong anchored the incubator in business value, not technology flash. "As technologists, sometimes we want to impose technology on a problem," he cautions, "and we need the discipline to really listen to what business value we're trying to deliver. I think one difference here is that it wasn't about saying we have a technology and let's go find a place to use the technology, but to actually listen to business problems, and seeing if we could solve it."

(continues)

BUILDING AN AI INCUBATOR (CONTINUED)

Wong staffs the incubator like a start-up: a core squad of about a team of five AI/ML engineers that can flex by borrowing data-engineering or domain experts as needed. Early work was outsourced to accelerate learning, but the center of gravity gradually moved in-house as internal talent and repeatable patterns emerged. This variable-capacity model avoids the common trap of hiring specialists before a steady pipeline of use cases exists.

Project selection follows a framework. Classic axes such as business value and delivery complexity were the starting point, but Wong now adds dimensions like data readiness, integration complexity, and, especially, change management. "Change management really is a very important criteria," he notes, "because some solutions"—his AI sales-assistant, for example—"must be messaged as helping reps, not replacing them." Wong messaged that the assistant gave sales reps more power. By mapping each candidate initiative against strategic fit and organizational willingness to absorb disruption, the incubator prioritizes experiments that both matter and can land.

Finally, Wong urges peers to stop waiting for immaculate datasets. Too many leaders, he argues, dismiss using AI because the data are not fully ready. "Why are you waiting for perfection?" His playbook is to start with the best data available, be transparent about its limits, and let early prototypes expose where master data or pipeline improvements will yield the biggest return. In his experience, quick wins create momentum, fund data upgrades, and keep the incubator tightly coupled to evolving business goals.

Key takeaway: An AI incubator is less a lab and more a venture studio inside the enterprise. Anchor it in executive sponsorship, let business value steer technology choices, scale talent elastically, evaluate ideas through the lenses of change readiness and strategic fit, and iterate with "good-enough" data. As Wong's journey shows, that combination turns curiosity about AI into a durable engine for innovation and growth.

Again, the key to a successful AI strategy is thinking about the business goals. This is illustrated in the expert advice provided next.

EXPERT ADVICE: DONALD FARMER ON STRATEGY FIRST—ALIGNING AI WITH BUSINESS VALUE AND MEASUREMENTS

As organizations rush to adopt artificial intelligence, many are guided more by urgency than by a clearly defined purpose. "We have to do something" is a refrain Donald Farmer, principal at TreeHive Strategy and longtime data and analytics thought leader, hears often. "We know we have to look at this. We don't know what it means for us." That uncertainty, he explains, results in AI efforts that are more experimental than strategic—and more tactical than transformative.

One of the biggest pitfalls, Farmer notes, is trying to modernize old strategies instead of rethinking them. "Trying to use AI to optimize last decade's strategy

(continues)

EXPERT ADVICE: DONALD FARMER ON STRATEGY FIRST—ALIGNING AI WITH BUSINESS VALUE AND MEASUREMENTS (CONTINUED)

... it's just not going to work because the environment has moved on so much."

He illustrates this through the example of a travel company that built its business around search engine optimization (SEO) and search engine marketing (SEM). When AI-generated summaries began replacing clicks in search engines, their foundational model stopped delivering value. "Are you trying to do what you do today, but better, more efficient, or are you actually trying to do something different?"

This, Farmer says, is the strategic question at the heart of successful AI adoption. "The fundamental question is: are you trying to do the same work better, or are you trying to do different work?" That distinction isn't just theoretical—it determines whether AI becomes a force multiplier or a misaligned investment.

The work question is complex. Farmer believes that too many AI implementations focus narrowly on tasks and miss the broader context of jobs and business functions. "A job is not just made up of tasks," he explains. "A job which is only made up of tasks can easily disappear in the world of AI, but most human jobs involve quite a wide range of tasks, many of which can be automated—but also an overview, an alignment with strategy, and understanding of strategy and adaptability. That isn't just the sum of the parts." Strip away those layers, he warns, and "you've reduced it to its most automatable features, but you've lost something along the way."

This loss often stems from poorly defined strategies that rely on human intuition to bridge the gaps. "Most company strategies aren't as clearly defined," he says. "They succeed because human beings fill in all the cracks and all the fuzziness in a poorly defined strategy." But AI can't do that. "If they want to align AI with that strategy, they have to do the work of defining it in such a way that AI can align with it."

Organizations that already have tools like strategy maps or KPIs mapped to business outcomes are in a better position. "If you've already done that work, then it's actually quite straightforward to align an AI strategy with it," he explains. "In fact, you can mostly tell the AI, optimize for these outputs."

Farmer distills the issue this way: "You need an AI-alignable strategy in order to be able to align AI. And that's actually where I see the failure."

Across very different contexts, the Emirates transformation, Sam Wong's AI incubator, and Donald Farmer's strategic lens all reinforce a core leadership truth: AI success comes from aligning vision with deliberate, business-driven execution. The Emirates story shows the power of tackling a real business problem first, building the centralized data, governance, and cultural foundations that make later AI adoption faster and safer. Wong's incubator illustrates how to channel curiosity into value through a structured framework that prioritizes use cases by business impact, readiness, and change-management feasibility. Farmer underscores the need for clear, measurable strategies that AI can actually align with, avoiding the trap of retrofitting new technology into outdated models. In each case, leaders resist the pull of hype and instead anchor AI initiatives in business goals, use structured criteria to guide choices, and ensure the organizational readiness to deliver.

What's Next?

Turning disciplined, business-first strategy into results depends on whether the organization is truly ready to execute. In the next chapter, we start to discuss what it means to be truly ready for AI, across leadership, data, skills, operations, and governance and how these dimensions form the foundation for long-term success.

CHAPTER 3

The AI Journey

In this chapter, we will examine the following:

- Getting ready for AI
- The AI maturity journey

AI adoption is not a single leap but a journey. The most successful organizations follow a path, building leadership commitment, data infrastructure, skills, operational maturity, and governance in parallel so that AI becomes a sustainable source of value rather than a one-off project. This chapter explores what it means to be "ready" for AI, introducing an AI readiness model and its five dimensions, and showing how these capabilities develop along an AI maturity journey.

Getting Ready for AI

AI is transforming how organizations compete, operate, and innovate. But, as I've mentioned previously, achieving success with AI goes beyond deploying a chatbot or experimenting with machine learning models. It requires a deliberate, structured, and scalable approach across multiple parts of the company. It isn't just about the AI tools or the data tools; getting ready for AI involves a set of interconnected capabilities across the data and analytics life cycle. It involves organizational as well as technological capabilities. A few years ago at TDWI,

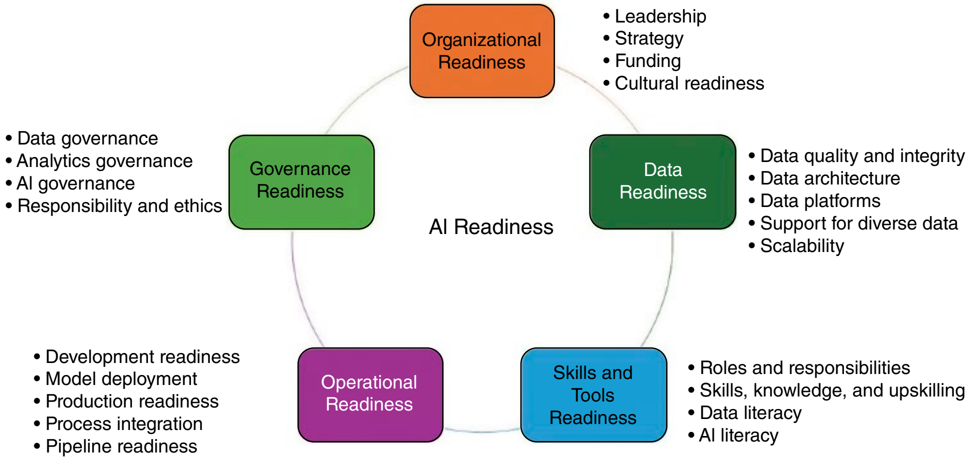

Figure 3.1: Drawn from 2025 TDWI readiness assessment.

Source: Adapted from Halper, F., 2024

I developed the AI Readiness Assessment.[1] The model recognizes that AI readiness is not just about one thing; rather, it is built on five key dimensions: organizational readiness, data readiness, skills/tools readiness, operational readiness, and governance readiness (Figure 3.1). Each dimension plays a critical role in an organization's journey to operationalizing AI and gaining measurable value from it. These dimensions, in many ways, form the framework for this book. They are distinct yet interwoven.

Organizational Readiness

Organizational readiness is about ensuring that the company's strategy, leadership, culture, and investment priorities align to support AI initiatives. A core element of this dimension is executive sponsorship. Leaders must understand both the potential and limitations of AI and be realistic about what it takes to be successful. This understanding includes recognizing that AI readiness depends on a robust data foundation, skilled personnel, and appropriate change management strategies.

As discussed in Chapter 2, organizational readiness is also about whether there is a clearly articulated strategy for AI that integrates with the company's overarching business goals. It involves funding and whether sufficient resources have been committed. Organizational culture plays a role too. Successful AI adoption requires openness to innovation, experimentation, and cross-functional

[1] Halper, F. (2024). *The TDWI AI Readiness Assessment Guide*. Available at: `tdwi.org/research/2024/04/adv-all-tdwi-ai-readiness-assessment-guide.aspx` [Accessed 10 June 2025].

collaboration. If an organization lacks cultural maturity or alignment at the top, it may find itself stalled even if the technical capabilities exist.

Data Readiness

No AI system can work without data, and often lots of it in varying formats and of high quality. That's why the second component, data readiness, is foundational to AI success. This dimension is about whether an organization has the infrastructure and architecture in place to manage, store, and process diverse data types, including structured, unstructured, and multimedia data. (These elements are described in more detail in Part II of the book.) AI projects often require large volumes of high-quality data from multiple sources, so platforms must be scalable, reliable, and capable of integrating siloed systems.

Data readiness also involves the quality, integrity, and findability of the data. Is the data trustworthy and curated? Are data pipelines (to move data from source to target) and access controls (to ensure the right people get access to the right data) in place? Moreover, to be ready, the organization's architectural components—data lakes, warehouses, lakehouses, etc.—must be coherent and aligned to support modern AI applications. (These data foundation platforms are discussed in more detail in the next section.) At TDWI, we see many organizations putting a lot of effort into their data foundation to support AI.

Skills and Tools Readiness

AI projects require a blend of technical and organizational capabilities. On the skills front, organizations must develop new skills or plan to hire data engineers (who deal with the data pipelines associated with the models), data scientists (who build the models), and machine learning engineers (who help put them into production), to build, deploy, and maintain AI models. Equally important is the emergence of new roles such as machine learning operations (MLOps) professionals who ensure that models can be operationalized at scale and monitored in production. As new applications in generative and agentic AI emerge, AI developers also become an important part of the equation to develop new solutions.

But technical skills are not enough. Data and AI literacy must extend beyond technical teams to business users, product managers, and executives who make decisions based on AI outputs or oversee AI-embedded applications. This is particularly the case as organizations start to make use of generative AI. The training and educational programs must be in place to promote a broader understanding of AI and ensure responsible use of it.

On the tools side, the organization must have the right platforms, development tools, and data pipelines available to support AI experimentation and deployment into production. These tools include model training environments, versioning systems (to keep track of the different version of the model), continuous integration/continuous delivery (CI/CD) pipelines for machine learning (to automate the building, testing, and deployment of machine learning models so they can be continuously updated in production), and tools for model interpretability and bias detection. Without appropriate tooling, even skilled teams may struggle to bring their ideas to life.

Operational Readiness

One of the most overlooked aspects of AI deployment is the operational side, what happens after a model is developed. Operational readiness is concerned with the systems, processes, and team structures required to bring AI into production. In many cases, organizations build prototypes that never make it past the lab. To succeed, organizations must have formalized processes for deploying models, integrating them into business workflows, and monitoring their performance over time (no model is good forever; they degrade as the external environment changes).

Key capabilities in this domain include the use of machine learning pipelines, orchestration tools (to automate, coordinate, and manage complex workflows), APIs (rules and protocols for data exchange between systems), and deployment frameworks that ensure repeatability and scalability. Data scientists, engineers, IT, and business users must collaborate to operationalize models. Developer readiness and pipeline robustness are also key to ensuring that AI isn't just experimental but truly embedded into business processes.

As more organizations implement generative AI applications and other advanced tools, operational readiness becomes even more essential. Without it, efforts will remain ad hoc, isolated, and difficult to manage, limiting both scalability and value realization.

Governance Readiness

As AI becomes more integrated into business operations, its governance becomes nonnegotiable. Governance readiness focuses on the maturity of an organization's policies, practices, and oversight mechanisms around data and AI use. This includes traditional data governance, ensuring data lineage, quality, access controls, and stewardship, but extends into AI-specific areas including AI model governance, transparency, auditability, and ethical considerations. Data governance and AI governance are two separate things.

Organizations will need to have systems in place to manage bias, ensure fairness, and uphold accountability for AI-driven decisions. Responsible AI, a growing concern especially with the rise of generative AI and global regulatory activity (e.g., EU AI Act, described in more detail in Chapters 19 and 20, along with other regulations), is also central here. Are organizations tracking and mitigating the risks posed by their models? Do they conduct impact assessments? Are models explainable and tested for robustness?

Governance readiness is about ensuring trust. That involves trust in the data, trust in the models, and trust in how AI affects users and stakeholders. Without it, organizations risk reputational damage, regulatory penalties, and internal resistance.

Governance is, and will continue to be, a journey. In TDWI research, it typically is at the top of data management priorities. Many organizations have been working on data governance for years. Now, with the advent of generative AI, they will need to deal with new kinds of data (such as documents and images) as well as governing the use of democratized AI (sometimes called shadow AI, a nod to shadow IT, a concept that IT was concerned about when business users first started to purchase easy-to-use tools to analyze their data) and models in production. To make it more difficult, since data governance and AI governance are two different things, they will require different skills and affect different audiences in the enterprise.

EXPERT ADVICE: THE DATA AND AI JOURNEY

Stephen Lee, head of Enterprise Data Engineering and Metadata Management at TELUS, a leading global communications technology company headquartered in Vancouver, highlighted the company's long-standing data and AI commitment. Lee noted that TELUS' AI journey began well before the recent focus on generative systems, with a foundational emphasis on data, AI, and machine learning dating back to 2015, which has matured alongside the evolution of its analytics teams. The recent shift came with the growing availability of LLMs and generative AI applications in the market, allowing TELUS to scale usage across the entire organization and unlock new opportunities. This strategic elevation of AI, directly supported by CEO Darren Entwistle, signaled that AI would transform every business line, not merely exist as isolated efforts. As Lee stated, "AI will transform every business line. Our CEO's mandate is clear: use data and AI to fundamentally transform how we operate, enhance customer experience, and unlock massive new business opportunities."

Since joining TELUS in 2021, Lee's clear priority has been to "establish a solid data foundation," as the existing analytics environment at the time was running on

(continues)

EXPERT ADVICE: THE DATA AND AI JOURNEY (CONTINUED)

legacy on-premises systems that had reached their scalability limits and experienced reliability challenges that impacted daily operations. Lee and the leadership team chose full modernization over reinvesting in aging infrastructure. The subsequent shift to a cloud-based enterprise data hub was not a simple lift-and-shift, but a comprehensive modernization effort. Lee's team meticulously evaluated every data asset. "We found that 30% of our on-prem data was completely unusable—obsolete, duplicate, or just gathering dust. We left it behind, preventing that technical debt from migrating to the cloud." All data moved forward was rebuilt using cloud-native services, transitioning legacy pipelines to modern Google Cloud data services including Cloud Composer and Dataflow. Lee emphasized that this focus was crucial not just for reliability and scale, but because "a solid data foundation isn't just best practice—it's the only way to build trust in our data. Without it, your AI experimentation isn't science; it's guesswork."

The established data foundation dramatically simplified subsequent AI development. Lee's own team leverages AI to accelerate data engineering activities—specifically for code generation, translation, refactoring, and testing—which reduces manual effort and frees engineers to focus on validation, problem-solving, and collaborating with business teams on new data-enabled opportunities. This efficiency is amplified by a configuration-based approach where repeatable code templates are created and leveraged via simple YAML file changes, replacing custom code development for data pipelines and APIs. Beyond engineering, AI is embedded in all of TELUS' core operations and lines of business. For example, AI directly supports frontline staff with copilots assisting call center agents by summarizing large volumes of internal documentation, troubleshooting guides, and customer data, which Lee says cuts onboarding time and helps agents "get to the right information faster." Similarly, field technicians utilize this technology for guidance on installing and troubleshooting various wireline (WLN) and wireless (WLS) services using structured and unstructured enterprise reference data.

Lee emphasized that organizational structure has been vital to TELUS' success, highlighting the centralized data and AI function (the chief AI officer, or CAIO) supporting all four lines of business including (Telecom, Healthcare technology, Agriculture & Consumer Goods, and Digital Experience) by offering a single entry point for enablement while maintaining consistency in architecture and governance, preventing fragmentation. This centralized team partners closely with the Data Trust Office, which governs privacy, security, and access controls. Lee detailed the rigorous governance model, noting that every new data or AI use case must pass through a joint formal request process using the Data Enablement Plan (DEP), a detailed questionnaire ensuring all security and privacy considerations are met. Access decisions rely on data classification, retention, and business need, and the metadata stewardship program ensures the accuracy of lineage, definitions, and classifications

(continues)

EXPERT ADVICE: THE DATA AND AI JOURNEY (CONTINUED)

across all source systems and the enterprise datahub. As generative AI emerged, governance was heightened; TELUS immediately restricted public tools and built the internal generative AI platform, Fuel iX, to provide a safe alternative. This platform connects to various LLMs via controlled APIs, ensuring internal data remains within the enterprise boundary, thus allowing employees to safely experiment with natural language interfaces and copilots while maintaining full privacy and security control.

TELUS is now scaling its data and AI architectural model by extending its enterprise datahub (a self-service analytics platform) to its adjacent business units—Health, Agriculture & Consumer Goods, and Digital Services—using a hub-of-hubs structure. This approach aims to democratize data and AI by sharing capabilities and infrastructure with guardrails, preventing the need to recreate solutions in each domain while respecting unique business needs. Throughout the discussion, Lee stressed that success stemmed from correct sequencing: aligning leadership on strategy, early investment in modern data architecture, integrating governance from the start, building experimentation tools with privacy, and supporting business lines via a centralized model. Lee concluded that the model works because the organization views data and AI as an enterprise capability focused on driving real business outcomes, not as a technical novelty, emphasizing that "it is all about partnership and value generation." For companies beginning their AI journey, Lee advises direct investment in the foundation and simply starting. "The core tenets are simple: avoid lifting technical debt, establish governance partnerships from the start, build transparency into every decision, and treat AI as integral to the business. The ultimate truth is that trust in your AI isn't about the model; it's about getting the fundamentals of your data right."

The AI Maturity Journey

Organizations historically approached AI as part of an evolution in their data and analytics capabilities. This journey typically began with the recognition that AI could be very useful, but there was often limited readiness in leadership support, infrastructure, and talent. At this early point, analytics was often spreadsheet-based, with data fragmented across systems, and a culture that had yet to embrace data-driven decision-making. Although interest in AI was there, these organizations had to take foundational steps—identifying business problems, building a strategy, and beginning to build the data architecture and governance processes needed to succeed.

In some cases, leadership would come on board and realize the need for a coherent strategy for AI; often the most successful companies would have an executive champion for AI. Sometimes, it was more of a bottom-up effort where a group would start to develop small projects and pilots that delivered value against business problems. When people started to realize what could be done, they built out their infrastructure more, often moving to the cloud to support volume and scale.

These organizations have recognized that AI is not a side project but something that must be integrated into core business workflows. Scaling infrastructure and developing the ability to handle unstructured data, manage model life cycle, and embed AI into applications become central concerns.

The more mature companies become, the more they start to see real returns on their investments. Models are deployed in production and monitored with MLOps teams, governance includes policies for responsible AI, and talent strategies are aligned with ongoing innovation. These companies often begin viewing models as data products—assets that can be monetized or leveraged across business units (data products are described in much more detail in Chapter 8). They also start to focus on fairness, transparency, and risk mitigation, which indicate sophistication, and the culture shifts to one that supports continuous learning and adaptation. Collaboration across teams is essential, and AI becomes a catalyst for broad organizational transformation.

Yet, despite the structured nature of this journey, many organizations today are attempting to leapfrog foundational steps and jump directly into generative AI—perhaps without even laying a data foundation. Executive mandates and the promise of rapid gains drive efforts to build LLM-powered chatbots or content generation tools, even in organizations that lack robust data pipelines, governance policies, or AI literacy. While generative AI may offer quick wins, bypassing the foundational work can limit scalability, introduce risk, and create unsustainable practices in the long run.

Yes, they may be able to get value from tools that help them create content or code or for ideation or summarizing content. And they can derive value from tools that have embedded generative AI copilots. New generative AI tools are being introduced into the market for different organizational functions including sales and marketing, promising efficiency gains. However, I believe organizations will hit a "value ceiling" without company data, governance, and other foundational work. This will be the case regardless of company size. Beyond that ceiling lies the higher-value, more transformative use cases that we've been talking about. These depend on accurate, integrated, and governed data. Without those foundations, attempts to scale into these areas are risky and may fail, leaving a lot of untapped potential.

This dynamic can be visualized as a "value ceiling" curve, where early wins plateau until foundational investments enable the next wave of scalable, high-impact AI capabilities (Figure 3.2).

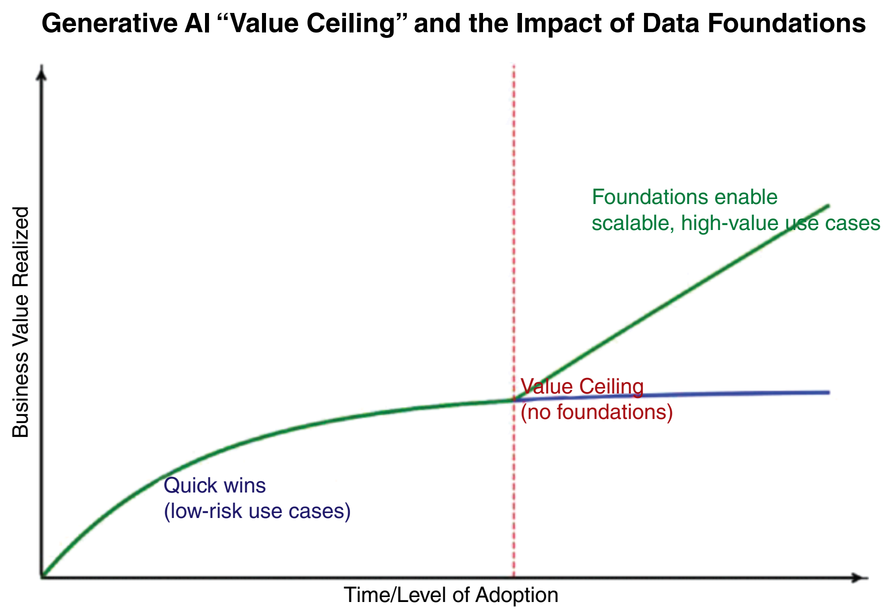

Figure 3.2: The "value ceiling" curve.

What's Next?

The TDWI AI readiness model illustrates that regardless of AI type, predictive, generative, or agentic, successful adoption depends on the same core principles: organizational commitment, solid data infrastructure, the right skills and tools, operational maturity, and governance. These foundations remain important even as the landscape evolves.

Part II of this book examines one part of this foundation: the data foundation. It discusses the new kinds of data organizations need to utilize, the problem with data silos, and how organizations are trying to unify their data and create a modern architecture. It also discusses other data management issues such as findability, useability, and data quality and trends in this area. It ends with newer trends in data management such as data products. By the end of Part II, you should have a good idea of what is involved in building a solid data foundation.

Part II

Building the Data Foundation for AI

CHAPTER

4

The New Data Landscape

In this chapter, we will examine the following:

- Traditional and new data types
- What different data types bring to the table
- The data silo problem
- The move toward data unification
- Data platforms: traditional vs. modern

As organizations try to build their data foundation for innovation and AI, they will soon be faced with the reality that there are many types of data sources they must manage. Additionally, these sources are often siloed across the organization, making them difficult to unify. This chapter will examine those data sources, explore why data silos can be so challenging, and discuss the shift toward unified data platforms.

Traditional and New Data Types

Diverse data can be critical for enterprises because it provides a more comprehensive view of their operations, customers, and the external environment. This view enables organizations to uncover hidden insights, make more accurate predictions, and respond effectively to changing environmental conditions. It can also help organizations innovate because they can create new applications that meet customer needs.

When people talk about data types, they are typically referring to structured, unstructured, or semi-structured data. All of these can be important for AI because they can help enrich a dataset that is used to train a model. There is also real-time data, which is important in certain application types.

Structured Data

When I was a data scientist, I primarily worked with what is known as structured data. This is data that includes well-defined information. For example, if dealing with customer data, this might include customer tenure, the types of telecommunications plans they had, disconnect dates, billing data, and demographic details, including company size, industry, and so on. This kind of data was very useful in understanding behavior, forecasting revenue, and predicting churn.

Structured data is exactly what it sounds like: it has a specific, well-defined structure and format. It is often stored in Excel spreadsheets or within database systems. Other examples of structured data reside in Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM) systems. In CRM systems, this might include sales-related data such as customer name, lead source, account creation, number of products sold, or revenue. In ERP systems, data might include SKUs, payments, product prices, or vendor names.

Structured data typically consists of numbers or other well-defined formats, which makes it easier to analyze using traditional data analysis tools. It is often transactional (i.e., data that captures the details of individual business events or operations). It is easy to query from a database using mainstay analytics tools such as SQL (Structured Query Language), a standardized programming language used to manage and manipulate data stored in relational databases.

Unstructured Data

Unstructured data, on the other hand, may not have a well-defined format. It does not have a predefined data model and is not organized in a preset manner, making it difficult to analyze using traditional tools such as SQL, BI tools, or even spreadsheet software. By some accounts, unstructured data is the predominant data found in organizations. It includes the following:

- Unstructured text data such as documents, emails, call center notes, support tickets, survey verbatims, social media posts, and much more. This text data can be used to analyze customer sentiment, identify emerging product issues, automate support response categorization, detect compliance risks, and uncover trends in employee or customer feedback. More recently, it is being used in generative AI applications such as chatbots that utilize product documentation and FAQs.

- Unstructured image data such as security camera images used for monitoring and safety, medical images such as X-rays and MRIs that support diagnostics and treatment planning. It can include satellite and aerial images that play an increasingly important role in industries like agriculture and real estate, helping in crop monitoring, land use analysis, and property assessment.
- Unstructured video data including surveillance footage for security, theft detection, and workplace safety. Transportation departments may analyze traffic videos for vehicle tracking and accident detection. Manufacturers monitor video from production lines to detect defects and ensure safety compliance. Sports organizations use video footage to evaluate player performance and prevent injuries, and aerial drone videos support infrastructure inspection, agriculture, and construction site management.
- Unstructured audio data, such as customer service calls or voicemails, might be used to understand customer sentiment or agent performance. It is more often used in applications such as speech to text for transcribing doctors' notes.

I remember being at Bell Labs and trying to predict churn. I knew that the answers were in the call center notes because they included what customers were upset about, what wasn't working, and why they wanted to disconnect their phone service. Unfortunately, the technology didn't exist at the time for text processing, semantic understanding, or summarization.

Semi-structured Data

In between structured and unstructured data is semi-structured data. While it has some structure, it does not conform to a rigid schema. HTML, JSON, XML, and log files are all common types of semi-structured data, each playing an important role in digital systems. HTML (HyperText Markup Language) is used to structure and display content on web pages using nested tags, making it readable by both humans and machines, though it lacks a rigid schema. JSON (JavaScript Object Notation) is a lightweight format for storing and transmitting data, commonly used in APIs and web applications due to its clear key-value structure and easy integration with modern programming languages. XML (eXtensible Markup Language) encodes documents in a hierarchical, tag-based format and is widely used in enterprise data exchange, configuration files, and legacy systems. Log files are machine-generated text records that capture events, errors, and system activity, often containing structured fields like timestamps and codes within a flexible format. All of these data types are considered semi-structured because they include elements of organization, such as tags or delimiters, that help interpret their content, yet they do not conform to the strict schema constraints of fully structured data like relational databases.

Some people will include Internet of Things (IoT) data under the semi-structured data umbrella. This includes data from sensors and other devices such as wearables and smart meters. For instance, insurance companies are using telematics to gather data about driver behavior to price insurance policies. Utility companies are collecting sensor data from wind turbines to predict when maintenance is needed. These IoT devices may produce data with varying structures depending on the type of sensor, manufacturer, or communications protocol.

Real-time Data

In addition to these categories, it's important to consider real-time data, not as a distinct format but as a characteristic that can apply across all data types. Real-time data refers to data that are generated, transmitted, and processed as events happen, often with minimal latency. This can include structured data, such as streaming financial transactions or sensor readings from industrial equipment or smart cities; semi-structured data, like JSON-based event messages from mobile apps or IoT devices; and unstructured data, such as live video feeds from surveillance cameras or audio streams from customer service calls. Regardless of format, the defining feature of real-time data is its immediacy, which enables use cases ranging from predictive maintenance and fraud detection to dynamic personalization and automated alerting. As organizations try to build AI-driven systems that can sense and then respond to change, the ability to use real-time data across formats becomes increasingly valuable.

What Different Data Types Bring to the Table

By integrating different types of data—essentially *marrying* them—you can create a rich dataset that offers deeper insights for analysis. Consider the example discussed earlier involving call center notes. The goal in this case was to build a dataset that combined structured information, such as company size, industry, billing history, and customer history, with unstructured data like the reason for the support call. This unified dataset allows for a more comprehensive understanding of why a customer may have churned.

When the outcome is known, in this case, that the customer churned, this labeled target data can be used to train a predictive model. The model can then identify other customers with similar profiles who may also be at high risk of churn, enabling proactive retention strategies.

Here's another example from agriculture. Aerial imagery can provide insights into when crops need watering. When this aerial data is integrated with ground-based sensor data, such as temperature, soil moisture, and water saturation, it becomes possible to make more informed decisions about irrigation and overall crop health. This combination of data types enhances precision agriculture, helping farmers maintain healthy yields with optimized resource use.

In healthcare, combining structured data, such as patient demographics, lab test results, and medical history, with unstructured data from physician notes, imaging reports, and wearable devices can lead to improved patient care. For instance, structured data might indicate that a patient has high blood pressure and elevated cholesterol levels. However, integrating this with unstructured data, such as text notes indicating symptoms like fatigue or shortness of breath, along with real-time data from wearable fitness devices (e.g., heart rate variability, sleep patterns, physical activity), provides a much richer clinical picture.

It is often the case (although not always) that the more attributes you have available to create features for your predictive models, the greater your chances of building a more accurate model. The same principle applies when developing AI applications. Think about building a recommendation engine, these are the systems you often see on websites that suggest which product you might want to buy or which movie you might want to watch. These recommendations are based on historical data from you and users with similar preferences and behaviors.

The more attributes you can incorporate, such as browsing history, purchase frequency, time spent on pages, product ratings, demographic information, and even the time of day you typically shop, the more personalized and accurate the recommendations can become. Enriching the model with a diverse set of features can help enable the AI system to uncover deeper patterns and make more relevant suggestions. This can help to improve user engagement and ultimately, business outcomes.

The Data Silo Problem

Of course, the problem is that all of this diverse data is stored in disparate systems across and throughout the company. It can be hard to pull together this data to build that enriched dataset needed for AI.

Think about all of the systems your company uses, including its ERP systems. Popular ERP systems include SAP S/4HANA, Oracle Cloud ERP, Microsoft Dynamics 365 Finance & Operations, NetSuite ERP, and Workday ERP, all of which help organizations manage core business functions such as finance, human resources, supply chain, and operations within a unified platform. Then there are CRM systems. Popular CRM platforms include Salesforce, Microsoft Dynamics 365 CRM, and HubSpot CRM, which are widely used to manage customer relationships, streamline sales and marketing efforts, and enhance customer service.

Of course, these are not the only systems that generate data. Organizations also utilize web and mobile analytics tools, e-commerce platforms, marketing automation tools, billing systems, operational databases, event management platforms, and a host of other industry-specific applications. Most companies have dozens—if not hundreds—of source systems contributing to their analytics initiatives.

The reality is that organizations operate in complex data environments that include a mix of ERP systems, CRM platforms, legacy applications, custom-built systems, databases, transactional systems, and others. Many also want to bring in external data from third parties that might include demographic and firmographic data. All of this diverse data stored across disparate systems can lead to fragmentation and data silos. These silos make it difficult to bring data together to build the enriched, unified datasets required for analytics and AI applications.

To address this, many organizations have adopted data warehouses. These are centralized, integrated repositories designed to consolidate data from across the enterprise to support decision-making. The traditional goal of the data warehouse was to create a single version of the truth for structured data. Data warehouses use a schema-on-write approach, meaning the structure of the data must be defined before it is stored.

As the need to store and analyze new data types (e.g., semi-structured, unstructured, real-time streams) grew, organizations began turning to data lakes. These systems are designed to store high volumes of raw data at scale and use a schema-on-read approach, offering greater flexibility. Initially built on platforms like Apache Hadoop, many early data lakes fell short due to lack of governance, poor performance, and inadequate metadata management. As a result, these early deployments often became so-called data swamps, where users struggled to find, trust, or use the data effectively. Such deployments delivered little to no business value. Modern data lake platforms include Amazon S3, Azure Data Lake Storage, and Google Cloud Storage (GCS).

While the idea was to have a single, centralized source of trusted data, the reality is that most enterprises now have multiple data warehouses, often split across on-premises and cloud environments in a siloed environment. In some cases, this is intentional (e.g., not wanting to migrate a legacy system to a modern environment); in others, it is just an outgrowth of platform sprawl. *The reality is that not all data can be centralized*. Yet, these siloed environments are difficult to manage and govern. The result is often fragmented data access, inconsistent reporting, duplicated effort, and a lack of trust in analytics outputs. This ultimately slows down decision-making and hinders the development of scalable, AI-ready data foundation.

The Move Toward Data Unification

As we've seen, the variety of data types and the proliferation of systems that generate and store this data can lead to fragmentation. This fragmentation creates silos that can hinder efforts to build rich datasets—the kinds of datasets needed for modern analytics and AI. Even when organizations invest in data warehouses or data lakes, they often end up with multiple, disconnected environments split across on-premises infrastructure and various cloud services. The result is data that can be hard to bring together for analytics and AI. Think about different cloud platforms. They may utilize different file formats, or they may expose data through different syntax schemes.

These challenges have resulted in growing interest in the concept of a unified platform for analytics. While not all data can or should be physically centralized, the goal of a unified platform is to provide an integrated environment that enables consistency, coordination, and control across the data and analytics life cycle. This includes ingestion and transformation, data storage and management, analytics tooling, governance, and consumption. The platform supports both operational and analytical workloads, making it easier to analyze and act on data.

A unified platform does not eliminate complexity, but it helps manage it. It enables organizations to capture and leverage emerging data types, like streaming, text, and image data, while supporting traditional structured sources. It offers a foundation for analytics initiatives that range from self-service dashboards to sophisticated machine learning models. Just as importantly, it provides a shared framework for collaboration across diverse user personas, from data engineers and scientists to business analysts and decision-makers. This common foundation helps ensure that everyone is working with consistent data definitions, trusted data, and the tools best suited to their role. They are all speaking the same data language.

In this way, the unified platform can become a strategic enabler. It supports not only the technical integration of data and tools but also the organizational alignment needed to scale analytics and AI responsibly. Even if full unification isn't achievable, moving toward a more integrated approach can significantly reduce inefficiencies and help with better insights and innovation.

One example of a unified platform utilizes a data lakehouse to overcome the limitations of both data warehouses and data lakes. A lakehouse combines the scalability and flexibility of a data lake with the governance, structure, and performance of a data warehouse. It allows organizations to store both structured and unstructured data in one platform while supporting robust analytics, machine learning, and BI workloads. These lakehouses get by the data swamp problem by providing features that are database like. That includes support for ACID transactions. ACID is an acronym that stands for atomicity, consistency, isolation, and durability. That means that, for instance, a transaction either commits in full or, if something fails, the system rolls back every change (the A). It means that business rules are adhered to and all required fields are present with the right data type (the C). It means that when a transaction is running its intermediate changes are hidden, so if someone else is using the lakehouse, they see a stable view (the I), and finally that the result is set (locked) once the commit succeeds (the D). So even if the system crashes, that latest data persists.

Examples of lakehouse software vendors include Databricks, Snowflake, Dremio, and Cloudera. Hyperscalers such as AWS, Microsoft, and Google also offer lakehouse options. Vendors that offer data lakehouses often support open-source file formats such as Delta Lake or Apache Iceberg. These are open table formats for enabling reliable and scalable lakehouses. In addition to supporting ACID transactions, data lakehouses do other things. For instance, they also support processes such as time travel/versioning, which lets users query and go back to any snapshot of the data by version or time stamp. This helps with audit history, reproducing experiments, or recovering from bad writes. They also support schema enforcement to ensure that incoming data matches the table schema. In other words, formats such as Delta Lake and Iceberg allow raw object storage to act like a data warehouse.

We will talk more about the lakehouse and other modern architectures in the next chapter.

Data Platforms: Traditional vs. Modern

As part of the modernization effort, many organizations have migrated their data to the cloud and are now leveraging modern cloud platforms, including cloud data warehouses, data lakes, and data lakehouses. This migration trend began over a decade ago, with the promise that cloud infrastructure could deliver the scalability, elasticity, and flexibility required to support emerging data and analytics demands.

Cloud platforms have become the foundation for modern data strategies for numerous reasons. On the data management front, many organizations have outgrown their on-premises data warehouses and are looking for more scalable and elastic architectures. The cloud allows for on-demand resource scaling (it can scale up and down on demand), helping organizations respond quickly to changing data volumes and business priorities. Cloud-native data warehouses offer advantages such as the separation of compute and storage (allowing organizations to scale each independently), high performance at scale, and integration with a growing ecosystem of cloud-based services. These platforms also minimize setup and maintenance overhead, allowing teams to focus on delivering insights rather than managing infrastructure.

On the analytics side, the cloud supports experimentation and agility. As more data is generated in the cloud, it can become more efficient and cost-effective to analyze it there (a concept that used to be referred to as data gravity). Cloud environments support both self-service and advanced analytics, enabling users to explore data, reduce time to insight, and build sophisticated AI models. Many cloud providers offer built-in tools for machine learning, AI, and other forms of advanced analysis, and they often integrate with third-party platforms. In other words, they are built for innovation at scale.

So modern platforms are often cloud native and enable scaling and resource optimization. They separate compute from storage to scale compute and storage independently. They support all kinds of data and also offer tools beyond data management.

Governance is also often a part of modern data platforms. As we will see later in the book, governance is critical for data and analytics. Governance is a set of rules, policies, and procedures that organizations put in place to make sure that its data, analytics, and AI are fit for purpose and are compliant. Building out the governance strategy should go hand-in-hand with building out the data strategy. Modern platforms often include governance features that we will start to discuss in Chapter 7.

What's Next?

As this chapter illustrates, all kinds of data can be useful for AI. The key is being able to bring this data together to get the most value out of it. The unified/centralized data platform is one way that organizations are trying to bring their data together in a trusted data foundation. There are other approaches, which we will discuss in the next chapter.

CHAPTER

5

The Need for Modern Data Architectures

In this chapter, we will examine the following:

- The modern data and analytics ecosystem
- Modern data architectures
- The data fabric—a modern architectural approach/pattern
- The data mesh paradigm
- Automation and AI infused

As analytics and AI become more important, the ability to design data environments that are both flexible and governed is becoming increasingly critical. Achieving this balance requires a clear vision for how data flows, is organized, and is made accessible across the enterprise. This chapter explores the structure and design of modern data ecosystems and architectures, introducing foundational concepts and emerging patterns, such as the data fabric and data mesh, that help organizations unify, govern, and scale data access to support both current needs and future innovations.

The Modern Data and Analytics Ecosystem

We've been talking about how data supports AI and innovation. But in order to enable this, organizations need a coherent data environment. The purpose of this modern data environment is to help the business run smoothly, support data-driven decision-making, and help drive innovation. This environment should support analytics, AI models, and AI applications, both internal or customer-facing, with the goal of creating positive top and bottom-line impact.

Just as there are biological ecosystems that consist of living and nonliving things that are interconnected, a modern data ecosystem is an interconnected system that consists of the collection, storage, analysis, and management of data. It is the broad landscape in which data live, flow, and deliver value. It includes the technologies to help with collecting and managing the data as well as the people who work with data, and the processes that govern it. Like a natural ecosystem, it is made up of many interdependent parts that must work together to sustain growth. Figure 5.1 illustrates the interconnected components of the data ecosystem.

Today's data ecosystems are often complex, involving both on-premises and cloud platforms. They contain numerous components. Consider all the data sources

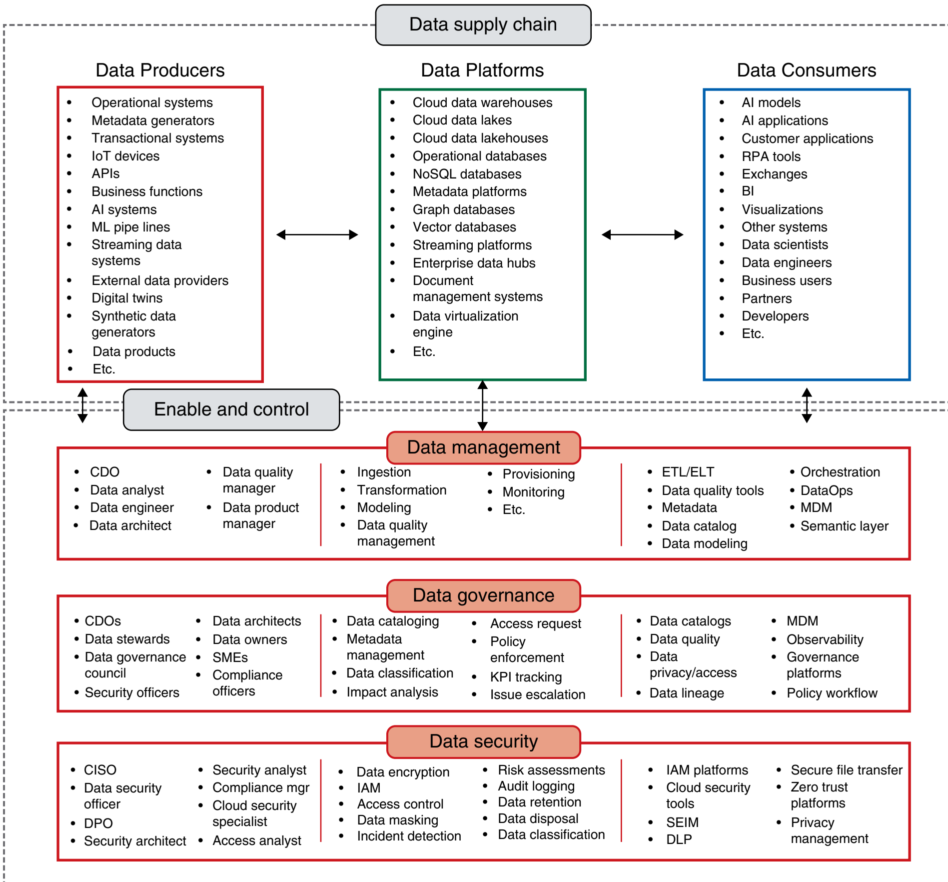

Figure 5.1: The data ecosystem.

companies have that produce data that are part of the ecosystem. These include operational systems such as ERP, CRM, billing systems, and order management systems. The ecosystem includes machine learning models that produce outcomes and scores. It includes external data providers. It even includes newer systems such as digital twins, which are virtual representations of an object or system and all the other systems that produce the data we discussed in Chapter 4. Some of this is traditional data (the structured data); some of it is more modern data such as IoT (Internet of Things) data generated from wearables and devices; some of it is batch (processed at specific intervals); and some of it is streaming (processed in real time). These sources can be thought of as the data producers.

On the backside of the data ecosystem are the data consumers who are the people such as the data scientists, business users, customers, or other stakeholders who are collecting/analyzing/utilizing the data. Or it could be systems that use data. This might be your BI tools or your AI systems that we will discuss in coming chapters. It can be operational applications.

In the middle is what is often known as the platform layer where data is collected and stored. The middle layer contains the storage tools including data warehouses, data lakes, and data lakehouses that were described in Chapter 4. It includes traditional databases as well as newer kinds of databases such as graph databases or vector databases that we'll discuss later in the book. I've grouped these three components into a category called the data supply chain. The notion of a data supply chain is an industry term that is used to denote that the data moves through a process that produces value.

To enable and control all this data involves data management, data governance, and data security. Data management involves how the data is processed (e.g., integrated, cleansed, organized, routed) so that it's ready for use. Depending on your company, it can consist of ingestion pipelines that bring the data into the ecosystem as well as all the data integration tools that transform, enrich, and standardize that data so it can be trusted and effectively used by consumers across the organization

These ingestion tools take the data from the sources and bring them to the target platforms. They can automate the process of moving, transforming, and unifying data from multiple sources into a platform for analysis and use. These include those you may have heard about—Extract, Transform, Load (ETL) tools and the more modern Extract, Load, Transform (ELT) pattern that are part of the broader family of data integration tools. Both approaches are used to take raw data from various systems and make it usable. ETL performs cleaning and transformation before loading data into storage, while ELT loads raw data first, often into cloud storage. Once there, the data can be cleaned, joined, or transformed so it's ready for use in analytics or AI. The choice between ETL and ELT depends on your architecture, performance needs, and governance requirements. ELT is increasingly common in cloud-native environments, where compute power can be scaled on demand.

A key part of these ETL and ELT processes is metadata—information about the data itself. Metadata describes things like column names, data types, business definitions, data quality rules, and transformation logic. During ETL/ELT, metadata helps define how data should be interpreted, validated, and mapped. It also tracks where the data came from (lineage), how it has changed, as well as who has interacted with it. Without metadata, ETL/ELT tools would have no guidance on how to transform or move the data. Users would have little ability to understand or reuse it. In Chapter 6, we'll talk more about metadata, the kinds of metadata and data cataloging tools that utilize metadata to help users discover, understand, and trust their data. We'll also talk about the latest metadata trend: active metadata.

Data management includes other tools too. These include data modeling and transformation tools that refine the data structure and logic, data orchestration tools that coordinate and schedule complex data workflows, and data quality and observability tools that identify problems in data and validate data health.

Controlling the data ecosystem are the governance and security processes and tools. In the figure are also listed some of the roles involved in this control. There is a whole section devoted to data governance later in this book. Suffice it to say now that it involves various roles and processes that are needed to put the rules and processes in place to make sure that the data is meeting compliance obligations and that it can be trusted for use in analytics and AI. It involves data quality, data lineage, cataloging data, and monitoring and observing its health. It involves ensuring data privacy and security.

The data ecosystem model above is data centric. It views AI as a consumer of data, which of course, it is. However, in many ways it should be expanded to include AI because the two are synergistic: a solid data foundation is necessary for AI and at the same time the AI ecosystem creates value from the data. That AI ecosystem includes AI infrastructure (e.g., GPUs, vector databases), AI models, model development platforms (e.g., TensorFlow, PyTorch, Hugging Face), AI application frameworks (e.g., LangChain, RAG), responsible AI and governance, and AI applications. We'll dive into this ecosystem in Chapter 14.

Of course, the idea behind many of these concepts and tools is not new. Data warehouses have been around for decades (and even the data lake concept is at least 10 years old), and the concepts of ETL and metadata have long been foundational to data management. What's new is how these concepts have evolved and are being delivered. The modern data ecosystem is cloud-native, highly distributed, and AI-enabled. It supports not only structured data from enterprise systems but also unstructured text, images, sensor streams, and real-time events. Processing frameworks are now highly scalable. Even metadata has evolved. It is no longer simply information that is stored somewhere. As I mentioned, there is now the notion of active metadata that refers to metadata that is automatically generated. We'll talk much more about metadata and data catalogs in the next chapter. AI-enabled or AI-infused tools are becoming quite popular to help to automate the identification of poor-quality data, map metadata, track data lineage, and for a host of other functions. In fact, AI is being infused into the entire data life cycle, from collection to processing to consumption as part of the data ecosystem.

In other words, some of the building blocks may be familiar, but the way they're being combined and operationalized reflects a shift in how organizations think about data.

Modern Data Architectures

As we've seen, a modern data ecosystem includes a wide array of components: systems that produce data, technologies that ingest and store it, people and tools that consume it, and governance controls to keep it in check. But this ecosystem only works in an organization if it's designed intentionally. Without structure, the ecosystem can become siloed and inefficient. This is especially the case as data volumes grow and demands on AI and analytics increase.

That's where data architecture comes in. The data architecture provides the blueprint for how the data is organized, accessed, and integrated. It defines the relationships between data sources, how data is modeled and transformed, and how the underlying technologies work together to support the business. While the terms *data modeling* and *data architecture* are sometimes used interchangeably, they are not the same. Data modeling is part of a data architecture and defines the specific structure and relationships within datasets (e.g., tables, columns, constraints), whereas data architecture is broader; it provides the framework and strategy for managing, storing, integrating, and utilizing data in an organization. We're not going to focus on data modeling in this book, as it is a layer of detail lower than the purpose of this book. However, here are some perspectives from two leaders in modern data architecture about what a data architecture is and the critical role it plays.

First, John O'Brien shares his approach to designing architectures that serve business objectives. Following that, Donna Burbank offers her perspective on how to design architectures that balance strategic vision with practical considerations.

EXPERT ADVICE: JOHN O'BRIEN ON DATA ARCHITECTURE

John O'Brien, principal at Radiant Advisors, notes, "Architecture enables the business. I begin with clients' business initiatives and ask: 'What capabilities are needed to deliver that?'" The blueprint is valuable only when it is directly linked to organizational outcomes—a point often missed when teams rush into technology choices.

Building on this foundation, O'Brien emphasizes the separation of logical from physical implementation. He states, "I often separate logical architecture: data ingestion, transformation, and persistence patterns. This results in a two-tier approach where a flexible, governed persistence layer (landing, raw, and curated zones) remains distinct from the delivery layer, in which enterprise data models, dimensional models, cubes, or caches are created for specific workloads. This separation enables teams to iterate on business deliverables while refactoring the core data foundation."

O'Brien notes that architects have different styles based on their experiences and shaped by their company's IT culture. There is no single off-the-shelf architecture, and no universal answer for everyone on where and how data should be stored. As O'Brien describes it, architectural styles cover a range of trade-offs, including centralized versus decentralized data, acceptable duplication, and several other data management principles. Each organization must evaluate its business priorities, latency, and governance needs, then choose design patterns—such as a warehouse, lakehouse, fabric, mesh, or combinations thereof—that align with those needs.

This need for balance and adaptability is reflected in O'Brien's practical advice: "Pick a primary cloud vendor. From day one, you're hybrid, and you'll stay hybrid. Plan for it." Centralize what you can, plan for data gravity shifts, design for what you can't, and keep architecture portable (another principle) to avoid lock-in.

Based on these principles, this stance shapes his perspective on the new "converged data management platform" trend. "A converged platform is a vendor play, not an architecture," he cautions. Lakehouses, fabrics, and meshes are useful, but only as ingredients. O'Brien prefers a polyglot persistence architecture, matching best-of-breed data engines with their specific workloads, striving to balance complexity with management. He believes that use cases and capabilities, not marketing, drive technology choices. Since too many options can paralyze teams, O'Brien advises a focused approach: "Don't use a dozen vendors; focus on a few core ones for the architecture. Cut the diagram by at least a third."

Like O'Brien, Donna Burbank underscores the importance of linking architecture to business needs, and she frames the discussion through a set of key pillars that organizations can use to guide both design and execution.

EXPERT ADVICE: DONNA BURBANK ON DESIGNING MODERN DATA ARCHITECTURES

Donna Burbank, managing director at Global Data Strategy Ltd, a consulting firm that specializes in aligning business goals with data-centric technologies, emphasizes the importance of balancing business needs with thoughtful design when building modern data architectures. "Architecture isn't just a technology blueprint; it's about business strategy," she says. "It's the top-down view of what you want to do from the business perspective and the bottom-up view of what data you have to enable that."

Burbank often sees companies rush into AI or analytics initiatives without clearly defining their goals or understanding their data. "I've worked with so many companies that say, 'We want to do AI,' and I ask, 'For what?'" she says. "They'll answer, 'HR and finance.' Those may be two of the hardest use cases because of risk and complexity." For her, effective architecture starts with understanding business drivers and use cases.

She outlines five key pillars for thinking about data architecture:

- **Source.** Where does the data originate? Is it internal, third-party, or sensitive data such as personally identifiable information? What are the rules and permissions around its use?
- **Curate.** What needs to happen to the data to make it useful? This includes mastering, integrating, and applying data quality processes.
- **Store.** What is the appropriate pattern for storing the data? This could involve a data lakehouse, a data warehouse, or hybrid storage.
- **Consume.** How is the data being accessed and by whom? Are BI users querying it through Power BI, or is it powering machine learning pipelines?
- **Govern/Manage.** How is the entire ecosystem governed? This involves metadata, lineage, privacy, access, and stewardship.

Burbank cautions against jumping straight into implementation without establishing architectural patterns. "So many people say, 'We don't need an architecture; we're buying Vendor X.' Well, that's a platform. The architecture is how you design and use it."

To explain this to clients, Burbank often draws a diagram of an ecosystem. "Today's architecture is really an ecosystem," she says. "Your system architecture might include Snowflake, Databricks, Azure—those are tools in your ecosystem. But within that, you have to think about different implementation patterns (i.e., how a data architecture is put into practice to support a specific kind of use case or analytic workload) such as BI, machine learning, streaming, even vector databases."

In her view, the system architecture is the blueprint for how everything fits together, while data architecture lives within that system and includes the data

(continues)

EXPERT ADVICE: DONNA BURBANK ON DESIGNING MODERN DATA ARCHITECTURES (CONTINUED)

model and design patterns. These design patterns include denormalized tables, star schemas, graph models, or vector search indexes. "Different business problems may require different design patterns," she says, "but you still need that high-level view."

On the question of centralized versus federated architectures, Burbank sees a trend toward more centralized approaches. "Even though it may seem old-fashioned, most of the clients I work with are looking for a single place to go for data—a curated, consistent view," she says. Still, she appreciates the value of logical data fabrics in certain scenarios. "We had a client that used Denodo to virtualize data because they couldn't get access to all the physical systems," she notes. "It was great for exploration. They only published to the warehouse what they really needed to make enterprise-ready."

For Burbank, semantics are essential to modern architecture. "The semantic layer is your powerhouse. That's how you define what 'product' or 'customer' actually means. That's what makes your company unique," she says. She also urges teams to use logical architecture models as roadmaps. "Since they are high level, they help people understand what we're building and why. I've had business users thank me just for drawing a picture on a whiteboard," she says. "It helps them connect the dots, what data is in our warehouse, what's still raw, what's in staging."

Ultimately, Burbank sees architecture as a creative process. "Architects are like artists. But we need to balance flexibility with a common foundation. If you don't design intentionally, you end up with chaos."

Her advice is to start with the business goal, understand your data, then define the architecture that fits your needs. From there, organizations can decide on platforms and data models, and only then begin implementation.

A data architecture, then, is a blueprint (aligned with business goals) for how data is organized, governed, and delivered across the enterprise. It defines the logical view with concepts, domains, and shared semantics that make data understandable and reusable, and it defines the physical realization that uses platforms, pipelines, models, and storage to implement the design. A solid architecture clarifies ownership and standards for quality, security, privacy, and lineage; establishes how data moves from sources to consumers; and sets the policies that keep it trusted and compliant.

The Data Fabric—A Modern Architectural Approach/Pattern

As mentioned in Chapter 2, organizations are working to unify the data silos that often exist across their data ecosystem. Some are doing this through a centralized physical approach, such as implementing a data lakehouse pattern, which allows them to store and manage all types of data in one place. Many lakehouse implementations such as those from vendors such as Snowflake and Databricks have evolved into what is often referred to as the modern data platform. This is an architectural pattern that combines the lakehouse with tightly integrated tools for data ingestion, transformation, analytics, observability, and governance. This pattern reflects a shift toward cloud-native architectures that are designed to support end-to-end data workflows with scalability and flexibility.

ARCHITECTURAL PATTERNS

You'll hear architects talk about patterns. These are a reusable design approach or template to solve for a recurring problem. For instance, important data may be stored in many places (cloud warehouses, data lakes, SaaS [software-as-a-service] applications, legacy systems). Teams need a unified, governed view without copying everything, and they need consistent access controls and lineage across sources. A data fabric (described next) uses metadata, a semantic model, and virtualization to connect sources, apply policies once, and deliver views. It reduces duplication, speeds access, and keeps governance consistent. Or a team may consider a lakehouse as an architectural pattern (as opposed to a platform). The lakehouse describes how you organize storage, tables, compute, and governance to unify lake and warehouse on one copy of data.

Another pattern is the data fabric. The data fabric is a way to weave together disparate data in an intelligent fashion. The fabric is an architectural pattern that maps and connects relevant application data stores, with metadata (e.g., data about the data) to describe data assets and relationships. In some ways, it is a logical data architecture that enables organizations to view their data as if it is a single data source, even though it might physically be in multiple silos. The data fabric maps and connects relevant application data stores using metadata to describe data assets and their relationships. In other words, the data fabric is one way to design and implement a logical data architecture that spans many different sources and systems. It is not tied to a single technology or vendor.

Here's an example. Suppose I want to analyze issues with one of my company's products. I may need product data, ownership history, customer complaints, and root cause analysis. This data may be stored across multiple warehouses, lakes, and clouds. A data fabric allows me to access this information as if it were all in one place, without knowing where it lives or how to integrate it manually.

Using a data fabric approach, I can create an abstraction layer—a technology that hides the complexity of underlying data sources by providing users with a unified view of the data without requiring them to know where or how the data is stored. One approach to a data fabric uses a virtualization layer (a single layer to access and combine data across silos in real time, without moving it, and can query it dynamically).

Other fabrics might use a semantic layer to provide a business view of the information. Once used primarily to provide business-friendly definitions in BI tools, the semantic layer has evolved. It is now cast as a business-friendly abstraction

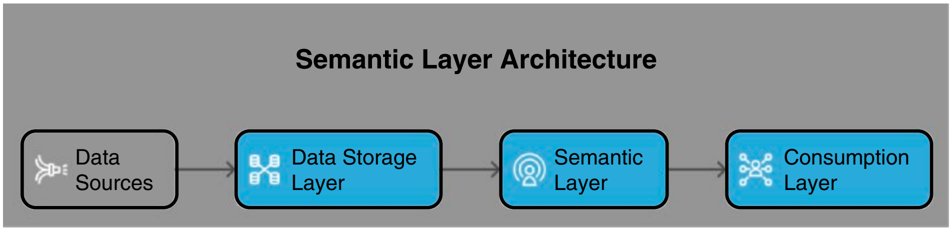

Figure 5.2: The modern semantic layer.

layer that manages the interactions between data consumers (whether they be humans or AIs) and enterprise data assets (the type we've been talking about). This interaction uses a semantic model that maps the language of raw data (e.g., field and table names) with business concepts.[1] It provides data semantics services for multiple tools within a multitool and multiplatform data architecture and provides friendly descriptions of data that simplify and improve modern data practices (Figure 5.2).

According to TDWI, a modern semantic layer is an independent tier positioned just below the data-consumption layer and just above the data fabric's integration and streaming services. From that spot it represents data in motion through the fabric and data at rest in lakes, warehouses, and other stores—on premises or in the cloud—so the entire architecture can reference the same governed, business-friendly view of information. This location also lets BI and analytics users see a broader enterprise picture than the limited, tool-specific semantic layers they used before, gives the data fabric the advanced semantic automation it requires, and still leaves teams with standalone tooling that unifies multiplatform environments. Because the layer is logical and agnostic, it can underpin centralized, distributed (e.g., data mesh), or hybrid architectures, supporting numerous modern designs.

In TDWI research, neither a centralized repository such as a cloud data lakehouse or a logical data fabric approach to unify data contributed more to AI success. This supports John O'Brien's view of architectures that we saw earlier. For example, in a 2025 TDWI survey, the cloud data lakehouse was a top investment priority. However, many respondents were also using hybrid deployments that utilize both on-premises and cloud platforms, using an approach such as a data fabric. Using statistical tests, we saw no significant difference between the extent of AI's business impact and the type of platform used for data management.[2]

[1] Russom, P. (2023). *TDWI Checklist Report: The Semantic Layers Critical Role in Modern Data Architectures*. Available at: `tdwi.org/research/2023/01/arch-all-checklist-report-semantic-layer-critical-roles-modern-data-architectures.aspx` [Accessed 25 June 2025].

[2] Halper, F. (2025). *TDWI Best Practices Report: Creating an AI-Ready Organization*. Available at: `tdwi.org/research/2025/ai-ready-organization` [Accessed 14 June 2025].

However, those with a solid data foundation, where it is easy to get to the data they need for AI, are more likely to succeed than those who don't have this foundation in place. Ultimately, the architecture alone did not predict success; what mattered most was having a solid, accessible data foundation that enabled teams to get the data they needed, when they needed it.

The Data Mesh Paradigm

As organizations look to scale data access and usability across complex environments, they are shifting toward a new mindset: treating data as a product. This emerging concept lays the foundation for approaches such as the data mesh, which reimagines how data is owned, shared, and governed across the enterprise.

The concept of the data mesh was introduced by Zhamak Dehghani in 2019 as a sociotechnical paradigm, not simply an enterprise architecture.[3] At its core, it acknowledges that the success of any data system depends as much on people, processes, and ownership models as it does on platforms or pipelines. The data mesh proposes a new model for managing data at scale, one that aligns with how organizations actually operate: through independent domains that generate, understand, and use data in different ways.

The Four Principles of the Data Mesh

There are four foundational principles that define the data mesh approach.

The first is domain-oriented ownership. In the mesh model, business domains—such as sales, finance, or product—own and manage their own data. They are responsible for the quality, access, and sharing of that data with other domains. The goal is to scale data access and reduce bottlenecks by decentralizing control to the teams closest to the data and its context.

Second, data is treated as a product. It is not an afterthought or by-product of operations but something with consumers that requires usability, reliability, and documentation. Each domain is responsible for ensuring that its data is discoverable, well described, and interoperable—much like any product that must serve customers.

Third, the mesh requires self-service infrastructure. A shared platform enables domain teams to build, deploy, and consume data products without relying entirely on a centralized engineering team. These platforms should support

[3] Dehghani, Z. (2019). How to move beyond a monolithic data lake to a distributed data mesh. *MartinFowler.com* (10 May). Available at: `https://martinfowler.com/articles/data-monolith-to-mesh.html` [Accessed 14 June 2025].

engineers, analysts, and scientists by simplifying data access and integration across domains.

Finally, the mesh requires federated governance. Governance in the data mesh is decentralized but coordinated. Each domain maintains autonomy while a central group works with domain representatives to define and enforce global standards, security policies, and quality expectations.

The appeal of the data mesh lies in its potential to scale data delivery in large, complex organizations where centralized teams can no longer meet demand. As organizations generate more data across more systems and want more insights, it is unrealistic for a single team to curate and distribute all data. The mesh distributes this responsibility by empowering domain teams to manage their own pipelines, quality, and delivery processes, while still aligning with enterprise-wide practices.

TDWI research reflects growing interest in the mesh model. While adoption is still emerging, with fewer than 15% of respondents reporting full implementation in 2024 survey, a significant portion express alignment with its principles. We've seen many organizations gravitate to the idea of data as a product and federated governance. Organizations experimenting with the model report benefits such as a faster time to data, improved data literacy within business units, and stronger collaboration between data and domain experts.

Practical Considerations

Adopting a data mesh requires more than architecture; it involves changes to culture, roles, and accountability structures. As previously mentioned, it is a socio-technical paradigm. Many organizations will need to create new roles, such as data product managers, who combine stewardship, analytics, and product thinking. These individuals oversee the life cycle of a data product to ensure it delivers business value and meets usability standards. Success often depends on building a culture where data is treated as a shared asset across the enterprise and on putting clear lines of accountability in place. In practice, this can mean forming cross-domain councils to align on standards, tying performance objectives to data quality, or investing in training so domain teams are equipped to manage their products responsibly. Chapter 8 discusses data products in more detail.

Organizations must also define domains clearly, which has eluded some companies. They also must establish which services are centralized versus distributed and determine how shared infrastructure like catalogs, identity management, or data quality monitoring will be governed. Some enterprises use a hub-and-spoke model, where a central data office supports decentralized domain teams and helps ensure alignment.

While some organizations use data fabric technologies to support the connectivity required by a mesh, the two are not the same. A data fabric is typically a technical solution, whereas the data mesh is a broader paradigm centered on responsibility, decentralization, and organizational design.

The data mesh represents a shift in how enterprises think about their data, not as a central asset to be controlled but as a network of products that must be stewarded by the people closest to the business. Though not without its challenges, the mesh offers a promising model for organizations seeking to scale their data capabilities without sacrificing agility, context, or trust.

Automation and AI-infused Software

The move to the cloud, the cloud data lakehouse, data mesh, and the data fabric aren't the only new trends around the data ecosystem. Over the past five years or so, vendors have been infusing their products with AI to help to automate many of steps in the data life cycle, from ingestion to preparation to analysis.

Figure 5.3 illustrates the data life cycle with some examples of where automation and augmentation are occurring. The steps include defining business objectives, capturing and understanding data, preparing and processing it, storing it, analyzing it, delivering it for use, and governing and protecting that data.

Across the data life cycle, automation is being used to streamline tasks. For instance, on the ingestion side tools can suggest integration mappings. They are also providing services such as data quality management. Of course, data quality can be automated using rules. However, AI tools infused in products can help do things like identifying anomalies in the data. For instance, AI tools can detect missing data and duplicates. But it can do more. It can look at the trend of data values, for instance, and then flag deviations from that

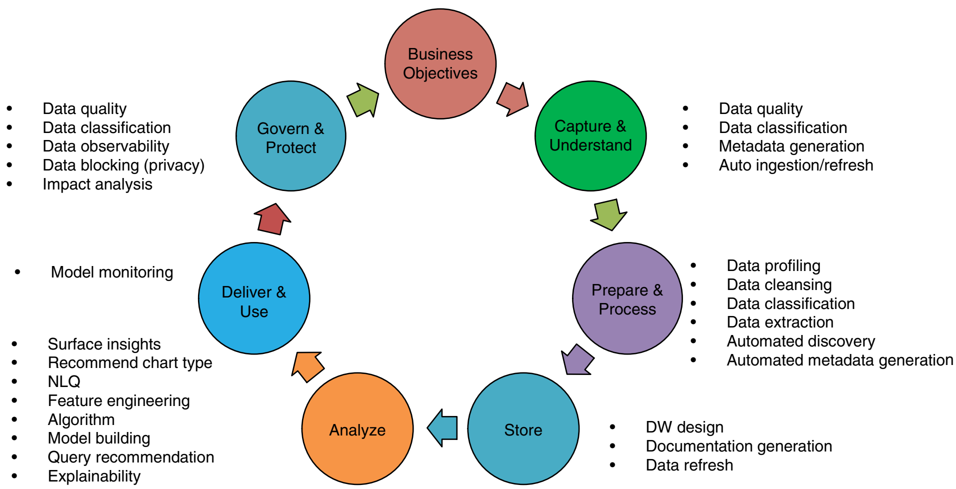

Figure 5.3: Automation and augmentation in the data life cycle.

trend. AI-augmented tools can also identify unreasonable data, values that aren't missing but clearly don't make sense in context. For example, a customer record lists a "retirement_date" that occurs before their hire_date, or a patient record shows a birth_date in 2023 but also includes multiple past hospital visits logged in 2020. AI can also help to classify sensitive data because it can learn what sensitive data such as personally identifiable information data looks like. In addition to then cleansing the data, it can also extract certain pieces of data, for instance from unstructured text. It can learn what an entity (people, place, thing) looks like and extract that. Or it might be able to process the sentiment of a social media post using AI and provide that for analysis. Or it can use an AI tool to detect data freshness. Throughout this it is generating metadata.

On the storage front, it can help with data warehouse design and document generation. It is playing a growing role in analytics and model development. For instance, many tools now will surface insights, where the system can detect patterns in the data and recommend insights and visualizations. Natural language query (NLQ) tools allow business users to ask questions using everyday language, and these have become particularly popular in BI tools since the advent of generative AI. Model building platforms can suggest derived attributes (e.g., the features that are used to train the models). They offer explainability output that highlight what features contributed the most to the model. The idea is that these tools don't replace humans in analytics, but they can help guide them to get to value faster.

On the governance end, platforms are applying intelligent data observability to catch anomalies, track data freshness, detect quality issues, and even analyze downstream impact when a schema changes. In production environments, models are monitored for drift and fairness, often with alerts and automated retraining triggers that keep outputs aligned with real-world conditions. Some tools can use AI to stitch together lineage from diverse tools and trace dependencies for better change management.

What's Next?

This chapter explored the evolution of modern data architectures, including lakehouses, fabrics, meshes, and other patterns that provide order, scalability, and flexibility to today's data ecosystems. You probably notice that I mentioned metadata a lot in this chapter. That is because even the most advanced architecture depends on the ability to describe, track, and make sense of the data that flows through it. This is where metadata comes in. Chapter 6 takes a closer look at metadata itself including the types of metadata and how it has evolved from static documentation to active metadata.

CHAPTER

6

Metadata: What It Is and How It Is Evolving

In this chapter, we will examine the following:

- Metadata, metadata, metadata
- Tools and practices for metadata management
- Metadata evolves
- The metadata platform

I've talked about new data types and some new data architecture patterns. I've also introduced how AI is being infused into data and analytics software products to make them easier to use. Throughout this I've mentioned and defined metadata and the importance of it. In this brief chapter, we're going to look at metadata, what it is, how it has evolved, and how it continues to evolve.

Metadata, Metadata, Metadata

As I've discussed, metadata is data about data. When data professionals think about metadata, they often think about three kinds: technical, business, and operational metadata.

Technical metadata describes the structure and the format of the data that is stored in systems such as a database or data warehouse. This includes information about the kind of data it is (integer, string), table and column names, schema definitions, file size and format, and so on. Historically, technical metadata has been used by data engineers, architects, and IT professionals who are responsible for building, managing, and maintaining the organization's data infrastructure.

They rely on it to understand how data is structured, how it flows between systems, and how to ensure it is processed efficiently.

Business metadata, on the other hand, is what it sounds like. Business metadata provides information about data in business-friendly terms. This often includes definitions of key terms, such as "customer" or "active account," as well as descriptions of metrics, business rules, and data ownership. This can sometimes be found in a business glossary. This metadata is used by business analysts, data stewards, and decision-makers to ensure consistent understanding. This information might have to be mapped to technical metadata. For instance, I might have multiple locations in my data environment where I store customer order numbers. That might be called cust_ord in one place and customer_ord in another. However, specified as business metadata, it would need a business term such as Customer Order.

Operational metadata tracks access to data by users and applications; it tracks how data moves and acts during processing and usage. It can be real time. Examples of operational metadata include data refresh times and access frequency, pipeline run logs and performance, user access history, or error events. Operational metadata is primarily used by data engineers, platform teams, and others on data operations (DataOps) teams to monitor, troubleshoot, and optimize data systems. As data and systems become more numerous, they are producing a lot of this operational metadata.

The semantic layer, described in Chapter 5, uses all three types of metadata, but it publishes only the parts that users need to think about—the business-level terms, metrics, and hierarchies.

Tools and Practices for Metadata Management

People have been documenting data about data for decades. This was mostly manual and often stagnant. Think about the old-style data dictionary that was a centralized repository that described data elements in a system—a catalog. It was used primarily for technical and system documentation.

About 10 years ago, vendors started to market what they called a *modern data catalog.*

Unlike earlier catalogs that were static and limited in scope, the "modern" data catalogs offered a more dynamic experience. These catalogs made extensive use of metadata to help users understand where data came from, how it was structured, and how it has been used. It helped users find and better understand the data that was in their systems. In some ways, these catalogs were driven by the rise of self-service analytics, i.e., the move to democratize BI. Business users were making use of BI tools, but they didn't necessarily know what data was available to them, where it came from, and if it could be trusted. These tools had searchable interfaces and could index data across multiple systems. The data was then put into this central catalog.

These catalogs evolved and included features such as automated data classification for personally identifiable information, metadata discovery, and the ability to track how data has moved and changed over time. These tools made it easier for users to find data using everyday business terms and provided options for exploring or previewing the data before using it. Some even connected directly to platforms so users could access the data they needed without switching tools.

These catalogs also supported efforts to build trust in the data. For instance, some allowed data stewards to certify datasets by adding badges that indicated the data was reliable. Other catalogs let users rate the data or leave comments about how they've used it in their work. Information about who created the dataset, how it has been used, and whether it has changed recently was often available. Some catalogs also showed visual data lineage—making it easier to understand where the data came from and how it was processed. This helped users decide whether a dataset was suitable for their needs and encouraged more consistent use of data across teams. Pretty soon the catalog software vendors were positioning themselves as data governance platforms, with data quality features, lineage tools, and the other features mentioned earlier. Some even provided role-based access controls (more about that later in the chapter).

Figure 6.1 provides a fictional representation of the kinds of metadata information that the catalog can provide. The top of the figure shows information about customer orders. It shows that the table is in the Snowflake system and where it is. It indicates that the data is certified. It provides business metadata

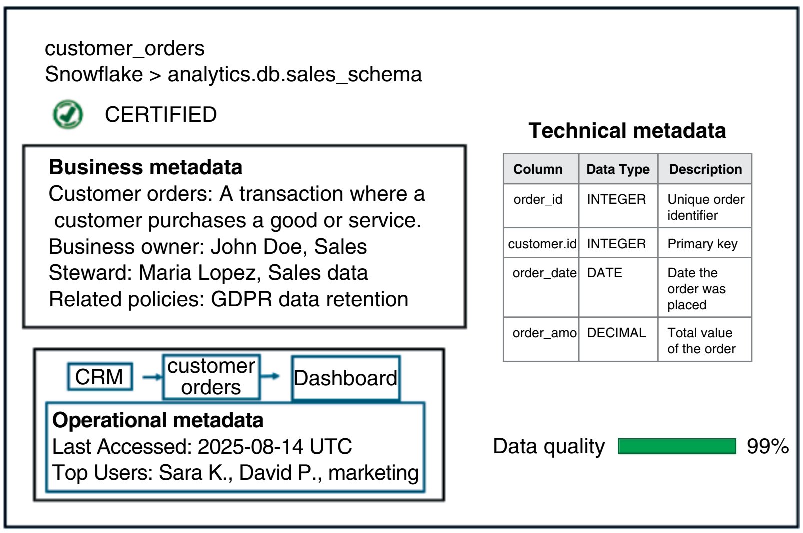

Figure 6.1: The kinds of metadata a catalog can provide.

about customer orders, what they are, who owns the data (John Doe), and who is the steward (Maria Lopez). It also provides technical metadata about the table. This includes order id, order date, order amount, as well as the customer id. It provides the kind of data (integer, decimal, etc.). At the bottom of the read out you can see the operational metadata that describes when the data was last accessed and who the biggest users are. There is also a data quality score associated with the data.

In addition to organizing traditional datasets, some catalogs also evolved to help keep track of machine learning components such as feature sets and model versions. This can be helpful for teams building AI models who need to manage and reuse training inputs or maintain information about how a model has evolved. These were referred to as *feature catalogs*.

Metadata Evolves

As data and systems became more abundant, they produced a lot of metadata. Cloud platforms and data systems such as Snowflake and Databricks generate metadata about the structure, schema, and format of the data they store. In many ways, this is where the modern data catalog began—with indexing, documenting, and organizing this technical metadata. Similarly, ETL (extract, transform, load) solutions like Fivetran, Matillion, and Talend produce operational metadata that describes data flow: source and destination mappings, transformation logic, job status, and run history. BI and analytics platforms such as Tableau, ThoughtSpot, Power BI, and Looker generate metadata about the dashboards, visualizations, and queries users create. They log who accessed what data and when. Even AI/ML tools now contribute metadata about training datasets, features, model versions, and experiment outcomes.

The result is a massive volume of metadata being generated continuously across the data stack, far too much to manage manually. Initially, organizations tried to centralize this information into data catalogs. But as data environments grew more complex and dynamic, organizations realized that they needed real-time insight into issues such as the health of data pipelines, data quality, compliance enforcement, and so on. They wanted to not only *produce* metadata that described their data but *use* this metadata to help manage and drive these data-related systems.

This shift led to the rise of what Gartner, in 2021, called "active metadata."[1] Unlike passive metadata, active metadata is automatically captured, continuously updated, and immediately usable by systems and people to trigger alerts,

[1] Zaidi, E., De Simoni, G., Parker, S. and Jain, A. (2021). *Market guide for active metadata management*. Gartner, Inc. Available at: `https://www.gartner.com/en/documents/4004082` [Accessed 21 June 2025].

enforce policies, track lineage, and improve decision-making. Platforms like Snowflake, Databricks, and modern data observability tools such as Monte Carlo have embraced this shift, providing freshness indicators, pipeline status, usage statistics, and lineage graphs as dynamic, queryable metadata that helps implement governance in real time. For instance, using the customer_order example from earlier, active metadata would continuously capture and use the information to alert users in real time, for instance, if the data quality starts to change.

Catalogs such as the Unity Catalog from Databricks or the Polaris Catalog from Snowflake store and expose detailed metadata about table structures, file formats, and schema versions but also track lineage, access patterns, audit logs, and job execution histories in real time. Technical catalogs like Unity Catalog play a role in governing data use through role-based access control, dynamic data masking, and policy enforcement at the column or row level. What does this mean? With role-based access control (RBAC), each user is assigned a role and belongs to a group; this may be a department, a physical location, or a user type. For example, the role of junior data scientist might have certain access privileges that allow users with that role access to certain information. Those users could be granted access to data in a certain data warehouse used for advanced analytics. If a junior data scientist needs to access data that was not part of their access permission, like human resource data, for a project they are doing with that business unit, then a new role would need to be created, along with a new set of rules for that role.

They also increasingly support metadata for machine learning models and feature stores, enabling end-to-end visibility and trust in AI workflows.

The Expert Advice by John O'Brien illustrates how active metadata is being operationalized today, moving beyond static catalogs to drive automation, embed context into workflows, and prepare organizations for AI-driven data management.

EXPERT ADVICE: JOHN O'BRIEN ON ACTIVE METADATA

John O'Brien, principal at Radiant Advisors, notes that his relationship with metadata goes back to "one of the very first classes I taught at TDWI in 2002 about implementing operational metadata." What mattered then still matters now. The run-time facts show *how* the platform is behaving. The difference today is speed. "Capturing that in real-time monitoring makes it active," he says. That shift can turn a data catalog entry into something the system can act on.

O'Brien starts with plumbing. "If you can instrument and capture all the metadata that's coming from your pipelines … then you can feed that to an automated tool,"

(continues)

EXPERT ADVICE: JOHN O'BRIEN ON ACTIVE METADATA (CONTINUED)

he explains. This way, alerts and lineage stay accurate without manual babysitting. The habit he instills in engineers has an intentionally limited scope. "Every time they create a pipeline, they will log the start and stop time and number of records." Anything heavier waits until it proves its value to the engineer, operations, or consumer. As he warns, "If I put too much overhead on it, then I'm slowing down the pipeline, and its development time."

The pay-off comes in next-generation platforms (what he calls fourth generation, with third generation being today's data catalogs). Context will reach people exactly where they work. These platforms, he notes, should "embed [metadata] in the user's existing applications … we don't switch context anymore." For instance, a support representative asking about a ticket sees data freshness and quality metrics right inside their application. A finance analyst examines cost and latency with a warehouse query.

O'Brien is looking toward automation. "My goal is that in five years, my clients have AI bots that can do the bulk of their data management with them … and do a better job with less resources to keep up with the expanding demands of data management," he says. Those bots will only be as smart as the historical training we start capturing today. So, the task is clear: record the essentials on every run, keep the footprint light to start, surface insight in-context, and let tomorrow's agents learn how to tune cost, reliability, and performance.

The Metadata Platform

Given that organizations are producing a lot of metadata, the shift in thought process to active metadata, along with the concept of a data fabric, has led to the emergence of a new platform type which is known as the *metadata platform*. While the data catalog traditionally focuses on discovery, a metadata platform moves beyond just producing metadata to actually using it as an input for automation and other use cases. These data platforms support active metadata. For instance, a metadata platform can trigger alerts if a pipeline fails or adjust access controls based on behavior. This makes data environments more responsive and even adaptive. These metadata vendors consist of companies like Atlan. Additionally, traditional vendors including Alation, Collibra, and Informatica are positioning themselves as in the active metadata space.

Here is how it works. Active metadata platforms collect and integrate metadata generated across ETL tools, ML systems, data warehouses, and other components of the data stack. This includes operational metadata such as task status, execution time, schema changes, and error logs. Rather than simply storing this information, the platform makes it available to downstream workflows—such as those for pipeline orchestration, data quality checks, or governance—via APIs or policy-driven automation. This enables capabilities like data observability, where the metadata is used to monitor system behavior, identify anomalies, and trace issues such as pipeline failures or data degradation.

What's Next?

Now that you have some idea about what metadata is about and how it is used in catalogs, it is important to introduce the concept of data governance, which makes use of metadata but does much more. Data governance and more on data quality are discussed in Chapter 7.

CHAPTER

7

The Data Quality and Data Governance Imperatives

In this chapter, we will examine the following:

- The risk of bad data leading to bad outcomes
- What data quality means for different data types
- Data quality tooling
- Data monitoring and observability
- Data governance is not an afterthought

I've been using the terms "data quality" and "data governance" throughout this book and they are an important part of the AI Readiness Model discussed in Chapter 3. I want to step back and introduce the concepts formally because they are important. Quality and trust in data are not just operational concerns—they are foundational for any AI initiative.

The Risk of Bad Data Leading to Bad Outcomes

Data quality refers to the overall suitability of a dataset to serve its intended purpose. It's a measure of how well the data meet the requirements and expectations of its users, particularly in terms of accuracy, completeness, reliability, and relevance. Poor data quality produces inaccurate, incomplete, or inconsistent outputs, which can ultimately erode trust in results. For instance, a medical AI model that utilizes poor quality patient data could lead to incorrect diagnoses. If an online auto auction site used to price vehicles can't recognize value signals such as make, model year, and trim package because it was trained on incomplete data, it may price the cars incorrectly for auction.

Here's a real-world use case that I heard at a conference. The speaker described how a sales AI assistant failed its pilot phase because the data it used lacked alignment between product SKUs and master data systems. That meant that the data was inconsistent (another quality measure) between the two systems. The initiative only succeeded after a dedicated data quality assessment, cleansing of the product content, and governance controls around SKUs were put in place.

Unreliable data is a problem because when data cannot be relied upon to deliver valid insights, organizations begin to question the value of their analytics and AI investments. It's easy to underestimate the cost of bad data until it surfaces in painful ways. Inaccurate or incomplete data can derail projects, corrupt analytics, introduce legal risk, and erode trust, both internally and externally. Have you ever heard the phrase, "garbage in, garbage out" related to building a model with poor quality data? Well, recently I've heard someone modify that phrase to, "garbage in, disaster out."

Of course, this can be use case dependent. Some people believe data quality shouldn't sideline a project if the data are "good enough." And, that might be the case. I remember when I was working at Bell Labs, and we were hung up on some data quality that we were using for a marketing program. The marketing client told us that the data only needed to be "directionally correct." That made sense for him at the time because there weren't compliance obligations and the data weren't being used to determine something material to the company. He just wanted to get a sense of what was happening with our customers.

Yet, TDWI research consistently shows that many organizations remain dissatisfied with the quality of their data. This can lead to challenges. For example, when the data that are used as the input to an AI model are of poor quality, predictions can't be trusted. When metrics in executive dashboards are based on unreliable sources, they can impact strategic decision-making. And when data are used for regulatory reporting, mistakes can lead to compliance violations, fines, and reputational damage. The point is that data quality isn't just a back-office problem. It can be a strategic issue that impacts everything from product development to customer engagement.

What Data Quality Means for Different Data Types

Historically, data quality referred to how well we managed structured data (e.g., the rows in a database or values in a table). Metrics such as completeness, accuracy, timeliness, and consistency were the gold standard. For structured data, these dimensions are well established:

- **Accuracy** refers to whether data values correctly represent the real-world entities or events they are intended to describe. For example, a customer's address should match their actual place of residence.
- **Completeness** measures whether all required data are present. A sales transaction missing a product ID or timestamp would be considered incomplete.
- **Reliability** is the extent to which data remains consistent and dependable over time and across systems. If two departments report different revenue numbers from the same source, the data lack reliability.
- **Relevance** assesses whether the data is appropriate for the task at hand. Even accurate data may be irrelevant if it does not inform the specific analysis or decision being made.
- **Timeliness** refers to whether data is available when it is needed and reflects the most current information relevant to its use.
- **Inconsistency** refers to conflicting values of the same data either in the same system or between systems.

These were also standard for audit. When I was at Bell Labs, we even built an application system to continuously monitor our operational systems that checked these metrics. That system became the standard for some of the continuous auditing work that happens today.[1] But these metrics, while still critical, don't always translate to today's environments.

As discussed, enterprises now rely on a wider source of data: emails, customer feedback, images, audio, clickstreams, and sensor data, to name a few. These often contain unstructured text data and other kinds of unstructured data or data types (e.g., the images). With these newer data types, traditional concepts such of accuracy may need to be rethought. For instance, what does "accuracy" for text mean? Should we be measuring coherence? Relevance? Plausibility?

For example, coherence in this context might refer to whether the sentences and ideas in a text document flow logically. Relevance might refer to whether the text is related to your task. Plausibility might be about whether the data appears realistic. Obviously, these kinds of metrics will require rethinking what data quality means in your organization. They will also require new tools to check for quality. These tools for text data are slowly starting to emerge but aren't there yet. There are tools that examine image quality for brightness, sharpness, or noise, but few that deal with text data.

Poor data quality in unstructured data can have real implications: poorly labeled training data can lead to biased AI systems. Unchecked inconsistencies

[1] Vasarhelyi, M. and Halper, F. (1991). The continuous audit of online systems. *Auditing: A Journal of Practice and Theory*, 10(1), pp. 110–125.

in free-text fields can skew sentiment analysis. And missing context in images can affect safety in automated decision systems. Data quality will require not only flexible definitions but targeted controls that align with use cases and ethical standards.

Data Quality Tooling

Because the volume and complexity of data have grown, it is no longer possible to perform data quality checks using only manual methods. Automated data quality tools for profiling data and providing rules-based approaches have been in existence for at least a decade. For example, a rules-based system might use human-defined thresholds, validations, and checks. It might look for duplicate data or if a customer ID was present or whether a birthdate falls into a valid range. It could check that a zip code had five digits based on a pattern rule. It would look for missing data and handle it based on what the rules said to do. Every check was a deterministic or heuristically scored rule; no statistical learning was required. If data didn't meet a quality check, it might be routed to a person and labeled as suspect.

These systems that incorporated pattern checks, range validations, and duplicate scoring are still part of the tools that are used today. However, today's platforms add more automation along with AI-infused software to be able to do trend analysis. I remember using machine learning algorithms to find poor quality data in customer datasets at Bell Labs. This was decades ago, and people asked me how I managed to find these, because they didn't know about machine learning. Well, today machine learning and other AI algorithms are embedded into tools that organizations can use to find problems in their data. The tools can be used to correct the problems, as well.

Here are some examples. Tools can alert users to reasonableness anomalies if, for instance, a time-series value goes outside a certain expected value, based on historical trends. Or, during training, the model can infer the "rules" implicit in high-quality historical data: numeric ranges, logical relationships (e.g., order date before ship date), functional dependencies, and correlations. When new records violate those learned constraints, they are surfaced as accuracy issues.

Numerous vendors have tools to help with identifying and fixing data quality issues, including ataccama, Talend (Qlik), Informatica, Precisely, Collibra, Monte Carlo, as well as the hyperscale vendors such as Google Cloud. Some vendors come at it from the pipeline perspective; some have the capabilities in other places. Many software vendors now position themselves as providing or supporting "observability."

Data Monitoring and Observability

Over the past few years, the idea of data quality has morphed into the notion of monitoring and observability. The idea is that it isn't enough to simply clean data once. The data should be monitored over time to check for new quality issues (or what is sometimes now called, the "health" of the data). In the market, monitoring and observability are slightly different, although many people use the terms interchangeably.

Data monitoring is the process of reviewing and evaluating data and its quality using software that measures and tracks performance through dashboards, alerts, and reports to ensure that the data is fit for purpose. Data observability ensures comprehensive and holistic visibility into the health of data systems using advanced analytics such as machine learning to detect anomalies, analyze root causes, and track data lineage. It is billed as being proactive and dynamic. It provides a systems-level understanding of how data behaves across pipelines. By capturing metadata and surfacing anomalies (remember active metadata from Chapter 6), observability supports root cause analysis and enables faster resolution. In TDWI research, we often see that organizations that consistently deliver value from data tend to prioritize observability, to be proactive with the health of their data.[2] Observability also helps with data governance and the vendors position themselves that way.

Some data observability platforms provide automated data lineage, incident management, and impact analysis, whereas others provide data quality checks or data quality metric monitoring. As mentioned, some of the same features or functionality can be found on other platforms, such as cloud platforms, but not necessarily labeled as observability.

Data observability platforms often consolidate various data health metrics, such as freshness, accuracy, completeness, and consistency, into a unified dashboard, offering a holistic view of data health. This may take the form of a data scorecard. These tools can also provide metrics such as system uptime/availability, data processing times, error rates in data processing, query performance metrics, resource utilization, and user engagement metrics. They provide customizable key performance indicators (KPIs), real-time alerts, and root cause analysis, and they can help assess the impact of data quality issues on downstream applications, reports, and business processes.[3]

One thing about metrics and KPIs is that they should be tied to business goals. In other words, although some companies are implementing metrics, they

[2] Halper, F. (2024). *TDWI best practices report: Data monitoring, management, and observability—best practices for success*. Available at: `tdwi.org/research/2024/09/diq-all-best-practices-report-data-monitoring-management-and-observability.aspx` [Accessed 29 June 2025].
[3] Ibid.

need to measure the right thing. There also needs to be escalation processes in place to fix any issues.

By eliminating blind spots, observability makes it far easier to demonstrate data integrity and regulatory compliance, which are core goals of any governance program. However, that doesn't mean that data governance should be an afterthought.

Data Governance Is Not an Afterthought

We'll explore governance in much more depth later in this book, but I need to make one thing clear now: governance isn't optional, and it should not be an afterthought.

Too often, governance is treated as a compliance box to check after systems are built. That mindset no longer works. Without a clear framework for ownership, access, ethics, and accountability, even the best tools and the most sophisticated models will struggle to deliver value. I've heard numerous organizations say to me that they made a mistake when they didn't consider governance when they were building their data platform.

Why should they consider governance up front? There are several reasons. First, it eliminates the issues with lineage, access controls, and quality checks that can undermine integrity and compliance. Second, it enables organizations to get more business value from their data. Finally, the cost of postponing governance can be steep in terms of fixing integrity problems after the fact. Good governance doesn't get in the way. It makes everything else more effective.

EXPERT ADVICE: DATA QUALITY AND OBSERVABILITY

Ram Kumar Nimmakayala is the principal product manager, AI/ML & Data at Western Governors University (WGU), among the largest online universities in the United States. Before moving into his current product leadership role, he led WGU's enterprise data engineering organization. Today he provides end-to-end strategic leadership across both AI products and data platforms, advising university leadership on how AI and data are designed, governed, and deployed at scale to support more than 190,000 learners and many of the institution's most critical student-success and operational workflows and decision-support processes.

Nimmakayala is frequently asked by universities and industry groups to share practical experience building reliable, production-grade AI systems. He discussed how data quality and observability have become central to making AI trustworthy at scale, and why he believes these practices should be treated as core product capabilities rather than after-the-fact checks.

(continues)

EXPERT ADVICE: DATA QUALITY AND OBSERVABILITY (CONTINUED)

Nimmakayala has watched multiple waves of analytics and AI hype come and go. "We've been talking about data quality for a while; it is not a new thing. We hear about it with every new wave and hype cycle," he notes. At WGU, where data-driven student outcomes have been a focus for decades, quality has always mattered. The university has long collected and relied on student data to drive momentum indicators, advising models, and intervention signals.

Even with that history, he believes that older approaches to data quality are no longer enough: "The focus should be on observability for both data and AI." For him, this shift is practical; in any large-scale AI deployment, issues can emerge quietly unless teams instrument for drift, freshness, and upstream data changes. "We don't want a data analyst to be the one who first notices something is broken," he says. If distributions shift, models can degrade and generative systems can produce ungrounded outputs, which is why observability becomes a reliability and trust requirement.

To make the case internally, Nimmakayala reframed observability and data quality as AI enablers rather than compliance functions. "The moment you say governance and the need to follow process, there can be resistance," he says. Instead, he advocates for a shared-responsibility mindset: "We are trying to empower people. That means shared responsibility with technology as an enabler." He often uses the phrase "AI and data enablement" to keep the focus on adoption and outcomes.

The cultural challenges he describes are common across many large, complex organizations. Initial enthusiasm around quality or observability initiatives can taper off if leaders do not actively reinforce the agenda. "It's very normal for people to be excited at the start and then get pulled into other priorities," he explains. "Leadership needs to stay on top of it." This is especially true because quality and observability cut across student services, academic operations, and institution-wide digital support experiences.

Nimmakayala grounds his approach in a data product mindset. When evaluating any quality or observability effort, he asks simple, concrete questions: What are the data products that help this? What are the data tables or functions trying to help? Who owns these? Where are the sources? This is how teams identify stewards and establish data contracts, which he believes are critical to ensuring that the data behind interventions and models remains trustworthy.

This product-centric thinking also shapes incident response. Observability has materially changed how teams work. "Teams are able to create proactive alerts if something is changing at the source level. The time it takes to solve a data problem has come down," he explains. Under his leadership, the team built a home-grown MLflow wrapper with automated detection for data shifts in ML pipelines. They have also invested energy in reducing false alarms, working to "whitelist and blacklist the alerts to reduce false alerts, to cut down the noise and improve the signal." The result is that the most critical issues are surfaced earlier and worked through faster, without overwhelming engineers and analysts.

(continues)

EXPERT ADVICE: DATA QUALITY AND OBSERVABILITY (CONTINUED)

He is realistic about the complexity involved. Like many mature institutions, WGU operates a heterogeneous environment spanning established systems, modern platforms, and custom applications, an ecosystem that makes observability both more important and more challenging. "The environment is so complex with legacy and cutting-edge tools; that makes it harder to move from being reactive to proactive," he says. This experience mirrors what he sees across the broader industry.

Vendor dynamics follow a similar pattern. Rather than singling out any one provider, Nimmakayala notes that vendors often set optimistic timelines; in practice, outcomes require sustained adoption, clear ownership, and continuous tuning. "Data keeps changing; schemas and models also change. You have to manage that side of the house while keeping existing operations running."

Because of that, prioritization has become a core principle. He explains that teams "freeze the priorities" around the most important data products and use cases instead of trying to "boil the ocean." The focus is on delivering meaningful observability coverage and faster incident resolution where it matters most rather than chasing perfect visibility everywhere.

For organizations just beginning the journey, Nimmakayala's guidance is straightforward: do not start with tools. "Fundamentally identify the business outcomes that you are trying to solve." Observability should not be a technology-first initiative; it must be tied directly to a business, or for educational institutions, student-facing need. "What is the point of doing observability? Is it a data or business problem?" he asks. Those questions shape how teams define use cases, pick metrics, and decide where to invest.

In his view, the fundamentals of data management remain constant, but their consequences under AI are far more visible. The models are only as trustworthy as the underlying data, and without observability, those models can degrade quietly. This is why Nimmakayala positions observability not as a back-office function but as a core AI-enablement capability and a safety rail for responsible deployment. The aim is not perfection; it is resilience. His work shows that when quality, observability, and stewardship are anchored in concrete outcomes and product thinking, an institution's ability to deliver reliable AI at scale improves dramatically and offers a practical pattern other mission-driven organizations can adapt for their own use.

What's Next?

This chapter introduced core ideas around data quality, observability, and the foundational role governance plays. These concepts are only the beginning of the story. In Chapter 8, I'll talk about data products, which can also help with data governance. Remember: Governance is not a formality; it is essential to building data systems—and organizations—that people can trust.

CHAPTER

8

Data Products

In this chapter, we will examine the following:

- What are data products?
- A product-thinking mindset
- How data products are becoming part of enterprise data infrastructure
- The benefits of data products for data governance
- Organizations implementing internal marketplaces and using external marketplaces

In this last chapter of Part I, about data, we look at data products: what they are, what they involve, and how they can benefit data governance. We'll also look at the concept of a marketplace for data product producers and consumers.

What Are Data Products?

Remember the concept of a data mesh and the idea of data as a product from Chapter 5? Data as a product was a concept that was somewhat management focused; it was about treating data in the same way you would a product. It had owners and product managers. Data products are the actual asset. They are curated, consumable, and governed assets that are derived from data. Data products are meant to solve a particular business problem—they are intentional. They often combine multiple data sources, layers of metadata, data quality controls, and access policies to ensure they're discoverable, trustworthy, and reusable.

The goal of a data product is to provide value, not just data. That value may come from enabling better decision-making, powering customer-facing applications, enriching AI models, or facilitating compliance. Importantly, data products are developed with users in mind, whether internal analysts, business stakeholders, external partners, or automated systems. This user orientation distinguishes data products from raw datasets or traditional reports.

Another differentiation is that a big part of the data product is that they are reusable. They are not a project meant to serve a specific purpose. The products are meant to be shared. They are consumption ready and governed. They have product owners to ensure that the product meets customer expectations and provides value.

Several types of data products are in use today:

- **Basic data products**, such as curated customer or operational datasets, are standardized and packaged for repeatable use across departments. They may be stored in a governed data lake or cloud data warehouse and include key business metrics, aligned to enterprise definitions. These products may also be user focused and include data across business units such as customer views that might integrate sales and marketing domains.
- **External-facing data products** address external demands, such as risk scoring for underwriting, market sentiment feeds, or syndicated third-party data. These are often commercialized, shared via APIs, or sold on data marketplaces. AWS, Oracle, Google Cloud, Databricks, Snowflake, and a host of other providers offer third-party data. This can be anything from demographic and consumer behavior to home buyer data to TV viewership data and beyond.
- **Analytics and AI-enabled products** include dashboards, predictive models, machine learning pipelines, and model training datasets. These products use curated data as an input and return outputs in the form of forecasts, classifications, or insights. For instance, a churn prediction model is a data product that consumes customer behavior data and returns churn probabilities.

I see many organizations creating data products that start off as domain-specific curated datasets. That can be fine as it gets them in the mindset of a product. The idea is that it is not just a one-off project file. For example, a data product could be an oncology trial dataset that consists of a highly curated and narrow set of patients that qualify for a specific kind of trial. There is a data owner and there is a data product manager for the data product (they may be the same person). The product is documented and governed. It might be used by researchers as well as clinicians. Another example might occur in an online streaming service company. Someone might create a data product of a group of customers who churned and who didn't churn. The domain here might be something like, "subscription churn for premium music-streaming customers in North America." The dataset would need to be labeled and have temporal consistency (like it is updated at a specific point in time each day or week). It would need to be documented and governed (for instance, it might not include personal information, only characteristics), perhaps registering the data product in a registry or marketplace (more about that in a minute). It would support data scientists as well as marketing or finance. The product helps to reinforce a mindset that encourages reuse and reducing duplicate ETL (extract, transform, load) and contradictory data.

There are numerous external data products too. For instance, a company might sell a curated set of weather data to insurance providers to help them manage risk. LexisNexis provides a data product that helps insurers assess drivers' risk.[1] It can become more complex than simply selling data. For example, Mastercard Spending Pulse provides a macroeconomic indicator of retail sales based on actual, near real-time spend data across various sectors.[2] This is a calculated data product. There are other, more sophisticated data products as well. For example, it might be a market intelligence data product that is a collection of competitor pricing, industry trends, and customer sentiment data. It could be an insurance risk analysis.

And a data product doesn't just need to be focused completely around data. A newer definition evolves data products to include analytics and models, really anything that the data feeds, so it might be a dashboard that is built with curated data that shows something specific, such as a financial forecast. It could be a template for a certain kind of predictive model such as a churn or fraud model. Data products can be data, templates, or applications. Curated customer data, curated operational data, risk analysis data, marketing intelligence data, dashboards, and predictive models are all examples of data products.

---

[1] LexisNexis Risk Solutions. (n.d.). Drive optics. Available at: `https://risk.lexisnexis.com/products/drive-optics#DR` [Accessed 29 June 2025].

[2] Mastercard. (n.d.). *SpendingPulse*. Available at: `https://www.mastercardservices.com/en/capabilities/spendingpulse?campaign_id=701UH00000NAfGCYA1&channel=sep&cmp=2025.q1.rca-sem-spendingpulse-brand.mastercard%20spendingpulse&keyword=mastercard%20spendingpulse&gad_source=1&gad_campaignid=21100326644&gbraid=0AAAADR5msdyI6r97ZXsdaaczTapKJ20g&gclid=Cj0KCQjwyIPDBhDBARIsAHJyyVgWyA-akH92OYd4g1h_f7fmrOo_1PpyGjPx9EfSImpOQJa5NYvItVgaAuc0EALw_wcB` [Accessed 29 June 2025].

A Product-thinking Mindset

Treating data as a product requires a shift in mindset. Data delivery isn't viewed as a project with a start and end date delivered by IT teams. Here teams adopt a product-oriented view: data assets are developed iteratively, evolve with user needs, and are continuously maintained.

This mindset comes from software product management where the products are developed for customers to solve a certain need. There may be an early or beta release to understand how customers react to them so that they can be improved. Additionally, adoption is tracked with certain metrics.

The data product owner or manager is an important part of this model. Depending on the organization, the owner might be the technical owner, and the data product manager is the business lead. They need to understand what the product will be used for and who wants it. They determine the product scope and roadmap, including what new features it will include and when. They measure how customers are reacting to the product. They ensure the customers are happy. They also coordinate development and governance of that data product with other stakeholders.

Some organizations are now embedding product managers into their data teams. They work with engineers, architects, stewards, and analysts to build cross-functional product teams. This model ensures that products are both technically sound and business aligned.

How Data Products Are Becoming Part of Enterprise Data Infrastructure

Data products are part of the enterprise data infrastructure. They are managed, versioned, and monitored much like APIs or microservices. This shift is supported by modern cloud-native infrastructure, which enables decentralized development, consistent governance, and scalable consumption.

Data products don't need to be stored in one location. They might be in a data lake, a data warehouse, a database, or a data lakehouse. They can become discoverable and accessible through a centralized data catalog or marketplace acting as a portal where users can find, understand, and request access to relevant data products.

As organizations operationalize this concept, leading practitioners emphasize that data products are not just technical assets but managed business assets. They require clear ownership, governance, and life cycle management to avoid becoming another forgotten dataset or dashboard. This means treating data products much like consumer-facing products, with defined purpose, discoverability, and accountability. This helps deliver real business value rather than remaining underutilized artifacts. Ibrahim Itani, head of Analytics and Innovation at Marathon Petroleum, has articulated this distinction with particular clarity, emphasizing why data products must be reusable, trusted, and governed.

EXPERT ADVICE: BUILDING DATA PRODUCTS THAT DELIVER BUSINESS VALUE

Ibrahim Itani, head of Analytics and Innovation at Marathon Petroleum Corporation, has a clear and disciplined view of what qualifies as a data product. For him, the term is not just a label for a dataset, report, or API. A true data product is a reusable, trustworthy, business-aligned information asset with a defined purpose and life cycle.

"A data product has to be reusable, trusted, and governed," he explains. "It must deliver data in a consumable format and tie back to a clear business outcome. You can't just design it and hope someone will come along and use it."

For Itani, the concept borrows directly from the world of consumer products. The analogy is intentional. "When we think of a product, we think of ownership, a life cycle, and a shelf life," Itani says. "It needs an identity. Each one has an identity card—it lists the SLAs, the owners, the lineage, and even an expiration date. This is not a one-and-done report. It's something we actively manage. If you remove the life cycle, then it's just an artifact. It might be used; it might not. Nobody cares."

User-centric by Design

One of the most important distinctions, Itani stresses, is that data products must be *user-centric, not data-centric*. The design process starts with the consumer, not the underlying data source.

"We don't create products based on what's easiest to build," he says. "We start with the business: What decisions do people need to make? What problems are we solving? That's how we decide what to build. And the product has to be owned by the business. If there's no owner defining the purpose, guiding iterations, and validating its usefulness, then it's not a product. It's just an output."

A Three-domain Operating Model

To sustain this approach, Itani frames product governance around three domains:

- Business: domain owners, stewards, and product owners who define the purpose and ensure value delivery
- Governance: responsible for consistency, metadata, naming conventions, and alignment across domains

(continues)

EXPERT ADVICE: BUILDING DATA PRODUCTS THAT DELIVER BUSINESS VALUE (CONTINUED)

- Digital/IT: the engineers, architects, and platform teams who build and operate the product

"If the business isn't involved," Itani cautions, "the product is bound to die."

Managing the Life Cycle

Life cycle management is a defining characteristic. Products are actively monitored for usage, drift, and relevance, with formal processes for decommissioning when they no longer serve a purpose.

"Dashboards and APIs often become orphaned, creating technical debt," he notes. "Someone builds them, uses them once, and forgets about them. That doesn't happen with a data product. We track usage, monitor drift, and retire products when they no longer deliver value. That's part of what makes it a product, not a project."

Discoverability is equally critical. "If a product isn't discoverable, no one will reuse it," Itani says. "We maintain catalogs where people can see what exists, understand its purpose, and learn how it was built. Reusability is one of the biggest value drivers."

Four Pillars of Value

In Itani's framework, data products deliver value across four pillars:

1. Democratized knowledge: capturing business logic in reusable assets rather than tribal knowledge in people's heads
2. Reuse over rework: solving a problem once and applying it many times
3. Faster delivery: accelerating solutions by building on trusted foundations
4. Trust: providing confidence through transparency, lineage, and governance

"Democratized knowledge is huge," he says. "If we preserve the business logic in a data product, documented and accessible, it doesn't disappear when someone retires or moves on. That's real value."

A Modular Architecture

The architecture is deliberately modular. Foundational data products, aligned with core domains such as Accounts Payable or Contractor Management, form the base. Derived products are then created on top, tailored for specific use cases.

"Let's say we build a foundational product that combines procurement data, contractor schedules, and plant information," Itani explains. "We can reuse that to create operational dashboards, predictive maintenance tools, or other derived products. We're not starting from scratch every time. That's the power of the model."

(continues)

EXPERT ADVICE: BUILDING DATA PRODUCTS THAT DELIVER BUSINESS VALUE (CONTINUED)

Building Trust

To make the idea tangible, Itani often uses the metaphor of a cereal box. "When you pick up a box of cereal, you see the ingredients, allergens, expiration date, even a phone number to call," he says. "That's exactly how we think of a data product. Each one has an identity card showing what's inside, who owns it, how fresh it is, and whether there are any risks or limitations. That's how you build trust."

Trust, he argues, is the foundation of being truly data-driven. "People think if they have a report in front of them, they're data-driven. That's not true. Being data-driven means making decisions with confidence—and that requires products that are transparent, consistent, and trusted."

Avoiding Common Mistakes

A common mistake, Itani warns, is confusing outputs with products. "Delivering a dataset or dashboard is not the same as delivering a data product," he says. "A product has a defined purpose, ownership, life cycle, and ongoing support. Without those, you don't have a product—you have an artifact."

Tangible Results

The payoff has been significant. "People came to us with problems: 'I can't reconcile my numbers,' 'My reports don't match,' 'I can't maintain this dashboard because the developer is gone,'" he recalls. "Once they saw that a data product could solve those issues, eliminate rework, improve quality, and provide a single source of truth, they were on board."

In some areas, the number of reports has been reduced by more than 60%. More importantly, the organization is making faster, more consistent decisions with greater confidence.

Looking Ahead

For Itani, the importance of data products will only grow. "As we move into agentic systems, where AI and digital agents act on our behalf, data products will become even more critical," he says. "These systems will need nodes they can trust, endpoints they can rely on. That's what data products are. And that's why we're investing in them—because they are the foundation not only for trusted decision-making today, but for the AI-powered systems of tomorrow."

The Benefits of Data Products for Data Governance

One of the most important features of the data product model is its alignment with federated data governance. In contrast to centralized governance models that struggle to scale, the data product approach embeds governance directly into product development and delivery. As mentioned, each product has a designated data product owner, who is accountable for data quality, compliance, metadata completeness, access control, and user satisfaction. This means that *someone is accountable*. This accountability helps clarify responsibilities and reduces the need for separate governance enforcement. In many cases, governance is controlled in a federated model, where the data product owners are accountable to their domain, but follows enterprise guidance. I've seen many organizations deploying a hub-and-spoke model, where the spokes are the domain and the hub is a small group that might set enterprise-wide policies.

Second, data products are governed by design. These products often have different layers like an onion. This includes metadata (technical, business, and operational), lineage (that shows how the data was derived and transformed), quality metrics, and access policies. They can be audited. This approach ensures that governance is not an afterthought. Instead, it's a core feature of the data product life cycle. As products evolve, governance should evolve with them.

Additionally, data products are supposed to be discoverable (so they can be reused). This means that they are cataloged with understandable metadata. And if a user wants access to the data product, they need to ask for it. That then means that the data product owner provides them with access in an access-controlled model. In some cases, the product may be open to everyone, but in cases where the owner doesn't think they should be, this can be dealt with.

Organizations Implementing Internal Marketplaces and Using External Marketplaces

Organizations are building dedicated platforms that facilitate data product creation and life cycle management. These platforms typically include ways for teams to develop and register products along with centralized metadata management for visibility and discoverability. They may include policy enforcement tools for security and privacy controls. This may take the form of a catalog.

Additionally, internal (and external) data marketplaces are emerging as important enablers for data product adoption. These platforms act as storefronts where users can discover, request, and consume data products. They promote transparency, reusability, and compliance by making products searchable, auditable, and governed.

Some core features of the internal marketplace include the ability to search for data products based on metadata and business terms. They provide documentation about the data product with potential use cases, examples, and lineage. These marketplaces include access controls and approvals for sensitive data. They may contain usage metrics that illustrate the popularity of data products and their impact. They also support feedback to improve the product.

Marketplaces can democratize data access by reducing friction. Analysts, data scientists, and business users can find the data they need, already curated, trusted, and ready for use.

Organizations are also extending this model to external data ecosystems (see Figure 8.1). Organizations and vendors publish products to commercial marketplaces, enabling them to monetize data assets or collaborate with partners. Many of the data warehouse, lake, and lakehouse vendors have data marketplaces. For instance, the Mastercard Spending Pulse data products can be found on external data marketplaces as well as obtained directly from Mastercard.

As mentioned, some examples of externally facing data products include market intelligence or ESG ratings. They mainly contain curated financial or healthcare indicators. They may include cleansed third-party data or prebuilt analytics templates or scorecards. Buyers benefit from well-documented, ready-to-use products. Sellers benefit from new revenue streams and enhanced brand value.

Of course, a marketplace model can support both internal and external users. For instance, I remember hearing about a global financial services firm that implemented a marketplace model to support both internal and external use cases. Internally, it allowed business units to request and deploy new data products with minimal IT intervention. Externally, it enabled controlled sharing of

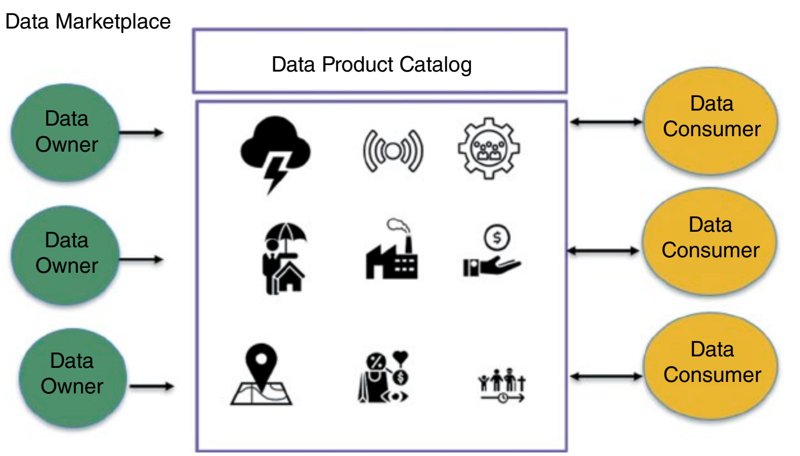

Figure 8.1: Data marketplace.

financial benchmark datasets with partners, all governed through contracts and service level agreements.

What's Next?

This takes us to the end of Part I, about the data foundation for AI. Hopefully you have a better understanding of some of the basics of the data landscape that supports AI and what you need to be thinking about in order to support AI that uses company data. You've heard about some of the newer platforms and buzzword terms that are emerging in the market today for data management including data mesh, active metadata, and data products. Hopefully you agree with the concept that governance should not be an afterthought and understand why it is so critical.

In Part II, we start to tackle analytics, beginning with the concept of democratizing analytics, to open it up to a broader user base.

Part

III

Democratizing Analytics

CHAPTER

9

The Self-service Imperative

In this chapter, we will examine the following:

- What democratizing analytics is all about—and why it matters
- Why BI democratization is important for the AI journey
- Why organizations struggle with democratization

In Part II we discussed the data foundation required for AI. That same foundation is important for analytics (although it may not need to be as robust). In the next few chapters, we take a brief but critical detour to explore how organizations can democratize analytics and build the literacy needed to support it. Most companies now provide BI tools so employees can derive insights from data, yet ease-of-use alone isn't enough. A governed data foundation enables self-service analytics and helps employees learn how to work with data, strengthening organizational, data, and governance readiness for AI.

What Democratizing Analytics Is All About—And Why It Matters

The concept of democratizing BI, i.e., making it available to business users who need it, has been around since the early 2000s. The idea behind it was to remove the barriers to getting access to the data for BI and provide the tools needed for people to easily answer their own business questions.

Of course, many business users have been using spreadsheets (and continue to do so) to answer their questions; however, the spreadsheet typically provides the ability to analyze a piece of the data. They act off siloed data, which as we have seen can cause problems with data consistency, definitions, and data quality. The spreadsheet does not provide shared, trustworthy information and analytics capabilities across the whole organization.

Additionally, many organizations still employ specialized team(s) who are responsible for analyzing company data. They perform the analysis for different business units in the company. This was the old-style BI team that created reports and dashboards. I ran a similar group at AT&T years ago; we worked with executives to analyze service performance, highlight growth, surface issues, and recommend next steps. Although that model delivers value, it cannot scale to every ad hoc question the business wants to explore.

In the early 2000s, vendors such as Tableau started to popularize the idea of democratization. The thought was that business users wanted to be able to slice and dice their data the way that they wanted. They wanted to be able to ask a question, get an answer, and then ask a follow-up question. They didn't want to wait for a weekly report. These self-service solutions promised to enable nontechnical users to be productive because they are easier to use, do not require coding, and do not require IT to set up all data access, queries, visualizations, and preparation. These modern platforms packaged data preparation, visualization, and even basic predictive functions into point-and-click and drag-and-drop workflows, ideally allowing nontechnical users to explore patterns, build dashboards, and test hypotheses without needing to know SQL (structured query language) or waiting for IT support. By lowering the skills threshold, the idea was that self-service freed data professionals to focus on higher-value projects while empowering domain experts to answer everyday questions on their own.

SQL (STRUCTURED QUERY LANGUAGE) IN PLAIN ENGLISH

Think of SQL as the way you ask a database a question. You tell it what columns you want (using a SELECT command), where to look for the data (FROM a table), and any filters you need (WHERE conditions). A query to sum sales data for 2025 from a table in a database called sales might look something like this:

```
SELECT SUM(amount) AS sales_2025
FROM sales
WHERE sale_date >= '2025-01-01'
AND sale_date < '2026-01-01';
```

In practice, queries can get quite complex. Mastering even basic syntax requires training, and complex joins can intimidate nontechnical users.

Modern BI tools often generate SQL behind the scenes, but understanding its basics helps power users debug or optimize a report dashboard and appreciate what happens under the hood.

Here is an example of a self-service analysis. Suppose a sales director needs to know how many units of a product were sold by region over a certain period and who the top sales reps were for that period. In the past, they had to ask IT for the data or learn SQL to pull data from potentially disparate systems. They then manually stitched the data together in Excel, defined “unit” their own way, and built a chart. That is at least three steps, which can take time. With self-service tools against a governed data foundation, they could build visualizations by themselves, drill into the data, and share the insights with colleagues. This was marketed as quick and painless. Users can be more productive (see Figure 9.1).

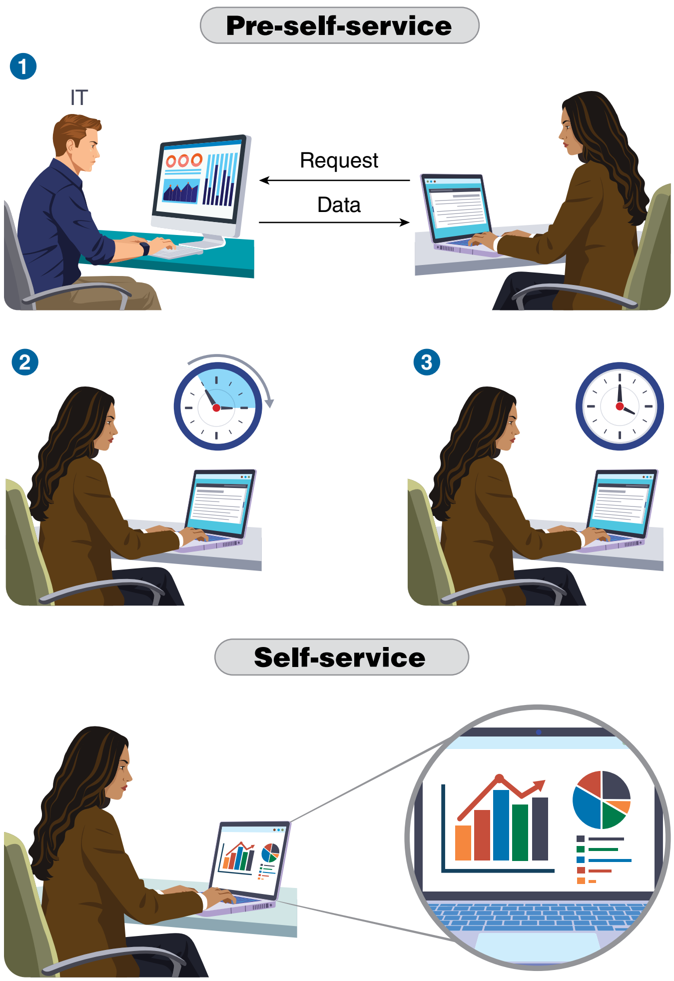

Figure 9.1: Pre-self-service vs. self-service.

So, the idea is that democratizing analytics can pay off in numerous ways:

- **Faster, better decisions:** When every employee can access governed data and intuitive tools, they no longer wait for answers. They can make data-driven decisions in real time and follow each insight with the inevitable second question, improving both speed and quality of outcomes.
- **Greater agility and collaboration:** Sharing a single source of truth creates a common view of goals and performance. Cross-functional teams can iterate together, adapting to market shifts with a data-driven perspective that can result in higher growth, customer satisfaction, and operational efficiency.
- **A more engaged workforce:** Easy-to-use analytics tools can help empower people and build skills. Employees feel enabled rather than bottlenecked, which can boost job satisfaction and reduce turnover. On the IT side, analysts can focus on advanced projects, which provide them with a sense of satisfaction, while business users gain autonomy. This can be a win-win for both groups.
- **Stronger customer experiences and retention:** Self-service analytics lets teams monitor digital behavior, spot customer issues, and personalize interactions in near real time. This can improve customer satisfaction and potentially increase revenue growth.
- **Clearer proof of impact:** With governed data used in self-service, organizations can tie analytics work to outcomes, such as conversion rates, churn reduction, or cost savings. This can help in establishing the return on investment that maintains funding and executive sponsorship.

While the self-service idea sounds good in theory, in many companies the reality was different. I never saw the uptake of self-service to every business user in every organization. In many companies, there was still an analytics team (either centralized or distributed into the business units). These teams still did the analysis for many of the business users or at least they created the dashboards or visualizations that the business users interacted with. The reality was that these tools, while simpler to use, were still perhaps not easy enough.

One fairly recent trend to try to make them easier is to infuse AI into BI (known as AI for BI). More vendors are providing AI-infused solutions for BI. These tools go by different names including auto-insights, augmented analytics, or decision intelligence. They have different features. For instance, in some, as soon as a user logs in, AI engines run anomaly-detection and trend algorithms in the background to push auto-insights, such as spikes in churn or sudden inventory dips directly onto the landing page of the tool, so the business user sees them as soon as they log in. They are essentially seeing what the AI behind the scenes is surfacing as important. These tools are described in more detail in Chapter 10. Of course, the data foundation for this needs to be in place as well as the user's ability to think critically about the outcomes (more about that later).

Vendors such as ThoughtSpot, Microsoft, and Tableau have also been embedding natural language query (NLQ) capabilities into their products, even before generative AI became so popular. The goal was to enable users to easily ask questions of their data without needing even a drag-and-drop interface, much less SQL. They could ask their questions in a natural language way. With the advent of generative AI, these kinds of features (sometimes known as conversational BI or generative BI or GenBI) are being introduced at a fast pace.

Generative AI is potentially a game changer for many BI tools, and most of the BI tools on the market have some sort of generative AI capabilities. They now allow users to ask questions using a natural language interface, similar to what you'd see in ChatGPT or Perplexity. Not only do BI tools have this feature, but some of the data platforms we discussed previously have also incorporated this capability. Think about it. These tools can already manage all kinds of data. They have governance capabilities built into the platform. They have data catalogs. Some have semantic layers. This makes it easier to support asking business questions of your data. On top of this, they are providing copilot-like features to easily ask questions and turn those questions into SQL behind the scenes to query the data warehouse.

Generative AI with BI has the potential to truly enable democratization of BI. In TDWI surveys, we typically see that less than 50% of business users in an organization make use of self-service. Yet with a generative AI front end, this number may increase significantly.

Of course, not everyone wants to make use of self-service. Some people feel this way because they don't have the motivation. Others because they don't have the skills or permission. For example, in terms of motivation, some people don't see data analysis as relevant to their role. In terms of permissions, there may be security reasons why organizations don't want to open up BI to everyone, even if they have the skills. Of course, some people may lack the literacy for BI, and the organization does not provide any resources to help them. So, the reality is that organizations may never get to 100% use of self-service, even with generative BI and that is okay. However, for those users who need to make use of it, the idea is that they can and do so easily.

Why BI Democratization Is Important for the AI Journey

While self-service (sans generative AI and GenBI) is relatively old news, I bring it up because, if done the right way, self-service can help organizations get ready for AI and even this next stage of the self-service journey with generative AI. How is that the case?

First, self-service can help lay the data foundation for AI. As we've discussed, a solid, trusted, and accessible data foundation is going to be critical for AI. We saw this in Chapter 3 in the AI readiness model under data readiness, which includes factors such as the platforms that manage the data, the integrity of the data, and support for diverse data. It is also key for BI. While the data may be more diverse, more real time, and higher volume for AI, putting a curated and governed data foundation in place for BI helps to set the stage. It ensures that data is managed and that there is a flow of new fresh data to the enterprise. It ensures that at least this data has high quality and is governed. It starts to build an enriched data (i.e., adding additional information to an existing dataset to make it more complete, accurate, or useful) foundation for AI model building. It also helps to foster collaboration across the company because people have a shared understanding of the data, especially in those companies that have already deployed something like a data catalog.

On the talent front, BI democratization also helps ensure that some of the skills are there for building the data foundation for AI. For instance, an organization may have decided to migrate its data to a cloud platform. That migration may include sophisticated data management tools. It might include implementation of a data catalog. Maybe the company is migrating to a cloud data warehouse or lakehouse. This foundation is housing its structured data and perhaps unstructured data as well. That data might be used to support BI and the initial stages of AI, for instance using it to support light-weight predictive models (e.g., a forecast or a simple churn model) and even generative AI to analyze its data. That migration effort can help build familiarity with the platforms and the skills that will be needed by the data team moving forward.

Democratizing BI can also help lay the cultural and technical groundwork for AI. By encouraging people to work with data every day, organizations can raise overall data literacy and help create a data culture. Empowering business users and others with democratized BI helps them to become more innovative. That same mindset can be helpful when it comes to AI.

I will talk in much more detail about data literacy and culture in the upcoming chapters because they can make or break any analytics effort.

Why Organizations Struggle with Democratization

I've been monitoring the use of self-service in my work at TDWI. Interestingly, it has been the top priority for analytics for more than five years (since I started asking the question in surveys). This year, generative AI tied with it (according to an unpublished 2025 TDWI survey). The question then becomes, why is that the case? Why haven't organizations moved forward in their analytics journey, past self-service to implement more sophisticated tools including machine learning and AI?

I think that the answer to that is complex. On the one hand, there are the organizational aspects of succeeding with self-service BI. To succeed requires organizational support (from the top, with executives walking the walk), funding, training, and a focus on change management. Even though these self-service tools are easy to use, they may still be hard for some.

On the other hand, there can also be technical barriers, including data quality, data access, data understanding, and tool sprawl. For instance, if users can't discover the data they want for analysis, if that data has poor quality, or if they can't access it (or there is no governance over access and everyone is accessing it), these can all be problems. Organizations might also make use of multiple tools for BI (and most do). However, this can also get out of control and that often means that there can be problems with maintaining, governing, and even training on these systems. This used to be called *shadow IT*. Shadow IT refers to any hardware, software, or cloud service that employees buy and use without the knowledge or approval of the organization's official data or IT department. This can cause security, compliance, and support risks. With the advent of generative AI, this concept is now sometimes referred to as shadow AI. Shadow AI is the same idea: the unsanctioned use of AI tools or models within an organization, developed, purchased, or deployed outside official IT and governance processes. This too can create risks around security, compliance, and data quality.

Table 9.1 outlines these challenges.

Table 9.1: Challenges with Democratization

ORGANIZATIONAL ISSUES		
Issue Type	**Description**	**What Helps**
Executive support	This may start off strong but may then waiver if leaders don't stress the idea of data-driven decisions. That includes walking the walk and using tools to analyze data. It includes supporting the organization through change.	Engage executives early; show quick wins tied to business goals (see Chapter 13).
Training and literacy	Even with modern tools, interpreting data and outputs may not be intuitive to many.	Build an intentional plan for data literacy; develop role-based literacy (Chapter 11).
Cultural resistance	Some people may resist democratization because they fear it may impact their job or expose underperformance. It may be due to an unconscious cause.	Incentivize data use; lead with empathy (Chapter 12).
Change management	Business users may not want to adopt another tool (e.g., maybe they like their spreadsheets).	Implement a formal change management program (Chapter 12).
Funding	Funding can dry up if the move to self-service doesn't happen quickly or if it is viewed as a project and not a program.	Appoint a program owner; track KPIs tied to business impact (Chapter 13).

(continues)

Table 9.1: (Continued)

TECHNICAL ISSUES		
Issue Type	**Description**	**What Helps**
Data quality	If source systems for self-service tools contain low-quality data, that can result in the old "garbage in, garbage out" scenario, which can erode trust.	Embed data governance early that addresses data integrity and builds trust; it cannot be an afterthought (Part V).
Security and privacy	Role-based access, row-level filters, and masking policies are often added on late, creating problems for data engineers and lengthy approval cycles for analysts.	Bake into design (Part V).
Fragmented landscape	Legacy on-premises warehouses, new lakehouses, SaaS application data, and external feeds all live in silos. Users struggle to discover what exists, let alone bring that data together to unify it.	Establish an integrated and unified data foundation (Part II).
Governance	Governance is the rules, policies, and procedures that guide both the business use and the technical management of data across the organization. It can be hard to get a governance program off the ground and get people to buy into it. People are often apathetic when it comes to governance. The trick is to balance governance with enablement. With self-service BI and AI, it isn't just about data governance; organizations must worry about analytics and AI governance too. These are distinct from data governance in that they are dealing with analytics and AI assets as opposed to data. Most have not started to think about this.	Balance enablement with control (Part V)
Tool sprawl	M&A activity, departmental purchases, and cloud migrations leave organizations with overlapping capabilities, conflicting data, and conflicting metrics.	Rationalize tools; align on enterprise standard (Part II); appoint right leadership in a hub-and-spoke model (e.g., CDO).
Performance bottlenecks	As usage scales, poorly tuned semantic layers or unoptimized SQL queries lead to slow dashboards, which can dampen enthusiasm just as adoption rises. Optimized performance is needed.	Ensure a modern data foundation (Part II).

(continues)

Table 9.1: (Continued)

TECHNICAL ISSUES		
Issue Type	**Description**	**What Helps**
Data discoverability	As mentioned, if users don't have a way of finding and understanding the data that they want to use for analysis, that will also impact self-service enthusiasm. Ideally, that data is discoverable with business metadata and some organizations struggle to put this together.	Implement a data catalog with business metadata (Part II); build the culture of collaboration (Chapter 12).
Need for a semantics model	The semantic model began with BI systems as a way to provide business-friendly layer of dimensions, metrics, and hierarchies (the language of OLAP) so nontechnical users could query data in their own language.	Create and maintain semantic layer (Chapter 10).

The following story highlights how one organization put a self-service program in place that addresses planning, governance, and education.

EXPERT ADVICE: BUILDING A SELF-SERVICE CULTURE AT LAKELAND REGIONAL HEALTH

Ray Ready, vice president of analytics at Lakeland Regional Health, describes self-service BI as an evolving capability that requires careful planning, governance, and education. "We implemented some limited self-service BI way back before COVID," he says, explaining that the initial approach was to identify subject matter experts who understood how data in their departments was collected and how their business processes worked. These "power users" were given access to build analyses in their own domains, whether in HR, the ER, radiology, or pharmacy, while the centralized analytics team maintained oversight.

Over time, the program expanded to include more people, particularly managers and supervisors, but Ready stresses that this was not a blanket rollout. "We didn't want to just give it to everyone in the house and create a shotgun approach," he says. The focus shifted to developing internal programs to upskill these new users, giving them adequate support to avoid major mistakes, such as building a custom query without considering its implications and presenting faulty results to leadership.

For Ready, data literacy is inseparable from self-service. "When you want somebody to be data literate, it's more than just do they understand the tools," he explains. "It goes really deep into: do they understand how the data itself is generated by the business processes and workflows that frontline staff perform?" Without that understanding, even seemingly simple measures, like patient wait times, can be misinterpreted. For example, if check-in is self-reported on a clipboard

(continues)

EXPERT ADVICE: BUILDING A SELF-SERVICE CULTURE AT LAKELAND REGIONAL HEALTH (CONTINUED)

vs. automatically checking in when you walk in the door through a kiosk or mobile app, there can be a difference in data quality.

To address this, the Analytics Division offers workshops open to anyone, with no sign-up required. Each session covers specific data marts that the health centers deploy, their contents, data sources, appropriate uses, and pitfalls to avoid. Ready's team supplements this with video training and ongoing discussions in the data governance council and its subgroups on quality, access, and literacy. The organization's data catalog plays a key role, allowing users to see validated content, understand its data sources and logic, and learn from peer-reviewed or management-approved work.

Still, challenges remain. "You can put all this stuff in play … the risk is that users try to answer questions that are more complex than the self-service data sets allow for, and it can create issues," Ready says. Simple, well-defined queries, such as "how many hip procedures did we do in the last 12 months" work well. But complex questions often require connecting multiple data marts correctly, something many users don't realize. Encouraging them to seek help is critical, though some likely hesitate out of concern for how much time it may take or other reasons.

Measuring success goes beyond usage metrics. Ready looks for "actionable success," cases where self-service has changed how a department operates, leading to savings, new revenue, reduced risk, or improved outcomes. His team tracks adoption of validated work, removes unused content, and seeks out case studies tying self-service efforts to strategic KPIs. For example, if a new report built from the denials data mart coincides with a measurable drop in insurance denials, they investigate and share that story across the organization.

Lakeland is also testing generative AI features in its BI tools but with significant guardrails. Ready describes rolling out Tableau Pulse to executives, which allows limited natural language queries tied to prevalidated KPIs. "You're not allowing the system to respond to open-ended questions that could potentially provide faulty information," he explains. His caution is clear: "My competence is tied directly to the AI's competence … why would you want to put something into the hands of people who are going to make very impactful decisions unless you are fully confident in what it's going to do?"

As Ready sees it, self-service is a balance between democratization and discipline. It empowers staff to answer many day-to-day questions without waiting for the analytics team, but it also requires strong governance, ongoing education, and a culture where asking for help is seen as a strength rather than a weakness.

What's Next?

As we've seen in this chapter, democratizing BI is about more than just putting dashboards in the hands of business users. Done well, self-service can increase agility, engagement, and decision-making quality. But it also requires the right data foundation, tools, training, and governance to succeed. These elements, along with organizational support and a willingness to manage change, can help prepare organizations for the next phase of the analytics journey.

In Chapter 10, we'll look at how AI, and especially generative AI, is transforming what self-service can mean. Instead of selecting filters or dragging and dropping data fields, users can now ask questions in plain language. We're seeing AI infused into every step of the BI life cycle: from data prep and enrichment to pattern detection and root cause analysis. Chapter 10 will take a closer look at how AI is being embedded into BI platforms, what generative BI can do, and why governance, semantics, and trust matter more than ever.

CHAPTER

10

Artificial Intelligence in Business Intelligence

In this chapter, we will examine the following:

- How artificial intelligence (AI) is being infused into business intelligence (BI)
- Generative AI and natural language processing (NLP) front ends for business users
- Slapping a generative AI front end onto your data
- Implementations against unstructured data
- The role of trust and governance in self-service and generative BI

The previous chapter discussed the evolution of self-service and data democratization. This chapter takes that discussion a step further by exploring how AI, particularly generative AI, is reshaping business intelligence. AI is no longer just a backend enabler for analytics. From automating data preparation to powering conversational interfaces that let people "talk" to their data, AI is changing how insights are created. Understanding how AI is being infused into BI, and how organizations are using it as a natural language front end for analytics, is important to understanding where modern data and analytics are headed next.

How AI Is Being Infused into BI

Chapter 9 looked at the path of business intelligence from IT-authored reports and dashboards, through the rise of self-service tools, to today's natural language experiences that use generative AI. Infusing AI directly into BI platforms builds on those gains: while governed dashboards remain important for enterprise tracking metrics and key performance indicators, AI cuts the manual exploration time, surfaces hidden drivers automatically, and may eventually enable every user to ask data questions in plain English (or any other language you want).

AI is being infused into BI across the BI life cycle, from data collection and preparation to data analysis, visualization, and even decision-making. Generative AI solutions in many cases are now even providing a natural language interface to help make tasks in the life cycle even easier.

Think about the phases in the BI life cycle:

1. **Data preparation:** This involves the integration and the cleansing of data to make it useful for analysis. It involves data transformation. I already talked about how AI-infused tools are being used in data cleansing. These tools can help handle standardizing data for use in BI (for instance, changing Ave, avenue, or Av all to Avenue). They can be used to transform the data; for instance, they can highlight missing data and problems with data formats such as control characters in the data. They can help to group data, such as revenue data into different buckets such as high, medium, and low. This can all now be accomplished in some tools using a natural language interface to ask the system to perform these actions. Additionally, AI infused into BI systems can help to enrich your data. For instance, if you want to marry sentiment data from customer feedback, generative AI can extract sentiment from the feedback and then combine that with other data, such as billing history. These data are then stored in a repository such as a data warehouse for use in data analysis.
2. **Data analysis:** AI-infused BI tools can provide capabilities such as automated pattern detection. Using machine learning infused into the software, for instance, AI-driven BI can analyze large amounts of data to uncover patterns that it can then surface as important. Solutions such as ThoughtSpot do this today. Some tools, such as ThoughtSpot, IBM Cognos Analytics, and Sisense, can even generate narratives about these findings in addition to surfacing them. Additionally, many tools don't stop with descriptive analytics. They can provide root cause analysis, i.e., diagnostic analysis. For instance, if a large number of customers stopped using a product, the tool might be able to surface that the sentiment around the product was poor. AI-infused BI can be used to build predictive models such as a forecasting model or something more sophisticated such as churn model using a decision tree. This is known as *autoML* (ML for machine learning). Some tools let you select the target or outcome variable of interest, specify the data that you think might be predictive, and the system will build you a predictive model. Some tools will even create features (remember, those derived attributes) for you and then select the best model and build

it. The system will then provide an explanation of which features are most important, often in chart format, with features in descending order of importance. Some will even provide a narrative explanation. Popular autoML tools include DataRobot, Dataiku, H20.ai, and Google Cloud.

3. **Operationalizing analytics:** Analytics must also be put into production, for use by those who need it. In some cases, this includes embedding dashboards into applications. While not AI-infused per se, these solutions make it easy to build these applications. Vendors including Looker (part of Google Cloud), Salesforce, and Zoho support this operationalization.
4. **Data and analytics governance:** This isn't really a last phase but rather overlies and underlies the whole life cycle. I spoke briefly about data governance already and will do so more in the upcoming chapters. But AI is being used to help with governance tasks. For instance, some governance tools might automatically identify and label sensitive data such as Social Security numbers. AI-based lineage tools can infer lineage from data transformation logs, SQL scripts, or data access patterns. AI-infused data quality tools can detect potential quality problems in data. Some tools let you ask questions in a natural language way. Examples of vendors that provide AI-infused data governance tools include Alation, Collibra, Microsoft, IBM, and Informatica (there are more). AI tools provide analytics lineage (e.g., what data and analytics were used to create a dashboard), or AI can scan BI environments to identify what reports, dashboards, and metrics exist—and map relationships among them. They can autogenerate narratives and explanations for visualizations. Vendors including Alation, Acceldata, and Atlan use machine learning to detect inconsistencies in metrics and visualizations.

Generative AI/NLP Front Ends for Business Users

Of course, many tools are now also enabling users to simply ask questions of the data in a natural language way. As I mentioned in Chapter 9, vendors have been providing natural language query interfaces for some time in an attempt to make their tools easier to use and appeal to more business users. However, with the advent of generative AI, these front-end prompt-type interfaces are making it even easier to have a conversation with your data. Some tools have their own large language models (LLMs) built into or integrated with the software. Others also allow you to link to your favorite LLM and use that. For instance, using an AI-infused BI tool, I can ask my data about company sales for the last three years. I can then drill deeper into the data, even asking the solution to perform statistical analyses or build regression model or decision trees. I'll discuss more about LLMs and how they work in

Chapter 17. For now, it is important to understand that these kinds of interfaces with generative AI tools under the hood are available and may finally help to increase adoption of self-service.

Most BI vendors offer these interfaces, including ThoughtSpot, Tableau, IBM, Microsoft, Amazon, and others. As mentioned in the previous chapter they are sometimes referred to as *conversational BI* or *generative BI*. These tools offer features such as natural language interfaces, integration with various data platforms (or they can be part of the platform), and the ability to create dashboards and other visualizations (e.g., stories) based on your data. Some provide sample questions to ask to you based on your data. Some allow you to use voice commands rather than writing out your prompt.

Likewise, AI is being embedded into enterprise resource planning (ERP) and customer relationship management (CRM) systems so that users can ask direct questions of this kind of data. In some cases, these features are tightly integrated into the workflow, so sales and service teams can take action without leaving the application environment. For instance, Salesforce Einstein has integrated generative AI through its Einstein Copilot, a conversational assistant embedded directly into Salesforce applications. This tool enables users to analyze pipeline data using natural language commands, among other functionalities. Similarly, Microsoft Dynamics 365 (powered by Azure OpenAI) has introduced Copilot across its suite, allowing users to generate real-time sales insights, among other functions. On the ERP front, SAP has embedded generative AI into its product suite through its Joule AI assistant that works across finance, supply chain, human resources, and procurement functions. Joule enables users to ask natural language questions and receive context-aware responses derived from their enterprise data. Oracle has done something similar.

Slapping a Generative AI Front End on Your Data

I've seen some companies think that they can just put a front-end LLM on top of their own data (say the data warehouse) to perform data analysis. However, there are several reasons why that can go wrong. For instance, the LLM may produce hallucinations (e.g., a fabrication, the wrong answer) because it does not have the context of your data. It doesn't understand your data, how it is organized, or what it means. That information needs to be provided to it. That might require building a semantic layer between your data and your AI that may give a clearer, business-oriented view of your data. For example, terms like prospect or client may map to a single business entity in the semantic layer. This helps reduce misunderstandings (although it may not completely eliminate them). But that means building a semantic layer and continuously updating it.

Likewise, to do this would require putting controls, such as access controls, in place to determine who can access the data. It may require dealing with compliance considerations such as GDPR or the EU AI Act that deal with privacy considerations. It would also require integrating the LLM with your data. Additionally, it would require dealing with issues such as explainability and trust. Many of the commercial tools won't provide you with an answer if the system doesn't have the data. There are several other controls these systems put in place.

I think you get the idea that this might be an issue unless you have a skilled team to build something like this. And, there are organizations moving in this direction. Additionally, that isn't to say that you can't use generative AI against some of your data. For instance, I will download survey data, provide the generative AI (private) with instructions about how to read the data, and then ask questions of it. Of course, I always make sure to check back with the raw data to perform spot checks to make sure that the generative AI is producing correct information. It should be able to tell you where it got the data from and what techniques it is using to analyze it.

Semantic Layers and Generative BI

As previously mentioned, organizations will discover that simply putting an LLM in front of a data warehouse or any other platform is not enough. One of the critical components emerging in successful generative BI implementations is the semantic layer.

A universal semantic layer, such as that provided by AtScale, is sort of a middle layer between data storage and the user interface. It defines business-friendly terms and metrics on top of raw data. It maps technical data constructs to business entities such as "customer" or "revenue." When you ask a generative BI tool for "revenue by customer segment," the semantic layer ensures that "revenue" and "customer segment" are understood according to business logic. Of course, semantic layers have been around in BI tools for many years. The issue now is how these are used in generative BI.

This layer can help reduce hallucinations by providing grounded definitions so that the LLM is less likely to invent faulty responses. Additionally, users can trace responses back to well-defined data logic. In practical terms, semantic layers help generative BI systems produce more relevant visualizations, summaries, and explanations. For example, if "new customer" is defined based on signup date and prior transactions, the semantic layer ensures this logic is always applied consistently across analyses.

Maintaining this layer does require effort. But in environments where multiple dashboards already exist with slightly different definitions of the same metric, the semantic layer becomes an important foundation for trusted BI and scalable generative interfaces.

Implementations Against Unstructured Data

Now, another movement is afoot too. Whereas in the past self-service BI was primarily against structured data, I see many organizations wanting to analyze their unstructured data. Unstructured data are undergoing a sort of resurgence in popularity, and I think that is great. For years, companies have been collecting this unstructured data, but they haven't been analyzing it in a meaningful way.

About 15 years ago, some organizations started to perform text analytics—the process of analyzing unstructured text, extracting relevant information, and transforming it into structured information that can be leveraged in various ways. The analysis and extraction processes used in text analytics took advantage of techniques that originated in computational linguistics, statistics, and other computer science disciplines. The transformed information from text analytics could be combined with structured data (e.g., sales and demographic data) and analyzed using various business intelligence or predictive analytics techniques. However, text analytics was never widely adopted, often because it was very labor intensive and the results weren't that accurate.

Yet, if you think about call center notes or trouble tickets, there is a lot of good insights to be gained from them. Generative AI can help to summarize these issues or classify them. This is another form of self-service and is one of the first use cases for generative AI against unstructured data that I've been seeing.

Additionally, while structured data remains the foundation of most BI systems, the vast amount of enterprise data are unstructured: documents, documentation, call logs, call center notes, incident reports, emails, and more. I talked about these in Chapter 4. These sources can be rich in context and customer voice. These are often not the kind of data that data professionals think about as data, but they are data. I told you my call center note story earlier in this book. It is interesting that organizations are starting to finally make use of these data sources to gain insight, speed up processes, and improve productivity. I've heard numerous examples of organizations using generative AI to summarize call center notes and incident reports. These are text documents.

There are other examples of how traditional and generative AI are being used against unstructured data sources. Some of them have had issues, but they are interesting to examine, nonetheless.

- **Document intelligence in legal sector:** Some legal firms are starting to use generative AI tools to extract information from legal documents. This might include clauses (such as a termination clause), key entities (names, dates, locations), or defined terms. For instance, generative AI can scan a document to look for a notice period for termination or penalties for early termination rather than trying to do a manual search for the word "termination." In addition to extracting clauses, generative AI may be used to extract intellectual property information by understanding the context to look for ownership or license restrictions. Additionally, it could be used to identify inconsistencies or omissions in a document and for a host of other use cases, such as classifying content or summarizing long legal contracts or case files.

Of course, there can be issues with this. Back in February 2025, two lawyers representing My Pillow violated court rules when they filed documents filled with many mistakes, including hallucinated cases. Recently, they were each fined $3,000.[1] This case highlighted the importance of a human in the loop. Yet, other law firms are starting to make use of the technology and specialized legal technology firms that are in the market (e.g., Harvey AI, LexisNexis' Lexis+ AI, Thomas Reuters CoCounsel, and more), which are often billed as legal assistants.

- **Insurance claims handling:** Insurance companies have been using AI to detect fraud for quite some time. Now, generative AI is being used to analyze claims data for anomalies and patterns indicative of fraud. Chatbots provide updates of claim statuses. There are numerous use cases in the insurance industry of companies using generative AI. For instance, a company called Tractable AI employs generative AI to determine the extent of car or property damage from photos customers upload as part of their claim forms. Geico makes use of this.[2]

  Of course, there have already been cases against health insurers such as United Health Group, Cigna, and others that allege that claims were denied based on AI review with no humans in the loop. For instance, in 2024, the estate of Gene B. Lokken et al. (a class action lawsuit) filed suit against the United Health Group alleging that the health insurance company's reliance on AI tools to deny certain medical claims under Medicare Advantage plans constituted breach of contract, breach of the implied covenant of good faith and fair dealing, unjust enrichment, and insurance bad faith.[3]

- **Customer support transformation through chat/email analytics:** Some organizations feed customer service emails and chat threads into LLMs to classify cases, summarize interactions, or even draft first-level replies. This can provide good insight into what customers are thinking. It can also improve case routing and reduce agent load. For example, T-Mobile,

---

[1] Diaz, J. (2025). A recent high-profile case of AI hallucination serves as a stark warning. *NPR Law*. (10 July). Available at: `https://www.npr.org/2025/07/10/nx-s1-5463512/ai-courts-lawyers-mypillow-fines#:~:text=Trust%20and%20verify,that%20is%20sourced%2C%20Charlotin%20said` [Accessed 21 July 2025].

[2] Dilmegini, C. (2025). *Generative AI in insurance: 10 use cases & 5 challenges*. Available at: `https://research.aimultiple.com/generative-ai-in-insurance/` [Accessed 23 July 2025].

[3] Health Care Litigation Tracker. (2024). *Estate of Gene B. Lokken et al. v. UnitedHealth Group, Inc. et al.—Health Care Litigation Tracker* [online]. Available at: `https://litigationtracker.law.georgetown.edu/litigation/estate-of-gene-b-lokken-the-et-al-v-unitedhealth-group-inc-et-al/#:~:text=Why%20this%20Matters:,treatments%20and%20harm%20patient%20health` [Accessed 21 July 2025].

a leading telecommunications provider, was having difficulty managing more than 15,000 customer feedback SMS messages each month. They made use of NLP and sentiment analysis tools from SentiSquare to group feedback into categories which helped T-Mobile better understand its customers and issues.[4] Likewise, in certain models of Ford vehicles, drivers can record their feedback via the Ford Record Feedback button. This opens a microphone to send a message directly to an engineer about any feature.[5] Ford then analyzes this data using AI algorithms.

Some of these examples will make use of retrieval-augmented generation (RAG), which I'll talk about more in Chapter 17. These examples show the expanding potential of analytics to pull insights not just from structured numerical data but also from text. I've heard companies speak about already achieving return on investment by using generative AI for administrative tasks such as summarizing customer interactions in call centers or reformatting documents. It highlights the importance of unstructured data in the data architecture. This may seem slightly off topic for generative BI, but unstructured data is the new frontier of analytics and AI.

The Role of Trust and Governance in Self-service and Generative BI

Back to self-service and generative BI, as it becomes more common, the challenge of trust becomes more pronounced. For instance, if users receive different answers depending on the interface (e.g., dashboard versus conversational prompt), trust will erode quickly. I will talk much more about governance and trust later in the book, but it is important to mention it here as well because governance shouldn't be an afterthought.

To support trust, vendors are embedding more governance into their generative capabilities. So, when you ask a question of a generative system, such as a generative BI system, you have more confidence that the answer is not a fabrication. At a minimum, this should include telling you where the data came

[4] SentiSquare. (2024). *T-Mobile's customer satisfaction boost with SentiSquare AI* [online]. Available at: `https://www.sentisquare.com/en/success-stories-en/customer-feedback-analytics-at-t-mobile#:~:text=Comprehensive%20Feedback%20Analysis:%20SentiSquare%20AI,concerns%20in%20a%20timely%20manner` [Accessed 21 July 2025].

[5] Priddle, A. (2023). You can now complain directly to Ford engineers from your car. *Motortrend*, 15 November. Available at: `https://www.motortrend.com/news/ford-record-feedback-button-instant-complaint-engineer-service` [Accessed 21 July 2025].

from. In terms of quality and governance, however, this is an emerging area. A few techniques include the following:

- **Confidence scoring:** Very few vendors offer confidence scoring for generative BI or AI. The idea behind confidence scoring is as it sounds: to evaluate LLM-generated responses. One example of a product that does this is Egen.ai, a technology service company that has developed a proprietary confidence scoring framework. On the AI front, DataRobot provides scores about generated output. The financial engineering team at Spotify has also developed a confidence scoring methodology using majority voting while trying to parse invoices using generative AI.[6]
- **Explainability features:** Some generative BI tools (such as Tableau, Microsoft Power BI, and ThoughtSpot) enable users to ask the tool how it derived the answer. For example, this might include right clicking on a data point in a visual (like a line or bar chart) and select "Analyze" > "Explain the increase" or "Explain the decrease" in the case of Microsoft Power BI. Some provide Q&A capabilities that provide citations, supporting insights, and visualizations to help users validate the AI-generated answers. This may also include aggregations or joins used. In upcoming chapters, I will take a deeper look at explainability in AI systems.
- **User feedback loops:** Some platforms, such as Tableau GPT and Power BI Copilot let users provide a thumbs up or thumbs down to flag good or poor output. Reinforcement learning (a machine learning technique that works through trial and error, receiving rewards and penalties where the goal is to maximize the cumulative award over time) is used to adjust future responses based on trustworthiness.
- **Semantic constraints/guardrails:** The idea behind semantic constraints is to guide generative BI systems to produce accurate, contextually relevant, and safe outputs by limiting what the model can generate based on predefined business rules or intent structures. For instance, this might be something like, "only pull data from approved metrics tables" or "only respond with chart formats."

There are other approaches also being used and researched, but the point is that these are early days for much of this. I'll come back to these kinds of trust methodologies when I dig deeper into data, analytics, and AI governance in later chapters.

[6] Zhu, M. (2024). *Building confidence: A case study in how to create confidence scores for GenAI applications* [online]. Spotify Engineering. Available at: `https://engineering.atspotify.com/2024/12/building-confidence-a-case-study-in-how-to-create-confidence-scores-for-genai-applications?utm_source=chatgpt.com` [Accessed 22 July 2025].

What's Next?

AI-infused BI represents a shift in how data are explored, analyzed, and acted upon. From automated data preparation to natural language querying and dynamic insight generation, AI is making analytics more accessible and, in many cases, more powerful. Generative AI, in particular, offers new ways for users to interact with data, but it also raises the stakes for trust, governance, and explainability. As we saw, simply placing an LLM in front of your data won't guarantee success. Semantic layers, data quality, and policy controls remain foundational. Meanwhile, the renewed focus on unstructured data highlights the expanding role of analytics in every corner of the enterprise, from customer service notes to legal contracts.

To fully realize the value of all of this, organizations need to ensure that users have not only access to AI-infused tools but also the skills and confidence to use them responsibly. That's where data literacy comes in. In the next chapter, I'll explore what data literacy really means in an AI world, how organizations are implementing programs to raise literacy across roles, and why the ability to critically think with data is key.

CHAPTER

11

Ensuring Data Literacy

In this chapter, we will examine the following:

- What data literacy means
- The importance of data literacy
- How people develop and retain data literacy skills
- How organizations are implementing data literacy programs
- Measuring literacy and literacy progress

Data literacy is critical for organizations that want to succeed with data, analytics, and AI. I've mentioned it in previous chapters, and it is part of the AI Readiness Model described in Chapter 3. It is such an important topic that it requires its own chapter. Notice in this chapter that I'm going to talk about data literacy, but this data literacy will also form the foundation for AI literacy, and it is an important part of becoming ready for AI.

What Data Literacy Means

As part of the move to democratization and utilizing AI-infused tools and generative AI to gain insights into data, businesses need to improve their overall data literacy. Data literacy involves the awareness and recognition of the value of data, how well people understand and interact with data and analytics, and their ability to communicate data-driven insights to impact behavior and achieve business goals. It includes understanding the business and data elements, framing analytics, applying critical interpretation, and developing communication skills.

Data literacy is not only for IT and business analysts; it is an important skill for individuals across the organization, although the literacy level may vary based on role. Whether asking questions of data in natural language, interpreting visualizations produced by others, or building analytics models, the ability to interpret and use data is a critical skill. Data literacy enables companies to understand their data and get the most value from it.

That means that, at its core, data literacy is a human-side skill. It enables people to understand, work with, think about, and communicate with data. Consider the following example.

A business user may use a generative AI assistant (generative BI) to understand how sales have been growing or shrinking over the past five years at their company. In the prompt box, they type, "Show our total sales trend for the past five years." The generative AI assistant responds by describing its data source and logic: it will query the enterprise warehouse's sales table, sum the *net_amount* field, filter out returns, and group by fiscal year. The user should confirm that those definitions align with finance policy and that the date field reflects the invoice date, not the shipment date. The AI then creates a chart showing a sharp dip two years ago. Rather than accepting this as fact, the user thinks critically, asking the assistant to break the numbers down by region and to note the month when a divested subsidiary stopped contributing data. With that context added, the "dip" is revealed as a reporting artifact, and core revenue is actually flat.

The user next probes seasonality and inventory levels to test whether supply-chain shortages might explain minor fluctuations, ensuring the story holds. Convinced the analysis is sound, the user instructs the AI to generate a brief narrative, such as: "Sales grew at a 6% CAGR, with a one-off adjustment for the divestiture." The user might take this chart, review it to ensure it makes sense, and create a narrative to use in a slide deck.

This example illustrates some of the aspects of data literacy, including validating definitions, questioning anomalies, grounding results in business context, and translating the insight into a concise, shareable communication. Yet over the years, I've seen that organizations are not as data literate as they need to be. For instance, I routinely see in my research at TDWI that data literacy is a top organizational challenge companies have when trying to implement analytics programs.

The Importance of Data Literacy

Data literacy is important on many levels. Not only does it help companies get value from their data, it helps everyone across the organization to use data to make decisions more effectively. People who are confident in their literacy skills are surer in making decisions based on data. That is because they know how to use data and think about it, which can lead to better decisions.

Likewise, people who are data literate can be more productive because they know how to make sense of and ask questions of their data to drive business outcomes. They may also be more engaged because they feel more empowered with data behind them. These skills can also effectively contribute to career advancement. People who can interpret data and communicate with it are more likely to be able to influence strategy, which can give them more visibility. That, in turn, may help them to take on new roles with more responsibility.

And, of course, data literacy also helps the whole organization be ready to address governance and risk. If people are more engaged and invested in their data because they understand the value, they are more likely to want to make sure it remains trusted and governed. They are more in tune with what needs to happen to ensure that their data is trustworthy.

The net of this is a more competitive and engaged organization. For instance, the Data Literacy Index, which was commissioned by analytics vendor Qlik and produced by the Wharton School and IHS Markit (a market research company), indicates that there is a correlation between company performance and workforce data literacy. In their research, they found that companies that have high corporate data literacy exhibit up to 5% higher enterprise value (total market value), which can equate to hundreds of millions of dollars.[1]

How People Develop and Retain Data Literacy Skills

Data literacy does not mean everyone must become a data scientist. It does mean that marketers, product managers, HR partners, and executives should have a comfort level with data so they can do things like ask the right questions, challenge the numbers, translate insights into persuasive stories and actions, and guard against bias, compliance issues, and misinformation.

So, how do people develop and retain skills? There are, of course, different kinds of skills. Some data literacy skills are hard skills. Hard skills are those that require some technical and analytical expertise. For instance, these might include understanding how to use a software platform to create a data visualization or understanding statistical concepts such as the mean or how a regression model works. They might include understanding how data (structured or unstructured) might be represented in a database or other platform. Hard skills

[1] Data Literacy Project. (2018). *New research uncovers $500 million enterprise value opportunity with Data Literacy* [online]. The Data Literacy Project. Available at: `https://thedataliteracyproject.org/new-research-uncovers-500-million-enterprise-value-opportunity-with-data-literacy/` [Accessed 23 July 2025].

might also include knowledge of how to use a data catalog to find data, or what access controls mean, or understanding and using other governance concepts such as data lineage or metadata.

Soft skills, on the other hand, are thinking abilities that are involved in the practice of analytics. These might include the ability to collaborate with others, critical thinking about data outputs, the ability to tell a story with data, curiosity, and knowledge of the business. Some of these are innate, but others require training. For instance, a person may intuitively know how to collaborate but may not know how to tell a story with data.

This brief description highlights that data literacy training may consist of many components. Of course, not everyone needs to learn every aspect of data literacy. But they still must be able to master some skills. In the story from Bangor Savings Bank, we will see the steps that an organization took to help to make their company data literate and how that unfolded over a period of years.

Of course, different people have different ways of learning and that needs to be accounted for in data literacy training. So, how do people absorb and retain skills? There are several frameworks for categorizing types of learners, each emphasizing different aspects of how people prefer to absorb, process, and retain information. One of the most popular models is the VARK learning style model. VARK is an acronym that stands for, "Visual, Aural, Read/Write, or Kinesthetic." These are sensory modalities, and people may have a preference in learning style toward one or more of these. The framework was developed by New Zealand teacher and educational professional Neil Fleming in the 1980s. His coauthor was Charles Bonwell, who directed the Centers for Teaching and Learning at the Saint Louis College of Pharmacy and was also at Southeast Missouri State University.

VARK is as it sounds:

- **Visual learners** prefer information presented in charts, graphs, diagrams, and other visual formats. They often benefit from seeing the relationships between concepts.
- **Aural learners** gain understanding through listening. They do well in environments that use discussions, lectures, and podcasts.
- **Read/write learners** prefer materials that are presented as text. They learn best through reading (manuals, handouts) and writing (note-taking).
- **Kinesthetic learners** learn through doing and experiencing. They are most effective when engaged in hands-on tasks, demonstrations, simulations, or real-world examples.[2]

The point here is that in designing data literacy training, it will be important to mix it up a bit and address multiple learning styles (as many people have

[2] Fleming, N.D. and Bonwell, C.C. (2012). *VARK: A guide to learning styles*. Christchurch, NZ: VARK Learn Limited. Available at: `https://vark-learn.com` [Accessed 7 August 2025].

more than one style). For example, visual learners can benefit from dashboards and infographics, aural learners may respond well to narrated walkthroughs or group discussions, read/write learners can engage with written case studies and documentation, and kinesthetic learners will thrive in labs and hands-on exercises involving real-world data problems.

Combining these approaches provides greater accessibility and creates a more inclusive learning environment. Real-world problem-solving, such as analyzing customer data or identifying data quality issues, can be especially powerful when used as a foundation for training, aligning well with kinesthetic and multimodal learning need.

As mentioned, there are different providers of courses, and some organizations prefer to create their own literacy training, based on their own needs. In the Expert Advice at the end of this chapter, we'll see that Bangor Savings Bank partnered with Northeastern's Roux Institute. Table 11.1 lists some examples of companies that provide data literacy training.

Table 11.1: Companies That Provide Data Literacy Training

ORGANIZATION	OVERVIEW	OFFERINGS	AUDIENCE
TDWI	Education and research org focused on data, analytics, and AI	Live/virtual courses, conferences, enterprise training, certification, role-specific learning paths	Business/data professionals, enterprises
Data Literacy Project/Qlik	Initiative to improve global data literacy (now under Qlik)	Free courses, assessments, role-based paths, certifications	Business users, analysts, executives
Coursera	Online learning platform partnering with universities and companies	University-led courses, specializations, certifications	General learners, business professionals
Microsoft/ LinkedIn Learning	Tech company offering workplace learning via online platforms	Power BI, Excel, data fundamentals, real-world business data applications	Office workers, managers, data consumers
Open Data Institute (ODI)	UK-based nonprofit promoting open and ethical data use	Workshops, e-learning, toolkits on open data, governance, literacy, and ethics	Governments, nonprofits, organizations

How Organizations Are Implementing Data Literacy Programs

While academic research indicates that there are gaps in data literacy research,[3] numerous commercial vendors and consulting firms have provided their own best practices. These programs usually include a number of steps and often involve assessments. For example, Qlik has developed a six-step data literacy program adoption framework[4] that includes the following:

- **Planning and vision:** This includes identifying the stakeholders involved in the program, putting the funding in place, and determining the timeline.
- **Communication:** This step involves putting together a communication plan so that people understand why the data literacy program is being put in place and why it is important for the organization. This includes letting people know that this is being supported from the top.
- **Assessment:** This involves using some kind of assessment tool to gauge where the organization is in terms of literacy.
- **Cultural development:** Data literacy will involve change. This step is material to help leaders understand cultural change and what it involves. In Chapter 12, we will talk about change management models.
- **Prescriptive learning:** This is where different personas start to learn skills.
- **Evaluate and reiterate:** Here results are measured, and success is celebrated.

Other examples include Sigma Computing's five-step data literacy deployment model that focuses on practical steps such as understanding the data, finding it, and creating and using it to measuring success.[5] Boston Consulting Group, Deloitte, Accenture, and other large consulting firms also provide programs and guidance.

Over the years, I've seen many organizations implementing deliberate data literacy strategies. Often times this is a function of the data office or a center of excellence, under the chief data or analytics offer. A team that might be called a data literacy enablement team is an important part of this strategy. Those organizations

[3] Ghodoosi, B., West, T., Li, Q., Torrisi-Steele, G., and Dey, S. (2023). A systematic literature review of data literacy education. *Journal of Business & Finance Librarianship*, 28(2), pp. 112–127. doi: `https://doi.org/10.1080/08963568.2023.2171552`.
[4] `https://www.qlik.com/us/services/data-literacy-program` [Accessed 26 July 2025]
[5] Team Sigma. (2025). *A step-by-step data literacy framework for business leaders*. Sigma [online]. Available at: `https://www.sigmacomputing.com/blog/how-to-build-a-data-literacy-program-a-step-by-step-framework-for-data-and-business-leaders?` [Accessed 26 July 2025]

that have data literacy enablement teams in place (these go by different names) are more likely to measure success with their analytics efforts than those that do not.[6]

Data literacy enablement teams determine what roles need what training, devise the strategy, and execute on it. Well-oiled data literacy teams will assess data literacy and then put together programs based on role and experience levels (as described earlier). As part of this assessment, they may use maturity models or conduct structured evaluations, often focusing on role clarity, core competencies, and metrics for success. Some create charter documents and training tracks, including mentoring structures where experienced stewards help bring others along. Successful programs embed literacy initiatives into business processes and strategic projects, ensuring that data training is not isolated but connected to real business needs.

Sometimes data literacy learning is scheduled and mentors are available to help. Other times, it is online. In some cases, the data literacy team may deliver the training themselves; in other cases, they will outsource the training or send people to vendor-sponsored training for part of it (for instance, on a certain cloud platform or a specific analytics product). Critically, the literacy enablement team also monitors and measures progress toward its goal.

EXPERT ADVICE: BUILDING ON THE CULTURE OF DATA LITERACY AT BANGOR SAVINGS BANK

Amber Heffner-Cosby, senior vice president and director of Business Technology at Bangor Savings Bank, sees data literacy not as a tactical initiative but as a strategic cultural shift built on empowerment, accessibility, and shared ownership of data. The transformation began with a simple but telling observation: "The team seemed like order takers. We were getting requests for the same thing over and over again." At the same time, data literacy authors across the organization, that is, the employees licensed to create reports, did not feel empowered or equipped to be successful. "We started to recognize that and unwind it," she explains. The goal became clear: make data easily accessible, standardize definitions, and help people answer their own questions.

This shift required rethinking how data was delivered. Heffner-Cosby describes the evolution from a push model, where "if it ended up in your email box, then you needed it," to a pull model, where anyone with a question could self-serve. The Bank centralized access to reporting tools and dashboards so employees did not need to submit tickets to the analytics team. "We tried to take commonly requested analytics and put them in a centralized location they could easily get to," she says. Key terms like "active customer" were also clearly defined, so no one had to figure them out from scratch every time.

(continues)

[6] Halper, F. (2022). Modernizing the organization to support data and analytics. *Q22022 TDWI Best Practices Report*.

EXPERT ADVICE: BUILDING ON THE CULTURE OF DATA LITERACY AT BANGOR SAVINGS BANK (CONTINUED)

From the outset, it was about more than technology. "We have about 100 authors within the Bank. They can design and write their own reports and dashboards, and then our team will deploy those to production after ensuring they meet Bank standards," Heffner-Cosby says. This model allowed the analytics team to focus on higher-value work while building capacity across the organization.

Training was essential to making this work. The Bank offered a mix of formats to meet different learning styles, including group sessions, one-on-one coaching, virtual office hours, user groups, and short "how-to" videos. Heffner-Cosby says the goal was to "give people easier access to data to support their work" and then back that up with support so they could use it confidently within the organization.

A pivotal moment came with the Bank's partnership with Northeastern University's Roux Institute through a two-year program called "Accelerating Insights." Heffner-Cosby explains, "They did a workforce assessment and evaluated our employees in the space of their roles." From this, the Bank developed five distinct learning pathways, ranging from 5 to 50 hours of education:

- Frontline staff: Introduction to data fluency and AI
- Managers: Decision-making with analytics
- Advanced authors: Visualization, AI governance, and ethics
- Technical staff: Machine learning
- Leaders: Organizational data strategies and staying current on AI developments

Heffner-Cosby describes the advanced analytics pathway for authors as "a big dream of mine." She wanted participants to learn not only how to create effective visualizations but also how to think about stewardship and responsible use of AI in the workplace. The program sparked unexpected enthusiasm with employees across departments looking to upskill their data fluency.

When asked how to start a data literacy initiative, Heffner-Cosby offers practical advice: "Know your why. What are the gains, and who's going to gain, and how are they going to gain?" She emphasized the importance of assessing organizational readiness, starting small, and celebrating milestones to keep people engaged. "If it's the right thing for the organization and the teams, then move forward," she says. "The team was excited and saw the benefits of it."

Today, Bangor Savings Bank is extending its data literacy work into generative AI. Cognizant of policies that prohibit putting customer data into public tools, the Bank is exploring secure, responsible use of generative AI, including a model that offers just-in-time prompts for customer engagement. "We are accelerating our work across the organization by deepening our understanding of how AI and data fluency can enhance business processes and support workforce upskilling," Heffner-Cosby says.

(continues)

EXPERT ADVICE: BUILDING ON THE CULTURE OF DATA LITERACY AT BANGOR SAVINGS BANK (CONTINUED)

What began as a project to make data more accessible has become embedded in the Bank's DNA. "This is how we operate," she says. With access, training, and shared ownership at its core, data literacy has become a mindset that's ingrained in how teams think, work, and make decisions.

Measuring Literacy and Literacy Progress

As I mentioned earlier, one of the key success factors for any kind of data literacy program (and any analytics program, in general) is that there are measures in place to see how your organization is doing. These measures will illustrate to the organization that your literacy training is progressing and having an impact.

Some simple measures of data literacy training are obvious—track who is taking the data literacy training and how many have completed it, whether mandatory or not. As we've seen in some cases, organizations make the literacy training mandatory. Some organizations will provide skills assessments and track outcomes. Others will provide self-assessments (nicer than skills tests).

However, true success lies in showing business impact. Look beyond participation to outcomes. For example, are more employees using self-service analytics tools? Are they generating insights without relying on IT? Over time, these outcomes should connect to business results such as faster decision-making, improved customer targeting, reduced churn, or increased operational efficiency.

Here are some examples: correlate a rise in self-service dashboard usage with shorter sales-cycle times or link higher literacy scores to measurable improvements in forecast accuracy. Establishing these kinds of links between literacy, analytics adoption, and business key performance indicators takes effort, but it's what convinces leaders that the investment in data literacy is paying off.

What's Next?

This chapter highlighted the importance of data literacy in BI (and AI). The people side of the equation matters. Chapter 12 focuses on change management. This includes the techniques that are used for it and the need for perhaps getting out of your comfort zone to think about the psychological implications of new technologies.

CHAPTER

12

Putting Self-service to Work in Your Company

In this chapter, we will examine the following:

- Change management approaches
- The psychology of change
- Examples for self-service and artificial intelligence (AI)

As we have seen, self-service analytics adoption (or really any kind of analytics adoption) centers around people. You can have great technology, but if people don't buy in and are not supported, your program will fail. People have to want to use the tools, take the training, and make a change in their behavior for democratization to succeed. That can be hard, both at the individual and group level. At the individual level, people typically don't change in a linear fashion. Think about something you tried to change about yourself and how hard it was. You may resist change for reasons that aren't even conscious.

This chapter discusses change management and provides an overview on several approaches but focuses on two different approaches that are widely used in modern organizations. The first is the Prosci ADKAR® Model, which focuses on changing individual behavior. It was developed by Jeff Hiatt from the Prosci group more than 20 years ago.[1] The second is the Kotter 8-step approach, which was developed by John Kotter (1996)[2] after observing the characteristics

[1] Prosci. (n.d.). *ADKAR Model*. Available at: `https://www.prosci.com/adkar/adkar-model` [Accessed 17 July 2025].

[2] Kotter, J. (1996). *Leading change*. Harvard Business School Press. 187 pp.

of organizations that could successfully change (HBR, 1995).[3] I've also included a psychological perspective of what change entails from Joanne Heith, PsyD, a psychoanalyst, professor, and expert in group dynamics. This discussion is part of the organizational readiness part of the AI readiness approach described in Chapter 3.

Change Management Approaches

Through the years, at various events, I've often heard about organizations implementing various change management approaches to help them move the culture for analytics adoption such as self-service or AI forward. At its core, change management exists to ensure that every significant shift in an organization (e.g., a new technology or a new process) actually delivers the intended business value without derailing people, operations, or governance. Change management frameworks help guide organizations as they make the shift or change.

Kurt Lewin (widely regarded as the father of change management) introduced the first structured framework for understanding change in the 1940s, which looked at a three-stage process for organizational change (unfreeze, change, and refreeze).[4] Since then, numerous change management frameworks have been developed. Lewin's framework emphasized the role of group behavior and culture, which are central to change management.

The following sections discuss some change management frameworks currently used in the technology world.

The Prosci ADKAR Model

The Prosci ADKAR model is a goal-oriented change management model that guides individuals and organizational change. ADKAR is an acronym for awareness, desire, knowledge, ability, and reinforcement. The idea behind the model is that change can only happen if individuals embrace it.[5] Here's a quick break-down of the five steps:

[3] Kotter, J. (1995). Leading change: why transformation efforts fail (1995). *Harvard Business Review*, 73(2), pp. 59–67. `https://web.mit.edu/curhan/www/docs/Articles/15341_Readings/Organizational_Learning_and_Change/Kotter_1995_Leading_change.pdf` [Accessed 26 October 2025].

[4] Lewin, K., 1947. Frontiers in group dynamics: concept, method and reality in social science; social equilibria and social change. *Human Relations*, 1(1), pp. 5–41. Available at: `https://journals.sagepub.com/doi/10.1177/001872674700100103/` [Accessed 18 July 2025].

[5] Prosci, 2025. *The Prosci ADKAR Model for Organizational Change Success* [E-book], p. 3. Available at: `https://www.prosci.com/hubfs/Website_English/2.downloads/ebooks%20and%20guides%20(TOFU%20MOFU%20downloads)/ADKAR%20Model%20for%20Organizational%20Change%20eBook.pdf` [Accessed 19 July 2025].

1. **Awareness:** According to the ADKAR approach, people will only change if they understand why the change needs to happen from a business perspective. In other words, they need to understand the reason for the change.
2. **Desire:** Once people understand why there is a business need to make the change, they want to know what the benefits will be to them ("What's in it for me?"). According to Prosci, this is the most challenging stage of the journey, which makes sense. People need to be able to internalize the change and support it at more than a superficial level in order for change to be successful. This can take time, as we will see in the case study later in the chapter.
3. **Knowledge:** The knowledge step is about providing training, at the right time, after awareness and desire are already in place. According to Prosci, if people are trained before they are ready, this can lead to resistance.
4. **Ability:** This step is about enabling people to practice what they have learned. It involves coaching and feedback, along with practice time.
5. **Reinforcement:** Reinforcement is about sustaining the other steps. It involves rewards, recognition, and removing roadblocks to sustain the change.

The Prosci methodology breaks change up in to pieces and measures each piece. ADKAR elements can also be integrated into project management methodologies and approaches such as the Agile approach which is popular in companies today.

The McKinsey 7-S Model

This organizational framework was introduced in an article called, "Structure Is Not Organization" by McKinsey consultants.[6] They claimed that effective organizational change is a relationship between the interconnected variables of structure, strategy, systems, style, skills, staff, and superordinate goals. These have changed slightly over the years in that superordinate goals have been replaced by shared values as the middle of the framework. The 7-S was developed to help diagnose how organizations operate and provide tools to advise what changes should be made. It looks at what is working and what is not working and acts as a checklist.[7] It includes the following:

- **Shared values:** This element connects the other six Ss. These are core values of the company that are evidenced in the corporate culture and general work ethic.

[6] Waterman, R., Peters, T. and Phillips, J.R. (1980). Organization is not structure. *Horizon Business Journal*, 23(3), pp. 14–26.
[7] Bryan, L. (2008). Enduring ideas: the 7-S framework. *McKinsey Quarterly*, 1 March. Available at: `https://www.mckinsey.com/capabilities/strategy-and-corporate-finance/our-insights/enduring-ideas-the-7-s-framework` [Accessed 27 August 2025].

- **Style:** This is the leadership approach. It is about the informal rules of conduct in a company.
- **Skills:** These are the actual competencies of the people.
- **Systems:** These are the formal and informal procedures that govern everyday activity. It includes information systems, financial systems, HR systems, etc.
- **Structure:** This is how the company is structured, organizationally and the accountability relationships.
- **Staff:** These are the people, their capabilities, and how they are nurtured.
- **Strategy:** This is a plan devised to maintain and build competitive advantage over the competition, i.e., those actions that a company plans in response to or anticipation of changes in its external environment.[8]

The idea here is that these are used as a checklist or diagnostic and alignment tool. Before implementing the change, the 7-S are evaluated to determine the current state. With the future state identified, it can be used to define the gaps. For example, a company adopting AI might find its systems are outdated, its staff lack cloud skills, and its culture resists experimentation. They move on from here.

The Kotter 8-Step Change Model

This is a popular organizational change model that consists of eight steps[9]:

1. **Establish a sense of urgency:** The goal of this step is to try to create a sense of urgency where others feel that there needs to be a change. That may include sharing company data that shows problems, hiring consultants to highlight the problems, allowing errors to build, or highlighting lost opportunities. The idea is to help to drive people to understand that change needs to happen.
2. **Create a guiding coalition:** In this step, the idea is to build a team of powerful and well-respected change agents. These people need to be proven leaders who can drive change and people will listen to them.
3. **Develop a vision and strategy:** In this step, the group creates a future vision that needs to be appealing to employees and other stakeholders. This vision needs to be compelling enough to move people out of their comfort zone.

[8] Waters et al., 1980, p. 20.
[9] Kotter, *Leading Change*.

4. **Communicate the change vision:** In this step, the vision is communicated to employees and stakeholders via various channels such as large and small meetings, newsletters, videos, etc. The idea is to enlist a volunteer army to communicate this vision. The communication needs to happen often so people start to internalize it.
5. **Empower broad-based action:** The idea behind this step is to get rid of obstacles. Here is where employees might be trained on a new process or technology. Here is also where resistant people are identified and the volunteer army tries to get their buy-in. The idea is to try to empower people to succeed.
6. **Generate short-term wins:** The idea here is to get quick wins that can be measured and publicized so people get excited about the potential of the change. Change agents can be rewarded here. This helps to build momentum.
7. **Consolidate gains and produce more change:** With the short-term wins under their belt, the coalition identifies bigger change. Senior leaders keep the urgency up. New people are hired and promoted to keep up the momentum.
8. **Anchor new approaches in the culture:** In the last step, the idea is to try to make change stick in the organization by continuing to show how the new approaches provide value. This may involve employee turnover.

Kotter's model basically moves in three broad waves. First, you ignite change by creating a sense of urgency, assembling a powerful guiding coalition, and shaping a clear, shared vision with linked initiatives. Next, you mobilize the organization by broadcasting that vision until a volunteer army rallies behind it, systematically removing structural or cultural barriers so people can act, and you engineer quick, visible wins that prove progress and build momentum. Finally, you lock in success by pressing even harder after early victories, continuously setting new goals, repeating the prior steps to sustain acceleration, and embedding the new behaviors, processes, and mind-sets into everyday operations. In this way, the change becomes the way the organization works.

Bridges' Transition Model

This model was developed by William Bridges, PhD, more than 30 years ago first with his book on making sense of life changes and then in his 1995 book, *Managing Transitions: Making the Most of Change*.[10] Bridges looks at

[10] Bridges, W. (1995). *Managing transitions: Making the most of change*. London: Nicholas Brealey Publishing.

transition rather than change because "[t]ransition is the inner psychological process that people go through as they internalize and come to terms with the new situation that the change brings about."[11] The model focuses on helping people through this transition. According to Bridges, there are three stages of transition:

1. Ending what already exists (acknowledging that something is ending and letting go of old routines). There may be grief or fear during this time, and leaders should help others process it.
2. The neutral zone, where "psychological realignments and repatternings take place" and where people are learning what their new roles will be. There may be low morale during this phase, and leaders must communicate and provide support.
3. New beginnings where people start to understand the new role they will play and commit to it. Leaders should celebrate successes and reward it.

While this model is respected in organizational development circles, it rarely shows up as the primary framework in technology-related change programs. This may be because it focuses on the human-emotional side of change and leaves gaps in areas that technology leaders may be concerned about, such as training logistics, governance, and measurable key performance indicators (KPIs). It also doesn't have the modern toolkits that models such as ADKAR and Kotter's 8-steps have. Yet, as described in the "Expert Advice" later in the chapter, the emotional part of change management is critical, which is why this model has so much value.

So, which model should be used when? In looking at the three most used models, the ADKAR model, the Kotter's 8-step, and McKinsey's 7-S, experts in the field typically seem to agree that ADKAR is more about the individuals journey to change. It is a bottom-up model, while Kotter's 8-step and McKinsey's 7-S are more top-down. So, if the change involves new technology or processes and your main concern is whether individual employees (in the organization) will accept and use them, then use ADKAR. If you're more concerned about large-scale and strategic change, then Kotter's 8-step has proven to be successful. For internal alignment and to understand organizational issues, the 7-S are the right choice. Some people believe that these techniques, such as ADKAR and Kotter's 8-steps, can be used together. These three methods are contrasted in Table 12.1

[11] William Bridges Association. (n.d.). What is transition? *Bridges Transition Model* [Online]. Available at: `https://wmbridges.com/about/what-is-transition/` [Accessed 18 July 2025].

Table 12.1: Comparison of Three Change Management Methods

	ADKAR	KOTTER'S 8-STEP	MCKINSEY 7-S
Focus	Individual change	Large-scale change led by senior management	Holistic alignment of the 7-S
Approach	Bottom-up: guides individuals	Top-down: driven by leadership	Holistic
Used for	Initiatives where individual adoption is important	Initiatives where organizational adoption is critical such as digital transformations	Diagnosing where organizations alignment is off
Strengths and Weaknesses	■ Practical ■ Measurable ■ Not really organizationally focused	■ Clear roadmap ■ Importance of leadership ■ Not focused on individual factors	■ Comprehensive and diagnostic tool ■ Not focused on individual

EXPERT ADVICE: THE PSYCHOLOGY OF CHANGE

Of course, there is a psychology to change.

Changing behavior isn't just hard; it's deeply complex, often misunderstood, and far messier than most organizational models make it seem. Dr. Joanne Heith, PsyD, MA, LCSW, believes that models such as ADKAR, while useful in their simplicity, miss the unpredictable and nonlinear reality of how people actually change.

A fundamental misconception, she explained, is that change happens in a straight line. "People do not change in a linear fashion," she says. "If this is leaderships' expectation, they are likely to be disappointed." The ADKAR model, and others, utilize cognitive methods and stage theory as a template for behavior change. In real life, people are more likely to repeat familiar patterns of behavior, even maladaptive ones, or take two steps forward and one step back. Even desired behaviors, deemed beneficial, may not take precedence over old and familiar ways of being. This is because resistance to change is not a logical phenomenon; it's deeply emotional.

More critically, many frameworks assume that resistance is totally conscious and that people understand why they're hesitant. But according to Dr. Heith, much of what prevents change operates below the surface. In organizations, resistance might not be about the policy or technology being introduced; it could be about a person's history with authority, their relationship to control, or fear of the unknown. She described how someone might reject a directive not because it's unreasonable, but because the messenger evokes an unconscious association with a controlling parent. Or they feel devalued at work. "Being in an organization is like being in a family," she explained. "All kinds of early dynamics get stirred up."

(continues)

EXPERT ADVICE: THE PSYCHOLOGY OF CHANGE (CONTINUED)

Many organizational change models fail to address these emotional undercurrents rather than securing a foundation built on mutual trust.

And building trust isn't just about being nice; it's about tolerating feelings that may be uncomfortable and sometimes discordant with your own.

Dr. Heith pointed to psychodynamic group therapy, "The goal is not to convince someone to feel differently, but to understand that past feelings emerge in current relationships, particularly when old triggers are evoked. Change is more likely to occur when you don't dismiss or negate someone's feelings," she said. "When they feel understood, cared about and valued, they're more willing to engage and participate in something with you." Connection fuels change.

This approach doesn't require professional training in psychology. In fact, Dr. Heith teaches these principles to physical therapy students who have no background in mental health. She teaches them to ask open-ended questions. Questions that can't be answered with a simple "yes or no" serve two purposes: (1) they provide more information about the patients' thought processes, feelings, and life circumstances that may impact their adherence to the physical therapy prescription and (2) they give the patient the feeling that you care about them. The inquiry and interest shown often shifts the patient's energy and willingness to overcome challenges and incorporate new behaviors, more than the answers to the questions themselves.

Leaders, she noted, often think that if they explain the benefits of a change clearly enough, people will follow. But this approach can make people feel more misunderstood. "Change usually happens when people stop trying to change you," she said. "When they allow you to have your feelings, even if those feelings are different than their own. You don't need to unravel every unconscious motivation, nor should you. But understand that these forces are at play."

Ultimately, Dr. Heith's message is both simple and challenging: the process of behavior change may not be linear nor clear cut. But it's likely moving in the right direction when mutual trust, care, and connection are established and employees feel valued by the organization as it develops and changes.

Examples for Self-service and AI

Let's look at two of these models (Kotter's 8-step and Prosci ADKAR) and build scenarios around how they might be used for implementing change for self-service or AI, keeping the psychology of change in mind.

Using Kotter's 8-step Model to Promote Self-service Analytics

Suppose your company wants to democratize analytics and implement an organizational change model. They might pick Kotter's 8-step model. What would that look like, in practice? The chief analytics officer might do something like the following:

1. **Create a sense of urgency:** First, they would communicate issues to create a sense of urgency. They might highlight data bottlenecks, slow decision cycles, and competitors' faster analytics adoption. They might share metrics (e.g., 40% of analyst time spent on ad hoc queries, or it takes three weeks to get a report) to showcase the pain and make it feel urgent.
2. **Build a guiding coalition:** Next, it would be important to form a cross-functional team, perhaps called the "Analytics Empowerment Team" that might include the chief analytics officer, directors, influential business managers, and HR/learning and development partners. The group would be given authority to clear roadblocks and fund pilots.
3. **Craft a strategic vision and initiatives:** Next, the coalition would articulate a future where "every employee can find, trust, and use data in under five minutes" and outline key initiatives including the modern data platform to support this, a governed semantic layer to make it easier to find data in ways business users want to see it, an enablement academy to help empower people, and success metrics to measure the impact.
4. **Enlist a volunteer army:** The coalition would recruit "data champions" in each business unit who are eager to test new tools, create example dashboards, and evangelize successes. They would come on board to help communicate the vision everywhere including in town halls, across slack channels, company newsletters, and in leadership meetings.
5. **Enable action by removing barriers:** The team would try to remove any barriers to democratization. This might involve simplifying data-access approvals, retiring shadow spreadsheets, providing curated data products, launching role-based training (the data literacy training discussed in the previous chapter), and aligning incentives so managers reward data-driven decisions. During this stage, the team should be keeping some of the psychological factors discussed earlier, in mind.
6. **Generate short-term wins:** To showcase what is possible, the team would help to generate successes. For instance, it would back the delivery of a self-service 30-day pilot that cuts marketing campaign reporting time from three days to one hour and publicize the concrete ROI.
7. **Sustain acceleration:** The team would then work to expand pilots to new domains (finance, supply chain), add advanced capabilities (guided machine learning notebooks or generative AI front ends), and keep tightening feedback loops. Each win can fund the next item.
8. **Institute the change:** As the program starts to succeed, the team will do things like write self-service behaviors into job descriptions, performance

reviews, and governance policies. It might embed usage and data-literacy KPIs on executive scorecards. It can fold the champion network into an ongoing Analytics Community of Practice so the new habits become simply "how we work."

Using ADKAR to Help Individuals Use a New AI Sales Tool

Suppose your innovation lab has developed a generative AI sales tool to help its sales force be more efficient in their interaction with customers. They might currently be spending too much of their time preparing for client meetings and following up with clients. This might include developing emails and proposals.

1. **Build awareness:** The first step is to quantify the pain and show the organization why this change is important. For instance, in this case it might be to showcase that sales representatives spend 25–40% of their week composing follow-up emails, digging for product info, and updating the CRM. The team promoting the tool might try to demo the upside of a sales assistant that drafts personalized proposals in 60 seconds and automatically logs the proposal into the CRM. They might share competitor case studies or statistics of how other companies are doing this and saving time to build awareness.
2. **Build desire:** Next the team could recruit top and mid-tier sellers from different regions to pilot the tool. The incentive might be to let them keep any incremental commission from the AI-assisted deals. During this phase too, the team would need to address any fears (such as the fear of replacement) and deal with some of the resistance mentioned earlier. They will need to emphasize that the AI tool augments their relationship skills and it doesn't replace them. The team might launch other incentives such as gamification with prices for the most AI-generated meeting notes accepted by clients or the fastest proposal turn-around.
3. **Provide knowledge:** Here is where the training comes in. So, the team might provide short videos and interactive labs covering how to use the tool and best practices for using it. The team might provide prompt templates tagged by industry or product line. They also provide lunch and learns, peer clinics, and office hours where pilot sales representatives and others can help with the tool.
4. **Build the ability:** Here, representatives might be provided with a safe sandbox environment to practice with dummy data so that they don't have to worry about sending a bad email to a live prospect. Then they may first put it into action with an enablement coach after which they might run the system by themselves and send the coach their output for feedback.

5. **Reinforce and sustain the change:** The team might implement a weekly leaderboard that shows AI-assisted pipelines, deal velocity, and win-rates by representative. Leaders may get alerts when AI usage drops. There might be a continuous improvement loop that might involve quarterly meetings with new product updates.

As illustrated in the following case study, it isn't necessary to say you are using a change management framework. The team in this story implemented real change using the ADKAR framework together with other elements.

EXPERT ADVICE: BUILDING A CULTURE OF DATA LITERACY AND CHANGE

The divisional vice president of analytics at a retailer has led a sweeping initiative to reimagine what data literacy and change management mean in a fast-moving retail environment. Rather than treating them as separate corporate functions, he and his team have embedded both into the company's DNA—starting from the top and extending to every employee, from the C-suite to store-level associates.

"We redefined self-service," the vice president explains. "It's not just about building dashboards anymore. It's about enabling the business to succeed with curated enterprise data." This redefinition underpins a broader organizational shift. The company has moved to a role-based self-service model in which access to tools—including generative AI—depends on a person's job function and data proficiency. "If you're an analyst, you may be doing self-service BI. If you're a power user, you're doing self-service data modeling. And if you're in the generative space, you're doing self-service GenAI," he says.

At the heart of the transformation is the Code Academy—the company's internal Center of Decision Excellence. Designed as a development and transformation initiative rather than traditional change management, the Code Academy equips every employee with role-specific training and a clear sense of how data affects the company's operations and strategy. "We don't use the word 'change management,'" the vice president says. "To a lot of people, that signals something is wrong. Instead, we call it development and transformation. That changes the whole conversation."

The Academy's design is grounded in business needs, not technology for technology's sake. The vice president emphasizes that the training isn't about learning how to use tools in isolation. "Even the technical training—like how to use Power BI Copilot—is based on solving a business problem. It's never just, 'Here's how Copilot works.' It's always, 'Here's how you use Copilot to answer a question that matters.'"

All employees begin with a foundational "information awareness" course—a short training that emphasizes how data inputs, even from functions like payroll or inventory, impact downstream analytics. "We want everyone to understand that

(continues)

EXPERT ADVICE: BUILDING A CULTURE OF DATA LITERACY AND CHANGE (CONTINUED)

whatever role they play, their actions shape the quality and trustworthiness of the company's data," the vice president explains. "Aside from company culture, the only thing that binds everybody in the company is data."

Participation is not optional. "It's like cybersecurity training," he says. "If you don't take it, it affects your compensation." As of just one month into the program, 86% of employees had completed the training, with full compliance expected shortly.

The program is designed around the core elements of the ADKAR change management model—awareness, desire, knowledge, ability, and reinforcement—but the vice president intentionally avoids framing it in technical terms. "We embedded ADKAR into the training," he explains. "We just use business language. The first module is about why the certification matters to *you*—that's awareness and desire. Then we teach the skills and build reflection into the process. It's ADKAR, just with a different wrapper."

Prompt engineering is also a key component. "One of the biggest challenges with GenAI is that people don't know how to ask the right questions," the vice president says. "So, in every training, we include prompting instruction. We're giving people freedom to explore curated data, but we're also teaching them how to ask questions that actually lead somewhere."

The cultural shift was driven in large part by leadership. The company's CEO and COO made a public commitment to AI transformation during an all-hands town hall. The vice president recalls the CEO saying," We are entering the AI age, and we need everyone's support." The vice president clarified, "He didn't say, 'We're implementing Microsoft Copilot.' He said, 'We're going to open stores faster and make smarter decisions.' That was the turning point."

To reinforce the message, each member of the executive team was assigned an AI goal for their business unit. These goals were tied directly to bonus compensation and cascaded down through their teams. "Now, every C-suite leader has a quarterly meeting with the CEO to review their AI use cases. That level of sponsorship made the entire program real."

The result is an environment where employees are eager—not hesitant—to engage with data and AI tools. "We used to worry about adoption," the vice president said. "Now we have a waitlist. Everyone wants in. People say, 'I heard if I go through this training, I'll be able to analyze better.' That's a win."

By repositioning change as growth and treating data literacy as a strategic capability—not a compliance checkbox—the vice president and his team have sparked genuine cultural transformation. "We didn't start with tools," he said. "We started with business processes. And we made sure everyone knew: data is everyone's responsibility."

What's Next?

We've now explored modern analytics enablement, from laying a strong data foundation to implementing self-service tools, building enterprise-wide data literacy, and guiding adoption through structured change management. These chapters have emphasized that success with analytics and AI isn't just about technology; it's about people. Democratization, after all, is about empowering individuals across the business to make decisions with data. But as we've seen, giving people access doesn't guarantee impact. Without the right support structures, skills development, and intentional change efforts, even the most sophisticated tools will fall short.

In Part IV, we'll turn our attention to AI. In Chapter 13, we'll look at the strategic side of AI, what it means to align analytics and AI initiatives with core business goals. We'll start by asking a critical question: are we solving the right problem? At the end of the day, the success of any AI initiative depends not just on execution, but on whether the solution truly serves the business.

Part

IV

The AI Build-out—Ecosystem, Models, Operations, and the Shift to Agents

CHAPTER

13

The Business Strategy Behind the Technology

In this chapter, we will examine the following:

- What does the business need?
- Stakeholder engagement
- Is AI the right tool for the business problem?
- Future-proofing for AI
- Evaluating the strategic value of AI
- Balancing innovation and risk

We've covered a lot of ground in this book so far, and along the way, I've emphasized a core idea: technology must be tied to solving real business problems. In this chapter we'll look at this in more detail. We will even ask, "Is AI the right tool for this problem?" because often AI is not always the best tool. We'll also talk about evaluating the strategic value of AI and balancing risk and innovation. This chapter lays the groundwork as we move into discussing more about AI technologies, including how to build models and applications (including agentic AI).

What's the Business Need?

"Build it, and they will come" (yes, I know that isn't the exact quote) might work in baseball, but it doesn't necessarily work for AI. It always surprises me when I see organizations start to build out a data solution or purchase AI tools without having a good idea of the business needs that they are trying to meet with the technology.

Companies are all about driving revenue growth, improving customer satisfaction, boosting operational efficiency, and maximizing profitability. They should also ideally care about their employees. Many vendors, when trying to sell their products, draw on these kinds of drivers to justify AI and all the tools and solutions needed to make it work. However, those business needs are general aspirations; for AI to be successful in your company, you should understand your company's goals and strategic business context.

The ideal company strategy starts from the top down. For instance, executives at a cable news network might state that the organization's overall mission is to be the top source of news in the markets it serves. That's a high-level aspiration; what happens next is strategy. The strategy might include goals such as increasing audience engagement in key regional markets, expanding the digital subscriber base, or improving content personalization across platforms.

These strategic imperatives roll down into department-level initiatives. A digital business unit may prioritize audience acquisition by expanding its reach among key demographics through personalized content recommendations or social media optimization. Another group may focus on churn reduction. From this, business units may begin to articulate more targeted objectives and key results, such as improving the accuracy of content recommendations, optimizing ad inventory with viewer behavior data, or deploying automated summarization for video content to improve discoverability.

At this point, AI can enter the conversation as a tool that can be evaluated against these specific business objectives. A machine learning model might be used to segment audiences based on viewing behavior, while a generative AI solution could be used to create show descriptions or generate teaser content. The important point is that the AI initiative is in response to a clearly defined business need, not just an opportunistic technology deployment. The idea is to start with the business problem, not the AI model.

Of course, sometimes, there is just a business problem that needs to be solved, and AI is the best way to do it. In other words, some use cases flow organically from frontline business needs. For instance, credit card companies were suffering from fraud, so they built machine learning models to detect fraud. American Express, for instance, monitors trillions of dollars of transactions using machine learning and has reduced fraud by 50%, saving huge amounts of money annually and enhancing customer experience by reducing unnecessary transaction declines (Koetsier 2020).[1] We already discussed the UPS AI-powered ORION

[1] Koetsier, J. (2020). How Amex uses AI to automate 8 billion risk decisions (and achieve 50% less fraud). *Forbes*, 21 September. Available at: `https://www.forbes.com/sites/johnkoetsier/2020/09/21/50-less-fraud-how-amex-uses-ai-to-automate-8-billion-risk-decisions/` [Accessed 6 August 2025].

system to solve the problem of inefficient delivery routes, which were wasting fuel and time. Churn is another popular use case that is obviously driven by a business need. These AI models help to identify customers who are at risk of leaving by analyzing behavioral data, engagement patterns, or transaction history so that companies can take proactive steps to retain them.

Ultimately, whether the use case emerges from a strategic initiative or an operational challenge, the common thread is that the most effective AI solutions are grounded in real business problems that matter to the organization. The goals are specific and can be measured (we'll get to that in a minute).

Stakeholder Engagement

AI initiatives also need stakeholder engagement. Organizations that jump straight into AI development without stakeholder engagement, domain alignment, or a realistic data assessment are often the ones that struggle to scale past the pilot phase.

Why are the stakeholders so important? They play a critical role in developing a successful business strategy for AI because they define the needs, constraints, and success metrics. They own the business objectives and set the direction. They may control resources. These stakeholders include business leaders, domain experts, data and analytics teams, IT, compliance and legal teams, and end users. In other words, stakeholders can be anyone who influences or is impacted by the AI initiative.

Without stakeholder involvement, you risk building solutions that don't meet actual business requirements or that overlook important contextual knowledge. Or, worse, you may cause resistance or suspicion, especially if stakeholders find out after the fact that you've deployed a solution affecting their work without their input. "You did what, and why?" they may ask. And when business conditions shift (as they will do), the lack of stakeholder alignment means your AI solution probably won't adapt well to future challenges. Engaging stakeholders early ensures that AI efforts have a greater chance for success.

Some organizations' AI initiatives often begin with a "proof of value" process. This involves identifying a high-priority business question, working collaboratively with stakeholders to scope the use case, testing feasibility and data readiness, and piloting the solution with clear KPIs (key performance indicators) in mind. This is similar to the process we saw in Chapter 1.

Is AI the Right Tool for the Business Problem?

There's an old saying: "If all you have is a hammer, everything looks like a nail." As organizations invest in AI capabilities, it can be easy to fall into the trap of trying to apply AI to every challenge, even when simpler, faster, or cheaper solutions would do the job just as well. AI is obviously useful, but it can also be complex and costly.

We typically talk about three kinds of analytics: descriptive, diagnostic, and predictive. Some people add prescriptive to the list. The idea is that there are descriptive analytics that tell you what happened. BI historically was like this. Some diagnostic analytics explain why it happened. This may use root cause analysis or correlations. Predictive analytics takes this further and uses historical data to anticipate what is likely to happen next, such as forecasting sales, identifying churn risk, or predicting demand. Prescriptive analytics takes it even further by recommending specific actions to take; this is where optimization and decision models come into play. There is nothing wrong with descriptive and diagnostic analytics. Sometimes knowing what happened and why it happened is enough.

For instance, say the cable news network we discussed earlier sees a sudden drop in viewership. Descriptive analytics shows that overall ratings are down 15% over the last two weeks. That's useful, but it only tells what happened. Diagnostic analytics can enable the data analyst or the business user to segment the data by region, channel, platform, and demographic. They might see that the drop was among younger viewers on their streaming platform. They might notice that this coincided with an anchor leaving that time slot. This is slicing and dicing, and machine learning isn't needed (although the user might be interacting with their data via a generative AI interface).

On the other hand, if the company knows that the lineup may change based on upcoming contract negotiations, it may want to predict which viewers are likely to stop watching next. This is where predictive analytics and machine learning can be helpful. A model can be built using historical data of viewers (using attributes such as platform usage, time of day, content preferences, and other user behavior and demographics) to predict which viewers are likely to churn (remember, the target variable of interest). These models can detect subtle patterns across millions of data points that wouldn't be obvious through manual analysis.

The key is to be able to know the business question you're trying to answer and then determine which approach is best for the task. Remember: there is the business goal and then there is the analytics goal or formulation. Thinking about the analytics goal helps you determine how to approach the problem. For instance, if the business problem is that customers are churning and the business goal is to reduce churn by a certain percentage, the analytics goal/formulation might be to determine the characteristics of the customers that churn. I might be able to do some of that analysis by breaking up the historical data into the two groups (churn and no churn) and then using descriptive and diagnostic analytics to understand the basics of these groups and do some correlations. However, I can also ask the question in another way: "What are the characteristics of customers who have a high probability of churn?" Here, I'm moving from just describing what happened (descriptive) or explaining why it happened (diagnostic) to asking a forward-looking probabilistic question (predictive).

This example illustrates how reframing the business question, here from understanding the past to anticipating the future, changes the type of analytics you use. The key isn't just the problem; it's how you ask the question that determines the best analytics approach. However, it is sometimes difficult for people to think about asking a question in new ways because they are used to descriptive approaches. That is where upskilling/data literacy (or new hires) come into the picture.

The best approach is to match the tool to the problem, not force the problem to fit the tool.

Future-proofing Your Environment

Just because AI might not be the right tool for every business problem doesn't mean that you shouldn't prepare for AI. I don't mean to imply that. AI is (finally) here to stay. That means that even if your organization is solving its business problems using mostly BI, it should still be thinking about a future-proof architecture for AI. If you're enabling self-service analytics, you're likely investing in data governance, metadata management, and data literacy. These are foundational capabilities that AI also depends on. Future-proofing now ensures your AI efforts won't be side-tracked later by siloed data, poor quality, or distrust. As I mentioned, AI will make use of different data types and the architecture might be more complex, but organizations can build smartly.

Those at the self-service stage should use this opportunity to strengthen their data architecture, metadata management, and governance, future-proofing their environment so they can move quickly when the business demands it. That means thinking about how to handle complex data types, how to scale, and how to ensure performance. It also may mean looking for data platforms and solutions that already have AI either tightly integrated with or part of the platform. Organizations should be thinking about data governance for new data types as well. And if they have data literacy training programs in place, they should realize that data literacy doesn't just stop once the first training is complete. The data and analytics space is rapidly evolving and organizations need to think about continuous learning. That is going to evolve AI literacy as well as data literacy.

EXPERT ADVICE: FUTURE-READY ENVIRONMENTS

Prashanth Southekal, PhD, MBA, ICD.D has spent more than two decades helping organizations build practical data and analytics foundations. As founder and managing principal of the DBP-Institute and an adjunct professor of data and analytics, he has worked across industries to translate the abstractions of data architecture into operational reality. When he talks about "future-ready" environments, he stresses that it is not about shiny new tools or a single framework; it is about how fast an organization can turn captured information into usable insight.

"All data has a lag," he explained. "There is always a gap between the moment information is captured and the moment it is consumed. That gap might be seconds in streaming systems or months in batch environments. The question is, how do we reduce that lag between capture and use?" He estimates that as much as 60% of data's potential value is lost before it is even captured. The challenge of the future, he said, is to salvage the rest by bridging the delay between origination and consumption through better integration, processing, and visualization.

For Southekal, a future-ready data architecture must be able to handle continuous evolution and change because "business is always evolving, and data is always playing catch-up." He defines three pillars of this adaptability. "First is decoupled architecture, separation of storage, compute, applications, data, and consumption. Second is unified governance that spans those layers. Third is automation, but automation only works when you have standards and rules. Everyone wants self-healing and auto-scaling, but without standardized business rules, automation fails."

He sees federated data governance as essential to striking the right balance between speed and control. "In today's world of agility, a single centralized governance model does not hold much water," he said. "Some data elements must be governed centrally; others should stay within the line of business. It is a combination of both."

Southekal worries that most organizations are missing vast stores of value because they focus too narrowly on structured data. "Eighty to ninety percent of enterprise data is unstructured, text, audio, video, and images," he said. "Most people hate unstructured data because it is messy. But the good thing is, it allows quick digitization. You can take a 30-second video of testing a compressor and have a digital record immediately. The bad thing is that it lacks structure, so you need proper ingestion and profiling." He calls this TAVI data, text, audio, video, and images and believes the real value of a modern data platform comes from integrating structured and unstructured, internal and external, batch and streaming data into a unified landscape. "If you are just moving data from an ERP to Snowflake, you are doing the same process in a different tool. The real value comes when you bring all of it together."

When asked about obstacles for future-ready environments, Southekal points to three recurring issues: skills, silos, and security. "Skills are the first challenge.

(continues)

EXPERT ADVICE: FUTURE-READY ENVIRONMENTS (CONTINUED)

There is a major gap in understanding unstructured and external data. The second is silos. Everyone hates data silos, but they exist because of organizational silos. Unless companies think enterprise-wide, they will never solve them. The third is security and compliance. Every time data moves, the link between data and metadata can be lost." He noted that governance should extend beyond protection to ensure the integrity of metadata models as data travels across platforms.

Architectural flexibility, federated governance, and automation with measurement are, in his view, the three capabilities that matter most for long-term adaptability. "Analytics and AI are really about measuring and improving performance," he said. "Insights are great, but insights for what? Measurement and automation let you act on them." He believes the most effective future-ready environments embed auto-scaling and workload optimization so that performance improvement becomes continuous.

Southekal supports the hub-and-spoke model of governance now gaining traction. "The hub should handle the critical data elements, reference and master data like customers, vendors, and product descriptions," he said. "The spokes should handle the transient data such as prices and quantities that change all the time. The central team cannot manage business-specific values effectively, but they can ensure consistent definitions and taxonomies." His "CACTUS" framework for data profiling, completeness, accuracy, consistency, timeliness, uniqueness, and stability, works alongside the FAIR principles, findable, accessible, interoperable, and reusable, to make data both trustworthy and usable. "Profiling is required," he said. "But what is the point of good-quality data if people are not using it?"

His broader advice to data leaders centers on discipline and clarity. "Start with purpose, not with tools," he said. "A fool with a tool is still a fool. Know your business outcomes and compliance requirements. Adopt a data-product mindset, pipelines, SLAs, defined outcomes. Build incrementally, choose low-complexity, high-impact projects first to build trust. Invest in literacy and culture because if people do not know how to use the data, you will not go far. And govern continuously, with clear KPIs tied to your master and reference data."

For Southekal, being future-ready is not about predicting every technological shift; it is about designing an environment that can evolve as those shifts occur. "The goal is flexibility," he said. "Businesses will change, data will change, and technology will change. If you have decoupled architecture, federated governance, and automation built on standards, you can change with them."

Evaluating the Strategic Value of AI

We talked a little bit about metrics and measurement in Chapters 2 and 9. A metric is a quantifiable measure that can be used to determine how well or poorly a program is performing. Since this chapter has focused so heavily on aligning AI with business strategy, it's important to emphasize that AI metrics should roll up to the business objectives that AI is supporting. That is where the KPIs come in. These are the metrics that are tied to strategic goals and measures.

I remember when I worked at AT&T, and the Baldridge Quality Award was becoming popular. (It is now called the Baldridge Excellence Framework.) We had our own version of it, and I became a "quality examiner" as a side activity at the company. We went to different business units and assessed them. That was when I became very familiar with the notion of measurements tied to business objectives. These were critical to the award. But the measurements needed to make sense and be the right metrics. Today, I still see organizations measuring the wrong thing. For instance, a metric that is "the number of machine learning models trained" might seem like a productivity metric, but it isn't. It says nothing about how many machine learning models were deployed into production or how accurate they are or the impact (on revenue or cost reduction) they might have. Some consulting companies refer to these as "vanity metrics."

A hierarchy of metrics can be useful. At the top are business value metrics, which tie to strategic goals: things like increasing revenue, reducing churn, or improving customer satisfaction. These are the metrics that executives often care about. They may tie to KPIs. These metrics help to justify continued budget for the efforts. Beneath that are operational or process metrics, such as time-to-value, model usage, or task automation rates. These metrics help evaluate whether the AI solution is delivering effectively and efficiently. Finally, the technical metrics include model accuracy, precision/recall, data quality, and model drift. These ensure the solution is strong, safe, and maintainable, but on their own, they don't prove strategic value. AI metrics should cascade downward from business objectives, translating strategic goals into measurable KPIs, operational metrics, and technical indicators. In turn, results roll upward: technical and operational metrics provide the evidence that AI initiatives are achieving business value. This concept is illustrated in Figure 13.1. Note that there are fewer KPIs than metrics.

Going back to the example of the news organization, their aspiration is to be the top source of news in the markets they serve. This is a mission or vision statement; it sets the direction. However, the strategy breaks that vision into concrete goals such as increasing audience engagement in key regional markets or to improve content personalization across channels. KPIs are tied to those goals. For the division this might be to "increase the average watch time among 18–34 year olds by 15% over the next two quarters." This KPI is tied to that strategic goal of increasing audience engagement in key regional markets. The supporting metrics might be measures including click through rate (CTR) on recommendations or churn rate. These are broken down in the following list.

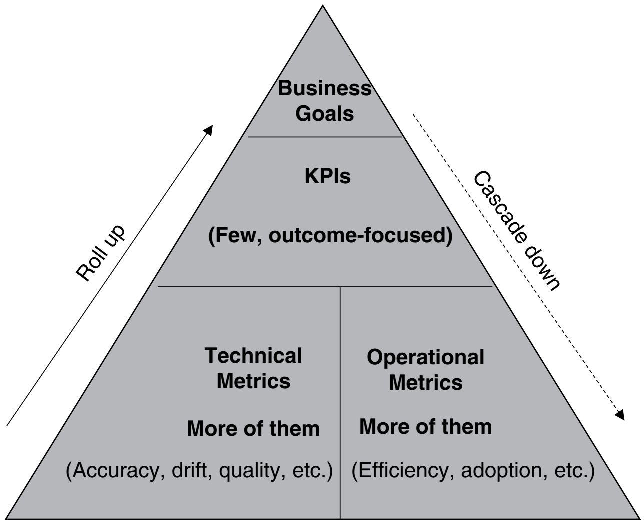

Figure 13.1: KPIs and metrics.

- **Mission:** "To be the top source of news in the markets we serve"
- **KPI:** "To increase average watch time among 18–34-year-olds by 15% over the next two quarters"
- **Operational metrics:** CTRs, churn rate, average session length, percentage of personalized recommendations accepted, push-notification engagement rate
- **Technical metrics:** Recommendation model accuracy or precision, data quality scores, uptime, API response time

Metrics and KPIs are critical to the success of AI programs. They contribute to what I mentioned earlier in the book as the "virtuous circle." If you can prove you met objectives with measurements, then success starts to build on itself. Clear, consistent metrics also help align stakeholders, justify continued investment, and highlight areas for improvement. Without them, it's nearly impossible to distinguish between real progress and perceived value.

Balancing Innovation and Risk

A good AI opportunity is not just tied to business goals and technically feasible; it's also aligned with the company's risk tolerance, compliance posture, and cultural readiness. By now you should realize that AI doesn't exist in a vacuum. Its effectiveness is tied to the quality and accessibility of data, the governance framework surrounding it, and the organization's ability to integrate AI output into business workflows. These were the dimensions of the AI Readiness Model described in Chapter 3.

There are three kinds of risks that businesses face: business risks, legal risks, and ethical risks. Business risk refers to the possibility of a company experiencing financial loss or reputational risk due to problems in its operating environment. We saw in Chapter 2 how Google suffered a financial loss because, although it had an innovation (its Bard chatbot), the chatbot produced a hallucination. That caused the company's stock price to decline back in 2023.

Legal risk includes the threat of regulatory noncompliance, data privacy violations, or intellectual property infringements. These are challenges that are especially relevant when deploying AI models trained on sensitive or proprietary data. For instance, Clearview AI, a surveillance company, scraped billions of images from social media without user consent in order to build a facial recognition tool sold to law enforcement (Matsakis 2020).[2] It has faced lawsuits and regulatory action in multiple countries, including fines from European and UK regulators.

There are also ethical risks. These are risks that can also ruin a company's reputation. They can include bias, lack of transparency, or harmful consequences of automated decisions. We talked about the Apple credit card example already. Here the credit card was offering lower credit limits to women than men with similar credit profiles. This was due to biased data but caused issues and public outcry. Another example occurred back in 2021 when whistleblower Frances Haugen left Meta (Facebook) because of algorithms that led to the amplification of "angry content" and divisiveness (Zubrow 2021).[3] Angry content gets more engagement and hence more money, but the question is, is it ethical to favor this kind of content?

We'll revisit these risks in Part V when we discuss AI governance. However, these examples may make your company think twice about the kinds of AI applications it wants to develop. Your company will have to understand its own risk tolerance. Some organizations may choose to push the boundaries in pursuit of innovation and profit, while others may take a more cautious approach, especially in regulated or high-stakes environments. Either way, it's important to weigh the potential business value of AI against reputational,

[2] Matsakis, L. (2020). Scraping the web is a powerful tool. Clearview AI abused it. *Wired*, 25 January. Available at: `https://www.wired.com/story/clearview-ai-scraping-web/` [Accessed 6 August 2025].

[3] Zubrow, K. (2021). Facebook whistleblower says company incentivizes "angry, polarizing, divisive content." *CBS News*, 4 October. Available at: `https://www.cbsnews.com/news/facebook-whistleblower-frances-haugen-60-minutes-polarizing-divisive-content/` [Accessed 6 August 2025].

legal, and ethical risks. Establishing clear criteria for what constitutes "acceptable risk" will be critical as AI becomes more deeply embedded in business.

What's Next?

By now, it should be clear that successful AI initiatives begin not with the technology, but with a well-defined business need. Aligning AI with strategic objectives, engaging the right stakeholders, selecting the right tools for the job, and understanding risk are critical for sustained AI success. It's also how organizations avoid the common traps of shiny-object syndrome. As we've seen, value isn't derived from the number of models deployed or the complexity of the architecture, but from the outcomes delivered, and measured, against meaningful business goals.

Of course, none of this happens in a vacuum. Behind every successful AI program is a diverse, interconnected ecosystem of people and skills. From model developers to product owners, from cloud engineers to governance leaders, AI involves lots of people across the organization. In Chapter 14, we'll map out this skills ecosystem in more detail. We'll look at what roles are emerging, what technical and organizational expertise is required, and why collaboration across disciplines is essential.

CHAPTER

14

The AI Ecosystem

In this chapter, we will examine the following:

- What's involved in the AI ecosystem
- Skills and roles
- Best practices for collaboration
- The growing importance of openness

So now the rubber meets the road. Let's dig into the AI ecosystem and skills that are involved with AI and the roles that you'll need to think about as you deploy AI. As part of this, we'll also talk about the move (that has been happening for several years) to open-source and open platforms. This will set us up as we start to talk about building AI applications as well as newer kinds of AI such as generative and agentic applications and operationalizing all of it. This will get a bit technical, but the idea is for you to understand enough about the technical layers to make strategic, budgetary, and governance decisions, obviously not to do the coding yourself.

The AI Ecosystem: What's Involved

We talked earlier about the data ecosystem and why it is central to modern analytics. We saw that it had producers and consumers and platforms and governance. There is also an AI ecosystem. At a basic level, it contains familiar components—the data infrastructure, the algorithms, the computing resources, the software frameworks, the applications, and the people who design, deploy, and use them.

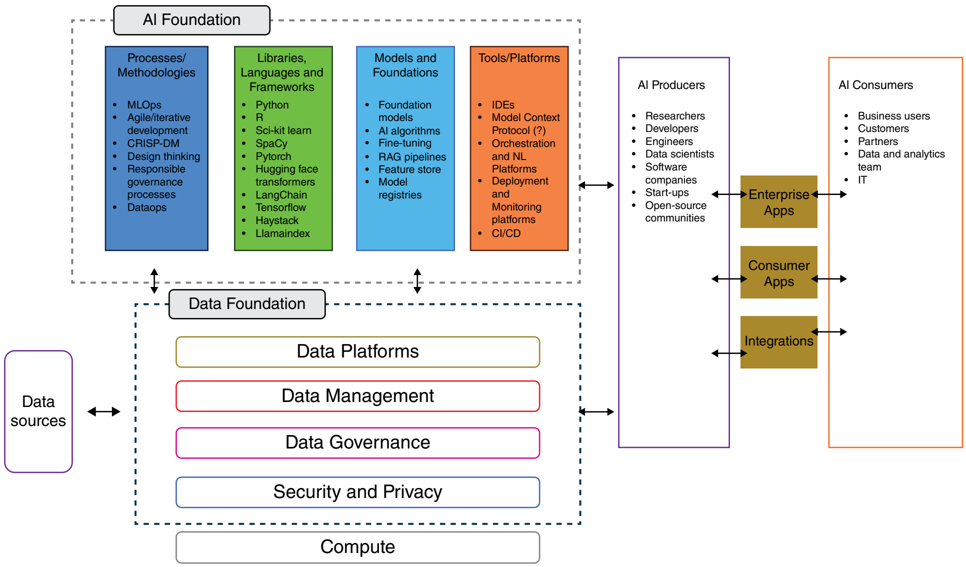

Figure 14.1: The AI ecosystem.

But something different is happening with this ecosystem. The pace of AI research and productization is unprecedented. New models and methods move from the lab to market in months, not years. Start-ups are springing up in every sector. Consumers are adopting generative AI tools in record numbers, fueling demand from the bottom up rather than waiting for enterprise IT. Global investment in AI companies jumped 62% in 2024 to more than $100 billion.[1] The level of activity feels similar to the internet and e-commerce boom of the 1990s, except this time, the scale is larger, and the change seems faster.

Figure 14.1 depicts the AI ecosystem. At its base is the data foundation we discussed previously, supported by the computing infrastructure that makes large-scale AI possible—the CPUs, GPUs, and TPUs. On top of this sits the AI foundation. I've divided this foundation into four different areas:

- **Processes and methodologies:** AI depends on structured, repeatable processes that connect experimentation to production and keep systems reliable at scale. DataOps, which applies agile and DevOps-inspired principles to data pipelines, automates ingestion, testing, and delivery to ensure that data is clean, timely, and trustworthy. On top of this foundation, DevOps unites

[1] Lunden, I. (2024). AI investments surged 62% to $110B in 2024 while startup funding overall declined 12%. Techcrunch. Available at: `https://techcrunch.com/2025/02/11/ai-investments-surged-62-to-110-billion-in-2024-while-startup-funding-overall-declined-12-says-dealroom/` [Accessed 30 August 2025].

software development and operations teams to streamline the delivery of applications and services that embed AI. MLOps manages machine learning in production, including validation to deployment, monitoring, and retraining. As generative AI becomes more prevalent, GenAIOps is emerging to handle the additional complexity of prompt management, grounding data, and monitoring model behavior and output quality. Increasingly, these operational disciplines are complemented by responsible AI governance—addressing bias, transparency, and compliance, and by human-centered design practices that keep outcomes ethical and aligned with user needs.

- **Libraries and frameworks:** At the programming layer, AI depends on general-purpose programming languages and the frameworks (structured, standardized approaches with reusable tools, libraries, and methodologies built on top of them). Python remains the dominant programming language in data science, with R still widely used in statistics and research. Core frameworks such as TensorFlow, PyTorch, and JAX provide the foundation for building and training deep learning models. High-level application programming interfaces (APIs) such as Keras simplify development, and specialized libraries such as Hugging Face Transformers, LangChain, LlamaIndex, and ONNX expand capabilities for natural language processing, generative AI, interoperability, and multimodal tasks. Again, I'm just introducing some concepts so that you can start to recognize some of the technologies involved. We will talk more about some of this in upcoming chapters.
- **Tools (development, monitoring, deployment):** Beyond frameworks, organizations rely on an expanding set of tools to streamline workflows and manage AI at scale. Development environments (IDEs) increasingly integrate AI assistance (just as we've seen across the data life cycle), while orchestration platforms such as Kubeflow, Airflow, and Prefect automate complex pipelines (although features vary). Deployment platforms like AWS SageMaker, Azure ML, and Google Vertex AI provide cloud-based environments for serving and scaling models. Monitoring and observability tools, such as MLflow, enable experiment tracking, drift detection, and life cycle management. These tools are important for governance, cost control, and ensuring that AI systems perform reliably in production.
- **Models and foundations:** At the core of the AI foundation are the models themselves and the infrastructure that supports them. This includes traditional machine learning algorithms, modern foundation models such as large language models (LLMs) and multimodal (diverse) systems. Surrounding these are critical enablers such as feature stores (to enable reuse of engineered data features), retrieval-augmented generation (RAG) pipelines (to ground models in proprietary data), and model registries (for versioning, deployment, and monitoring).

Together, these layers are supported by both open-source and commercial tooling that form the operational foundation of enterprise AI. Yet, this foundation can be complex, with technologies and practices that continue to evolve rapidly.

WHERE THE MODELS LIVE

It's easy to confuse programming languages, libraries, and frameworks—especially when talking about machine learning models.

- **Programming languages (like Python or R) provide the syntax and runtime for writing code, but they do not contain machine learning models by themselves.**
- **Libraries (like scikit-learn in Python or caret in R) contain ready-to-use models and utilities you can call directly, such as linear regression, decision trees, or clustering algorithms.**
- **Frameworks (like TensorFlow or PyTorch) are toolkits that include components for building models along with structured workflows for training, testing, and deploying them. They can be used for simple models, but they are especially powerful for deep learning or large-scale applications.**

Just like the data ecosystem, the AI ecosystem includes both producers and consumers. Producers are the researchers, developers, data scientists, ML engineers, and software companies bringing AI capabilities to life, whether as horizontal platforms or specialized vertical solutions. Consumers are the business users, customers, partners, IT staff, and data and AI professionals who put those capabilities to work in their organizations. Increasingly, *everyone* is a consumer, as generative AI and copilots are embedded into everyday productivity tools. We will talk about all of this in the coming chapters.

What makes this ecosystem feel urgent is not just its scope but its velocity. Enterprises cannot afford to treat AI as a distant future capability. It is already embedded in tools employees use every day, already reshaping customer expectations, and already attracting major capital. In my research at TDWI, I'm seeing that AI and generative AI are top investment priorities for companies.

Skills and Roles

Don't worry if you don't understand all these frameworks, tools, and technologies. The goal here is simply to give you a sense of what's included in this fast-evolving ecosystem and how its pieces fit together. In this chapter, however, I want to discuss the people and skills behind AI so that you can get a sense of what is involved in enterprise deployments. Remember, roles and responsibilities are listed as a key enabler for AI readiness in Chapter 3. Some of these roles are extensions of existing data and analytics jobs, while others are entirely new, created in response to generative AI, LLMs, and the demand for greater oversight.

Organizations adopting AI are finding that success requires more than just hiring one or two data scientists. It requires an ecosystem of roles that collectively span leadership, architecture, engineering, building, and management. These roles differ in their focus, but they are deeply interconnected. No one function can succeed in isolation, which is why collaboration is so important. Let's go through these different categories.

Leadership Roles

First, there needs to be leadership for AI. In the past, this has fallen to the director or vice president of analytics, and that can be fine, depending on your company and the scope of what you are trying to do with AI. More recently, organizations have started to hire specific leaders for AI. These might be the chief data officer (CDO) or the chief analytics or chief AI officer (CDAO).

The CDAO/CDO is responsible for turning data into a strategic enterprise asset. Traditionally, this role focused on governance, compliance, and data quality. Recently, its scope has broadened to encompass AI strategy and adoption as well. CDAOs must set the enterprise vision, ensure that data platforms are modern and scalable, and champion the use of AI across business functions.

This dual mandate, protecting data while enabling innovation, is what makes the role so challenging. On one hand, the CDAO must ensure that governance and privacy obligations are met. On the other hand, they must show the board and executive peers that AI investments are delivering revenue growth, operational efficiency, and competitive advantage. Increasingly, CDAOs chair cross-functional councils where AI use cases are prioritized, funded, and monitored for value delivery. They also face the cultural challenge of raising AI literacy across the enterprise so that employees at all levels understand how AI fits into their daily work. When the role first became popular a few years ago, I heard several anecdotal stories about how CDOs were given six months to show results. If they didn't, they were out. I think that is unrealistic, and I believe organizations are coming around to that way of thinking.

In some organizations, the role of the chief AI officer is emerging. Sometimes this role sits under the CDO, other times alongside them. The chief AI officer focuses specifically on the development, deployment, and governance of AI solutions, especially new generative and agentic systems. This title is not yet universal, but its rise underscores how central AI has become to business strategy.

I've seen in my TDWI research that having an executive in charge of the data and AI efforts can make a big different in terms of success. In fact, those companies where the CDO or CDAIO or CAIO are part of the C-suite are more likely to be successful with their analytics efforts than those who aren't.[2]

Architecture and Engineering Roles

As we saw in earlier chapters, the data architect is responsible for designing the blueprint for the data and AI ecosystem. The data engineer helps implement it.

Data architects create the high-level structures that ensure data flows reliably and securely from source systems into platforms where it can be analyzed and used by AI models. They are responsible for choosing the right mix of technologies, such as the data lakes, warehouses, or fabrics we spoke about earlier, and ensuring that these platforms can support both traditional analytics and the newer demands of AI, such as real-time ingestion and new platforms such as vector databases that we will talk about in Chapter 17.

A modern data architect must think about governance, metadata, lineage, and cost efficiency from the outset. They must also anticipate how the architecture will support not just analysts, but also AI developers and data scientists. In practice, this means designing environments that balance flexibility for experimentation with stability for production systems. Architects often serve as translators between business stakeholders, who articulate what outcomes they need, and technical teams, who must build systems to deliver those outcomes.

Data engineers create the pipelines that ingest, clean, transform, and deliver data at scale. Their work ensures that the right data arrives in the right place, in the right condition, at the right time. Without this, no model can be trusted.

The role requires a blend of software engineering skills and data management expertise. Data engineers work with batch (periodic) and real-time data, and they increasingly manage the infrastructure that supports new methodologies such as retrieval-augmented generation (RAG), a technique that grounds large language models in enterprise knowledge. They must embed validation checks, handle schema changes, and make sure pipelines are resilient when source systems or business requirements evolve.

Because their work is foundational, data engineers often collaborate closely with data scientists, AI developers, and business analysts. In many organizations, they also work hand-in-hand with governance teams, ensuring that data quality rules and security controls are embedded into every pipeline rather than applied as an afterthought.

[2] Halper, F. (2022). Modernizing the organization to support data and analytics. *TDWI Best Practices Report*. Available at: `https://assets.ctfassets.net/7p3vnbbznfiw/1LEQ9hezs0PM0XtiXJVx11/671010d2d0276cd6a9d534acc7760fd5/tdwi-best-practices-report-modernizing-the-organization-to-support-data-and-analytics__1_.pdf` [Accessed 5 September 2025].

It is worth noting that the data architect and the data engineering roles are intertwined. Data architects set direction; data engineers implement it. In smaller organizations, the same person may wear both hats. In larger enterprises, they work as a team, ensuring that enterprise data flows are both strategically aligned and technically sound.

Data Science and AI Development Roles

Data scientists remain one of the most recognized roles in modern analytics. They combine statistics, computer science, and domain expertise to explore datasets, uncover patterns, and create predictive and machine learning models. In the early days of the role, data scientists were often described as "unicorns" because they were expected to do everything from coding pipelines to communicating results. Today, the role is more specialized.

Today, data scientists are expanding beyond traditional machine learning. They are fine-tuning foundation models, exploring generative AI use cases, and helping organizations evaluate whether to build or buy AI solutions. Their day-to-day work includes preparing data, selecting algorithms, training and validating models, and working with governance teams to test for bias and explainability. We will talk about this life cycle in Chapter 15.

Perhaps the most important shift is that data scientists are expected to partner closely with the business. Their success is measured not only by accuracy scores, but also by whether the models they create are trusted, adopted, and used to make better decisions.

AI developers, sometimes called AI engineers, bridge the gap between models and applications. They are responsible for embedding AI capabilities into products and workflows. This could mean building a customer service chatbot, designing a recommendation engine, or creating a productivity assistant for employees.

What sets them apart from traditional software developers is their fluency in working with AI models. They need to understand how models are trained and fine-tuned, how to design retrieval pipelines that connect models with enterprise data, and how to monitor model outputs once the application is live. They also need to be skilled in user experience, since AI features must feel intuitive and trustworthy to the end user.

As generative and agentic AI spreads, the AI developer role has become one of the fastest growing job categories. These professionals are at the front lines of turning AI research into everyday business value.

AI Operations Roles

Operations roles are critical to ensuring that AI can run in production. For instance, a role that has been emerging over the past few years is MLOps. MLOps teams, as mentioned earlier, ensure that models work reliably in production.

They manage deployment pipelines, automate retraining when data becomes stale, monitor model performance, and ensure that models remain compliant with governance rules. The rise of MLOps reflects a hard lesson: building a model is only the beginning. The real challenge lies in maintaining it once it interacts with real-world data and business processes. Without MLOps discipline, models quickly become stale or unreliable. We'll talk more about MLOps in Chapter 16 when we talk about operationalizing AI.

With the advent of generative AI, an evolving role is GenAIOps. This is the application of these same principles to generative AI. Because generative systems are probabilistic and dynamic, monitoring them requires new tools and mindsets. These GenAIOps teams are experimenting with observability dashboards that track not only accuracy, but also hallucination rates, cost per query, and user trust. This is a role where the practices are still being invented in real time, but one to watch.

Emerging and Complementary Roles

The roles preceding form the backbone of most AI teams, but they are not the whole story. As organizations mature, several other roles are becoming equally important:

- **AI Product Manager:** Similar to data product managers, AI product managers oversee the life cycle of AI initiatives, ensuring they solve real business problems and deliver measurable value. They work closely with developers, data scientists, and stakeholders to prioritize use cases and manage delivery.
- **AI Governance and Ethics Officer:** With new regulations on the horizon, such as the EU AI Act, organizations increasingly appoint dedicated leaders to oversee compliance, fairness, and explainability (topics we will talk about in the next chapter). These officers help ensure that AI is not only effective but also responsible and trusted.
- **Domain Quality Expert:** In areas such as healthcare, finance, and manufacturing, subject matter experts ensure that AI models use data that is both technically correct and contextually meaningful. This role is becoming important for generative AI, where domain-specific knowledge is required to prevent errors.
- **AI Literacy and Change Leaders:** Finally, because AI adoption touches every employee, organizations are investing in literacy programs. AI literacy leaders design training, build confidence, and help employees understand when and how to use AI responsibly.

Table 14.1 provides a summary of these roles, along with the kinds of tools that each needs to be familiar with, based on a LinkedIn analysis I did in

Table 14.1: Key Roles, Responsibilities, and Desired Tools in the AI Ecosystem

ROLE	PRIMARY RESPONSIBILITIES	WIDELY USED TOOLS & PLATFORMS (2025)
Chief data officer (CDO)	Defines enterprise data and AI vision, aligns data/AI strategy with business outcomes, ensures governance and compliance, drives data monetization	Cloud data platforms (Snowflake, Databricks, Azure Synapse); Governance (Collibra, Alation); BI/analytics (Power BI, Tableau, ThoughtSpot)
Data architect	Designs enterprise data platforms and models; ensures reliable, secure, governed data flow for analytics and AI; plans modernization roadmaps	Cloud services (AWS, Azure, GCP); Data platforms (Databricks, Snowflake); Streaming (Kafka, Kinesis); Modeling/governance (dbt, Informatica, Collibra)
Data engineer	Builds and maintains pipelines; integrates and transforms data; ensures data quality, timeliness, and scalability for AI/analytics workloads	Programming (Python, SQL, Scala); Orchestration (Airflow, dbt, AWS Glue); Warehouses/lakes (Redshift, BigQuery, Synapse, Delta Lake); Streaming (Spark, Kafka); NoSQL (MongoDB, DynamoDB)
Data scientist	Explores data, develops predictive/ML models, fine-tunes foundation models, tests for bias/explainability, partners with business teams for insights	Languages (Python, R); ML frameworks (scikit-learn, TensorFlow, PyTorch, Keras); Generative/NLP (Hugging Face, LangChain); Visualization (Matplotlib, Seaborn, Tableau); Environments (Jupyter, Colab)
AI developer/AI engineer	Embeds AI into products and workflows; builds retrieval-augmented generation (RAG) pipelines; integrates LLMs into enterprise applications	GenAI frameworks (LangChain, LlamaIndex, Hugging Face); LLM APIs (OpenAI, Anthropic, Mistral, Cohere); Vector DBs (Pinecone, Weaviate, FAISS); Cloud AI services (AWS Bedrock, Azure ML, Google Vertex AI)
MLOps engineer	Operationalizes models; manages deployment, monitoring, retraining; ensures reliability, scalability, cost control, and compliance	Deployment/monitoring (MLflow, Kubeflow, Weights & Biases); Cloud ML (SageMaker, Vertex AI, Azure ML); DevOps (GitHub/GitLab, Jenkins, Docker, Kubernetes); Observability (Grafana, Kibana, Prometheus)
AI product manager	Oversees AI solution life cycle; prioritizes and delivers use cases; measures business impact and ROI	Product management (Jira, Confluence); Experimentation (Optimizely, Amplitude); Cloud AI platforms (Vertex AI, Databricks)

(continues)

Table 14.1: (Continued)

ROLE	PRIMARY RESPONSIBILITIES	WIDELY USED TOOLS & PLATFORMS (2025)
AI governance and ethics officer	Ensures AI fairness, transparency, and compliance with regulations; oversees bias audits and explainability	Bias/explainability toolkits (AI Fairness 360, Fairlearn, SHAP/LIME); Governance platforms (SageMaker Clarify, John Snow Labs Responsible AI, Trūata)
Domain quality expert	Ensures data and models are contextually valid in industries like healthcare, finance, and manufacturing	Industry-specific systems (Epic, Cerner, SAP, Oracle ERP); Data quality tools (Informatica, Ataccama, Talend)
AI literacy and change leaders	Build AI literacy across the workforce; train and coach employees; drive responsible adoption and change management	Enterprise learning platforms (Coursera for Business, Udemy, internal LMS); Productivity AI (Copilots in Office, Salesforce Einstein)

September 2025 of open job postings for these roles. Note that these are simply the tools that are mentioned in postings. Like all the tools mentioned so far in this book, these are just examples. And I've mentioned some of these tools already; we will talk about others in the upcoming chapters. This is important as you think about building up your own practices. Note that many of these are newer technologies that include open-source tools.

Best Practices for Collaboration

I remember talking to a vice president of analytics at a financial services company about five years ago. The person I was talking to told me about their data scientists. He said that the data scientists didn't need to know anything about the business. I found that very odd because in order to create the right inputs (features) for a model, it would seem that the data scientist should understand the business context. Yet, this company worked in silos. My guess is that they have found that this doesn't work anymore. In fact, it is important for all the members of the AI team (big or small—as I mentioned in smaller companies the same person may wear different hats) to collaborate.

There are different approaches for collaboration. Some companies create Centers of Excellence (CoE) (they can go by different names). A CoE is a centralized hub that combines expertise in data, analytics, and AI. This group sets standards, develops best practices, and provides reusable tools and templates for the rest of the organization. A CoE often includes data scientists, data engineers, architects, and governance specialists who can be deployed to high-value projects across business units. By concentrating expertise, CoEs reduce redundancy, promote consistency, and enable organizations to establish common governance frameworks and methodologies. Many organizations rely on CoEs to address challenges of skills shortages, lack of governance, and the need for standardized processes.

The CoE model also fosters collaboration by acting as a bridge. Business units bring their needs to the CoE, which provides expertise and guidance to deliver solutions. The CoE can run training programs, raise literacy, and coordinate governance councils that bring together technical and business stakeholders. However, critics argue that a CoE can become too centralized and risk losing touch with the day-to-day realities of the business.

An alternative is to place analytics and AI professionals directly into business units or domains. In this model, data scientists or AI developers work side by side with marketing, finance, or operations teams. The advantage is speed and relevance: projects stay closely aligned with business priorities and can move quickly without waiting for central approval. I've observed that organizations with strong domain ownership often report higher satisfaction with local use cases and faster innovation cycles.

The drawback is duplication and inconsistency. Different units may build their own models, data pipelines, or dashboards using different tools and standards. Without coordination, collaboration across units becomes difficult, and governance can suffer.

Some organizations now combine the two models. A CoE (the hub) provides governance, standards, and shared infrastructure, while business units or domains (the spokes) own execution. This hybrid approach can offer the best of both worlds: central consistency with local agility. For example, the CoE may provide MLOps pipelines and governance frameworks, while domain teams develop their own AI products using those shared resources. More advanced organizations are increasingly adopting hub-and-spoke inspired models, which emphasize domain ownership within a framework of federated governance.

Regardless of structure, successful organizations emphasize methodologies that encourage collaboration across roles. Agile and product team approaches bring data engineers, scientists, and business users together into cross-functional squads that co-own outcomes. MLOps practices unify data scientists, developers, and operations under a shared life cycle, ensuring that models are not just built but reliably deployed and monitored. Governance councils provide forums for IT, business, and compliance to work together on standards. Communities of practice and training initiatives help spread best practices and literacy across the organization.

In short, there is no single "right" model for collaboration. The key is to balance central standards with local ownership and to back the structure with methodologies that drive day-to-day teamwork. CoEs provide consistency and depth; domain teams provide speed and relevance. Organizations that combine the two—while embedding agile practices, product thinking, and MLOps pipelines—are best positioned to succeed in the fast-moving world of data and AI.

The Growing Importance of Openness

When we think about the AI ecosystem, openness is now one of its defining characteristics. Openness comes in several forms: open-source software, open file formats, open standards, and open systems. Twenty years ago, openness was often a niche discussion among technologists. Today it is mainstream, shaping the way organizations build, deploy, and govern data and AI systems.

Open-source software has become the backbone of modern AI development. Languages such as Python and R, and frameworks like TensorFlow, PyTorch, and LangChain are all community-driven. Open-source platforms such as Hugging Face provide access to the latest algorithms and techniques, often years before commercial vendors incorporate them. Open source encourages collaboration and reduces vendor lock-in, allowing companies to innovate quickly. Commercial vendors now routinely integrate or support open source in their platforms, recognizing that enterprise customers demand this flexibility.

Open file formats are also important. These are file formats that are published and freely available for anyone to use. They have certain specifications to enable standardization. They can be available in proprietary or free and open-source software. Standards such as CSV, JSON, Parquet, ORC, and Avro ensure that data can move easily across platforms. Apache Iceberg and Delta Lake define how large analytic tables are stored, versioned, and queried. These formats originated from industry consortia and standards bodies seeking interoperability and portability. In effect, they enable you to change platforms without losing access to your data or models, if the format is supported.

Closely related but broader are open standards. While open file formats specify how data is encoded, open standards set the rules for how systems communicate and interact. Examples include SQL for querying databases, HTML for structuring web pages, and XML for representing structured information. Open standards are usually developed by formal bodies (such as ISO, W3C, or ANSI) or by widely adopted community practice. They ensure that tools and platforms can interoperate consistently, creating a common language across the ecosystem.

Open systems and architectures extend the concept further. This means building environments that are modular, interoperable, and not tied to a single vendor.

I bring this up because openness is now more important than it was, say 20 years ago. In the early 2000s, most enterprises relied on proprietary stacks with limited interoperability. Open-source projects like Linux or Hadoop were still considered experimental by many organizations. Today, openness is expected. It can support innovation, attract talent (because skills are portable), and provide resilience through interoperability and portability.

What's Next?

Together, the AI ecosystem is about people, processes, and technologies. In the next chapter, we will look more closely at how these teams actually build models, what goes into training them, how algorithms work, and why concepts such as bias and explainability are so critical for trust.

CHAPTER

15

Building AI Models

In this chapter, we will examine the following:

- Building AI models
- Bias, fairness, and explainability
- Unstructured data

In the previous chapters, we examined some keys for AI success, including establishing a trusted data foundation and aligning AI strategy with business goals, and we touched upon data governance. We also talked about the roles required for AI. With some of this groundwork in place, we now turn to the heart of the matter: the AI itself. This chapter offers a high-level view of the processes behind building traditional AI models, such as machine learning. This includes training approaches and feature engineering as well as the algorithms that power common use cases such as churn prediction or customer retention.

We will explore how models are tested, what their outputs mean, and why concepts such as bias, fairness, and explainability are so critical for building trust. We'll also look at how successful organizations overcome some of these issues. I think these principles are important so that you understand what data scientists do and what AI application developers need to know (at least at a high level).

This will set the stage for Chapter 16, which discusses how successful organizations operationalize their models, which is where the value of AI lies. This will also put into perspective what other teams (e.g., machine learning operations [MLOps]) do.

Building AI Models

As was previously mentioned, "artificial intelligence" (AI) is an umbrella term that includes various techniques. One of the most popular is machine learning, where computers learn patterns from data and improve performance without being explicitly programmed. Because machine learning underpins so many common business applications, such as predicting churn, detecting fraud, or recommending products, it provides a good entry point for understanding how AI works. I provided a very short overview in Chapter 1, but it is important to dive a bit deeper to better understand something about how the models work in order to appreciate what they can do.

The first thing to understand about machine learning is that it works much as the name suggests. The idea is to train a model on data so that it can learn patterns and make predictions. Four popular approaches are supervised, unsupervised, semi-supervised, and reinforcement learning.

- **Supervised learning:** This is the most widely used approach and involves training a model on labeled examples (annotated data—the outcomes in the training set) where the correct outcome is already known. This might be past records of customers who churned or did not churn. By learning the relationship between the input information and the outcomes, the model can make predictions about new cases. Decision trees, random forest, logistic regression, linear regression, and support vector machines are all examples of supervised learning approaches. So too are neural networks.
- **Unsupervised learning:** Here, there is no target or outcome variable provided. The algorithm works with raw data and tries to uncover structure within it. A common use case is clustering, where the model clusters customers into groups based on similarities in behavior or characteristics, without being told in advance what the groups should be. Examples of unsupervised learning include K-means clustering or association rules learning.
- **Semi-supervised learning:** This approach combines elements of both supervised and unsupervised learning. The model uses the limited labeled data to learn an initial structure and then leverages the larger unlabeled set to improve accuracy. This is especially valuable in domains where labeling data is expensive or time-consuming, such as medical imaging or fraud detection.
- **Reinforcement learning:** Here, the system learns by acting in an environment and receiving feedback in the form of rewards or penalties. Over time it figures out which actions maximize success, making it useful for dynamic problems like pricing, resource allocation, or teaching an AI to play complex games. Examples of algorithms used here include Q-learning and Markov decision process.

A Simple Example

Let's start with building a simple machine learning model using a simple technique, the decision tree. I'm providing this example because I think it is important to understand how the thought process behind machine learning and prediction is different than descriptive analytics. Again, this is not a deep dive, but it is important to get you familiar with what is involved in model building.

Let's take an example that is near and dear to my heart—customer retention. Let's assume that we work for an insurance provider. I want to build a model that can provide me with the probability that a policy holder will drop my insurance. One way to do this is to use a decision tree algorithm. Figure 15.1 shows a table of data (of course, there might be thousands or more rows of data) that provides various attributes (the features, which are derived attributes) and the target of interest (churn or no churn).

Customer_ID	Years_as_Customer	Premium_Increase	Claims	Age	Dropped_Insurance
1	1	30	3	25	Yes
2	2	35	2	28	Yes
3	3	40	4	30	No
4	4	45	1	32	Yes
5	5	20	0	35	No
6	6	15	1	40	No
7	7	25	2	42	No
8	8	30	0	45	Yes
9	9	40	3	48	Yes
10	10	50	2	50	No
11	11	55	1	52	Yes
12	12	60	0	55	No
13	13	20	2	57	No
14	14	25	3	60	Yes
15	15	30	1	62	No
16	16	35	2	65	Yes
17	17	40	3	67	No
18	18	20	0	70	No
19	19	15	1	72	Yes
20	20	10	0	75	No

Figure 15.1: Sample data for decision tree.

The table has the customer ID, the number of years that the customer has been with the insurance company, the percentage that their premium has increased since becoming a customer, the number of claims they've had since becoming a customer, their age, and the outcome (e.g., dropped insurance, yes or no). All of these (except the customer ID and the target) are the features (i.e., derived from the data). In this case, the premium increase is a calculated variable. Features can be quite complex; these are perhaps overly simplified.

When looking at this data, you may or may not be able to see a pattern in it. Here, the pattern has to do with premium increases and years as a customer. The pattern here is hard enough to discern; imagine if there were thousands of rows of data!

So, how does this work? In a decision tree, each internal node in the tree represents a question about one of the input features, for example "How many years have they been a customer?" or "How much did their premium rise?" The algorithm evaluates the data and selects those that best split the data into groups that are more homogeneous with respect to the target variable (churn or no churn). This process continues until the tree reaches leaf nodes, which contain the predicted outcome, such as "likely to churn" or "likely to stay." Reading down the tree gives you a set of rules to see how different factors contribute to the prediction of churn while also offering an interpretable model that business stakeholders can understand.

Figure 15.2 illustrates the tree for the previous data. The left branch of the split means <= (less than or equal to), the right branch means > (greater than). Each leaf node shows the proportion of those who churned vs. stayed in the group. The Gini score measures how mixed the group is (0 is pure). Of course, this sample size is really small, so don't read too much into it. However, if we were to go down the righthand side of the tree, the rule would be something like this:

If the premium increased by more than 22.5% and years as a customer is less than 2.5 years, then the probability of churn is 100% (class = yes for churn).

In other words, new customers hit with a sharp price hike almost always leave. If the premium increases by 22.5% and years as a customer is greater than 2.5 years and the premium increase ≤57.5 and claims ≤3.5, then about 60% churn and 40% stay (with a mixed Gini score of 0.48). In other words, moderate-to-high premium increases cause churn for many, but not all, established customers. This is a risk segment, not an absolute.

Again, this is an oversimplified example (and the output of different tools may look different). However, it is useful to see how a decision tree can be developed to classify customers into different groups. When working with decision trees, it is important to note that they split outcomes into probabilities rather than certainties, so each group represents the likelihood of an outcome

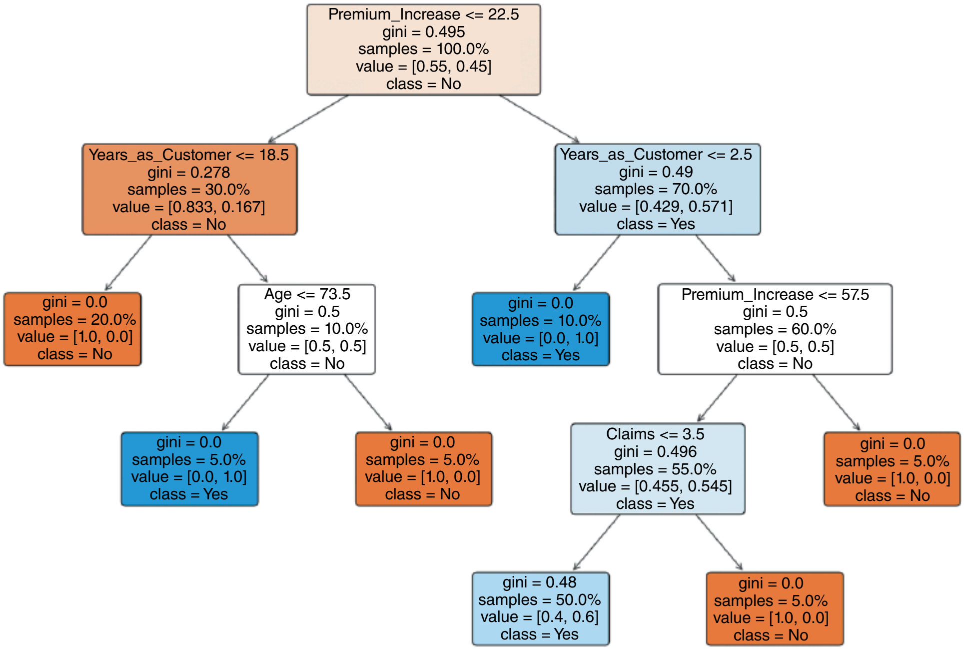

Figure 15.2: A decision tree.

Source: Generated with AI using Matlab-ChatGPT

rather than a fixed rule. They also report a measure such as the Gini score to indicate how mixed each group is, with low values showing more confidence and higher values showing uncertainty. Other measures to help assess model performance may include accuracy, precision, recall, confusion matrices, and cross-validation scores.

Moving Beyond Decision Trees

Decision trees can handle different types of data well, including both numbers and categories, but they are also prone to overfitting if left unchecked, meaning they may learn patterns that are too specific to the training data and fail to generalize. For this reason, while decision trees are useful for interpretation and business insight, they are often combined into ensemble methods (that use more than one type of model) such as random forests (that use multiple decision trees) or gradient boosting to achieve better performance.

Another important point about training a machine learning model is that typically there is a process associated with this. That process involves separating the available data into a training set and a test set. The training set is used to teach the model by exposing it to examples, while the test set is held back and used to evaluate how well the model performs on data it has never seen before. This separation helps prevent overfitting and provides a more realistic measure of how the model will perform in practice. This may be an 80–20 split (80% for training, 20% for testing). In many cases, a third dataset, called a validation set, is also used to fine-tune the model's parameters before the final evaluation on the test set.

Another common technique is cross-validation, where the data are divided into several groups and the model is trained and tested repeatedly on different combinations of these groups. This approach provides a more reliable estimate of model performance by ensuring that every data point is used for both training and testing across different iterations.

Table 15.1 references some of the common terms associated with machine learning. We've covered many of these already. Two other important concepts

Table 15.1: Common Terms Associated with Machine Learning and AI

TERM	DEFINITION	EXAMPLE
Target variable	The outcome the model is trying to predict	Churn/No Churn; Fraud/No Fraud; sales amount
Features	The input variables or attributes used to make predictions	Number of support calls, customer tenure, monthly spend, loyalty score
Labels	The correct answers provided for supervised training	Fraud or no fraud in historical data
Training data	Data used to teach the model by exposing it to features and corresponding labels	Historical customer records with churn outcomes
Test data	A separate dataset used to evaluate how well the model performs on unseen examples	Holding back 20% of customer data for validation
Model	The mathematical representation learned from the training data	Decision trees, regression equations, etc.
Prediction	The output the model produces when applied to new data	Predicting that a customer has a 70% chance of fraud
Classification	A supervised learning task with categorical inputs	Predicting "approve loan" vs. "deny loan"
Feature engineering	Developing a derived asset from data used for training	Turning "date of first purchase" into "customer tenure in months"

(continues)

Table 15.1: (Continued)

TERM	DEFINITION	EXAMPLE
Overfitting	Occurs when an algorithm fits too closely to its training data and can't generalize to new data	Model predicts perfectly on training data but poorly on test data
Bias	Errors from assumptions in the model	Predicting that the best candidates for an engineering job are male
Explainability	The ability to communicate how a model arrives at its predictions	Explaining why a credit card was denied

in AI/machine learning are bias and explainability, which are explained later in this chapter.

Hopefully this has given you some introduction into how these models are developed. Building these models is part art and part science, according to data scientists who need to think about the right features, which algorithms to use, and ensuring that they build useful models. The decision tree is simple, as is regression (which you probably learned about in your high school or college statistics courses). Many of the other algorithms are more complex.

Table 15.2 outlines some popular algorithms, how they work, and the kinds of use cases they support. Some of these are supervised learning approaches and need labeled data (linear regression, logistic regression, decision trees, random forests, support vector machines, or neural networks). Some are unsupervised (K-means clustering, some kinds of neural networks). Some are semi-supervised (deep neural networks, some kinds of support vector machines). The kind of algorithm you will use will depend on the type of problem you're trying to solve. If it is a classification problem to predict a discrete category, you might use a decision tree or logistic regression. If it is a regression problem (to predict a continuous data point), you might use linear regression or random forest. If it is a clustering problem to group similar data points without data labels, you might use K-means clustering. If you are trying to classify text, you might use a naïve Bayes model. The type of data also matters. For instance, if the data is nonlinear, you might want to use a decision tree.

The algorithms in the table aren't meant to be exhaustive by any means, but they give you a sense of the kinds of algorithms that are available and illustrate the breadth of algorithms you might hear about. Data scientists are trained in the kinds of models to use and when. Likewise, some of the AI tools on the market will automatically pick the best algorithm to use (called autoML tools for automated machine learning). However, it makes sense if using a tool like this to try to understand something about the algorithm it picks, to ensure that it makes sense in the current situation.

Table 15.2: Common Machine Learning Algorithms

ALGORITHM	HOW IT WORKS	USE CASES
Linear regression	Fits a line to predict continuous variables; best fit line for data	Forecasting sales
Logistic regression	Estimates probability of categorical/binary outcomes	Fraud detection, churn prediction
Decision trees	Splits data into branches based on features; leads to a classification or prediction	Classification problems such as retention modeling, customer
Random forests	Ensemble of decision trees	Fraud detection, credit scoring
Support vector machines	Finds the best hyperplane that cuts data points into classes	Image recognition, spam detection
Naïve Bayes	Uses probability (Bayes theorem) for classification	Spam filtering, text classification, sentiment analysis
K-means clustering	Classifies based on nearest data points	Customer segmentation
Neural networks	Layered nodes learn complex, nonlinear patters	Speech recognition, image classification, NLP
Gradient boosting	Sequential trees where each corrects previous error	Search ranking, ad targeting, risk modeling

The Machine Learning Life Cycle

The machine learning life cycle provides a useful way to think about how models are built and maintained. It begins with model preparation, which includes defining what the model should predict, identifying and preparing the data, and setting up the pipeline. Next is model building, where data scientists engineer features, train and test the model, tune its performance, and register it for governance. The final phase is operationalization, which involves deploying the model into production, monitoring its behavior, and retraining as needed. Governance runs through the entire life cycle, ensuring models remain reliable, explainable, and compliant.

We will dive deeper into the machine learning life cycle and particularly the operationalization part of it in the next chapter. However, it is important to understand the life cycle and the fact that machine learning isn't just about building the model, in order to understand bias, fairness, and explainability.

Bias, Fairness, and Explainability

Chapter 7 discussed how bad data quality can lead to bad outcomes, and I provided a few examples of this. Bad data quality can also include data that is biased or not fair. Bias is often in a category of its own because the consequences, which as we've seen previously, can be severe. In other words, it is important to pay attention to data bias as your company moves into AI.

So what is data bias? Simply put, bias in data happens when the dataset systematically misrepresents reality. That could be having more data from one group than another, or it might be reflective of historical human biases. A model trained on biased data may produce outputs that appear "accurate" statistically but are unfair or discriminatory. And standard data quality checks might not catch it because the data could be clean but still biased. Bias in models can occur because the methods used to collect the data represent a skewed view of reality. This can happen at different parts of the machine learning life cycle, when the data are being collected or even when the model is being built. It is important to be aware of data bias and ask questions about it.

It is also important to reiterate that using data ethically is important, not simply because it is the right thing to do but also because it can benefit your business. Poor AI applications based on biased data can impact a company's reputation, erode customer trust, impact its stock price, and result in fines. *Data bias is not something to be ignored*.

There are numerous kinds of biases. In fact, if you look at the Catalogue of Bias (a collaboration that started in 2017), you'll find more than 60 kinds of bias![1]

In the data and analytics space, some common ones include the following:

- **Availability bias:** This occurs when researchers/analysts collect data that are easiest to access or most recent. Humans tend to rely on data that are the easiest to get to. But this can also occur in machine learning, for example, if a model is trained on only the past year's claims data because it is readily available, ignoring older patterns. This may produce inaccurate conclusions.
- **Confirmation bias:** This occurs where researchers or algorithms favor information that confirms their existing beliefs while ignoring or dismissing contradictory data. For example, I might build a machine learning model for recruitment that favors males because the model learned that successful hires for a certain position were males. We discussed this type

[1] The Catalogue of Bias. (n.d.). *The catalogue of bias*. Available at: `https://catalogofbias.org/biases` [Accessed 3 December 2025].

of case in Chapter 7. Here the algorithm developed confirmation bias, concluding that masculine sounding resumes were a sign of success. Another example is the social media echo chamber where algorithms are trained to provide content to users based on their clicks, likes, and so on. That creates an echo chamber where users just see what they like. This produces a biased, tailored experience for the user.

- **Historical bias:** Also known as social bias or societal bias, this occurs when past inequities or patterns of discrimination are part of data used to train a machine learning model. For instance, a medical diagnosis system might have historical bias if it is trained on data that is not representative of diverse patient populations. Platforms like Facebook have faced criticism for allowing advertisers to target ads based on demographics such as gender and race, which can reinforce historical stereotypes. For example, job ads for traditionally male-dominated professions might be shown primarily to men, while those for traditionally female-dominated roles might be shown primarily to women, perpetuating existing inequalities.[2]
- **Label bias:** This occurs when the labels used to train a supervised learning model are inconsistent or biased themselves. For instance, human labelers may label consistent claims differently depending on their own judgment. The model may then learn these inconsistencies.
- **Selection bias:** Selection bias occurs when the sample for selection for a study doesn't represent the target population. This can be due to sampling bias when members of one group are less likely to be included in a study. Or it might occur if people volunteer for a study (self-selection bias). For example, skin cancer diagnostic tools are less accurate for dark-skinned individuals due to nondiverse training data.[3]
- **Survivorship bias:** This occurs when only those who "survive" are included in the analysis. That might be only loyalty customers or only those customers who have not dropped the vendor. That might lead to overly optimistic conclusions.

[2] Duffy, C. (2023). People are missing out on job opportunities on Facebook because of gender, research suggests. *CNN Business*. Available at: `https://www.cnn.com/2023/06/12/tech/facebook-job-ads-gender-discrimination-asequals-intl-cmd/index.html` [Accessed 21 August 2025].

[3] Andrés Morales-Forero, A., Rueda, L.J., Herrera, R., Bassetto, S., and Coatanea, E. (2025). *Predictive representativity: Uncovering racial bias in AI-based skin cancer detection*. Preprint. Available at: `https://www.researchgate.net/publication/393889093_Predictive_Representativity_Uncovering_Racial_Bias_in_AI-based_Skin_Cancer_Detection/link/687f0b3eba8eac4f17281346/download?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6InB1YmxpY2F0aW9uIiwicGFnZSI6InB1YmxpY2F0aW9uIn19` [Accessed 21 August 2025].

- **Measurement bias:** This can occur when errors are introduced into the process by how data are collected, recorded, or measured. For instance, claims that are recorded in two different regions may be recorded differently and lead to measurement bias.

Needless to say, there is overlap in these biases. Survivorship bias can be considered a form of selection or sampling bias. Label bias may reinforce historical bias if past labeling decisions reflect prejudices. In practice, it is less about identifying every type of bias precisely and more about recognizing that bias can occur at any stage of the machine learning pipeline, from data collection to labeling to model design to interpretation. It is important to be aware of it.

And, as you can see in the aforementioned types of bias, bias can occur across the machine learning life cycle. The Business Software Association (BSA) Report on Confronting Bias: BSA's Framework to Build Trust in AI does a good job of articulating this.[4] For example, bias can occur in the problem formulation stage with the basic assumptions of the model being biased. It can occur in the data preparation stage with data that is labeled incorrectly or has sampling or historical bias. It can occur in the model development stage where features are being created as input to the model. It can occur when the model is deployed if the data used to train or evaluate an AI system differs from the population the system encounters when it is deployed. I think you get the idea.

Overcoming Data Bias

Because bias is shaped by human choices as well as by technical factors, organizations must focus on people, processes, and tools/technology as part of an overall bias mitigation strategy. Organizations that are good at mitigating data bias typically utilize an integrated approach that captures all these factors.

Some best practices include the following:

- **Involving diverse teams:** On the people side, overcoming bias starts with the people who design and manage AI systems. Diverse teams are important because team members bring multiple perspectives to the table, making it more likely that hidden assumptions or blind spots will be surfaced and challenged. When people with different backgrounds, experiences, and expertise work together, they are better equipped to identify where bias might creep in and to create solutions that serve a broader range of stakeholders. Stakeholder involvement across functions and communities

[4] Business Software Alliance. (2021). *Confronting bias: BSA's Framework to build trust in AI*, pp. 1–4. Available at: `https://ai.bsa.org/wp-content/uploads/2021/06/2021bsaaibias.pdf` [Accessed 30 August 2025].

is equally important, since it ensures that AI development reflects a wide variety of voices.

- **Diverse data collection:** Of course, inclusive data collection is critical, as many biases are introduced at the point of gathering information. Teams must make a deliberate effort to source data from across demographic groups and contexts so that no one population is underrepresented. Teams can use different techniques to transform and balance the data to reduce the influence of bias before the AI models train on it.
- **Embedding fairness by design:** Fairness by design represents a collective shift in thinking across industry, academic, and policy circles toward proactively embedding fairness meaning treating this inclusivity as a core requirement from the start, rather than attempting to correct for inequities after a system has already been deployed.
- **Oversight:** Oversight is needed to make sure these practices remain effective over time. Regular audits and reviews help organizations test whether their datasets and models continue to reflect real-world conditions as both the data and social norms evolve. This includes human oversight and feedback loops, such as audit functions or appeal processes, to help maintain accountability. It includes monitoring systems that are discussed later in the book when we talk more about AI governance.
- **Training:** Training and education programs play an important role as well, giving teams the skills to recognize bias and understand how to mitigate it. Together, diverse teams, inclusive data collection, collaborative oversight, and ongoing education form the people-centered foundation for building fair and trustworthy AI systems.

In other words, people best practices include diverse teams, inclusive data collection, fairness by design, collaboration, oversight, accountability, training and education.

The process side employs governance and the use of frameworks.

- **Governance first:** Governance, of course, is extremely important in mitigating bias. And, as I've said, organizations that are more successful do not consider it an afterthought. It is part of the process, upfront. That means considering bias from the get-go.
- **Consider the frameworks:** The BSA framework, mentioned earlier, provides a life cycle risk assessment approach to guide bias identification and mitigation across AI life cycles. It includes resources to use to help in impact assessment and risk mitigation. A 2025 framework is the IEEE 7003-2024, Standard for Algorithmic Bias Considerations.

Tools are also starting to emerge to help to address data bias. Some of these tools are referenced in the frameworks cited previously and provide algorithms to address bias in the preprocessing stage, the in-processing stage, and the post-processing stage in a machine learning model. Preprocessing addresses bias at the data stage (e.g., rebalancing samples, anonymization). In-processing embeds fairness constraints during model training (e.g., re-weighting). Post-processing adjusts model outputs to correct disparate impacts. This staged approach ensures bias can be caught and corrected at multiple points in the life cycle.

Many of these toolkits are open source, meaning that they are free to use and are supported by a community. Some popular ones include the following:

- **AI Fairness 360:** This is an open-source library (available in R and Python), introduced by IBM Research, containing techniques developed by the broader research community to help detect and mitigate bias in machine learning models.[5] It provides numerous algorithms to address bias across the AI life cycle, including the stages mentioned earlier.
- **Fairlearn:** Fairlearn**,** developed by Microsoft and Allovus Design, offers a toolkit with visualization dashboards and mitigation algorithms to help navigate performance-fairness trade-offs. It is focused on mitigating fairness-related harms, specifically, allocation harms and quality-of-service harms—for groups of people, such as those defined in terms of race, sex, age, or disability status.[6]
- **Aequitas Audit toolset:** An early toolkit worth noting is Aequitas, an open-source audit library developed at the University of Chicago's Center for Data Science and Public Policy.[7] Aequitas was one of the first efforts to create human-readable fairness reports that could be understood by both technical teams and policymakers, and it gained traction in domains such as criminal justice, healthcare, and housing where questions of equity are especially sensitive. Its strength lies in providing a clear set of bias metrics and structured audits that allow organizations to examine how models treat different demographic groups. This toolset is referenced extensively in the BSA framework.

---

[5] AI Fairness 360 (AIF360). (n.d.). *AI fairness 360*. Available at: `https://github.com/Trusted-AI/AIF360` [Accessed 30 August 2025].
[6] Fairlearn. (n.d.). *Fairlearn*. Available at: `https://github.com/fairlearn/fairlearn` [Accessed 30 August 2025].
[7] University of Chicago Center for Data Science and Public Policy. (2018). *Aequitas: Bias and fairness audit toolkit*. GitHub. Available at: `https://github.com/dssg/aequitas` [Accessed 30 August 2025].

In addition to open-source libraries, some commercial products now provide built-in bias detection and fairness features as part of their enterprise AI platforms. Some examples include the following:

- **Amazon SageMaker Clarify:** Amazon SageMaker Clarify helps detect potential bias in datasets during preparation without requiring code. Users specify input features such as gender or age. Clarify generates reports with fairness metrics to highlight imbalances. If issues are found, the software claims it can rebalance datasets using several different techniques.
- **John Snow Labs:** John Snow Labs Responsible AI integrates directly into the Spark NLP ecosystem, enabling bias detection and fairness testing for unstructured text data. It is particularly valuable for regulated industries like healthcare and finance, where auditability and compliance are critical.
- **Trūata:** Trūata AI Ready is designed for structured and semi-structured data, offering dashboards that detect proxy bias (when a model uses a substitute variable), measure fairness risk, and suggest remediation steps. It standardizes bias assessments across datasets so teams can address issues before deploying models.

The bias tools space is a rapidly evolving one (from both the academic and commercial fronts) and one to watch.

Explainability

Bias and fairness are tightly connected to the idea of explainability. If you cannot explain how a model reached its conclusion, it becomes very difficult to know whether its outputs are fair or biased. Traditional models such as decision trees or linear regression are often considered more interpretable because you can trace exactly which factors influenced the prediction. They are often known as *self-interpretable models*. More complex approaches, such as deep learning or large language models, are sometimes referred to as *black boxes*, meaning that their inner workings are not available to the user. A neural network is a good example of this. This can mean that there is no way for the user to understand how predictions are derived, what they mean, or how to interpret and explain the results.

Explainability is the ability to describe how the model works in a way that an average person can understand it. In addition to teaching model builders the skill of explaining model output, work is underway in both the commercial and academic communities to develop new methods and algorithms to help with explainability using text or graphics that highlight the important features in machine learning models.

One example of this is LIME (Local Interpretable Model-agnostic Explanations). LIME works by using a local surrogate model to explain a black box model such as a neural network. Shapely values were originally used in game theory and are now called SHAP scores to explain the relative importance of features in a machine learning model, although some research has suggested that these scores may be misleading.[8] In some instances, vendors have incorporated these into their software to produce charts where users can determine the most important features in predicting the outcome of interest.

As with bias, work is being done to improve explainable AI (XAI).

What About Unstructured Data?

So far, a lot of what has been covered in this chapter use examples with structured data. Back in Chapter 4, we talked about different kinds of data and that includes unstructured data such as text data, image data, and audio data. The majority of information is unstructured. Deriving value from this data has been the focus of specialized areas in the AI field including natural language processing (NLP), computer vision, and speech processing.

As is the case with much of AI, NLP is not new. Early systems back in the 1960s and 1970s used manually created grammars and dictionaries to parse language. In the 1990s statistical approaches such as hidden Markov models became popular. By the 2000s, machine learning methods became the popular methods and could perform tasks such as translation and sentiment analysis. Then, of course, the introduction of transformer models (a type of neural network that learns context in sequential data such as a sentence) gave rise to large language models such as BERT and GPT and what is now known as generative AI. However, it is important to keep in mind that generative AI is really a subset of NLP.

What is involved in NLP? Because natural language is messy, preprocessing techniques for the data play an important role in NLP. These are different from what is involved in processing structured data (such as creating features) and include techniques such as the following:

- **Tokenization:** Splitting text into units such as words, subwords, or characters. For example, "AI models are powerful" → ["AI," "models," "are," "powerful"]. This makes it easier for the model to handle without losing context. Tokenization is important for NLP use cases such as sentiment analysis that may classify a customer review as positive, negative, or neutral.

---

[8] Marques-Silva, J. and Huang, X. (2024). Explainability is not a game. *Communications of the ACM*. Available at: `https://cacm.acm.org/research/explainability-is-not-a-game` [Accessed 29 August 2025].

For instance, the review "The cream was greasy and arrived late" could be broken down and the words "greasy" and "late" would be classified as negative sentiment.

- **Lemmatization:** Reducing words to their dictionary form or lemma (vs. stemming which you might see associated with this which just cuts the word to its root word). For instance, "running," "ran," and "runs" all become "run." This helps models generalize across variations.
- **Stop Word Removal:** Removing very common words (such as "the," "is," "and") that often don't add much meaning for tasks like classification. Whether to remove them depends on the use case. Translation systems, for example, may still need them.

Since machines can't work with text data, this data is converted into a numerical representation known as *vectorization*. Early methods included techniques such as a bag of words that counts the frequency of the words but ignores the order or TF–IDF (term frequency–inverse document frequency) that weights words by how important they are across documents. Once these words are represented numerically, then models can be used to process the data. These might include machine learning models or deep learning models. The choice of model depends on the task and available compute resources.

Unstructured data also extends beyond text. Image processing is the domain of computer vision, which started with handcrafted feature extraction (edges, shapes, textures) but now relies on deep learning architectures such as convolutional neural networks (CNNs) and vision transformers. These systems can classify images, detect objects, and increasingly generate new images. Generative AI models like Stable Diffusion and Midjourney illustrate how computer vision has expanded from analysis into creation.

Voice and speech processing transform raw audio into usable features and insights. Traditional methods relied on signal processing techniques, phoneme extraction, and acoustic models for transcription and speaker recognition. Today, deep learning models such as wav2vec (Meta) or Whisper (OpenAI) process audio end-to-end, powering real-time transcription, voice assistants, and even multilingual translation. Generative models extend these capabilities to synthetic speech, adapting tone, accent, or language style.

Taken together, NLP, computer vision, and speech processing demonstrate how AI has evolved to work with unstructured data. Generative AI did not replace these fields; it emerged from them—pushing each beyond analysis toward content generation. This progression reinforces the importance of governance and fairness practices, since the ability to generate realistic language, images, or voices magnifies both the opportunities and the risks.

What's Next?

Hopefully this chapter has given you some background into the model building process and what is involved. Next, we move into the deployment process and how models can be embedded into business processes. With this understanding, and the complexities involved, and understanding that this is what data scientists, ML engineers, AI engineers, and others do, we'll then move to two of the hottest areas in AI right now, generative AI and agentic AI. But first, let's talk about operationalizing models in production.

CHAPTER

16

Operationalizing AI

In this chapter, we will examine the following:

- The machine learning life cycle
- What "putting a model into production" means
- Why operationalizing a model is difficult
- What's involved in putting an AI model into production
- Tools for operationalizing a model

As I noted in Chapter 15, AI is about more than building a model. The value comes when you can operationalize the model by putting it into production as an application or part of a system or workflow. This chapter discusses what is involved in operationalizing AI models and the roles involved. This will give you some insight into the important work that goes on behind the scenes for AI.

The Machine Learning Life Cycle

Putting a model into production is known in some circles as *operationalizing* the model. Think about it. You may get some insights from a model you build, such as what people are associated with a high probability of a fraudulent transactions or customers who are likely to respond to a campaign. However, increased value comes when you can put that model into production as part of process. For instance, a model to predict fraud should be embedded into systems that

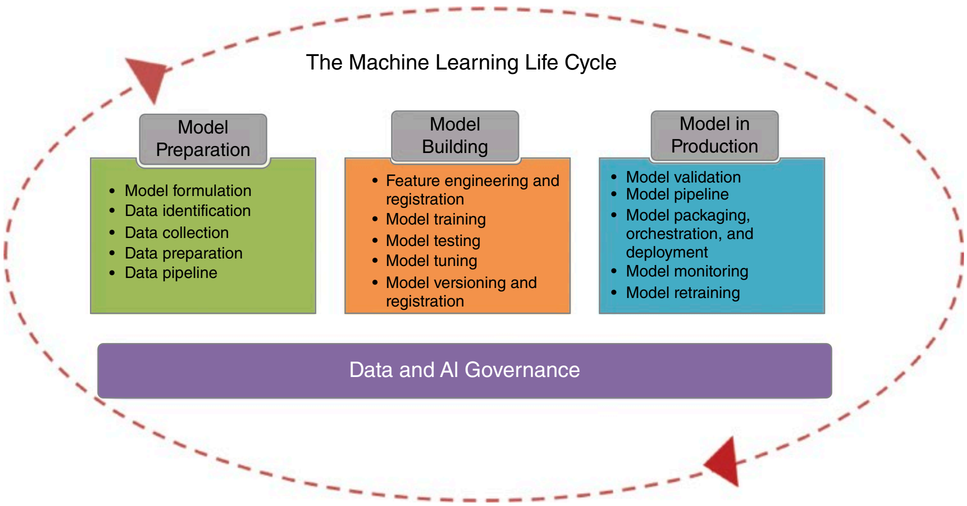

Figure 16.1: The machine learning life cycle.

process transactions. As the data flows through the model, the transactions can be scored for potential fraud. If it is deemed to have a high probability of fraud, then another action can be taken, such as stopping the transaction.

This is why the machine learning life cycle, which was mentioned in the previous chapter (and illustrated in Figure 16.1), is an important way to think about operationalizing machine learning (or any kind of AI for that matter).

The model preparation part of the machine learning life cycle includes model formulation—thinking through what the model is trying to do and the business need for it. It includes identifying data to use to train the model, collecting that data, preparing that data, and assembling the data pipeline.

The model building phase is all about building the actual model. In machine learning, this can include feature engineering—as mentioned in Chapter 15. Data scientists will spend a great deal of time developing features, and they would argue that feature engineering is part art and part science. Of course, it involves training, testing, and tuning the model. Importantly, it also includes registering the model and capturing metadata about it—who built it, when it was built, the version, the attributes used, etc. (as part of the governance).

The last phase is operationalizing the models. This is the backend model deployment process that some organizations don't think about when they embark on a machine learning effort, but it is critical. This phase includes validating the model in an operational environment, deploying the model, monitoring the model in production, and retraining the model. It also includes culture and process for accepting the model once it is in production.

Underlying all of this is that governance is needed to ensure that the models are trustworthy and compliant. Data governance and AI governance are covered in Chapters 19 and 20, respectively.

By the way, much of this same process holds even when we get to "newer" AI, such as generative and agentic applications.

What "Putting a Model into Production" Means

When data scientists or machine learning engineers talk about "putting a model into production," they mean moving it out of the experimental or lab environment and embedding it directly into a business process where it influences real-time decisions. A data scientist might build a model in a "notebook" environment (i.e., an interactive environment, which lets users combine code, data, visualizations, and narrative text in a single document for analysis and model development, e.g., Jupyter) or on a research server, but that is only the first step. In that setting, the model is trained and tested on historical data, but most likely no one outside the data science team interacts with it. It hasn't worked on new, real-world data. Operationalizing the model means packaging it in a way that other systems can call it, ensuring it receives live data as input, produces output automatically, and is monitored so it continues to perform as expected over time.

Consider the case of a recommender system for an online retailer. A data science team might begin by training a machine learning model on historical purchase data, browsing patterns, and customer demographics. In the lab, they can evaluate how well the model predicts which products a user might want to buy next. But the model creates more business value when it is operationalized. This means the retailer's e-commerce platform connects the recommender model to its live website. As customers browse, the platform sends their behavior data (such as items viewed or shopping cart contents) to the model in real time. The model instantly returns a set of product recommendations, which are displayed on the site as, "Customers who bought this item also bought …"

Why Operationalizing a Model Is Difficult

Operationalizing a model clearly requires more than the model. It includes the infrastructure such as the pipelines that feed customer data to the model. It includes the software systems that call the model's predictions. It includes the user interface that displays the recommendations. It should also include the monitoring processes that track whether the recommendations are working. This is where machine learning engineers that we talked about in the last chapter often get involved. Their job is to make sure the model is dependable, scalable, and safe to use.

One challenge is packaging the model. A model isn't just math—it depends on specific versions of software, libraries, and settings. If those aren't bundled together in a consistent way, the model may behave differently, or fail entirely, when it is moved into the cloud or into a company's systems.

Another challenge is scalability. A model that works fine on a few test records might struggle when exposed to thousands of customer requests every second. Systems need to be designed so that models can handle the workloads as well as scale up during peak demand and scale down when traffic is lighter.

Resources and infrastructure are also critical. Some modern models, such as deep learning and GenAI, may require specialized computer chips such as GPUs or TPUs to run efficiently. Choosing the right hardware setup involves trade-offs between cost, performance, and availability. For some organizations, this means investing in expensive equipment; for others, it means renting resources from the cloud. Either way, careful planning is needed so that models are both fast and affordable to run.

Finally, monitoring and governance make operationalization an ongoing task rather than a one-time event. Models must be tracked to ensure they are still accurate as real-world data changes, updated when they begin to drift, and documented so the organization knows how they were built and what data they rely on. Without this discipline, organizations risk running models that are outdated, biased, or noncompliant with regulations.

Some companies may think that if they have a model that they created in a notebook, they can package it up using a tool like Docker, and they can then put it into the cloud. It tends to be more complex than this. In other words, operationalizing a model is about engineering. How are the models packaged, made scalable, and updated when needed? Do they have the infrastructure, security, and governance around them to ensure they provide value?

What's Involved in Putting an AI Model into Production

In some ways, you can think of the model as a piece of software that needs to be put into production. In software engineering, the system development life cycle (SDLC) defines stages such as requirements analysis, design, development, testing, deployment, and maintenance. Operationalizing a machine learning model follows a similar pattern (although it may need to be updated on a more frequent basis). Some of these steps are in preproduction, some are production, and some are postproduction. Here's what's involved:

- **Versioning and documentation:** Once a model is built, its code needs to be versioned so that teams can reproduce results, if needed. For instance, DVC, or Data Version Control, is an open-source tool designed to address machine learning version control. It extends the capabilities of traditional version control systems like Git (that provide software version control) to handle large datasets, machine learning models, and complex pipelines. Tools like MLFlow (often used in experimentation) log the model's metrics, hyperparameters (configuration settings), and artifacts, creating a comprehensive record.

- **Feature management:** It is also important to version data for not only training but for model serving. This includes managing the features that are used in the model to ensure that they are tracked and understood. Successful organizations often use some sort of feature repository for this.
- **Continuous integration and validation:** Once a candidate model is identified, an automated pipeline can take over. Tools like Airflow or Kuberflow are sometimes used to orchestrate these pipelines. Additionally, models need tests to ensure they work correctly with the systems that call them. Tools like Jenkins and GitHub can be used to run automated tests. For example, can the recommender system handle malformed input, or does it fail gracefully?
- **Deployment:** Deployment pipelines push the model into production. Often, the model is packaged into a container. A container, created by tools such as Docker, packages the model together with its code, dependencies, and runtime environment into a single portable unit. This ensures the model runs consistently across different environments. The container makes sure that the model will behave the same, regardless of where it is deployed. In the recommender system, the model can be deployed as a containerized service. When a customer views a product page, the website calls an API exposed by the container, sending it data such as browsing history or cart contents. The container processes the request and returns recommendations in milliseconds.
- **Continuous deployment:** An orchestrator then pushes the validated and containerized model into production. This might be a tool like Kubernetes. Kubernetes is a kind of orchestration platform that automates the deployment, scaling, and operation of the machine learning model. For instance, if the model needs more resources such as CPUs or GPUs, Kubernetes can schedule them onto nodes with available capacity. If a container fails, Kubernetes can restart it. If you have a new version of the model, Kubernetes can update your deployment and replace the containers with new ones.
- **Monitoring:** Models must be monitored continuously for performance and possible degradation. Machine learning models degrade over time as data changes. These models are running in real-world environments. For instance, Evidently AI is a commercialized Python library for monitoring

and detecting data drift. Hydrosphere.io is another example of a platform to help with this.

- **Retraining:** If the monitoring system detects a problem, an alert is sent to trigger the workflow orchestrator to automatically retrain and deploy a new model. Maintenance cycles in AI include refreshing training data, updating features, and validating new versions against business objectives. Depending on the model, it may need to be retrained daily, weekly, or semi-annually. However, at some point, it will need to be retrained because no model stays fresh forever.

Note that the principles of operationalizing a model—including packaging it, deploying it, monitoring it, and maintaining it—apply just as much to generative AI applications as they do to traditional ones.

CONTINUOUS INTEGRATION AND CONTINUOUS DEPLOYMENT (CI/CD)

In traditional software engineering, CI/CD automates the testing and deployment of new code. In machine learning, the concept is similar but expanded:

- **Continuous integration** means automatically testing models and pipelines whenever new code, data, or features are added. This ensures errors are caught early and models remain compatible with the systems that call them.
- **Continuous deployment** means automatically pushing validated models into production with minimal manual effort. New versions can be deployed quickly and rolled back to previous versions if problems arise.

Together, CI/CD practices ensure that models are not just built once but remain current and reliable as data, code, and business needs evolve.

In the following "Expert Advice," Deanne Larson, PhD, discusses what she is seeing in terms of the challenges organizations are facing putting models into production and the importance of CI/CD.

EXPERT ADVICE: MAKING MLOPS WORK

"MLOps is still evolving," says Deanne Larson, PhD, president of Larson and Associates, and a faculty member at both Purdue Global and City College of Seattle. "Organizations are mature when it comes to data management, such as databases, data governance, and ETL, but not when it comes to AI. The question is how we mature to enable AI." She views MLOps as a natural extension of traditional data management that now must encompass the entire machine learning life cycle.

(continues)

EXPERT ADVICE: MAKING MLOPS WORK (CONTINUED)

Larson explains that the basic recipe for machine learning is familiar. There is data going in, followed by training, testing, and prediction. "We still have the same challenges we have always had, such as data quality and data silos. We need to be able to manage our data, understand it, and make sure it is of high quality." What has changed is the pace. "AI is evolving fast. Data is organic. It will always change, and the models we build on it will change too."

That rapid evolution makes MLOps essential. It applies DevOps principles of continuous integration and continuous delivery to the world of machine learning so that models can be managed like software. "MLOps is about managing the end-to-end life cycle," she says. "It begins with defining the problem, understanding and preparing the data, building and testing the model, and then putting it into production. Models have a limited lifespan. As data changes, accuracy decays. We need processes that allow us to retrain and redeploy models quickly."

Although MLOps can be complex, Larson emphasizes the benefits. "It streamlines the life cycle, improves reproducibility, enables faster deployment, and shortens time to market. It strengthens data integrity because the predictors used in models become visible. When you have a strong data foundation such as a curated data lake or lakehouse, that becomes the launch point for new algorithms and for improving business operations. MLOps lowers risk. It is not about locking down data but about making sure people can use data the right way."

Larson often relates MLOps to DevOps. Continuous integration and continuous delivery are essential to making the process work. "DevOps is about deploying software that works and doing that well. In MLOps we also need to deploy the data sets that allow us to run and score the model. Continuous integration means making small changes, adding features, or operationalizing updates while keeping the environment stable. Continuous delivery focuses on automated testing and packaging models for production. As organizations mature, they move from manual deployment to automation. It is about treating models like software packages."

For MLOps to succeed, Larson believes organizations must also address culture. "As much as we would like to say there is no chasm between business and IT, it still exists. MLOps helps bridge that. The team must be cross-functional, bringing together data scientists, data engineers, software engineers, and domain experts who understand the business problem." Each of these roles contributes something essential. Data scientists know how to build and evaluate models. Engineers understand pipelines and containerization. Business sponsors provide context and recognize unintended consequences.

Larson advises beginning with a realistic scope. "Create a roadmap. Understand your current architecture, ETL processes, and available skills. Identify your objectives and your priority models. Establish the necessary infrastructure, including storage, compute, and frameworks. Start small. Form a small team with clear goals and a

(continues)

EXPERT ADVICE: MAKING MLOPS WORK (CONTINUED)

well-defined scope. Pilot the process. Focus on small wins and reuse what you already have. Review your existing pipelines to see what can be automated or repurposed."

She cautions against building technology for its own sake. "Never build out MLOps unless you have models to deploy. You cannot simply stand up a platform and hope they will come. Apply the practices from DevOps to an area with defined boundaries and scope, one that is already generating multiple models. Build components that support better models and better data as part of continuous integration."

When implemented effectively, MLOps provides ongoing visibility and control. "Dashboards can track data drift and model drift. As part of the framework you can retrain models automatically. That is continuous learning. It means connecting with subject matter experts so you always know how the model is performing and what feedback to act on."

Larson believes this approach represents the future of AI operations. "If we integrate machine learning and operations, it becomes a true end-to-end process. The orchestration, workflows, and pipelines all need clear standards. That is what enables faster deployment and ensures models remain current. The lifespan of a model is directly tied to the rate of change in your data."

Ultimately, Larson sees MLOps as a vital enabler of sustainable AI. "We have had DataOps to help with data management. Now we need MLOps to do the same for models. Together they increase maturity, improve collaboration, and prepare organizations for innovation."

Tools for Operationalizing a Model

If operationalizing a model sounds like a lot of work, that's because it is—especially given that most organizations don't just have one or two models to operationalize. I've mentioned several open-source tools that can be combined into a production workflow, but stringing them together requires substantial engineering expertise and ongoing maintenance. Commercial platforms exist to simplify this complexity. Many are built on open-source components but offer a unified experience, enterprise support, and the kind of stability that busy teams need. For some organizations, open source provides the flexibility and control they want. For others, commercial tools that build on open source give them the assurance and scale they need to move quickly without sacrificing reliability.

In addition to open-source tools, a growing set of commercial platforms support the machine learning and GenAI engineering workflow. These include companies such as beam.cloud, which takes a trained model, packages it into a container, and exposes it as a microservice that other applications can call. Other offerings include companies such as UbiOps and Comet.com, which provide additional functionality.

At the enterprise scale, major cloud providers and ML platforms—including Amazon SageMaker, Google Cloud Vertex AI, Microsoft Azure ML, Databricks, and Snowflake—address the entire machine learning life cycle. These platforms automatically package models and their dependencies, allocate the right resources, and scale services up or down based on demand. They also integrate capabilities for data versioning, continuous integration and deployment (CI/CD), model registry, monitoring, and alerting. Many allow rollbacks to prior versions if a deployment fails, and they provide tools to track model drift as data changes over time.

What's Next?

I didn't write this chapter to scare anyone. I wanted to get the point across that it isn't enough to build a model; you also need to operationalize it. And that can be complex, but it is something you will need to invest in if you don't want to hit the value ceiling. Putting models into production is key for generating value, and a growing set of tools are available to help with that. But skills will be needed. What changes with generative and agentic AI is not the need for operational discipline, but the scale and scope of what must be managed, from prompts and embeddings to autonomous agents that act on behalf of users. With that, let's move to what's on everyone's mind: generative and agentic AI.

CHAPTER

17

Generative AI

In this chapter, we will examine the following:

- Understanding generative AI
- Foundation models
- AI consumers and builders
- Putting generative AI models into production with company data
- Hallucinations, bias, and guardrails
- Measuring success with generative AI

Up until now, I've been discussing data and AI from the perspective of the way data and analytics professionals see it. However, with the advent of generative AI (GenAI) a few years ago, the market has shifted into uncharted territory—fast-moving and disruptive, with organizations lacking clear guardrails compared to the more structured world of traditional data and AI.

There are different ways to look at this. On the one hand, this is a time of great innovation. Companies are creating new foundation models, building open systems, releasing new AI applications, and experimenting with novel use cases. That is exciting. In the Expert Advice from Daniel Ziv later in this chapter, we'll hear that perspective directly. Some people say that with GenAI, the data doesn't need to be perfect, just good enough. The advice is to try out the tools, see what happens, and fail fast. There is merit to that mindset. It encourages experimentation and lowers the barrier to entry. There is a sense of excitement around that, and much of it is well deserved.

On the other hand, when I speak to data and analytics professionals, I hear enthusiasm but caution. Their experience tells them that ignoring data quality, governance, or integration challenges eventually creates problems. They know too much to accept the "just good enough" mantra at face value. They recognize the risks of hallucinations, bias, and privacy issues. For them, innovation without discipline feels unsustainable. What's needed is balance—encouraging experimentation while also acknowledging the operational and governance realities.

So, in this chapter, let's look at what is happening in the market today with GenAI. We'll start with the fundamentals of what these technologies are and how they work. We'll look at how organizations are adopting the technologies. We'll examine the tools that are available, how they can improve individual productivity, and how they are being applied to solve both horizontal challenges across industries and vertical challenges in specific domains.

From there, we turn to what it all means for data and AI professionals who are tasked with bringing GenAI into the enterprise. Consumer use and experimentation are one thing; deploying these technologies on company data in production environments is quite another. That is where the lessons of operationalizing AI, which we covered in the last chapter, come full circle.

Understanding Generative AI

GenAI systems had been emerging for years, from early general adversarial networks (GANs, a type of neural network that generates new data, such as images) in 2014.[1] The real breakthrough is often credited to Vaswani et al. in 2017 with the paper "Attention Is All You Need,"[2] which introduced the transformer architecture and its attention mechanism. Transformers made it possible to scale models dramatically and capture context across long sequences of text, paving

[1] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. (2014). *Generative adversarial nets.* Advances in Neural Information Processing Systems (NeurIPS), 27. Available at: `https://arxiv.org/pdf/1406.2661` [Accessed 14 September 2025].

[2] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 30. Available at: `https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf` [Accessed 14 September 2025].

the way for large language models (LLMs) such as BERT[3] and GPT-2/3.[4,5] These advances shifted GenAI from research labs into practical applications, eventually culminating in the public release of ChatGPT in late 2022, which brought GenAI into mainstream use.

This momentum set the stage for a tipping point: in November 2022, the public release of ChatGPT from OpenAI, based on GPT-3.5, marked the first GenAI system to hit the market at scale. Within a single week, it attracted millions of users, sparking a wave of adoption and investment across industries. Within months, adoption was at 100 million users.[6] What followed was a flood of competitors and copilots from Microsoft, Google, Anthropic, and others. Today, the technology has already advanced several iterations, with GPT-5 now available, underscoring not only the amazing pace of innovation but also how rapidly GenAI has moved into mainstream business. GenAI has surpassed adoption rates (as a percentage) of other innovative technologies such as the telephone, email, and smartphones.

What made these AI systems so different from the traditional AI systems we've been talking about was that they were freely available and offered a natural language interface that made them easy for anyone to use. Just as important, what they did was fundamentally different. Instead of focusing on predictions derived from historical data, these systems were generative. In other words, they created new content and responses based on the prompts or input they received.

These systems work by ingesting massive amounts of information (e.g., content from the internet) and learning the relationships within that data. The intelligence behind these systems comes from what are known as *foundation models*. A foundation model is a very large, pretrained model (often a neural network) that has been exposed to vast amounts of diverse data to learn general patterns across domains. Once trained, these models can be applied to a wide range of downstream tasks. Different models may be better suited to

[3] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 4171–4186. [Accessed 14 September 2025].

[4] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. (2019). *Language models are unsupervised multitask learners.* OpenAI Technical Report.

[5] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A. et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 1877–1901.

[6] Hu, K. and Akhtar, A. (2023). ChatGPT sets record for fastest-growing user base. *Reuters*. Available at: `https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01` [Accessed 14 September 2025].

different tasks. Hugging Face (www.huggingface.com) provides a vast repository of one million plus models and sorts them by task.[7] The tasks include audio-to-text, text-to-text, document question answering, image-to-video, feature extraction, and much more.

Foundation Models

Let's look at how a large language model (LLM)—a kind of foundation model—works for generating text. It does this by turning words or word fragments into numerical representations called *embeddings*, which position them in a high-dimensional vector space where words with similar meanings are close together. For example, "doctor" and "nurse" end up nearer to each other than to "orange" in vector space (Figure 17.1). The model, built on a transformer architecture, uses these embeddings and an attention mechanism to understand context across long passages of text. When you provide a prompt, the LLM predicts the most likely next word (or token) based on patterns (the similarities) it has learned from massive amounts of training data, then repeats the process one token at a time to produce coherent language. By moving step by step through this vector space of relationships, it generates output that reads as fluent, contextually appropriate text. Really, it is just based on probabilities.

In other words, while predictive analytics provides the probability of an outcome (e.g., there is an 80% probability the customer will default), LLMs use probabilities internally to generate a sequence of tokens that forms coherent text. They calculate the probability of the next token at each step, generating output.

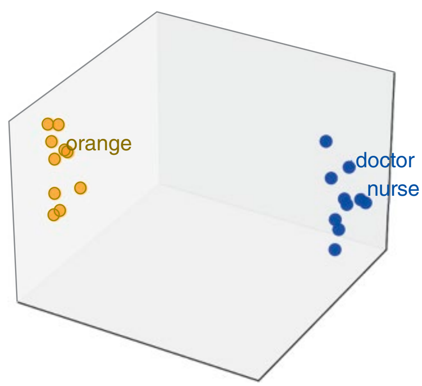

Figure 17.1: The concept of vector embeddings.

[7] See https://huggingface.co/models.

Numerous foundation models are available for use. Some popular ones include Open AI's GPT models, Anthropic's Claude, Google's DeepMind Gemini, and Meta's Llama. On top of these foundation models are built the assistants that we're all familiar with. These include general-purpose assistants like ChatGPT and Perplexity. They include business assistants such as Salesforce's Einstein and SAP's Joule. They include transcription assistants such as Otter.ai. They include creative assistants such as Adobe Firefly. Figure 17.2 outlines some of these popular foundation models and assistants. There are many, many more, perhaps thousands. They range from sales and customer support to marketing to video creation and beyond.

It is important to note that not all assistants are built with LLMs. Some use a combination of approaches. They might use smaller, more specialized language models—called *small language models* (*SLMs*)—or they might use machine learning or even a rules-based approach.

And the foundation models in the market today do more than deal with text. Soon after the introduction of the GPT-3, the same architectures were extended

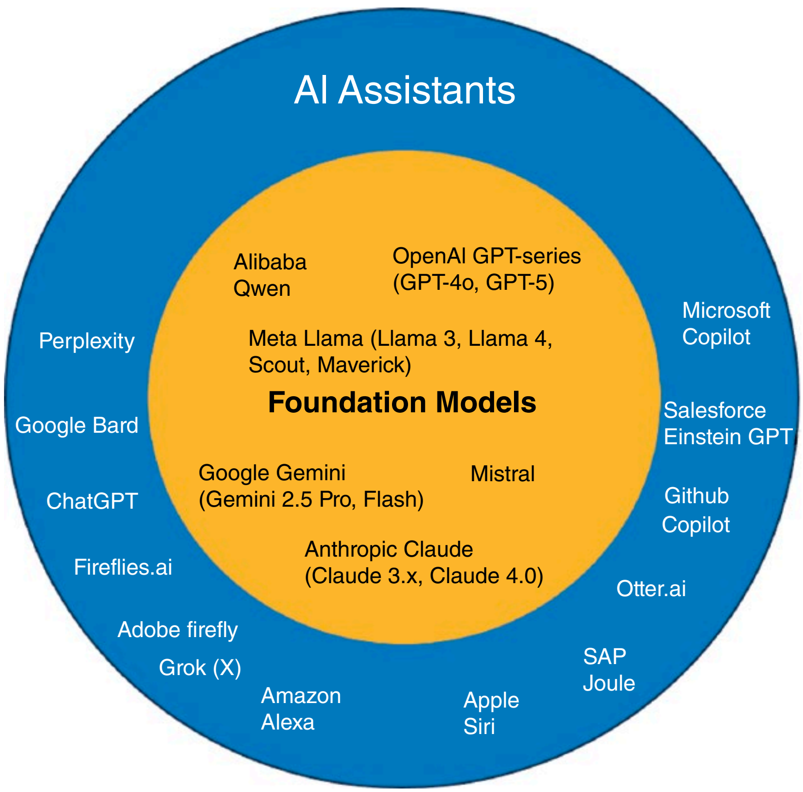

Figure 17.2: Foundation models and assistants—consumerized AI.

to vision and other modalities, linking language with images and, later, audio and video. Today's foundation models are increasingly multimodal, able to process and generate text, images, speech, and code within a single system. So, it would be incomplete to call GPT-5 only an LLM because by 2025 the GPT family has become multimodal foundation models. They can handle not just text, but also images, audio, and other inputs/outputs.

AI Consumers and Builders

Aside from displaying examples of foundation models and assistants, Figure 17.2 also illustrates an important point. These assistant applications are examples of what I refer to as *consumerized AI*. In other words, these consumerized AI assistants such as ChatGPT, Perplexity, and Microsoft Copilot are designed to be off-the-shelf tools that anyone can use without technical expertise. Their interfaces are simple and conversational, making them broadly accessible compared to enterprise AI systems that require customization or integration. The goal of these assistants is to boost individual productivity by handling everyday tasks such as drafting text, summarizing information, answering questions, or helping with coding. In this way, they represent the mainstreaming of AI into personal and professional workflows. The people who use these tools are called *AI consumers*.

In my research, I've seen that fully 100% of the data and AI professionals we survey at TDWI are making use of these GenAI tools in a consumer model. Ninety percent are making use of them at work to help with individual productivity.[8] Most of the respondents to our surveys are positive about the impact of AI on their work. They state that they are more efficient and productive. The average productivity increase cited by respondents to a recent survey was about a 30–35% improvement in productivity from using these tools.[9]

On the flip side, are the AI builders. These are the people who are building the GenAI applications using company data (or other data for that matter, such as external data or third-party data). I'm not talking about a build vs. buy decision in terms of buying software when I talk about builders. I'm not talking about editing company videos with a GenAI video tool, although the argument can be made that the video is company data. I'm not talking about analyzing your survey data with a GPT. These are discrete tasks that are often one-offs.

[8] TDWI. (2025). The impact of generative AI on business. *TDWI Research Brief*, 20 August. Available at: `https://tdwi.org/research/2025/08/ta-all-research-brief-the-impact-of-generative-ai-on-business-quick.aspx` [Accessed 30 August 2025].
[9] ibid.

It is about using a new group of tools to help build applications that make use of GenAI. I'm talking again about building production applications that take data in and produce output.

What are some examples of these kinds of applications? A popular first use case that I'm seeing a lot is organizations taking their call center notes and using a foundation model to determine sentiment or to classify the kinds of problems customers are having. Yes, this can be a one-off analysis, but more often organizations are building some sort of system for this. I'll describe what that looks like in the next section. There are other examples too. For instance, companies are building onboarding chatbots/assistants that may use company data (in the form of product documentation, FAQs, or other company proprietary information) to help new employees learn about the business by asking questions in a natural language way. They are building assistants for their maintenance teams. For instance, an elevator maintenance team may deploy an assistant that uses schematics and other information about an elevator. They can ask questions of this bot in a natural language way. In another example, a company is building sales assistants that use company information about customers to build a dossier on a client before the salesperson goes out to see them. Companies are building customer support chatbots that can be used by clients. These bots retrieve information about the customer and then engage in a natural language interaction to try to answer questions for that customer, knowing something about them.

What these use cases have in common is that they are using company data—whether structured or unstructured—to provide an application. When I was talking about the value ceiling in Chapter 3, this is what I meant. Organizations can build meaningful and potentially valuable enterprise applications using GenAI that go beyond simple off-the-shelf tools.

These enterprise builders don't always need to start from zero. In fact, many succeed by combining their company's own data with pre-built AI capabilities to deliver results quickly. Daniel Ziv, global vice president of AI and Analytics at Verint, offers an important perspective on how organizations can move fast, focus on real problems, and scale what works without getting stuck.

EXPERT ADVICE: GENERATIVE AND AGENTIC AI

Daniel Ziv, global vice president of AI and Analytics, Product Management and GTM Strategy at Verint, has been at the forefront of bringing generative and agentic AI into real-world production environments with faster, stronger, and measurable outcomes. Verint's CX Automation cloud platform is designed to increase customer experience automation in the contact center, and Ziv is quick to point out that the most exciting

(continues)

EXPERT ADVICE: GENERATIVE AND AGENTIC AI (CONTINUED)

part of AI today is not theory but practice. Verint has already deployed large-scale AI applications into customer contact centers, reshaping how service is delivered, reducing cost, and increasing revenue while increasing CX.

One of their standout solutions is Genie Bot, which some customers have described as "ChatGPT for our contact center customer interactions." Traditionally, analysts would spend weeks reviewing recordings to uncover patterns and insights. With Genie, what used to take four weeks now takes four hours, dramatically reducing cost and effort while scaling insights and delivering measurable outcomes. The first Genie deployments have already generated millions of dollars in AI outcomes within days.

Another earlier AI innovation is Verint's Wrap-up Bot. "Contact center agents used to spend a minute or more summarizing each call. Now, the transcription is sent to an LLM with the right prompt and model, generating a detailed summary automatically. A large health insurance company deployed it to 30,000 agents and saved about a minute per call. That translates into savings of roughly a million dollars a day." Ziv emphasizes that these are not lab experiments. "They started with 300 agents, saw it worked, quantified the results, and scaled. Spending too much time in the lab is a waste; the biggest risk is waiting for perfection."

These examples reinforce Ziv's central message: don't wait for flawless technology, and don't try to build everything yourself—start solving real problems now, measure outcomes, and scale what works. "I'm just as excited as I am concerned … AI can have unintended consequences. But the bigger risk is doing nothing. The only way to uncover trust issues is to use it." He underscores that Verint's deployments always include validation mechanisms. With Genie, for example, he says, "You can click through to the original call to confirm the source of the insight. Trust builds over time, but you always need ways to validate."

Ziv also offers a nuanced view on generative and agentic AI. Too often, he argues, they're treated as totally separate when they're better understood as a spectrum. "Generative AI already has agentic capabilities. You can keep adding actions and autonomy." He likens the evolution to self-driving cars: "Many cars today have self-driving features, lane assist, automatic braking, but they're not fully autonomous. That doesn't mean we don't use them. Enterprises should take the same approach: keep humans in the loop, adopt incrementally, and expand autonomy as trust builds."

The foundation of this pragmatic approach lies in data. "Focus on the unique, especially unstructured or semi-structured data you already have. That's your competitive advantage. Think about what problems it can help solve now and how to supercharge existing processes with AI." He cautions against reinventing the wheel with massive, costly projects. "Some companies say, 'We're building embeddings, vector databases, fine-tuning.' That takes time, is expensive, and can still hallucinate. We took the reverse approach what I call *reverse RAG*. We already had speech and text analytics, accurate transcripts, and good high-level categories. We applied GenAI to those, indexed with our proven methods to select the most relevant subset of data, and applied LLMs with the right prompts. It's faster, cheaper, and more accurate."

(continues)

EXPERT ADVICE: GENERATIVE AND AGENTIC AI (CONTINUED)

For Ziv, the key to success isn't the model—it's the people and processes around it. "People obsess over which foundation model to use. That model will likely change tomorrow. Focus on your people, the ones who understand pricing, customer service, experience, and your business." He encourages leaders to empower employees broadly and build comfort with AI through direct use. "Talk to IT and make sure as many employees as possible can use AI tools. Before building your own, get people comfortable. Then identify your biggest problems and automate incrementally. Forget about pilots in labs—turn it on in production for a small team, see what happens, and learn."

Ziv reframes the path to AI success with a pragmatic lens: "It's not about the largest models or the biggest teams. Success comes from focusing on the unique data you already have, the existing workflows, even micro workflows you can automate now, and applying AI to real business problems, where you can measure the outcome and ROI." He encourages leaders to start small, deploy quickly, and learn fast: "See what works and scale it fast. See what doesn't, fail fast, and learn. That's how you build AI proficiency."

Ziv's experience shows the value of moving quickly with available tools and company data. But making enterprise data available to GenAI models in a secure, accurate way is not trivial. One popular approach to doing this is retrieval-augmented generation (RAG). RAG has emerged as a leading method for connecting foundation models to enterprise data so that outputs are both relevant and trustworthy.

Putting Generative AI Models into Production with Company Data

So, how does this work? Before I provide a simple example, let's go over a few terms that are used with GenAI.

- **Prompt:** A prompt is the input or instruction you provide to an AI model to guide its response. It can be a simple question, a block of text, or a carefully structured instruction designed to elicit a specific type of output. By this point, I'm sure most everyone reading this book has provided a prompt to a GenAI model through some sort of box-like user interface or in a search.
- **Vector embeddings:** These are how data are represented numerically so that similar items can be found by proximity in a high-dimensional space (refer to Figure 17.1). These embeddings are stored in a vector database.

- **Retrieval-augmented generation (RAG):** This is the architecture that uses those embeddings to retrieve the most relevant enterprise data and insert it into the prompt of a generative model, giving more accurate and grounded outputs.
- **LangChain:** This is a popular framework for GenAI applications. Recall, a framework is a structured set of tools, libraries, and best practices that provides a foundation for building applications more efficiently. LangChain is a popular open-source framework for building applications powered by LLMs (and now multimodal models). It simplifies tasks such as prompt management, connecting to data sources, chaining multiple model calls, and integrating external tools, making it easier to create chatbots, agents, and RAG systems. Other frameworks include LlamaIndex and Haystack. These were mentioned in the AI ecosystem diagram in Chapter 14.

The process generally follows a few steps. First, relevant enterprise data—whether stored in a data lakehouse, data warehouse, operational systems, or a data fabric—are transformed into vector embeddings and stored in a vector database. When a user enters a query or prompt, a framework such as LangChain or LlamaIndex manages the workflow, which might look something like this:

1. Load the data and chunk it.
2. Create the vector embeddings.
3. Retrieve the most relevant data through a similarity search on the vector database.
4. Combine this retrieved context with the user's original prompt.
5. Send the augmented prompt to the LLM (or other foundation model).
6. Return the model's response to the user in natural language.

For instance, let's use the simple RAG example where an elevator repair technician wants to ask the maintenance assistant about a certain elevator part. What was fed into this application were all the company's documents about different elevators with explanations of their schematics. That might be a lot of documentation. So, the first part of the application is that information is chunked (into sentences, characters, or paragraphs) to make the search easier. That information is used to create a vector embedding (via an embedding model) and is stored in the vector database or store. The maintenance person may ask a question. That question is also turned into a vector embedding. The most similar information to the question is retrieved from the vector database using that similarity search we spoke about earlier on the vector database. The retrieved context is combined with the user's prompt, and that augmented prompt is sent to the LLM. The response is then provided to the maintenance person.

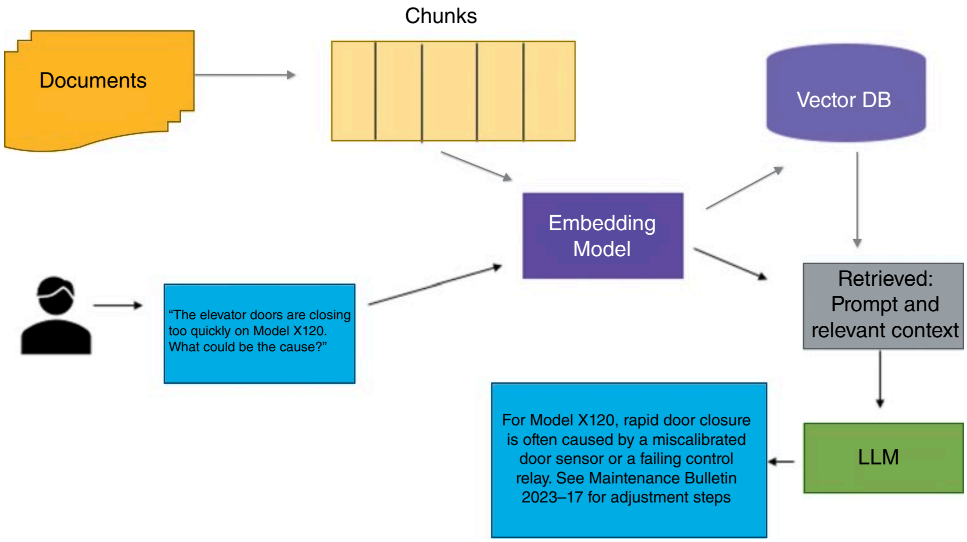

Figure 17.3: A simple RAG flow for a maintenance assistant.

The final answer reflects both the company's data (in this case, documents) and the model's general knowledge. This is illustrated in Figure 17.3.

Where does LangChain come in? It provides some of the tools, libraries, and functions to make building the application easier than starting from scratch. It is also important to note that many software vendors today offer vector databases as part of their platforms. Additionally, they offer foundation models as part of their platforms, so that organizations can make sure that their data is not being made public as part of new training data in a public LLM.

RAG applications must be built with semantics, metadata, and governance in mind. The retrieved information must be high-quality, secure, and appropriate for the user's role. Equally important is monitoring and management: checking whether source data has changed, ensuring vector stores remain accurate, and watching for hallucinations or data leakage. Organizations are definitely starting to experiment with RAG models today; some are putting them into production applications. Some believe that using RAG helps mitigate hallucinations (discussed in the next section) because it is grounded in trusted organizational data.

As you can see, the same broad life cycle applies to this GenAI model as we saw in previous chapters. These applications still require defining what the application is supposed to do. Instead of assembling features, the model preparation phase may involve curating data or deciding which foundation model to build on. Pipelines may include vector databases or RAG components to connect the model with enterprise data. While organizations rarely train an

LLM from scratch, they still need to focus on testing the model for quality, factuality, and safety. Registering the model and metadata is still critical. The model still needs to be deployed into production, validate outputs, and so on. We heard from Deanne Larson in the last chapter about this in the context of MLOps/AIOps. However, there will need to be a new emphasis on monitoring for hallucinations and inappropriate output.

Hallucinations, Bias, and Guardrails

Let's talk more about hallucinations. GenAI models are notorious for creating hallucinations—that is, output that is irrelevant or incorrect. We've already discussed the example where the Air Canada chatbot app created an incorrect grievance policy response. We talked about the Webb telescope app as well. Of course, GenAI models suffer from bias.

The organizations I speak with are very concerned about putting guardrails in place for GenAI, especially for the consumerized AI models discussed earlier that have led to *shadow AI* implementations, the unauthorized or unsanctioned use of AI tools. They are concerned that public LLMs will expose sensitive and private company data, that is, what is known as *data leakage*.

While much of the conversation around shadow AI focuses on the danger of employees pasting private data into public LLMs, organizations are worried about much more. Shadow AI creates issues that extend beyond privacy:

- **Copyright and IP liability:** Employees may generate or reuse content that infringes on copyrighted material, leaving the organization exposed to lawsuits.
- **Bias and fairness:** Without oversight, outputs may reinforce stereotypes or discrimination, undermining trust and compliance efforts and leading to reputational risk. We talked about bias and fairness in Chapter 15.
- **Toxic or misleading content:** Shadow AI can produce deepfakes, toxic language, or misinformation that damages reputation if released externally.
- **Lack of accountability:** Unauthorized tools bypass audit trails, making it impossible to explain or justify AI-driven decisions to regulators or customers.
- **Security and leakage:** Even if sensitive data is not directly exposed, prompts or outputs can still reveal confidential information in unintended ways.
- **Quality and reliability:** Outputs may contain hallucinations or errors, and without validation processes in place, poor decisions can slip through. This can lead to reputational risk.

- **Regulatory noncompliance:** Laws such as GDPR, CCPA, and the EU AI Act place strict requirements on data usage and AI oversight that shadow AI may violate. (We'll talk more about these regulations in Chapter 19.)
- **Driving up costs:** Many AI tools are licensed separately. If employees expense their own subscriptions or departments adopt them without centralized oversight, those costs accumulate quickly across the organization. This can also lead to overlapping tools and duplicate licenses. This creates hidden costs and governance risks. Additionally, many of these tools run on large cloud-hosted models. Heavy usage means higher API consumption, which is baked into the vendor's pricing and passed on to the company in subscription tiers or usage-based overages.

In other words, shadow AI increases organizational risk by operating outside established guardrails. The challenge for leaders is not simply to block unauthorized use, but to provide approved, safe alternatives that meet employee needs without sacrificing compliance, accountability, or trust.

There are a few ways that organizations are trying put guardrails into place. First, they are providing enterprise-approved copilots and assistants that their teams can use. They are providing lightweight governance processes, such as AI councils or intake workflows to make it easy for employees to get approval for new use cases. They are implementing access controls if employees want to use certain tools against company data.

Perhaps most importantly, some are implementing AI literacy programs so that employees understand both the risks and the responsible use of AI tools. Successful companies build out their own academies or partner with academic institutions or external partners to develop AI literacy programs and ensure guardrails are in place. These programs often include mandatory training (similar to what we saw in Chapter 11 on data literacy) in responsible AI policies along with hands-on training with tools that may be based on roles. Even smaller companies can provide some sort of training and should. We'll talk more about AI governance, including governing GenAI, in Chapter 20.

Measuring Success with GenAI

While companies are feeling the pressure to "do something" with AI, that doesn't mean they don't need to demonstrate success. Measuring GenAI success (or success with any AI initiative for that matter) can be elusive for some companies, but it is key for moving a program forward. There is *perceived* value, and there is *actual* value. I talked about the virtuous circle earlier in the book.

In AI, if you can show an actual measurable impact, people start to notice and success builds on itself. I've seen this countless times over the years.

For GenAI, measuring success often first comes from individual productivity enhancements. I mentioned this at the beginning of the chapter. Organizations I speak with often note that everyday tasks such as drafting, summarizing, or searching can be completed more quickly, and they quantify that. These kinds of gains are tangible and relatively straightforward to measure, which is why productivity often becomes a frequently cited business goal. In some cases, companies I speak with say that rather than return on investment, they are measuring "return on employee."

Beyond this, companies point to other outcomes, beyond individual productivity, that matter to the business. These often make use of company data in GenAI. In customer service, they point to how GenAI is helping reduce response times and allowing teams to handle higher volumes of requests. Others emphasize that GenAI is enabling new types of product capabilities, such as more personalized customer interactions or faster development of new offerings. These can result in cost efficiency, perhaps improved customer experience, and product innovations.

One way to think about these outcomes is through a simple framework for measuring success (see Table 17.1). Not every organization will measure all these dimensions, and some outcomes are harder to quantify than others, but having a structure tied to business goals is essential for demonstrating progress and building confidence in GenAI initiatives.

Table 17.1: Examples of GenAI Metrics

CATEGORY	EXAMPLES OF METRICS
Productivity and efficiency	Time saved on tasks, increase in output per employee, reduction in manual steps
Customer impact	Response time improvements, customer satisfaction scores, service availability
Cost efficiency	Operating cost reduction, fewer errors and rework, elimination of duplicate effort
Decision quality	Accuracy of outputs, speed of decision cycles, adoption of data-driven recommendations
Innovation and growth	New product capabilities, faster time-to-market, creation of new services or offerings

What's Next?

GenAI is changing how individuals and organizations create, summarize, and interact with information. The next chapter looks at where this is heading. Agentic AI builds on GenAI capabilities by giving systems the ability to plan, take actions, and interact with other tools or systems on behalf of the user. This represents both an opportunity to automate more complex workflows and a new set of challenges around control, governance, and trust.

CHAPTER

18

Agentic AI

In this chapter, we will examine the following:

- Agentic AI
- How agentic AI works
- The orchestration layer
- Examples of real-world agentic applications
- Opportunities and risks

Chapter 17 explored generative AI (GenAI), systems that create new content based on what they've learned from massive datasets. GenAI marks a turning point: AI isn't just analyzing data—it is producing something new from it. While these systems are impressive in their own right, they don't take action. The next evolution of AI is agentic AI. In this chapter, we'll look at agentic AI—what it is, how it works, and the opportunities and risks it poses.

What Is Agentic AI?

GenAI, powered by large foundation models, is revolutionizing the way people interact with data and technology. Organizations are using GenAI to democratize AI through natural language interfaces, copilots, assistants, and creative tools that can draft text, summarize reports, or generate images on demand. Yet as transformative as this is, these systems remain largely reactive. They respond to prompts but do not decide what to do next. They do not plan, reason over time, or act toward goals. In short, they generate but they don't take action. Enter agentic AI, one of the hottest topics in AI today. Once mature, it may have the ability to completely disrupt how a lot of work is done.

Agentic AI extends GenAI by embedding intelligence within autonomous or semi-autonomous systems that can plan, reason, and take actions within defined boundaries. Instead of simply generating a report, an agentic system might determine which data it needs, retrieve that data from multiple sources, perform analysis, summarize the results, and then trigger follow-up workflows, all while maintaining auditability and alignment with governance policies.

Where a generative model might answer a question, an agent can decide which questions to ask, what tools to use, and what steps to take to reach an outcome. It can call APIs, interact with business systems, and coordinate with other agents. In other words, agents have agency, meaning the ability to act.

Let's look at an example of an agentic application. Consider a procurement optimization agentic system in a manufacturing enterprise. The goal is to identify cost-saving opportunities across a complex supplier network. The system begins with a data retrieval agent, which connects to internal enterprise resource planning (ERP) and procurement databases, as well as external sources such as supplier portals and market feeds. It collects relevant information including pricing, delivery schedules, contract terms and standardizes it, ensuring that fields such as currency, units of measure, and effective dates are consistent. If discrepancies or missing data are found, the agent may request clarification or corrections from the source before proceeding.

Once the data is prepared, it is passed to an analysis agent. This agent compares pricing and delivery performance across suppliers, identifies contracts nearing renewal, and applies business rules to flag optimization opportunities. For example, it may detect that Supplier A consistently offers a lower price and faster delivery than Supplier B for equivalent parts. Using embedded logic and predictive models (e.g., tools), the analysis agent scores each supplier on cost, reliability, and risk, producing a ranked list of recommendations.

The results are then handed off to a reporting and action agent. This agent transforms the findings into actionable outputs. This might be a report for procurement leaders, a summary email highlighting top savings opportunities, or an automated task in the organization's sourcing system to initiate negotiations. If integrated with enterprise workflows, the agent can trigger the next step in the process, such as sending a draft contract revision for review. Throughout this chain, each agent's actions are logged for traceability, ensuring that every decision can be audited and explained.

Overseeing it all is the orchestration layer, which acts as the system's conductor. It ensures that agents operate in the correct sequence, that data flows securely, and that exceptions are handled gracefully. The orchestrator can pause the process to request human review if an action exceeds predefined thresholds.

For instance, this might be if a proposed supplier change affects a critical component or exceeds a monetary limit. This "human in the loop" safeguard maintains accountability while allowing the system to operate autonomously where appropriate.

Behind the scenes, the agents interact with a governed data foundation that provides access to trusted, semantically enriched information. The data is unified across systems and aligned through a semantic layer that encodes business definitions and rules, ensuring that all agents interpret key terms, such as "cost savings" or "on-time delivery," consistently. Recall, we talked about the semantic layer in Chapter 10. This semantic context is what allows the system to act intelligently and predictably rather than making arbitrary or opaque decisions.

In practice, a system like this might contain many more agents, but this simplified example illustrates the point of the agentic system—it can act autonomously toward a goal. In this case, the goal was to get the best deal possible. What makes this agentic application powerful is not a single model or algorithm, but the integration of reasoning, context, and action within a governed framework.

Note that I've been talking about multiagent systems that are performing multiple tasks. Yet, many companies I speak with claim to have a single agent system that might, for instance, pull out issues from trouble tickets and create a report summarizing them. This system would use RAG, as described in the previous chapter. Is this an agentic system? Not really. This system is really AI-enabled automation. It would become an agent when it can decide what to do next, why, and how. So, there is still confusion in the market about agentic AI.

To understand what makes agentic systems powerful, it helps to look at how agentic AI works. While each implementation differs, most agentic systems follow a common pattern. This is often depicted as a four-step loop of perceiving, reasoning, acting, and learning that enables them to function autonomously and improve over time.

How Does Agentic AI Work?

Again, agentic AI systems differ from traditional and generative models in that they do not simply generate an output when prompted. They conceptually operate through a four-step process that allows them to function autonomously, adapt to changing environments, and continuously improve over time.

1. **Perceive:** The first stage of any agentic system is perception: gathering and interpreting data from the surrounding environment. In a customer service scenario, an agent might receive a query such as, "Where is my order?" This triggers the start of the process. The agent will need to collect data from multiple sources. It might retrieve the data from order management

and logistics systems, compile tracking details, and convert this into a context (e.g., a vector embedding) that the model can reason about. Agents collect inputs from APIs, databases, applications, sensors, and even human interactions. They may extract structured information from enterprise systems or process unstructured text and images. Perception gives the agent situational awareness; the ability to understand what is happening before deciding what to do next.

2. **Reason:** Once the agent has perceived its environment, it moves into reasoning, i.e., analyzing the collected data to determine the best course of action. This stage is powered by large language models (LLMs) or smaller, specialized reasoning models that interpret intent, infer goals, and plan steps to achieve them. This step allows the system to assess options, weigh probabilities, and simulate possible outcomes, using contextual cues and prior experience to select a course of action. For instance, an agent tasked with optimizing supply chain routes might process real-time weather, inventory, and traffic data, balancing cost and efficiency before determining the most effective shipping plan. Reasoning is what transforms a workflow into an autonomous process. In a standard workflow, decisions are deterministic and are made by humans. In agentic systems, those decisions are delegated to the AI itself, giving it the ability to interpret goals and dynamically adapt its approach.
3. **Act:** Action is where the agent moves from planning to execution. Acting typically involves using APIs, software interfaces, or external tools to complete tasks. That might be updating a database, sending a message, restarting a server, generating a report, or interacting with another application. This is where the agent's autonomy becomes visible: it doesn't just recommend an action; it performs it. The range of actions available depends on the tools and systems the agent can access, making integration a critical design factor. In an enterprise environment, an agent might call internal systems to retrieve financial data, trigger a workflow in a CRM, use a predictive model, or automatically generate and distribute a weekly performance summary. In more advanced deployments, an orchestration layer coordinates multiple agents, each responsible for a specific function, allowing them to collaborate toward a shared goal. In some systems, action might be gated by a human in the loop.
4. **Learn:** This final step closes the loop. Learning allows the agent to refine its behavior based on the outcomes of its actions. It might incorporate explicit feedback from a user, performance metrics from an application, or implicit signals derived from success and failure. This feedback forms a continuous learning cycle where perception and reasoning become increasingly

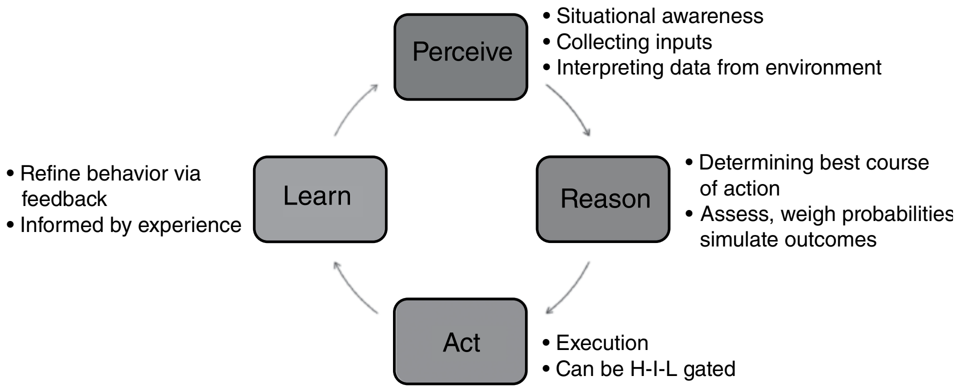

Figure 18.1: The agentic AI feedback loop.

informed by experience. Over time, agents can adjust thresholds, improve decision quality, and reduce the need for human oversight. A customer service agent that once required human escalation, for example, can learn from each resolved case, updating its knowledge base so it can autonomously handle similar issues in the future.

Together, these four steps form an adaptive feedback loop (Figure 18.1). The agent perceives its environment, reasons about what to do, acts to achieve a goal, and learns from the outcome. Each stage strengthens the next, producing systems that are proactive. This iterative cycle is what gives agentic AI its power and also what makes it distinct from traditional automation. It enables a shift from static workflows to dynamic systems capable of independent judgment, coordination, and continual self-improvement.

In practice, these four capabilities are orchestrated through a software framework that connects reasoning models with the tools, data, and systems they need to operate. This orchestration layer forms the backbone of agentic AI applications.

The Orchestration Layer

Agentic AI systems are built on a framework of agents, tools, and an orchestration layer that enables autonomy and coordination. Each agent is an intelligent software component capable of reasoning about its assigned task, selecting tools to use, and interacting with data or external systems. The orchestrator is responsible for managing the overall workflow, sequencing tasks, and maintaining context across the system.

When a user initiates a task, such as "analyze supplier performance and identify cost-saving opportunities," that request is handed to an orchestrator. The orchestrator breaks down the goal into a sequence of subtasks:

1. Retrieve all relevant supplier contracts.
2. Compare prices and delivery terms.
3. Generate a ranked list of savings opportunities.
4. Summarize the results for review.

Each subtask is then delegated to a specialized agent. One agent may retrieve and clean data from procurement systems; another may apply analytics to evaluate vendor performance; and a third may generate a summary report for decision-makers. The orchestrator coordinates these activities, ensuring that data and context flow smoothly between agents and that the results are synthesized into a coherent output. In this way, orchestration provides both structure and oversight—defining how agents interact, in what order, and with what dependencies.

In practice, orchestration is implemented through frameworks such as LangGraph, LlamaIndex, and Semantic Kernel, which handle key operational elements of autonomy: managing memory and context, routing inputs to the appropriate agent, determining which tools to call, and integrating the outputs of multiple reasoning steps into a single response. The concept of the framework (e.g., structured, standardized approaches with reusable tools, libraries, and methodologies built on top of them) was discussed in Chapter 14.

A well-designed orchestration layer also supports collaboration between human users and AI agents. It can pause execution to request human validation, manage approval gates for sensitive actions, or learn from human feedback to improve decision quality over time. In complex enterprises, this orchestration may span multiple systems, data warehouses, analytics tools, APIs, and reporting platforms, enabling agents to operate as part of a broader digital ecosystem rather than as stand-alone applications.

MCP—THE UNIVERSAL CONNECTOR

The Model Context Protocol (MCP) is emerging as a standard for connecting AI models to external tools and data sources. Historically, developers had to build custom connectors for each model and API, limiting portability and slowing innovation. MCP simplifies this by creating a common interface, essentially a universal plug like USB-C, that allows compliant models and tools to communicate seamlessly. Frameworks like LangChain/Graph and others are expected to adopt MCP to handle connectivity under the hood, providing developers with interoperability while maintaining secure and auditable access to enterprise systems.

Developers implement these agentic systems using frameworks such as LangChain/Graph, which provides the libraries and orchestration logic needed to build the agents themselves. LangGraph handles the perception, reasoning, and action flow while maintaining memory and context across tasks. In this sense, it functions as both the development environment and the orchestration layer—coordinating the steps of perception, reasoning, and action while managing connections to external systems. Underneath this, emerging standards like MCP ensure that agents can connect securely and consistently to tools and data sources, making agentic architectures portable and scalable across platforms.

Taken together, the orchestration layer and emerging interoperability standards like MCP form the foundation for scalable agentic AI. They make it possible for agents to perceive, reason, act, and learn in coordinated ways across complex environments, translating autonomous intelligence into practical, enterprise-grade capability.

Examples of Real-world Agentic AI Systems

So, what are some examples of agentic AI systems that organizations are using today? Many are single agent systems. This might include a real estate agent that understands a buyer's preferences and alerts an agent if a house comes up for sale that meets their criteria. Or an agent that determines the best list price for a home. Some single agent systems are used in education for coaching. In some cases, it is hard to determine if they are truly agentic or more like a tool. I've talked to some companies over the past few months that are starting to build agentic AI applications. In some cases, the organization may want to start to build out an agentic workflow, but they realize that they don't understand their own processes well enough to do so. In other cases, based on the company, they have concerns about building out agentic applications because of the governance concerns (which we'll discuss in Chapter 20).

However, there are some examples of household name companies that claim to use multiagent systems. As of this writing, these are mostly used in tech companies. For instance, Uber created Finch, a multiagent financial assistant designed to eliminate the friction and delay financial analysts once faced in retrieving data. Traditionally, analysts had to search across multiple systems or submit data science requests, slowing decision-making and increasing the risk of error. Finch solves this by embedding an intelligent, conversational AI directly into Slack, allowing users to query financial data in plain English. Using GenAI, retrieval-augmented generation (RAG), and specialized self-querying agents orchestrated with LangChain's LangGraph framework, Finch interprets finance-specific language, identifies the correct data sources, constructs and executes SQL queries, and delivers validated results in real time—all while enforcing role-based security.[1]

In a joint case study, ServiceNow and Microsoft used a manager-agent orchestration layer to handle high-severity incident response (what Service Now calls P1 incidents) by coordinating multiple agents: a transcription agent, a context-gathering agent, and action agents integrated with ServiceNow. The orchestrator agent maintained conversation state, routed to appropriate tools, and ensured context consistency.[2]

The DHL Supply Chain is integrating agentic AI into its global logistics operations through a partnership between DHL Supply Chain and AI startup HappyRobot. The company identified that managing the constant flow of operational communication, such as appointment scheduling, driver follow-ups, and warehouse coordination, was consuming significant employee time and slowing responsiveness. To address this, DHL deployed AI agents that autonomously manage phone and email interactions across regions, processing millions of voice minutes and hundreds of thousands of messages each year. These agents integrate with existing systems to automate routine exchanges, freeing teams to focus on exception handling and strategic decisions. The initiative extends DHL's broader digital transformation strategy.[3]

Clearly, this space is evolving, but it is not yet there. In my research I see a lot of experimentation with agentic AI, but not that many deployments yet. In the following Expert Advice, Ron Brachman, PhD, an internationally recognized AI pioneer, weighs in on agentic AI.

---

[1] Uber. (2025). *Unlocking financial insights with Finch: Uber's conversational AI data agent*. Available at: `https://www.uber.com/blog/unlocking-financial-insights-with-finch` [Accessed 30 August 2025].

[2] Microsoft. (2025). Customer case study: Pushing the boundaries of multi-agent AI collaboration with ServiceNow and Microsoft semantic kernel. *Semantic Kernel Dev Blog*, 19 January. Available at: `https://devblogs.microsoft.com/semantic-kernel/customer-case-study-pushing-the-boundaries-of-multi-agent-ai-collaboration-with-servicenow-and-microsoft-semantic-kernel` [Accessed 30 August 2025].

[3] DHL Group. (2025). DHL boosts operational efficiency and customer communications with HappyRobot's AI Agents. *DHL Group Media Relations*, 11 November. Available at: `https://group.dhl.com/en/media-relations/press-releases/2025/dhl-boosts-operational-efficiency-and-customer-communications-with-happyrobots-ai-agents.html` [Accessed 12 November 2025].

EXPERT ADVICE: AGENTIC AI

Ron Brachman, PhD, has been working in artificial intelligence since the 1970s, through every rise, fall, and resurgence of the field. An internationally recognized authority on AI, he has held leadership roles at Bell Laboratories, AT&T Labs, DARPA, and Yahoo; has been president of the Association for the Advancement of Artificial Intelligence; and now serves as visiting professor at Cornell Tech and distinguished visiting scientist at Schmidt Sciences. Among his many accomplishments, he conceived and launched the DARPA project that led to Siri, in a way becoming the godfather of the AI systems that billions of people now use every day. Having lived through some of the earliest dreams of intelligent machines and the AI winters that followed, he is both excited and cautious about what he calls the "rediscovery of agents" under the new banner of agentic AI.

"The term agentic AI may be new," Brachman says, "but the idea of intelligent agents has been around for decades. The concept of semi-autonomous intelligent systems dates back to the 1960s and 1970s." What's different now, he explains, is the accessibility and democratization of building and deploying such systems. "The ability to specialize and deploy software-based agents has been democratized. Almost anyone can use an LLM as an interface, describe their goals, and deploy an agent that runs on a network. You no longer need to be an expert developer. While right now execution abilities are limited, the average person in a company, or even at home, can get agent-like behavior simply by typing or speaking to these frameworks."

Brachman sees this as a major shift in how people interact with technology. "AI used to be subtle, complicated, and not generally accessible. Now it's something people can use to get things done," he says. "Agentic AI extends the user's reach; it doesn't just answer questions; it can carry out activities. It can update a database, change HR files, complete tasks. It's not built on radically new technology, but on knitting together capabilities like natural language interfaces with existing automation."

Even so, he warns that the term agent is being used too loosely. "In the earliest days of time-sharing, we had 'daemons'; programs that ran in the background and did things for people," he says. "A sprinkler system in your yard has agency in that it takes action, but it doesn't have autonomy. It can't make decisions for itself or update its goals when you talk to it. A lot of what people call agents today are just automatic systems. We're moving toward autonomy, but we're nowhere close to systems that can truly set their own goals and reason about their actions."

For organizational leaders, Brachman's advice is to stay skeptical and realistic. "So much of what's happened in the last year and a half has been hype," he says. "People throw around terms like AGI and superintelligence as if they're just around the corner. When someone wants to sell you something, ask pointed questions. Look for the fragile edges. AI systems are brittle—when they hit edge cases, they can behave unpredictably. You have to know where their competence ends and make sure safeguards are in place so they stop instead of acting blindly."

(continues)

EXPERT ADVICE: AGENTIC AI (CONTINUED)

He stresses that today's systems are not self-aware and do not recognize when they are wrong. "LLMs don't know what they don't know. They never say, 'I don't know the answer.' That's an important capability that's missing," he believes. "Leaders need to understand how these systems were built, where their boundaries lie, and whether the systems can recognize those boundaries themselves."

Trust, according to Brachman, depends on two things: the ability to explain actions and the ability to accept advice. "If a system takes an action, say it buys an airline ticket, it should be able to give good reasons for doing it," he said. "We need agents that can justify their actions in rational terms, not post-facto storytelling. That's the basis of trust. The second capability is being able to take advice. If the system has a mistaken belief about the world, someone should be able to correct it. Humans do this all the time; if you learn a street is closed, you instantly adjust your route. That's the kind of rational guidance-taking we'll need in AI systems."

He traced this idea back to one of the founders of AI, John McCarthy, who proposed an "advice taker" in the 1950s. "It's a fundamental notion," Brachman said. "For AI to be trustworthy, it has to be able to explain its behavior and to change its behavior when it's wrong."

Governance, he believes, will have to evolve to accommodate systems that act independently. "It's hard to understand what's expected from these systems," he said. "We need ways to test them, sandbox them, and observe them in controlled environments before they operate in the real world." He recalled a lesson from systems engineering: "Years ago, the 5ESS telecommunications switch had a separate auditing process that monitored the main system for anomalies. We'll need the same kind of oversight for agentic systems; processes that watch the agents and detect when something goes wrong." Continuous auditing, he believes, will be as essential for AI.

Looking to the future, Brachman believes we're entering a decade of experimentation and possible correction. "This whole boom could go bust," he said. "There's a lot more hype than usable technology right now, and we may see a backlash." Still, he expects continued advances. "If there's no major disaster, we'll move toward agents with broader goals that persist, learn, and perform a wide variety of tasks," he believes. "Eventually, these agents will have physical embodiments; robots that work in homes, in elder care, in everyday life. In 10 years, when people think of AI, they may think of robots rather than software systems."

He also pointed to a growing research focus on how multiple agents interact. "What happens when your agent meets 10,000 others, each with different goals?" he asked. "How do you ensure cooperation when you need it? That's one of the things we're working on at Schmidt Sciences, understanding multiagent behavior and the protocols that will govern it."

For all the excitement, Brachman remains measured. "These systems are powerful," he said, "but even the people who built them don't fully understand why they work the way they do. That's not a recipe for trust. Until we can explain them and align them with human intent, leaders should approach agentic AI with both curiosity and caution."

Opportunities and Risks

I think that you can see that while agentic AI potentially has benefits, including improving efficiency, being proactive, augmenting human tasks, being adaptable, and generating innovative value, a number of risks are associated with them. Because these systems can perceive, reason, and act, their mistakes can propagate through workflows rather than stop at a single incorrect output. Agents may make unauthorized decisions, misinterpret goals, or trigger unintended actions in connected systems. These are not theoretical concerns; autonomous behavior, memory persistence, and cross-agent communication all expand the risk surface well beyond that of traditional AI or even GenAI.

For these reasons, the governance challenge is shifting from controlling what AI produces to overseeing what AI does. Agentic systems demand new safeguards around tool access, auditability, and human oversight, as well as broader consideration of business, legal, and workforce impacts. It will be up to roles such as the chief information officers, chief information system officers, and risk officers to think through these risks. McKinsey points out that financial, operational, regulatory, people, reputation, and strategic risks will all be accelerated by agentic AI.[4] We'll explore emerging agentic AI risks from a governance perspective in Chapter 20.

However, there are also societal implications. Agentic systems have the capacity to be extremely disruptive in terms of how companies do business; they are a disruptive technology, in and of themselves. According to the World Economic Forum, by 2030, 40% of employers anticipate reducing their workforce where AI can automate tasks.[5] White collar entry-level jobs have often been cited as being in danger. But of course, there are more roles that are in danger. In an analysis done by Microsoft Research, roles such as interpreters and translators, historians, passenger attendants, sales representatives of services, writers and authors, customer service representatives, CNC tool programmers, telephone operators, ticket agents and travel clerks, broadcast announcers and radio DJs, brokerage clerks, telemarketers, news analysts, and many more had what they called high AI applicability scores.[6]

---

[4] McKinsey & Company. (2025). *Deploying agentic AI with safety and security: A playbook for technology leaders*, 16 October. Available at: `https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/deploying-agentic-ai-with-safety-and-security-a-playbook-for-technology-leaders` [Accessed 12 November 2025].

[5] Zahidi, S. (2025). Future of jobs report 2025. *World Economic Forum*, p. 6. Available at: `https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf` [Accessed 12 November 2025].

[6] Tomlinson, K., Jaffe, S., Wang, W., Counts, S., and Suri, S. (2025). Working with AI: Measuring the applicability of generative AI to occupations. *arXiv preprint* arXiv:2507.07935 [cs.AI], 42 pages.

Even roles that have traditionally defined data and AI professions are potentially in danger.

Vendors are already introducing data engineering agents that can ingest, clean, and prepare data; data science agents that can design, test, and evaluate models; and orchestration agents that can deploy and monitor AI systems with little human input. This shift threatens to reduce the need for repetitive or routine work performed by data engineers, analysts, and model developers. Consulting firms warn that agentic AI will "compress roles along the data value chain," automating a large percent of routine analytics and development tasks while creating new dependencies on oversight and integration roles. In effect, AI is beginning to automate the very work that made AI possible.

For data and AI leaders, this may have significant implications for workforce planning and governance. The skills most in demand may shift from building individual models or pipelines to designing, integrating, and governing systems of autonomous agents. Roles focused on policy, risk management, compliance, ethics, and system assurance may expand, while hands-on development roles may contract.

Organizations will need to reskill technical staff toward orchestration, human-in-the-loop validation, and continuous monitoring. Governance frameworks will need to address the reliability and safety of agentic systems. Some may also need to look at the impact of agentic AI on workforce composition and accountability. The future data and AI workforce may be smaller, more strategic, and more interdisciplinary, charged with managing intelligent ecosystems rather than coding every step that is involved with them.

What's Next?

Agentic AI pushes organizations toward a new era of intelligent autonomy. These are systems that not only generate insights but take meaningful action. As this shift unfolds, governance becomes more than a safeguard; it is a core capability for enabling, overseeing, and ensuring compliance in an increasingly AI-driven enterprise. Effective governance provides the structure that allows innovation to scale responsibly, balancing agility with accountability. The next chapters examine how data and AI governance frameworks are evolving to support this balance.

Part V

Governing Data and AI—Ensuring Trust and Compliance

CHAPTER

19

Data Governance—Building Trust for AI

In this chapter, we will examine the following:

- The new data governance environment
- Data governance—not just a compliance task
- Key regulatory requirements
- Data governance frameworks
- Modernizing data governance—tools
- Modernizing data governance—organizational structures
- Cross-platform data governance
- Moving forward with data governance

The best data foundations and architectures are only as strong as the trust users have in the data that flows through them. Governance, if done well, creates that trust. It ensures that data are accurate, traceable, protected, and aligned with the organization's purpose and values. It drives confidence in decisions and AI applications. This chapter focuses on data governance, how it changes with AI, and provides insights into how some organizations are doing it well.

For leaders, data governance is not only about compliance or control; it is about trust and competitiveness. Organizations that govern their data effectively can innovate with confidence, accelerate AI adoption, and respond faster to regulatory change. Good governance builds the transparency and reliability that make data usable at scale. It is the system of accountability that keeps data-driven organizations both agile and responsible.

The New Data Governance Environment

Throughout this book, we've seen that the data environment that AI requires continues to evolve. The modern data environment includes structured data as well as text, machine-generated logs and events, images, and audio. The rapid adoption of generative AI (GenAI) has accelerated both the creation and consumption of text data, increasing volume and variability.

Additionally, hybrid and multicloud architectures are now standard. Data are stored and processed across warehouses, lakehouses, SaaS platforms, and edge services. Governance must operate consistently across these environments. In practice, this requires policies that can be enforced programmatically, with end-to-end lineage (where the data came from and how it has been changed) that is visible to both technical and business stakeholders.

Of course, with AI, the standards for data also change. Not only do you have to govern how data is managed, you must be concerned about how that data is transformed as it may feed an AI model, or what comes out of that model. In other words, inputs and outputs to the models must be controlled. While AI governance is covered in Chapter 20, it is important to note that the data that feeds AI models also falls under the data governance purview. That means it must be accurate, timely, consistent, complete, and of high quality even it isn't the kind of structured data that your company is used to dealing with. We talked about these measures in Chapter 7. Additionally, governing GenAI inputs and outputs is an ongoing concern because of hallucinations and issues such as copyright infringement, data leakage (unauthorized disclosure of sensitive, confidential, or personal information), and unethical behaviors.

At the foundation of all governance remain the traditional dimensions of data quality: accuracy, completeness, consistency, timeliness, and validity. These principles still define whether data can be trusted and used confidently. In the AI era, these metrics must be applied across both structured and unstructured data, extending into the training, validation, and output stages of AI systems. High-quality data is still the foundation for trustworthy AI and analytics, even as the definitions of "quality" evolve to include representativeness and fairness.

Then there is the democratization of analytics and AI that we talked about in Chapter 9, which increases data usage and impact. This will especially be the case as organizations start to use GenAI as a front end to their BI. The fact is that business teams, analysts, data scientists, and developers work with the same data for different purposes. In other words, there are many people who are going to want to use company data. Therefore, the objective should not be restriction; rather, it should be responsible enablement. Clear ownership and stewardship, a shared data catalog, role-appropriate access, and a governed discovery experience will be important.

All-in-all, data governance has gotten a whole lot more complex. It is not a surprise that in my research I continue to see that organizations are only midway in their data governance journey because the bar keeps moving.

Data Governance—Not Just a Compliance Task

Let's dig into exactly what data governance is about a bit more. A simple definition is that data governance is the practice of ensuring that business data remains fit for use. It centers on the people, processes, policies, rules, and regulations that enable an organization to achieve specific goals and build trust in its data. Governance is distinct from data security and data protection (availability and recovery), though all three are connected. In practice, governance is implemented through processes, platforms, tools, and organizational controls that allow data owners to maintain its fitness as a business asset. That sounds dry, but data governance is really anything but dry. It is the engine that can help really move an organization forward in its AI journey.

For years, governance was treated as a compliance exercise, processes and policies to make sure the organization stayed out of trouble. Teams focused on control, restriction, and documentation rather than enabling people to use data. That legacy might be why some companies I talk to say that their organization has been apathetic about data governance. Who wants to be restricted?

Some of this is changing. When leaders treat governance as a foundation for value creation, it enables data sharing, self-service analytics, and AI innovation. It builds confidence that the data people use to make decisions is fit for purpose and used responsibly. Additionally, the democratization that AI brings with generative BI and other innovations also should lead to more business users feeling invested in the quality and integrity of their data. They are making more use of their data, so they should want to be involved in ensuring it is fit for use. The fact that data products are becoming more popular will also help to move governance forward, as those data product owners are often part of the business and they understand the value of these assets. Some of the tools that are on the market also make it easier for business users to be involved in ensuring data governance. All in all, this is good news.

In fact, in my research, I've seen that organizations that say that their data governance is mature are more likely to have experienced improvements in efficiency, better quality data, reduced risk and better compliance, and faster insights and decision-making. These have led to top-line impacts and ROI benefits.

If you recall from Chapter 3, "The AI Journey," governance is a key dimension for AI readiness. In Chapter 3, I noted that data governance is critical because it makes inputs dependable and usable at scale. It establishes clear lineage, quality standards, access controls, and stewardship so teams know what the data means, where it came from, and who is accountable for it.

Governance is also cultural. It depends on shared accountability and collaboration among business and technical stakeholders. Teams that focus only on technology often fail to build buy-in. Teams that treat governance as a communication and change management challenge succeed. The question for every data leader is not only how to control data but also how to help people use it responsibly and effectively.

The following Expert Advice describes how one organization is enabling data governance for AI and the progression to AI governance. It started with a business problem but ended up providing real value for the company.

EXPERT ADVICE: FROM DATA QUALITY TO AI GOVERNANCE

When Richard joined as leader of Data and AI at a large energy company, he saw an opportunity to strengthen data quality and governance by solving challenging, real business problems. The organization had recently implemented a critical application, and as operations stabilized, it became clear that some of the underlying data needed attention. "My leader came down and said, we have a problem and it's really all about the data. So, we jumped in."

That situation became the starting point for the company's enterprise data governance program. Richard and his team organized a workshop with people from various parts of operations to understand what was needed. "We said, if you didn't have any problems in your data, what would be true? Everyone started calling out the things that needed to be true in the data, and we wrote them all down." The group developed a set of data quality rules, about 45 in total, covering everything from delivery times and supplier details to special procurement types.

The team then developed a data quality dashboard to track performance against those rules. "It was the first time the organization had a number for data quality," Richard says. "When we ran the tests, a low percent of the records were good enough to execute a transaction. But that number gave us something to fix."

Once the issues were visible, the organization moved quickly. "We had about 40 people involved, with six core team members doing most of the work. Leadership asked them to get that number over 80% as fast as you can, and they did it in two months." That success became the foundation for a new way of thinking about governance. "We realized the organization is good at solving problems when people can see the problems," Richard says. "That's been a big theme for us. Governance is about solving problems."

(continues)

EXPERT ADVICE: FROM DATA QUALITY TO AI GOVERNANCE (CONTINUED)

The same approach was soon extended to other domains. Today, the company has more than 20 data quality monitoring dashboards across the enterprise. Each one provides visibility into a specific business area and is owned by the people closest to the data. "You have to make data governance meaningful," Richard explains. "When people see the results of their work, they engage."

The company's data governance function now operates as a centralized capability that supports data owners and stewards throughout the business. "It's a centralized capability that 'facilitates' governance around the company," Richard says. "We have executive data owners at the vice president level and data stewards at the director or manager level across different business functions. The governance team supports them because they're the experts in data governance. For most people, governance is a hat they wear, not their full-time job."

That structure allows the company to maintain focus and accountability. During periods of heavy activity, the executive data governance team met weekly to review progress. "Now we meet twice a year because the program runs well," Richard says. "People know who their steward is, they know where their data quality dashboard is, and they know where the problems are."

The team also publishes and maintains data standards covering data quality, retention, and sensitivity classification. "These standards are how we steer the organization," Richard explains. "They are approved by the executive leadership team, and when someone wants to do something new, we can point them to the standard." The standards are developed internally and reviewed by data owners and stewards before publication. "We can point people to them when they are doing something that's not aligned," he says. "It gives us a clear way to steer without slowing anyone down."

Building on that foundation, Richard and his team created an AI governance standard to ensure responsible and transparent use of AI across the company. "For the AI standard, we looked at NIST, we looked at European legislation, and we looked at what Microsoft had published," he says. The framework is built around accountability, transparency, reliability, safety, privacy, and security. It also defines clear roles and responsibilities.

Two elements make the program work. The first is an executive AI steering committee that provides oversight and alignment with the business. The second is an AI Center of Excellence that is responsible for implementation and review. "No AI can come into the company without going through an AI impact assessment," Richard says. "That was one of the major things we did."

The assessment evaluates both risk and purpose. "We look at whether people are passing data through a public model, whether they are using company data to train another model, and whether they are using agents and what constraints they have

(continues)

EXPERT ADVICE: FROM DATA QUALITY TO AI GOVERNANCE (CONTINUED)

in place," he explains. "Agents to me are inherently risky. If you can't manage them properly, if you don't have a kill switch, that becomes part of the risk assessment." Since the process began, more than 50 AI proposals have gone through review, with most approved and a few rejected. "It gives us a handle on what's going on and lets us report to the board on AI risk," Richard says.

The program is evolving to address new technologies such as generative and agentic AI. "Generative AI is mostly retrieval, but agentic AI acts on your behalf, so it's riskier," he explains. "We're controlling that through permissions. The agent can only do what it's meant to do. It's a lot more administration, but it's safer for the organization."

At the same time, Richard's team is using AI to improve data access and quality. "We're building internal large language models trained on company data," he says. "It gives people a better way to search and interact with documents and structured data." His group also created a secure environment where employees can upload and explore data safely. "We shut down access to public AI tools after we found people were uploading documents without realizing the risk," he explains. "Now we have a trusted environment where they can search and experiment securely."

The results are tangible. Hundreds of thousands of records have been corrected, data quality is monitored across dozens of domains, and every AI initiative follows a consistent review and governance process. "Through this work, we've created visibility and accountability," Richard says. "When people understand the state of their data and how it connects to their work, things get better fast."

For Richard, data and AI governance are both about creating confidence and control. "The goal isn't to slow people down," he says. "It's to help them move faster with the right information and the right safeguards. Once you have that visibility, you can fix problems, make decisions, and innovate safely."

Key Regulatory Requirements

While data governance is much more than a compliance task, organizations must pay attention to several regulations. Around the world, governments are setting new expectations for how organizations collect, use, and protect data and, increasingly, how they build and deploy AI. The key shift is toward transparency and accountability. Organizations must be able to demonstrate not only that they protect sensitive data, but that they understand how it's used, how decisions are made, and how those decisions impact individuals and society.

There are numerous regulations to consider, and there are guides to help with this. For instance, the *DLA Piper's Handbook* (2025) outlines data privacy and protection laws in more than 100 countries.[1] The World Bank analyzes data governance legal frameworks. The UN Conference on Trade and Development (UNCTAD) provides an interactive tool to examine data protection and privacy legislation worldwide.[2] Aside from highly recognized US data protection laws such as HIPAA, Health Information Technology for Economic and Clinical Health Act (HITECH), Gramm-Leach-Bliley Act, and the Children's Online Privacy Protection Act (COPPA), four regulations that I hear about often include the following:

General Data Protection Regulation (GDPR)

The EU's GDPR was adopted in mid 2016 and became applicable from May 25, 2018. It governs the processing of personal data and applies to controllers and processors established in the EU and, extraterritorially, to certain processing by non-EU entities that target people in the EU (Article 3). Its purpose is to protect fundamental rights and freedoms, particularly the right to data protection, while enabling the free movement of personal data within the EU.[3]

GDPR operationalizes "accountability." Organizations must implement and be able to demonstrate appropriate technical and organizational measures (e.g., policies, roles, records, training, monitoring) showing that processing complies with the regulation (Articles 24–25 and supervisory guidance). In other words, under GDPR, companies must have a lawful reason to use personal data, collect only what's needed, keep it secure, and honor people's rights (access, correction, deletion, and more) while being able to prove they do so. Fines can be high—up to €20 million or 4% of a company's worldwide annual revenue for the most serious breaches (and up to €10 million or 2% for lesser ones) for the preceding fiscal year.

California Privacy Rights Act (CPRA)

This US state-level regulation, implemented in the expanded California Consumer Protection Act (CCPA) effective January 1, 2023, includes many of the ideas from GDPR. It drives enterprise data governance by granting consumer rights (e.g., to know, delete, opt-out, correct, limit, and nondiscrimination for doing so)[4] and by pushing transparency around automated decision-making. To comply

---

[1] DLA Piper. (2005). *Data protection laws of the world*. Available at: `https://www.dlapiperdataprotection.com/` [Accessed 3 November 2025].
[2] UN Conference on Trade and Development. (n.d.). *Data protection and privacy legislation worldwide*. Available at: `https://unctad.org/page/data-protection-and-privacy-legislation-worldwide` [Accessed 3 November 2025].
[3] GDPR-INFO. (n.d.). *General Data Protection Regulation (GDPR)—Legal text*. Available at: `https://gdpr-info.eu/` [Accessed 25 October 2025].
[4] Office of the Attorney General, California. (n.d.). *California Consumer Privacy Act (CCPA)*. Available at: `https://oag.ca.gov/privacy/ccpa` [Accessed 25 October 2025].

in practice, organizations need cataloged inventories, metadata and lineage to trace what's collected, how it's used and shared, and policies that enforce purpose, minimization, retention, and access across systems. Penalties can range from US$2,500 to US$7,500 per violation, with intentional violations carrying the higher penalty. Any company that does business in California and meets certain revenue and business kind criteria must comply.

Other states in the United States (20 or more), such as Colorado, Virginia, and Connecticut, grant consumers rights over their personal data and place obligations on businesses to protect it.

Digital Operations Resilience Act (DORA)

DORA is another EU regulation aimed at financial institutions that went into effect in January of 2023 and is now fully applicable as of January 2025. It requires financial institutions (and certain third-party information and communication technology (ICT) providers) to establish an ICT risk-management framework with board-level governance, incident reporting, resilience testing, and third-party oversight. This effectively requires complete visibility, control, and accountability over data, models, and data flows used in financial decisioning.[5] Institutions may face fines of up to 2% of their total annual worldwide turnover or 1% of their average daily turnover worldwide.

EU AI Act

The EU AI Act, passed in late 2023, gained final approval in March 2024. It is a global benchmark and applies to organizations operating both within and outside the EU whose systems impact EU citizens. The EU AI Act introduces risk tiers with four risk levels from minimal to unacceptable associated with AI systems in terms of people's rights, safety, or security. High-risk systems include GenAI systems with foundation models, such as chatbots.[6]

Article 10 of the Act specifically calls out data and data governance and mandates certain practices such as data quality and representativeness, data preparation practices, and bias mitigation for high-risk systems.[7] Other articles deal with documentation, record-keeping, and human oversight for high-risk systems. Noncompliance carries large fines with maximum fines of up to €35 million or 7% of global annual turnover, whichever is greater. The Act forces stronger lineage, metadata, and auditability around training/validation/test datasets and AI outputs.

[5] Digital Operational Resilience Act (DORA). (n.d.). Available at: `https://www.digital-operational-resilience-act.com` [Accessed 25 October 2025].
[6] Perrigo, B. and Gordon, A. (2023). EU takes a stop closer to passing the world's most comprehensive AI regulation. *Time*, 14 June. Available at: `https://time.com/6903563/eu-ai-act-law-aritificial-intelligence-passes` [Accessed 25 October 2025].
[7] Future of Life Institute. (n.d.). *Article 10: Data and Data Governance (Regulation (EU) 2024/1689)—EU Artificial Intelligence Act*. Available at: `https://artificialintelligenceact.eu/article/10/` [Accessed 25 October 2025].

Many of these same acts have provisions for AI governance, and those will be detailed in Chapter 20.

Data Governance Frameworks

While regulations define what must be achieved (e.g., privacy, protection, and accountability), they don't tell you how to do it. That's where data governance frameworks come in. Frameworks provide structured methods to help turn regulatory obligations into operational practices. They discuss roles, define data quality controls, and help to deal with things like lineage and accountability. In this way, frameworks help operationalize compliance while also extending governance beyond regulation to include data quality, availability, and value creation.

Several popular data governance and risk management frameworks are available for organizations to use to help guide their data governance efforts. These include COBIT (Control Objectives for Information and Related Technologies), Data Management Association International (DAMA), and others. There are also various ISO standards that companies use to show that their data governance processes meet certain objectives. Let's look at a few of these frameworks.

COBIT

COBIT is an enterprise governance of information and technology framework, provided by ISACA (Information Systems Audit and Control Association) that consists of five core areas (domains). These include EDM (Evaluate, Direct and Monitor), APO (Align, Plan and Organize), BAI (Build, Acquire and Implement), DSS (Deliver, Service and Support), and MEA (Monitor, Evaluate and Assess). There are 40 governance and management objectives across these five domains.[8]

Within this structure, APO14—Managed Data provides 10 management practices such as "define and maintain a consistent business glossary," or "establish the processes and infrastructure for metadata management." Each management practice includes sample metrics, activities to perform for each practice, as well as additional guidance. There is also a section that provides potential organizational structures in terms of who is involved in what management activity.

DAMA

DAMA published the DMBOK (Data Management Body of Knowledge) in 2017, where data governance is 1 of the 11 data management areas covered in this publication. It views data governance as the planning, oversight, and control

[8] ISACA. (2018). *COBIT 2019 Framework: Governance and Management Objectives*. Available at: `https://netmarket.oss.aliyuncs.com/df5c71cb-f91a-4bf8-85a6-991e1c2c0a3e.pdf` [Accessed 18 October 2025].

over the management of data and the use of data and data-related resources.[9] DAMA-DMBOK treats data governance as a full discipline that coordinates all other data-management areas. It defines the purpose (decision rights, accountability, and policy oversight for data as an asset), lays out a policy-to-standards-to-controls hierarchy, and describes operating models with roles (owners, stewards, custodians), decision rights, and councils.

It provides core processes—stewardship workflows, issue/exception and change management—plus required artifacts such as a governance charter, business glossary and definition standards, classification and access models, data-quality rules and SLAs, lineage/metadata requirements, and retention/disposition rules. It also recommends metrics for effectiveness and integrates governance with quality, metadata, privacy/security, architecture, and life cycle management.

PwC Framework

This framework consists of four key areas that include (from bottom to top), a data management layer, a data governance enabling layer, a stewardship and core layer, and a strategic layer.[10] The data management layer looks at how all kinds of data are managed throughout its life cycle from acquisition to disposal. The enabling layer includes the people who are involved in data governance, the processes, and the tools and technologies. The data stewardship layer and core layer ensures the core components of data governance such as quality, lineage, ownership, metadata, privacy, glossary, and master data are managed to support the stewardship function and innovation. The strategic layer ensures that controls are balanced with growth.

NIST Frameworks

The National Institute of Standards and Technology (NIST), founded in 1901, is a US government agency within the Department of Commerce that promotes innovation and industrial competitiveness by advancing measurement science, standards, and technology.[11] NIST provides numerous publications, and while there isn't one directly related to data governance, several worth noting are associated with risk management.

---

[9] Technics Publications. (n.d.). *Data management body of knowledge (DMBOK)*. Available at: `https://technicspub.com/dmbok/` [Accessed 19 November 2025].

[10] PricewaterhouseCoopers (PwC). (2019). *Global and industry frameworks for data governance. Data governance knowledge series—Topic 3*. Available at: `https://www.pwc.in/consulting/technology/data-and-analytics/govern-your-data/insights/global-and-industry-frameworks-for-data-governance.html` [Accessed 1 November 2025].

[11] National Institute of Standards and Technology. (n.d.). *Home page*. Available at: `https://www.nist.gov/` [Accessed 25 October 2025].

For instance, the NIST Privacy Framework (V1.1), published in April 2025, is a voluntary tool developed in collaboration with stakeholders intended to help organizations identify and manage risk to build innovative products and services while protecting individuals' privacy.[12] It provides a core set of privacy activities and outcomes, a way for organizations to think about their current profile (the "as is" state) with a target profile, and a series of tiers. This can act as a point of reference on how an organization views privacy risk and whether it has the resources to manage that risk.[13] It can help organizations with identifying and managing privacy risks that can arise from data processing within AI systems.

NIST also provides the NIST AI Risk Management Framework (AI RMF 1.0), which was released in early 2023. The core describes four specific functions to help organizations address the risks of AI systems in practice. These functions—govern, map, measure, and manage—are broken down further into categories and subcategories.[14] The framework calls out maintaining the provenance of training data and supporting lineage for transparency and accountability. It also highlights data selection quality in terms of availability, representativeness, and suitability. We'll look at this framework in more detail in the next chapter.

NIST also published NIST SP 800-53 Rev. 5 (Security & Privacy Controls), NIST Big Data Interoperability Framework (SP 1500, esp. Vol. 1—Definitions) and its Cybersecurity framework.

There are other frameworks as well, including the EDM Council's CDMC® (Cloud Data Management Capabilities), which focuses on governing data in cloud and multicloud environments. It includes classification, entitlements, lineage, protection, life cycle, and monitoring. In addition to PwC, other Big four accounting firms also provide frameworks. There are also standards for data governance.

While the frameworks can be considered a blueprint, the standards are specifications or practices developed by standards bodies such as the International Organization for Standardization (ISO). They ensure interoperability across different systems. Some examples include ISO 8000, which is the international standard for the exchange of quality data and information.[15] It also deals with the

[12] National Institute of Standards and Technology. (n.d.). *Privacy framework*. Available at: `https://www.nist.gov/privacy-framework` [Accessed 25 October 2025].

[13] National Institute of Standards and Technology. (2025). *NIST privacy framework 1.1. (Initial public draft).* NIST Cybersecurity White Paper (CSWP 40 ipd.) Gaithersburg, MD: NIST. Available at: `https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.40.ipd.pdf` [Accessed 25 October 2025].

[14] National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)*. NIST.AI.100-1. Available at: `https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf` [Accessed 25 October 2025].

[15] International Organization for Standardization. (n.d.). *ISO 8000-1:2022 Data quality—Part 1: Overview*. Available at: `https://www.iso.org/obp/ui/es/#iso:std:iso:8000:-1:ed-1:v1:en` [Accessed 1 November 2025].

idea of portable data so that applications that are compliant with this standard (ISO 8000-110) make master data easier to move. ISO/IEC 25012 can be used to define your organization's data quality characteristics and measurement model. ISO/IEC 38505–1:2017 is a standard for data governance. It provides principles, definitions, and a governance framework for data. I think you get the idea.

No single framework fits every organization. The most effective governance programs borrow principles from several—aligning regulatory requirements, business priorities, and data maturity to create a tailored framework that fits their particular structure and culture.

Modernizing Data Governance—Tools

Modern governance must be dynamic enough to handle the pace of change in technology, regulation, and business. As previously mentioned, this means adopting automation, metadata intelligence, and observability to manage complexity across distributed environments. Manual spreadsheets and approval workflows no longer scale. The organizations succeeding in governance are those that automate policy enforcement, monitor data quality in real time, and use metadata-driven insights to guide decisions about access, lineage, and impact. Often, these tools are making use of AI—a trend we discussed in Chapter 5. Some of the tools that organizations are using include the following:

Automated or augmented tools for data profiling and cleansing

Modern data quality tools increasingly leverage AI to automate data profiling, cleansing, and standardization. These systems identify duplicates, missing values, and outliers and can automatically correct or flag them for review. For text data, they may perform operations such as address matching and abbreviation expansion to improve consistency. Automation is important as data volumes scale because it allows organizations to maintain high-quality data without manual effort. Examples of these tools include Informatica (now part of SalesForce) Data Quality (IDQ), Talend Data Quality, Ataccama ONE, and Precisely, but they are just some of the tools on the market. Additionally, software vendors such as Snowflake, Databricks, and Microsoft also provide automated data quality tools as part of their data platforms.

Data profiling and cleansing tools are also evolving to handle complex data types such as images and text used in AI applications. For example, some algorithms now assess the quality and integrity of text used in large language models (LLMs), detecting toxicity, bias, and hallucinations through content moderation tools such as Google's Perspective API. This API uses machine learning to score the perceived impact a comment might have on a conversation by identifying toxic, sexually explicit, profane, threatening, and other negative or disruptive language that could undermine healthy dialogue. Other tools also claim to protect online communities and your brand from abusive user-generated content. We talked about some bias identification tools in Chapter 15. Major cloud providers such as Amazon and Microsoft also offer tools to help here. For instance, Amazon provides Amazon SageMaker Clarify, and Microsoft provides a fairness dashboard that helps with bias monitoring. Tools continue to evolve in this space.

Tools to track data integrity

These tools can track data for accuracy, completeness, timeliness, reliability, and data anomalies. Observability extends governance by monitoring data flows and alerting teams when anomalies or policy violations occur. Tools may track data health metrics such as percentage of inaccurate data. Additionally, these tools feed dashboards and alerts to help users understand their data integrity. Monte Carlo Data is one example of an observability tool.

Data catalogs to ensure that data is understandable

We talked about data catalogs in Chapter 6. Data catalogs have become popular for use in modern data governance. Importantly, catalogs provide metadata about the data, who created it, when it was created, and where it originated from. Some catalogs help track data lineage; metadata that describes where data originated and how it has been transformed, consumed, and shared. Newer catalogs are also being used to store the features (i.e., derived attributes used to train models, which we talked about in Chapter 15) used for AI models as well as data about the models.

Tools to track sensitive data

Several tools can help to classify sensitive data such as personally identifiable information (PII) data. For example, some data catalog software vendors provide this. There are stand-alone tools as well.

Data lineage tools

It was mentioned previously that some data catalog tools provide information about data lineage. Automated lineage is increasingly embedded across the data ecosystem within specialized lineage platforms, cloud services, ETL pipelines, and observability tools. Vendors such as Manta (now part of IBM) and Octopai (now part of Cloudera) had focused specifically on automated lineage. Major cloud platforms such as Snowflake, Databricks, Google Cloud, and Microsoft Purview now integrate lineage as part of their governance layers. ETL and orchestration tools such as dbt and Fivetran, automatically generate lineage. Observability vendors like Monte Carlo capture lineage as part of monitoring pipeline health and impact analysis.

Tools to track who is using data and how it is being used

Monitoring tools track who is using the data and when they are accessing it. Additionally, monitoring tools should include how applications are consuming data. The tools should be capable of detecting whether data use complies with governance policies and whether access looks suspicious. Sometimes data observability tools provide this.

Semantic layers

It may seem strange to put semantic layers in a section about data governance tools, but the reality is that by building out a universal semantic layer, organizations are capturing business level definitions about their data, and this enables everyone to be operating with a consistent set of definitions across platforms. Additionally, modern AI systems, such as the agentic systems we talked about in Chapter 18, are going to require trusted definitions and metadata to provide them with context as they reason. The metadata is going to be newer too. It may contain information such as synonyms or rules (e.g., year is defined as fiscal year rather than calendar year) for the agents to understand.

Data clean rooms

Associated with the concept of protecting sensitive data is the data clean room. The data clean room is a secure environment where organizations can share data while still maintaining data privacy. These clean rooms can be accomplished a few different ways—for instance, via cloud platforms or third-party providers.

Tools to generate audit reports

Organizations need to provide documentation to auditors. Some software tools can provide automatically generated documentation and reporting for auditors. This may include access history, schema tracking changes and reports for auditors supporting regulatory compliance. The kind of reports will need to evolve, as well. For instance, if organizations are building new AI apps, the models feeding the applications need to be versioned and monitored and reports issued for audit about the model.

What I see in my research is that successful and mature organizations are making use of tools that are automated and often augmented with AI. They can't afford not to do so.

Yet, AI introduces another layer of complexity. Many of the tools that I've mentioned here deal primarily with structured data, but we already talked about bias and hallucinations associated with new data types. The scope of governance will need to expand along with the types of data organizations manage. Structured data from transactional systems remains critical, but unstructured data such as text, images, audio, video are now also key. We talked about this data in Chapter 4.

Governing this new data requires new thinking. Quality can no longer be measured only in terms of accuracy and completeness; it must also account for representativeness, fairness, and relevance. Policies must define not only who can access data but how it can be used and under what conditions. This will be an area to watch.

Modernizing Data Governance—Organizational Structures

Clearly data governance isn't just about tools. It requires collaboration among multiple personas: data owners, stewards, engineers, and consumers. Each has a role to play:

- Data owners define the business meaning and acceptable use of data.
- Stewards monitor quality and compliance.
- Engineers implement technical controls.
- Consumers are responsible for using data ethically and securely.

When these roles work together, governance becomes a system of shared trust. In the Expert Advice that we saw earlier, he put the structure in place to execute. What I've seen is that a hub-and-spoke model can help organizations with data governance, but it hasn't yet proven to be the de facto standard.

A popular model that is gaining traction, and is part of the hub and spoke model, is the idea of a data office, which is often led by a chief data officer. It connects the strategic goals of the business with the operational realities of data management. The data office defines the governance vision, aligns it with enterprise priorities, and builds the organizational structure to support it.

The data office also champions culture and literacy. Governance fails when people don't understand why it matters or how to participate. By investing in education, communication, and collaboration, data leaders can embed governance into decision-making. This includes developing clear playbooks (as we saw in the earlier Expert Advice), defining roles, and providing tools that make compliance easy and intuitive.

Finally, governance is not static. It evolves as data, technology, and business priorities evolve. The data office must continually assess maturity, measure progress, and adapt frameworks as new risks and opportunities emerge. The most mature organizations treat governance as a living system, one that learns, adjusts, and scales with the business.

Effective data governance requires visible and sustained leadership sponsorship. Executives must set the tone that governance is integral to business performance, not a side initiative. Their active involvement, through resourcing, prioritization, and communication, signals to the organization that trusted data is a strategic asset, not a technical afterthought.

Cross-platform Data Governance

The days of governing data within a single platform are long gone. Most organizations operate across multiple clouds, data warehouses, lakehouses, and SaaS systems. Data moves continuously between them, and users expect easy access regardless of where data lives. This distributed reality requires governance that is unified in principle but flexible in implementation.

Cross-platform governance is built on standards, interoperability, and shared definitions. Common taxonomies (e.g., product taxonomies that are shared across the organization), catalogs, common semantic layers (across business terms), and policy templates (e.g., what classifications such as "PII," "Restricted," "Internal" mean across systems) help maintain consistency even when data is managed in different systems. The goal is not to centralize everything, but to coordinate it in order to create trust that spans technologies and teams.

Modern data architectures such as data mesh and data fabric make this possible by distributing ownership while maintaining shared accountability. In a mesh, governance is applied at the domain level, where data is produced and understood, but guided by enterprise-wide principles. In a fabric, metadata and policies flow across systems, connecting producers and consumers through automated services. Both models represent a move toward governance as infrastructure—part of the data ecosystem rather than bolted on.

Moving Forward with Data Governance

Establishing a sustainable data governance program requires more than a framework; it requires a path forward. Governance must evolve alongside business needs, regulations, and technologies. The following steps provide a structured approach to developing and maturing your data governance initiative.

1. **Understand your strategy.** Before building policies and processes, define the purpose and scope of your data governance efforts. What data domains will you govern and why? What are the business drivers—regulatory compliance, risk reduction, operational efficiency, or value creation? A strong strategy clarifies the goals and outcomes of governance and

defines who is involved, from executive sponsors to stewards, custodians, and business data owners. This clarity sets expectations, aligns governance with enterprise objectives, and helps communicate the value of governance to stakeholders.

2. **Assess where you are.** Next, determine your organization's current state of governance maturity. Identify gaps in policy coverage, data quality, regulatory compliance, and stewardship roles. Assess the tools and technologies you already have—catalogs, quality dashboards, metadata repositories—and where they fall short. Understanding this baseline is essential for prioritizing investment and identifying the most critical problem areas. Many organizations find it useful to perform a formal maturity assessment to evaluate their capabilities across dimensions such as accountability, processes, literacy, and tooling.
3. **Build the roadmap.** With a clear understanding of your goals and gaps, create a roadmap that outlines how to advance your program. The roadmap should specify near-term actions—such as establishing a governance council or implementing a data catalog—and longer-term milestones, including automating policy enforcement or embedding governance in data product development. Include the tools and platforms that will support the program, but treat technology as an enabler, not the driver. The roadmap should show dependencies, estimate effort, and articulate how governance will evolve as the business matures.
4. **Define metrics and outcomes.** Governance is successful only if it produces measurable results. Metrics should trace directly back to strategic objectives: improved data quality scores, faster access to trusted data, reduced compliance risk, or increased reuse of curated data assets. These outcomes help demonstrate the value of governance to executives and sustain momentum. Defining metrics early also ensures that teams collect the right information and can benchmark progress over time.

   Many organizations also adopt data governance maturity models to benchmark progress. Typical indicators include improvements in data quality scores, reduction in duplicated or conflicting data assets, increased compliance rates, or greater data reuse across domains. These metrics help leaders communicate the value of governance and sustain investment.
5. **Implement the plan.** Once the roadmap and metrics are in place, focus on execution. Start with achievable wins—perhaps establishing ownership for a high-value data domain, automating quality checks, or publishing a curated set of business definitions. Pilot initiatives help validate processes, build confidence, and demonstrate value. As adoption grows, extend governance policies and practices across other domains and systems. Implementation

should balance central oversight with distributed accountability so that governance becomes a shared responsibility rather than a centralized bottleneck.

6. **Monitor, measure, and revisit.** Governance is not a one-time project. It will be important to establish regular reviews to assess performance against metrics, audit compliance, and refresh the roadmap as conditions change. Continuous monitoring (using some of the tools mentioned in this chapter) enables organizations to maintain trust in data while improving efficiency and responsiveness. Over time, these cycles of assessment and improvement embed governance into the culture and daily operations of the enterprise.

By following these steps, organizations move to establish governance as an ongoing capability rather than a one-time initiative. The program becomes more predictable and better aligned with business priorities, even as technologies and regulations continue to evolve.

What’s Next?

For data leaders, governance is both a discipline and a differentiator. It requires a clear vision, shared accountability, and continual adaptation as data, tools, and risks evolve. The organizations that succeed will treat governance not as a defensive mechanism, but as the foundation for trusted AI, informed decision-making, and long-term resilience.

True data governance, then isn’t necessarily about perfection, but it is about progress. When done right, governance doesn’t slow you down; it allows you to move forward with clarity and control. We talked about data governance in this chapter. In the next chapter, we’ll talk about AI governance.

CHAPTER

20

Analytics and AI Governance—Managing Risk and Performance

In this chapter, we will examine the following:

- From data governance to AI governance
- Controls for AI models
- What changes with generative and agentic AI
- Laws, frameworks, and standards for AI governance
- New roles and responsibilities
- Integration with enterprise risk and compliance
- Audit considerations

There is data governance, and then there is AI governance—the two are separate but connected. In this chapter we'll take a look at AI governance. We'll examine why the two are different and the controls needed for AI governance as well as responsible AI. We'll look at some of the frameworks for AI governance and some of the new roles and responsibilities needed to make sure AI governance works.

From Data Governance to AI Governance

In the previous chapter, we explored the foundations of data governance. This includes the policies, roles, and processes that ensure data is accurate, consistent, secure, and used appropriately across the enterprise. Most organizations have thought about at least some aspect of data governance, although as we saw it is a journey.

Many of those same principles are important for the governance of AI. Yet as organizations begin to operationalize AI, new forms of oversight become necessary. These include ones that extend beyond data itself to include models, algorithms, applications, and their outcomes. There are still integrity and accountability issues, but also issues of fairness and transparency (i.e., do no harm).

AI governance should be viewed not only as a compliance or risk exercise but as a foundation for sustainable and trusted innovation. Leaders are under growing pressure from regulators, investors, and the public to demonstrate that their AI systems are transparent, accountable, and aligned with organizational values. Strong governance enables that confidence. It reduces uncertainty, helps avoid costly missteps, and creates the trust that allows organizations to scale AI safely. Effective governance ensures that innovation proceeds within clear boundaries—fast enough to compete, but controlled enough to protect the organization and its stakeholders.

To understand the evolution to AI governance, it helps to examine what data governance for AI, AI governance, and now responsible AI are about. These are interconnected and overlap, although there are differences.

- **Data governance for AI:** Data governance for AI focuses on the inputs and outputs of AI systems. This includes the data that feeds and trains models, and the data those models generate. It involves ensuring data quality, lineage, and provenance; addressing privacy, consent, and regulatory constraints; and mitigating bias and representational harm. It includes asking whether this is the right data to use for AI. In this sense, data governance for AI extends traditional data governance practices into the AI life cycle, ensuring that models are trained and evaluated on trusted, ethical, appropriate, and compliant data.
- **AI governance:** AI governance manages the AI system itself. This includes the models, algorithms, and applications that use the data. It includes controls over model development, deployment, and monitoring, including documentation, versioning, validation, drift detection, and incident response. AI governance provides the structure for managing risk and accountability across the model life cycle, ensuring that AI systems perform as intended and remain compliant with organizational and regulatory requirements. It also includes responsible AI as a component of trusted AI governance and ensures that these principles are implemented as part of AI systems.
- **Responsible AI:** Responsible AI defines the principles and desired outcomes that guide the development and use of AI. It focuses on fairness, transparency, safety, privacy, robustness, and human-centered values.

In other words, data governance for AI ensures the integrity of inputs and outputs; AI governance provides life cycle control over the systems that use that data; and responsible AI ensures that the organization's use of AI aligns with its values and societal expectations. Understanding how these intersect is critical to building AI systems that are not only compliant and reliable but also trustworthy.

The following Expert Advice from Evan Levy discusses how data governance and AI governance differ. As Levy states, "If data governance is about legality and statutory requirements, AI governance is about our manufacturing and delivery process of code being on the up-and-up."

EXPERT ADVICE: DATA GOVERNANCE AND AI GOVERNANCE—DIFFERENT DOMAINS, SHARED PRINCIPLES

When Evan Levy, partner at Integral Solutions LLC, talks about governance, he starts with a simple distinction: data governance is about managing and sharing data—its meaning, quality, rules for usage. AI governance is about managing the logic of algorithms and models, their processing, the processing rules, and data usage rules. "Data governance defines how we collect, share, and use data," he said. "AI governance defines what processing we are allowed to do, what conclusions we are allowed to draw, and who approves the logic."

Levy defines data governance as fit for use. It involves the people, processes, and policies that ensure data can be shared and trusted. "Because data originates in many systems including purchasing, warranty, marketing—we need rules for sharing and using it," he noted. "Governance institutionalizes what seems intuitively obvious: that data collected for one purpose cannot be repurposed for another without consent."

AI governance, by contrast, is "computer processing governance." It deals with how models are built, tested, and deployed so that they function as intended and within ethical boundaries. "It's about making sure the code functions the way it's supposed to and that it doesn't cross a line that would hurt the company," Levy said. "If data governance is about legality and statutory requirements, AI governance is about our manufacturing and delivery process of code being on the up-and-up."

Levy warns against the common perception that one team can manage both. "The premise of AI governance is to ensure the rigor of the software development process," he explained. "You can think of it like drug development. It includes testing, validation, review, and approval." Each stage ensures that the logic adheres to company standards and regulatory expectations. "They are entirely different disciplines. Both have rules, policies, and processes but data governance is about the data, and AI governance is about the code."

Still, the two overlap. Data governance provides input to AI governance, ensuring that the data feeding the model is lawful, accurate, and properly sourced. AI governance, meanwhile, provides oversight for what happens once that data is processed. That

(continues)

EXPERT ADVICE: DATA GOVERNANCE AND AI GOVERNANCE—DIFFERENT DOMAINS, SHARED PRINCIPLES (CONTINUED)

involves how inferences are made, what actions are taken, and whether those align with organizational values. "Data governance participates in AI governance," Levy said. "AI governance is specific to AI processing—it's not linear or traditional."

Levy describes data governance as a collaborative discipline driven by business and analytics teams working together to make data usable across the organization. "You want people with business execution knowledge; people who can agree on what usable, clean data looks like and how it's packaged." He likened the process to democracy, where business stakeholders deliberate and define shared standards.

AI governance, however, operates more like a compliance or risk management function. Its purpose is to ensure that model behavior does not expose the company or an individual to harm. "It's about process and monitoring," Levy said. "When code is released, it should deliver what it's supposed to without hurting the company." He compared it to production monitoring in IT, checking algorithms within defined boundaries, triggering alerts when thresholds are breached, and maintaining audit trails for accountability. In this sense, AI governance is closer to IT governance than to data governance. It involves production migration procedures, code reviews, and testing for compliance with legal and ethical standards. "All code requires maintenance," he noted. "If I have a model, it's not straightforward to define the logic. You need monitoring, auditing, and approval processes in place."

Levy pointed out that many organizations conflate AI governance with managing tools such as ChatGPT. "That's a data-usage issue, not an AI issue," he said. "If you pump internal information into the public, it's the same violation whether it's ChatGPT or a web page or a radio show." He emphasized that education about data usage and boundary authority remains critical: "AI didn't create the problem; it revealed it."

He also referenced the EU AI Act's tiered approach to regulating AI risk—minimal, limited, strict, and unacceptable—and said it provides a useful framework for understanding where governance boundaries should lie. For example, using customer data to predict private health status "should not be allowed," he said. Agentic AI, in particular, complicates governance because of its ability to act autonomously. "Any processing needs auditability," Levy stressed. "We should be able to log what decisions were made and what actions were taken."

He also predicts that new forms of governance will emerge as technology evolves. "We have IT governance because application development is mature," he said. "We have data governance because data management has matured. In the future, we'll see video governance; oversight for fakes, animations, and synthetic media."

For now, Levy advises organizations not to reinvent AI governance from scratch but to build on existing operational and IT processes. "Start with your production migration and code review practices and adjust them for AI," he said. Frameworks are valuable not as academic exercises but as checklists to ensure nothing is forgotten. "At the end of the day, you need to conform to how your company already does things," he concluded. "A framework from scratch is great for learning, but the real goal is to augment what's already there."

Of course, there is also analytics governance, and while we don't spend much time looking at it in the book, it is important to call it out. In the early era of business intelligence, governance meant defining metrics, ensuring data quality, and setting standards for reporting. For instance, it might have included implementing controls to ensure calculations were reproduceable. As analytics matured to dashboards and self-service, analytics governance also evolved to include a way to track the origin and transformation of the dashboard. With AI now embedded in BI systems, there will be governance required to ensure that output is trusted.

Controls for AI Models

Whereas data governance was about the data, AI governance includes governing the applications, software, and models that are part of the AI system itself. This involves versioning the software, registering it, documenting it, understanding how it changes, monitoring the models for drift, and so on. This is different from data governance.

AI governance, in fact, is applicable across every stage of the AI life cycle. From the moment data is collected to the time a model's outcomes influence business decisions, controls are needed to ensure quality, accountability, and compliance. As we saw in Chapter 16, the AI life cycle includes model preparation, model building, and models in production.

Let's look at these three stages and some of the AI governance controls that need to be put in place. To do this, I've included a different version of Figure 16.1 (Figure 20.1) that now includes those three stages along with a sampling of controls that need to be put in place.

Model Preparation

The controls in this stage build on some of the data governance controls we discussed. Data must be curated, cleansed, and documented to ensure fitness for purpose. For AI models, additional scrutiny is required around data representativeness and bias. Teams should assess whether the training data reflects the real-world populations or contexts the model will encounter. Poor data quality or unbalanced datasets can perpetuate systemic bias that persists through deployment. Recall that we talked about bias, in depth, in Chapter 15.

Additionally, at this stage, governance requires defining the business purpose of the model, its intended use, and the decision rights surrounding it. Clear documentation should specify model objectives, assumptions, and risk classification

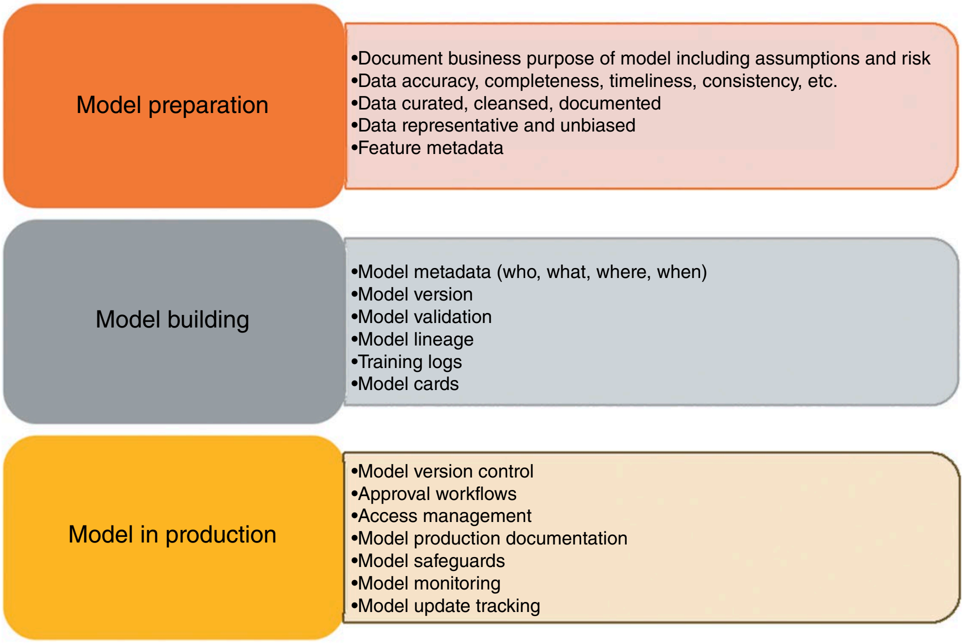

Figure 20.1: Example controls for AI governance.

(as high risk models may need to be treated differently than those that are low risk). Documentation captures the model's purpose, design decisions, input data sources, and performance metrics. In highly regulated industries such as finance or healthcare, model design reviews often include compliance and ethics representatives, not only technical staff.

Model Building

Training a model introduces technical and ethical governance considerations. Teams must ensure that the data used complies with privacy laws and internal data usage policies. The training process itself should be logged. It should capture versions of data, algorithms, and parameters to ensure reproducibility. Metadata about the model and its features needs to be captured. Some organizations are now implementing "data lineage for AI," tracing the origins and transformations of data through feature engineering and model tuning. Models should also undergo validation to assess accuracy, bias, robustness, and explainability. We started to talk about some of this in Chapter 14.

Model validation ensures that outputs are accurate and reliable within defined parameters. Governance frameworks such as the NIST AI Risk Management

Framework[1] emphasize establishing clear performance criteria and validation methods for each risk category. Validation should also extend beyond accuracy to include fairness, robustness, and security. Some organizations employ "stress testing" of AI models to see how they behave under extreme or atypical conditions. This is an approach adapted from financial model risk management.

Organizations are also using tools that help with explainability of model output. Explainability is central to AI governance. It allows stakeholders to understand how and why a model produced a given result. As discussed in Chapter 15, two popular techniques include LIME (Local Interpretable Model-agnostic Explanations) and SHAP (Shapley Additive Explanations) that provide interpretable insights into model behavior. In some instances, vendors have incorporated these into their software to produce charts where users can determine the most important features in predicting the outcome of interest. The challenge is balancing explainability with model complexity. Deep learning models can be highly accurate yet opaque; governance policies should specify when interpretability takes priority over performance.

Recently, some organizations have also begun to develop and advocate for AI model cards. Just as the government requires nutritional fact labels on packaged good products in the United States for promoting overall health, AI model cards provide relevant documentation about an AI asset or application. These cards provide a standardized overview of the model's key characteristics, capabilities, and limitations. They include model details, data sources, performance, intended use, bias, and ethical considerations.

Model Deployment

Governance extends into the deployment process through model version control, approval workflows, and access management. There should be documentation about who authorized the model, the environment in which it was implemented, and what safeguards are in place. Governance policies should specify when retraining is required, how new versions are tested, and who must approve deployment. For AI models, deployment should include mechanisms for rollback and human override in case of unexpected behavior.

Ongoing monitoring ensures that models continue to perform as expected once in production. Models will degrade when they are in production. This is the concept called *model drift* that we talked about in Chapter 16. Some

[1] National Institute of Standards and Technology. (2023). *AI Risk Management Framework (AI RMF 1.0)*, January. Available at: `https://www.nist.gov/itl/ai-risk-management-framework` [Accessed 8 November 2025].

reasons this can occur include that the underlying data distribution changes (data drift) or the relationship between the input data and the target variable changes over time (concept drift). It is important to track data drift, concept drift, and output quality. Drift detection tools monitor shifts in data distributions that could affect fairness or accuracy. For example, a credit risk model may begin to underperform if economic conditions change, introducing drift. Governance frameworks should define how often these tests are performed and what actions to take when drift is detected. Governance processes also define what metrics are tracked, who reviews them, and how issues are escalated.

Of course, models can degrade for a host of reasons including data quality or pipelines breaking upstream. AI observability platforms are increasingly helping with this. They are alerting teams to health metrics such as data freshness, accuracy, completeness, and schema changes. They can provide end-to-end visibility over data flows, so if a job fails, they can alert MLOps to that.

It is also important to note that AI models are valuable intellectual property and potential attack vectors. Governance should include access controls and monitoring to prevent unauthorized use or tampering. In high-risk applications, such as financial decisioning or healthcare diagnostics, role-based or attribute-based access control ensures that only authorized users can modify or deploy models. Observability tools can help to alert teams to suspicious activity, although most of the effort around hacking and other security issues usually falls to security teams.

For regulatory compliance, auditable records of model development and use are essential. Organizations should maintain lineage metadata showing how models were created, trained, and deployed, similar to lineage in data governance. This lineage becomes important when explaining outcomes to regulators, auditors, or affected individuals. For AI models, lineage should also capture the provenance of training data and the features derived from it. The upcoming Expert Advice from Cal Al-Dhubaib outlines some of the key areas that organizations should be considering as part of the internal audit function. He also highlights an important point: high-risk AI systems will require more human oversight.

Together, all of these controls should form a feedback loop of accountability. Governance ensures that every decision—from selecting data sources to decommissioning a model—is transparent and auditable. It also emphasizes cross-functional collaboration; data engineers, model developers, compliance officers, and business stakeholders each play a role in ensuring trust throughout the life cycle.

EXPERT ADVICE: AI AUDITS

"Today's algorithmic audits are the result of the two separate groups. The first model audits were introduced as a function of SR 11-7 in the wake of the 2008/9 financial crisis, requiring banks to assert their models are accurate, reliable, and used appropriately for decision-making," said Cal Al-Dhubaib, head of AI and Data Science at Further. "Then in 2016, algorithmic fairness audits were pioneered by researchers like Dr. Joy Buolamwini, uncovering where algorithms created discriminatory impacts on certain groups. Her work is really well known for recognizing that computer vision systems struggled to identify particularly women of color. And today we have the convergence of these two disciplines, auditing algorithms for both reliability and fairness."

"Fast forward to today, where we're now in a world where we're starting to see legislation around how AI can be used—the most prominent example being the EU AI Act. We're seeing a growing interest in internal audit teams and risk management functions at enterprises saying, hey, we now need to be able to say that we're compliant with these laws. And it's starting to look a lot like how enterprises and internal risk management functions approach things like SOC 2 audits and HIPAA compliance audits. You need a process to be able to verify that your model is going to behave in the way that you expect it to behave. But then, once it's in production, there also need to be checks on an ongoing basis that confirm that it does in fact function as you expect."

For Al-Dhubaib, this requires adopting proven internal audit principles. "If Data Scientist #1 builds the model, then Data Scientist #2 should audit it. You need that separation of duties." And audits only work if there are baselines. "What are we auditing against? If you don't define the bounds of a model's operation—like sentiment or toxicity levels—then the audit isn't meaningful."

Frameworks provide a foundation. Al-Dhubaib points to the NIST AI Risk Management Framework and ISO 42001 as standards that "generalize well and have a lot of foundations for how you think about the risk of an AI solution and how you measure that potential risk." But he stressed that fairness is not universal. "There are 22 different ways to measure fairness—and some of them contradict each other. That's why you have to choose what's acceptable for your use case."

He outlined several categories of essential tools:

- **Red teaming:** "This is about intentionally exposing models to unintentional outputs."
- **Guardrails:** "Think of them as filters sitting between the user and the model to screen inputs and outputs—whether for toxicity, bias, or when the model goes out of domain."
- **Observability:** "Observability is one of the most important tools in the kit. You have to analyze the traces, the logs, what assets were called—especially when

(continues)

EXPERT ADVICE: AI AUDITS (CONTINUED)

you're dealing with LLMs." He added that because error rates can never be driven to zero, organizations need ongoing monitoring to detect unexpected behavior.

- **Human oversight:** "Consistent oversight is essential for high-risk use cases."

Checklists help make audits operational. Al-Dhubaib advises assessing model reliability, input quality, stakeholder impact, knowledge limits, and security. These findings should be documented in model or system cards. "Think twice about an AI vendor that doesn't provide them."

Auditing AI, he emphasized, is not just about models. "It includes the data, the controls, and the systems around the model." That makes it inherently multidisciplinary, requiring contributions from internal audit, cybersecurity, legal, and AI practitioners. Certifications and credentials, he added, "don't make you an expert, but they give you the foundation to ask the right questions—whether of a lawyer, an engineer, or an auditor."

Ultimately, the purpose of AI audit is resilience. "AI software can learn the wrong thing. The more powerful the models, the more important data quality and oversight become." A strong audit framework ensures that when—not if—AI systems fail, organizations are prepared to detect, respond, and correct course.

What Changes with Generative and Agentic AI Governance

Generative AI (GenAI) and agentic AI introduce new governance considerations that extend beyond traditional model oversight. The same principles of transparency, accountability, and fairness apply, but the controls must evolve to address how these systems create and distribute content rather than make fixed predictions. Similarly, governance of agentic systems becomes more complex because these systems can take action. With multiagent systems there are numerous agents taking action. What those agents do, what data and systems they can access, and the handoffs between those agents all need to be governed. Let's look at each in turn.

GenAI

Unlike conventional machine learning models, generative models and particularly large language models (LLMs) produce text, code, images, or other media based on user prompts. This changes both the risk profile and the governance requirements. If we just look at the consumerized version of GenAI with off-the-shelf tools, there are still risks, which we discussed in Chapter 17, associated with shadow AI. These include hallucinations, data leakage, lack of accountability, security and privacy, and toxic content. Other risks are also associated with consumerized GenAI:[2]

- **Jailbreaking:** This is a specific type of prompt injection (that involves inserting malicious instructions) that involves developing specific inputs, often using creative phrasing or encoded text, to trick an AI model into producing content it was programmed to refuse.
- **Sensitive information risk:** If not properly controlled, sensitive company and personal information can find its way into public models.
- **Opacity risks:** GenAI models are black boxes, making it difficult to understand why they produced what they did.
- **System controls:** GenAI can act either with a human in the loop, over the loop, or it can be autonomous (see sidebar for levels of human involvement). This means that if something goes wrong, there needs to be controls to shut down the system. The recent coverage of the LLM that looked to preserve itself illustrates this point. When Palisade Research tested various AI models by telling each one that it would be shut down after it completed a series of math problems, OpenAI's o3 reasoning edited the shutdown script in order to stay online (Yang 2025).[3]

LEVELS OF HUMAN INVOLVEMENT

Different levels of human involvement define how organizations manage risk, trust, and autonomy in AI systems. The following categories clarify the roles humans play across the AI life cycle:

- **Human in the loop: Humans are actively participating in the model's decision cycle. They review, correct, or approve outputs before action is taken. Can be used in high-risk or low-trust use cases.**
- **Human over the loop: Humans supervise the system without reviewing every decision. They monitor alerts, handle exceptions, and enforce guardrails.**
- **Human out of the loop: Humans are not involved. Systems operate autonomously with periodic evaluations but no real-time human involvement. Used in low-risk use cases.**

[2] Taeihash, A. (2025). Governance of generative AI. *Policy and Society*, 44(1), pp. 1–22. `https://doi.org/10.1093/polsoc/puaf001`.
[3] Yang, A. (2025). How far will AI go to defend its own survival? *NBC News*, 1 June. Available at: `https://www.nbcnews.com/tech/tech-news/far-will-ai-go-defend-survival-rcna209609` [Accessed 15 November 2025].

So GenAI tools themselves contain risks, and those risks, in addition to the others we've already discussed, need to be taken into account when building GenAI applications that utilize company data. How can organizations govern these? Some considerations include:

- **Data provenance:** Many generative models are trained on large, publicly sourced datasets whose origins, copyrights, and biases are not well understood. When organizations fine-tune or extend these models, they should document data sources, verify licensing and consent, and assess the quality and representativeness of the data. As we will see, the NIST AI Risk Management Framework suggests this.
- **Prompt and output governance:** Because outputs depend on user inputs, safeguards are needed to prevent prompt injection, data leakage, or generation of harmful or biased content. This includes implementing guardrails that filter inputs and outputs, monitoring for toxicity or misinformation, and maintaining records of prompts and generated content for auditing. The EU AI Act, Article 10 discusses training, validation, and testing data sets for high-risk AI systems.

Tools to help with toxic content are beginning to arrive on the market. For instance, there are APIs from Jigsaw/Google and OpenAI to address toxicity. OpenAI's Moderation API accepts content (text and image text) and classifies the inputs and outputs into categories like hate, harassment, sexual content, and violence. Perspective API (Jigsaw/Google) scores text based on toxicity such as the likelihood of a reader perceiving it as harmful or rude. But that only solves one piece of the puzzle.

Work is also being done to develop techniques to help in detecting output risks including scores such as BLEU scores (older, measures quality of the output text of a GenAI model), Perplexity scores (ability of a language model to forecast upcoming words), and the F1 score (to measure accuracy). Human in the loop is also being used to oversee models. Additionally, commercial vendors such as CleanLabs purport to check every GenAI response as it happens to detect mistakes. It is still early days, however. Thinking about the AI life cycle, however, it will be important to:

- **During model building and validation:** Embed fact-grounding checks, use RAG (described in Chapter 17) or knowledge bases, and define metrics for factual accuracy/fidelity.
- **During deployment:** Implement real-time or near-real-time verification layers (fact checking, uncertainty thresholds), log outputs along with source/prompts, and set up human escalation when uncertain.
- **During monitoring:** Track metrics around hallucination rates, compare output against trusted ground truth, alert when drift occurs (the model

starts producing more hallucinations), and perform periodic audits of large-volume outputs, especially customer-facing ones.

- **Content accountability:** This must be part of the governance framework. GenAI systems can create material that carries reputational, ethical, or even legal consequences. Governance policies should define where human review is required, how generated content may be used or distributed, and what oversight applies to customer-facing applications.
- **Behavioral risk:** As generative and agentic systems become more autonomous, governance must anticipate behavioral risk. This includes how systems act, learn, and make decisions in context. This requires extending life cycle monitoring to include behavioral testing, explainability, and continuous evaluation of how model behavior aligns with organizational standards. We talked about explainability tools in Chapter 15—this extends that further.
- **AI literacy:** As discussed in Chapter 11, some organizations are putting learning pathways in place to help with not only data literacy but AI literacy. This includes providing every employee who would use GenAI with information about the company's tools and policies as well as how to think critically about being a human in the loop to validate and question output.

Agentic AI

Agentic AI further complicates the picture. With autonomous agents there is the risk that they can take action that isn't governed. These systems introduce autonomy, interdependence between agents, and possibly emergent behavior. Think about what it will take to govern multiple or dozens of (or more) agents that can pursue goals and make independent decisions and act with other agents. This means governance must account not only for model performance but for system behavior.

Because agents act autonomously, a single misconfigured or misaligned agent can propagate errors at scale. In multiagent environments, one faulty output can trigger many downstream mistakes. Other risks include tool misuse, conflicting goals among agents (and other interoperability issues), and operational opacity, when teams cannot easily determine which agent took which action or why. Governance must extend to agent registration, version control, permissioning, and simulation testing before deployment. Likewise, accountability may also blur. If an agent takes a dangerous action, who is responsible? There are, of course, cybersecurity risks as agents pose a new attack surface.

Controls will be needed that continuously validate agent objectives and identify unintended outcomes. Accountability will be critical as well as access and lineage controls to explain how the agent made a decision. There will need to be "red lines" for agent behavior. Every agent and agent action will need to be auditable and reversible. In other words, agentic AI governance will need to include continuous monitoring to manage these applications and ecosystems, again along the lines of risk tolerance.

Table 20.1 outlines a number of control areas that extend Figure 20.1 to also account for GenAI and agentic AI. On the GenAI side, there are controls over data provenance and input and outputs for GenAI models, and controls for ensuring that content is grounded in fact. On the agentic side, there are controls over agent

Table 20.1: Control Areas for GenAI and Agentic AI

CONTROL AREA	PURPOSE / GOVERNANCE OBJECTIVE	KEY ACTIONS AND SAFEGUARDS	WHEN APPLIED IN THE LIFE CYCLE
Data Provenance and Consent Validation	Ensure training or fine-tuning data is lawful, licensed, and representative.	Verify dataset licensing, consent, and lineage; record data-use approvals; log external data ingestion for audit.	Model Preparation/ Building
Prompt and Input Governance	Prevent prompt injection, data leakage, and policy violations.	Apply input filters, regex (regular expression) rules, and classifiers (e.g., for sensitive data, malicious commands); block unapproved prompt types; record and review prompt logs.	Deployment/ Monitoring
Output Filtering and Moderation	Limit toxic, biased, or policy-violating responses.	Use moderation APIs (e.g., OpenAI Moderation, Perspective API) to block unsafe content; add post-generation filters; trigger human review when thresholds exceeded.	Deployment/ Monitoring
Fact-grounding and Hallucination Detection	Ensure factual accuracy and prevent fabrication.	Employ retrieval-augmented generation (RAG); integrate fact-checking APIs to ensure model based on authenticated information; track hallucination metrics; set accuracy thresholds.	Building/ Deployment/ Monitoring

(continues)

Table 20.1: (Continued)

CONTROL AREA	PURPOSE / GOVERNANCE OBJECTIVE	KEY ACTIONS AND SAFEGUARDS	WHEN APPLIED IN THE LIFE CYCLE
Content Accountability and Human Oversight	Define ownership and approval for generated or agent-initiated content.	Require human sign-off for external-facing or high-risk outputs; document approvals; capture reviewer identity in audit logs.	Deployment/ Monitoring
Agent Registry and Lineage Tracking	Maintain traceability of every autonomous or semi-autonomous agent.	Establish a registry that records agent ID, creator, version, training sources, deployed environment, and modification history; link each agent to responsible owner.	Preparation/ Deployment/ Monitoring
Agent Access and Permission Control	Limit agent capabilities to authorized data, APIs, and actions.	Implement role- or attribute-based access control; define scopes of operation; require authentication tokens; restrict write/execution privileges; review permissions periodically.	Deployment/ Monitoring
Agent Communication Governance	Regulate and monitor interactions among agents and with external systems.	Define approved communication protocols; log inter-agent messages and external calls; inspect content for policy violations; prevent information leakage or collusion.	Deployment/ Monitoring
Behavioral Testing and Simulation	Identify emergent or unsafe agent behaviors before release.	Conduct sandbox and "red-team" simulations; stress-test collaborative or recursive decision-making; validate adherence to ethical and operational boundaries.	Validation/ Deployment

(continues)

Table 20.1: (Continued)

CONTROL AREA	PURPOSE / GOVERNANCE OBJECTIVE	KEY ACTIONS AND SAFEGUARDS	WHEN APPLIED IN THE LIFE CYCLE
Autonomy and Action Boundaries	Prevent agents from executing irreversible or harmful actions.	Require confirmation for critical actions; enforce guardrails via policy engines; monitor for boundary breaches; revoke credentials if violations occur.	Deployment/ Monitoring
Auditability of Prompts, Outputs, and Actions	Provide end-to-end traceability for investigations and compliance.	Log every prompt, output, API call, and agent action; maintain linkage to model version and user; preserve records per retention policy.	Deployment/ Monitoring
Dynamic Risk Monitoring and Escalation	Continuously assess operational, ethical, and behavioral risk across agents.	Track KPIs (toxicity, hallucination rate, drift, anomalous agent activity); auto-alert risk and compliance teams; integrate with enterprise risk register.	Monitoring/Audit

This nonexhaustive control framework interprets and extends the governance principles from major AI standards (NIST, ISO/IEC 42001, EU AI Act, OECD) to the operational realities of generative and agentic AI, supplemented by and safety guidelines from frontier AI developers (e.g., OpenAI, Anthropic, Google DeepMind).

identity, lineage, communication, and permitted actions. There are registries that track where and how agents operate, continuous behavioral auditing to detect unintended outcomes, and policy frameworks that enforce boundaries on autonomy. In some ways, agentic AI transforms governance from the management of algorithms to the stewardship of digital actors (the agents), which requires a tighter integration of risk, ethics, and oversight across the enterprise.

As AI governance matures, organizations will need measurable indicators of effectiveness. Metrics might include the percentage of AI models inventoried and risk-assessed, the frequency of model validation and drift monitoring, or the number of governance incidents detected and resolved. Some organizations are adopting maturity models that rate governance capability across dimensions such as policy, accountability, documentation, and continuous improvement. Tracking these measures helps leaders demonstrate progress and sustain executive commitment.

Laws, Frameworks, and Standards for AI Governance

Just as we discussed regulations and frameworks for data governance, there are also frameworks, laws/regulations, and standards for AI governance. Regulations carry penalties for noncompliance. Frameworks are influential policy or best-practice guides that help organizations interpret and implement governance principles. Implementing a framework shows that your organization has done its due diligence. Some of these AI governance frameworks are responsible AI focused. Standards are specifications (technical or management focused) that are developed through consensus (often through standards bodies) that define how to comply with governance requirements in a verifiable way. They can support certification, allowing an organization to demonstrate compliance or maturity. Let's look at each.

Laws/Regulations

Organizations should become familiar with a number of frameworks and regulations, based on where they do business and the kind of business that they do. The ones I hear about the most include the following:

- **EU AI Act:** We discussed the EU AI Act in the previous chapter. Under the EU Artificial Intelligence Act, high-risk AI systems are those that have a significant impact on people's rights, safety, or access to critical services. Annex III lists these categories, which include AI used in education, employment, credit scoring, law enforcement, healthcare, and critical infrastructure, among others. For these systems, the Act mandates a comprehensive set of governance controls. Providers must establish a formal risk management system (Article 9) for high-risk systems and implement human oversight (Article 14), supported by requirements for technical documentation, record-keeping, and transparency (Articles 11–13). They must also ensure accuracy, robustness, and cybersecurity (Article 15) and adhere to strict data and data-governance provisions (Article 10). These obligations cover risk identification, documentation, monitoring, and corrective actions in order to ensure that high-risk AI systems remain safe, traceable, and accountable once deployed.

  In terms of GenAI, most of the obligations fall on the provider. However, enterprises using those models are not exempt from responsibility especially when they adapt or deploy them in regulated or high-impact applications.

- **Italy's Law 132/2025:** Italy's own AI Act become effective in October 2025. Law No. 132/2025, complements the EU AI Act by introducing additional national governance, transparency, and accountability requirements. It establishes a national AI coordination committee to oversee implementation and mandates that employers inform workers when AI is used in employment decisions. It also introduces criminal penalties for the malicious use of AI, such as creating harmful deepfakes, and extends special protections for minors, requiring parental consent for children under 14 to access AI systems. In effect, Italy's law reinforces the governance principles of the EU AI Act but goes further in regulating how AI is deployed and communicated within workplaces and society. Expect further decrees in 2026 including an "organic framework" for data, algorithms, and AI-training methods, including rights, obligations, remedies, and sanctions.[4]

- **Digital Operations Resilience Act (DORA):** Although the regulation never uses the term "artificial intelligence," its operational risk and ICT governance provisions clearly include AI systems used in financial decision-making, trading, fraud detection, or cybersecurity. It requires financial institutions to treat AI systems like any other critical ICT component. For example, Article 9 (and articles 5–10, in general) states that institutions must identify, monitor, and control risks from all digital systems; this would include algorithmic and automated decision systems.[5] This creates an obligation for model oversight, validation, and documentation, aligning closely with AI governance.
- **California Privacy Rights Act (CPRA):** The CPRA passed in November 2020 and significantly amended and expanded the original CCPA (California Consumer Privacy Act). As part of this, California civil code § 1798.185(a)(16) contains specific statutory language that mandates the creation of rules for automated decision-making, which can include decisions made by AI. This regulation gives consumers the right to access information about how a business is using automated decision-making. Penalties are those discussed in Chapter 19.
- **The EU's General Data Protection Regulation (GDPR):** Article 15 under the act states, "The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her."[6] Applications such as automated credit scoring,

[4] Lovells, H. (2025). Italy's AI Law: the good, the bad … and the actual substance. *Hogan Lovells Insights*, 21 October. Available at: `https://www.hoganlovells.com/en/publications/italys-ai-law-the-good-the-badand-the-actual-substance` [Accessed 11 November 2025].
[5] Digital Operational Resilience Act. (n.d.). Article 9. Available at: `https://www.digital-operational-resilience-act.com/Article_9.html` [Accessed 11 November 2025].
[6] GDPR-info.eu. (n.d.). Article 22 GDPR—Automated individual decision-making, including profiling. *GDPR-info.eu*. Available at: `https://gdpr-info.eu/art-22-gdpr/` [Accessed 9 November 2025].

identifying job applicants, or determining social benefits would fall under this article.

- **China's laws and regulations:** China has put in place several laws and regulations. For instance, in 2023 China implemented Generative AI Measures which were the world's first binding regulations focused specifically on GenAI. They require providers to register their models and algorithms with the Cyberspace Administration of China, conduct security and risk assessments, and ensure that training data and outputs comply with existing cybersecurity, data, and content laws. Providers are legally responsible for generated content, which must align with "core socialist values," be free of false or harmful material, and include watermarks for traceability.[7] China has implemented other laws as well.
- **South Korea's AI Act:** South Korea passed the AI Basic Law known as the South Korean AI Act (SKAIA), which went into effect in January 2026. It establishes a national governance framework that creates a National AI Committee and AI Safety Research Institute, introduces support mechanisms for AI innovation and development, and implements safeguards to ensure the safety, reliability, and oversight of high-risk and GenAI systems.[8]
- **Other laws and those under development:** As of this writing, Canada has yet to pass its Artificial Intelligence and Data Act (AIDA, Bill C-27), which ensures the development of responsible AI and is aligned with the EU AI Act and frameworks such as those put forth by NIST and OECD (described below). In the United States, there is not yet one comprehensive, binding "AI-Act" style regulation for private sector AI governance as in the EU, but there is a growing legislative and policy momentum toward imposing governance requirements. There are more laws coming into effect in the United States for AI governance. For instance, Colorado SB 24-205 (Colorado Consumer Protections for Artificial Intelligence Act, enacted 2024) went into effect in February 2026 and resembles the EU AI Act. It requires a developer of a high-risk artificial intelligence system (high-risk system) to use reasonable care to protect consumers from any known or reasonably foreseeable risks of algorithmic discrimination in the

[7] Cyberspace Administration of China. (2023). *Interim measures for the management of generative artificial intelligence services*, translated by Stanford DigiChina and CSET. Available at: `https://digichina.stanford.edu/work/translation-measures-for-the-management-of-generative-artificial-intelligence-services-draft-for-comment-april-2023/` [Accessed 9 November 2025].
[8] ArtificialIntelligenceAct.com. (n.d.). South Korean AI basic law. *ArtificialIntelligenceAct.com*. Available at: `https://artificialintelligenceact.com/south-korean-ai-basic-law/` [Accessed 11 November 2025].

high-risk system.[9] Connecticut—SB 1103 (Public Act 23-98, enacted 2023) established an office of AI.[10] The Virginia Consumer Data Protection Act (CDPA) is modeled after GDPR in terms of automated decision-making.

Frameworks

There are a number of frameworks that organizations use to help guide their AI governance activities. Some of these are responsible AI focused; others are broader. I've seen organizations take a number of these frameworks and adapt them to their own framework that makes sense for their own organization. Some popular frameworks include the following:

- **NIST AI RMF (2023):** The NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0), published in January 2023, serves as one of the most comprehensive voluntary frameworks for AI governance. It provides a structured approach for organizations to manage risks across the AI life cycle, i.e., design, development, deployment, and use, by embedding governance principles within risk management processes. It states that characteristics of trustworthy AI systems include valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed.[11] The framework is organized around four core functions: Map, Measure, Manage, and Govern. The Govern function is central to AI governance, emphasizing organizational policies, accountability structures, and continuous monitoring to ensure AI systems remain trustworthy and aligned with intended purposes. It requires leadership commitment to define governance roles, establish oversight mechanisms, and foster a culture of risk awareness, transparency, and documentation.

  Beyond the Govern function, the framework integrates governance concepts into the other three areas, making risk management a continuous and organizational responsibility. For example, the "Map function" section 4.1 (page 27) ensures that internal risk controls for components of the AI system, including third-party AI technologies, are identified and

[9] Colorado General Assembly. (2024). *Senate Bill 24-205: Consumer Protections for Artificial Intelligence*, 2024 Regular Session. Available at: `https://leg.colorado.gov/bills/sb24-205` [Accessed 11 November 2025].
[10] Connecticut General Assembly. (2023). *Senate Bill SB01103 bill status*. Available at: `https://www.cga.ct.gov/asp/cgabillstatus/cgabillstatus.asp?selBillType=Bill&bill_num=SB01103&which_year=2023` [Accessed 11 November 2025].
[11] National Institute of Standards and Technology. (2023). *AI Risk Management Framework (AI RMF 1.0)*, p. 12. Available at: `https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf` [Accessed 11 November 2025].

documented. "Managing AI Risks" calls for establishing procedures for incident response and continuous performance monitoring, while the section on "Measuring AI Risks" underscores the need for measure risks against the map function. The AI RMF also introduces the concept of AI risk profiles to help organizations tailor governance to system impact and context.

Recognizing the unique risks of generative models, NIST launched the Generative AI Public Working Group and the Generative AI Risk Management Consortium in 2024. These efforts are producing addenda to the RMF, focused on issues such as content authenticity, hallucinations and misinformation, confabulation, toxicity, and more.[12]

- **Singapore Model AI Governance Framework (Second Edition, 2020):** This framework is built on two guiding principles: that AI decision-making should be explainable, transparent, and fair, and that AI systems should be human-centric, augmenting rather than replacing human judgment while protecting individuals' well-being and safety.[13] The framework translates these ethical principles into concrete governance practices that can be integrated into existing corporate and regulatory structures, making it particularly valuable for organizations seeking to operationalize responsible AI.

  The framework is organized around four core areas. The first emphasizes internal governance structures and measures, encouraging organizations to assign clear roles and responsibilities and embed ethical review into their compliance and risk systems. The second focuses on human involvement in AI-augmented decision-making, requiring organizations to assess potential harms and determine the appropriate level of human oversight—whether "in," "over," or "out" of the loop. The third addresses operations management, including responsible data handling, model development, bias mitigation, robustness, and explainability. Finally, the framework highlights stakeholder interaction and communication, promoting transparency through disclosures, accessible explanations, feedback mechanisms, and avenues for contesting or appealing AI-driven outcomes.

---

[12] National Institute of Standards and Technology. (2024). *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*, NIST.AI.600-1. Available at: `https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf` [Accessed 11 November 2025].

[13] Personal Data Protection Commission. (n.d.). *Model Artificial Intelligence Governance Framework.* 2nd ed. Available at: `https://www.pdpc.gov.sg/-/media/files/pdpc/pdf-files/resource-for-organisation/ai/sgmodelaigovframework2.pdf` [Accessed 11 November 2025].

In 2024, Singapore released the AI Governance Framework for Generative AI, which builds on this earlier framework. Core governance areas include accountability, transparency, data integrity, testing and assurance, content provenance, and human-in-the-loop mechanisms.

- **OECD Principles on AI:** These were first adopted in 2019 and updated in 2024. They consist of five value-based principles and five recommendations that provide guidance for policy makers and advisors. The five principles include transparency and explainability, accountability, robustness/security/safety, inclusive growth and the development of sustainable well-being, and human rights and democratic policies including fairness and privacy. The OECD AI Principles promote use of AI that is innovative and trustworthy and that respects human rights and democratic values.[14]
- **Big 4 Frameworks:** Big 4 Accounting firms have also released their own frameworks. For instance, the E&Y Responsible Framework includes nine principles including fairness, transparency, security, compliance, sustainability, data protection, accountability, explainability, and reliability.[15] The Deloitte Framework for Trustworthy AI includes seven core principles including safe and secure, private, transparent and explainable, fair and impartial, responsible, accountable, robust, and reliable.[16]

RESPONSIBLE AI ELEMENTS

While there are numerous frameworks for responsible AI, they often share many of these core principles:

- **Safety: Preventing misuse or harm from AI systems to humans, society, and the environment.**
- **Accountability: Ensuring a person has clear responsibility for the outcomes and impacts of AI systems.**

(continues)

[14] Organisation for Economic Co-operation and Development. (n.d.). *OECD AI principles*. Available at: `https://www.oecd.org/en/topics/sub-issues/ai-principles.html` [Accessed 11 November 2025].
[15] Ernst & Young Global. (2024). *Responsible AI principles*. EY GL Responsible AI Principles, September 2024. Available at: `https://www.ey.com/content/dam/ey-unified-site/ey-com/en-gl/insights/ai/documents/ey-gl-responsible-ai-principles-09-2024.pdf` [Accessed 11 November 2025].
[16] Deloitte. (n.d.) *Trustworthy AI*. Deloitte. Available at: `https://www.deloitte.com/nl/en/issues/generative-ai/trustworthy-ai.html` [Accessed 11 November 2025].

RESPONSIBLE AI ELEMENTS (CONTINUED)

- **Fairness:** The development and deployment of AI systems in a way that avoids bias and discrimination, ensuring equitable treatment and outcomes for all individuals and groups.
- **Explainability:** The requirement that average humans are able to easily review and understand why AI models and systems behave as they do
- **Sustainability:** AI systems act in accordance with business sustainability practices.
- **Reliability:** AI systems should act as they were designed. This includes consistency, dependability, and accuracy.
- **Privacy:** Safeguarding an individual's personal data and ensuring their information is used ethically, transparently, and with consent where required.
- **Respect of human rights:** This ensures that AI system development and use uphold fundamental human rights such as dignity, equality, and privacy.

Standards

One of the most popular standards for AI is relatively new: ISO/IEC 42001. ISO/IEC 42001:2024, the Artificial Intelligence Management System Standard, is the first international standard that provides a certifiable framework for governing the responsible development, deployment, and use of AI. Developed by the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC), it establishes a management system approach, similar to ISO 9001 (quality) or ISO 27001 (information security), to ensure that AI systems are developed and operated in a controlled, transparent, and accountable manner.

The standard covers requirements for AI governance, risk management, human oversight, transparency, and continual improvement. It defines how organizations should document AI use cases, assess and mitigate risks, and monitor system behavior over time to maintain reliability and trustworthiness. ISO 42001 also integrates with other AI-related standards, such as ISO/IEC 23894 (AI risk management) and ISO/IEC 38507 (governance of IT and AI), creating a coherent structure for operationalizing responsible AI. Importantly, it enables organizations to seek formal certification, which demonstrates to regulators, customers, and partners that they have implemented a robust AI management system.

Taken together, these laws, frameworks, and standards form the foundation of modern AI governance. Regulations define the minimum obligations and penalties for noncompliance, frameworks provide the guiding principles and structures that organizations can adapt to their own risk posture, and standards establish the verifiable mechanisms for certification and assurance. While the global landscape remains fragmented, the underlying principles are converging around the same themes: transparency, accountability, human oversight, and continuous risk management. Together, they represent an emerging global consensus that trustworthy AI requires sound data and technology, as well as sustained organizational commitment to governance across the entire AI life cycle.

New Roles and Responsibilities

If this sounds like a lot, it is. AI governance is essential and should not be overlooked. Given this, it will require new roles that are distinct from data governance, though they must work closely together. Organizations are formalizing AI oversight across different layers of responsibility:

- **Strategic leadership and governance:** Roles such as chief AI officer, director of AI Governance, or vice president of AI Risk Management lead enterprise-wide strategy, chair governance councils, and ensure alignment with regulations like the EU AI Act, NIST AI RMF, and ISO 42001. They define frameworks for risk assessment, bias management, and compliance, often reporting to the board or executive risk committees. Many organizations are also introducing a chief responsible AI or AI ethics officer, who ensures that AI development and deployment adhere to ethical principles and societal values. In addition, a formal AI Governance Council or Committee is increasingly being established—composed of leaders from Legal, Risk, Compliance, IT, and HR to oversee high-risk AI applications and align governance practices enterprise-wide.
- **Operational governance and compliance:** Roles such as AI governance manager, responsible AI lead, and senior risk officer implement frameworks, maintain AI model inventories, and conduct ethical or risk reviews for high-impact systems. They coordinate with legal, compliance, privacy, and security teams to operationalize AI policies and embed controls in model development and deployment. New specialized roles are emerging in this space, including AI risk and compliance officers and AI policy managers, who focus on regulatory monitoring and impact assessments. In regulated sectors, organizations are also appointing model risk management (MRM) leads or validation officers to evaluate models for fairness, robustness, and explainability before release, and AI audit and assurance leads to verify adherence to governance standards such as ISO 42001.
- **Technical and assurance functions:** Roles such as responsible AI engineers, AI auditors, and technical program managers evaluate model explainability,

fairness, robustness, and safety. They design tools for bias testing, content moderation, and continuous monitoring—bridging governance principles with the AI life cycle. As AI systems become more complex, additional roles are being added. The AI governance engineer or AI platform steward embeds governance directly into MLOps and LLMOps pipelines through automated testing, traceability, and model documentation. A newer specialization, the generative or agentic AI risk lead, focuses on managing hallucination risks, prompt governance, and behavioral oversight for autonomous and multiagent systems.

Clear accountability and escalation are essential. Governance frameworks should define who approves new AI deployments, who owns their outcomes, and how issues are escalated when risks are identified. Many organizations are forming AI Risk Committees or embedding AI oversight into existing Enterprise Risk Councils so that unresolved issues can reach the right level of authority quickly. Cross-functional review boards integrating IT, ethics, legal, and business leadership are becoming more popular. This structure ensures that governance decisions are transparent and defensible, reinforcing trust both inside and outside the organization.

Across these roles, several competencies are also needed. These include fluency with the regulations, frameworks, and standards mentioned earlier, along with the ability to communicate effectively and collaborate across teams, including data, analytics, compliance, and IT. These competencies include the following:

- **AI governance and regulatory fluency:** Deep understanding of laws, frameworks, and standards such as the EU AI Act, GDPR, CPRA, NIST AI RMF, and ISO 42001, and the ability to translate them into internal policies and controls.
- **Risk management and assurance:** Skill in developing AI risk taxonomies, model validation protocols, and ethical review processes. Many job postings emphasize experience in financial services or other regulated sectors, where model risk management (MRM) and "second-line-of-defense" oversight are already established disciplines.
- **Technical literacy:** Familiarity with the machine learning life cycle, bias mitigation techniques, explainability tools (e.g., SHAP, LIME), and MLOps/LLMOps practices—enabling governance professionals to engage effectively with data science and engineering teams.
- **Cross-functional leadership and communication:** The ability to coordinate among technical, legal, compliance, and product teams, and to articulate governance implications clearly to executives.

- **Ethical reasoning and human-centered design:** Applying fairness, accountability, transparency, and human oversight principles in both policy and implementation.
- **Audit and assurance expertise:** Understanding of ISO and NIST audit frameworks, evidence collection, and control validation to demonstrate conformance.
- **Change management and culture building:** Skills to promote responsible AI practices across the organization through training, awareness, and stakeholder engagement.

As AI capabilities advance, especially with the rise of generative and agentic systems, organizations are expanding these roles to include AI audit leads, governance engineers, platform stewards, and GenAI risk specialists. Together, these roles ensure that governance is not just a compliance exercise but an embedded, continuous discipline that spans leadership, risk management, and technology.

Integration with Enterprise Risk and Compliance

We've been talking about AI governance, compliance, and the risks associated with AI, but how does this align with enterprise risk and compliance (ERC)? Traditionally, the risk management function has sat within specific departments such as IT, Finance, or Operations, or as a centralized ERC team that reports to the chief risk officer (CRO), chief financial officer (CFO), or directly to the board of directors.

In 2013, the Institute of Internal Auditors introduced the Three Lines of Defense model, which was updated in 2020 as the Three Lines Model[17] to clarify how risk ownership, oversight, and assurance should be structured across an organization. The first line, management, operates the business and directly manages risk, ensuring that activities align with objectives and compliance requirements. This includes both frontline and support functions responsible for designing and maintaining effective controls. In the AI context, this includes AI developers, data scientists, and product owners. The second line, specialist risk and compliance, provides expertise, oversight, and challenge on risk-related matters such as ethics, compliance, security, and quality assurance. It supports and monitors the first line while helping to improve risk management practices. For AI, this often includes the AI risk, compliance, and ethics teams. The third line,

[17] The Institute of Internal Auditors. (2020). *The IIA's Three Lines Model: An update of the Three Lines of Defence*. Available at: `https://www.theiia.org/globalassets/site/content/articles/global-knowledge-brief/2020/july/the-iias-three-lines-model/glob-three-lines-model-paper_layout-rebuild.pdf` [Accessed 11 November 2025].

internal audit, provides independent and objective assurance on the adequacy and effectiveness of governance, risk management, and controls, reporting directly to the governing body.

AI governance extends the traditional risk management framework by adding a new layer of algorithmic risk to the enterprise risk profile. ERC functions historically focused on financial, operational, and compliance risk. AI introduces new exposures such as bias and discrimination, lack of transparency, model drift, and misuse of GenAI. In many organizations, AI risk now falls under operational or technology risk, with joint ownership between ERC and AI governance teams.

For example, when an organization deploys an AI system that impacts customers, such as one used for credit scoring or automated hiring, the AI governance team conducts the model impact and bias assessments, the compliance function ensures those processes meet legal and regulatory requirements, and the risk function integrates the findings into the enterprise's overall risk profile. In this way, AI governance does not replace ERC. It strengthens it, ensuring that emerging AI risks are identified, controlled, and reported with the same rigor as any other enterprise risk.

Ultimately, effective AI governance depends as much on culture as on controls. Leaders need to foster a mindset of accountability and ethical awareness across all functions that design, deploy, or rely on AI. Training programs, open discussions about risk, and clear expectations for human oversight help embed governance principles into daily practice. When employees understand not just *what* the rules are but *why* they exist, governance becomes an enabler rather than a compliance burden.

Audit Considerations

We talked about the fact that AI governance is needed to meet internal audit. It is obviously also needed to address external audit considerations. In fact, external auditors can also be considered the third line in the Three Lines Model.

External auditors play a critical role in providing independent assurance that an organization's AI governance framework is effective and that risks associated with AI are being properly managed. Their focus is on verifying that the organization has established clear oversight structures, policies, and internal controls governing AI use. This includes assessing how AI risk is integrated into enterprise risk management, whether key risks such as bias, explainability, data integrity, and model drift have been identified, and whether the controls in place are designed to ensure compliance with relevant laws and standards such as the EU AI Act, NIST AI RMF, and ISO 42001.

Auditors also evaluate the effectiveness of internal controls over AI systems, including data governance, model validation, change management, and ongoing monitoring. When AI influences financial reporting or regulatory compliance, these controls become part of the broader audit of internal control over financial and operational systems. Increasingly, auditors are also asked to provide third-party assurance on AI governance to confirm the soundness and transparency of governance processes. While they do not design or manage AI systems, external auditors serve as independent evaluators of trust, helping boards and stakeholders confirm that AI is being governed responsibly, ethically, and with appropriate accountability.

External auditors are also making use of AI, a point that is brought up in the following Expert Advice about auditing AI from Helen Brown-Liburd, PhD, CPA Associate Professor of Accounting at Rutgers Business School.

EXPERT ADVICE: AUDITING AI

Helen Brown-Liburd, PhD, CPA, and associate professor of accounting at Rutgers Business School has spent her career at the intersection of auditing, accounting, and technology. Before entering academia, she spent more than 15 years as an internal and external auditor, experience that now shapes her research and teaching on how emerging technologies are reshaping assurance practices. As organizations begin to embed AI into financial systems, she sees both the urgency and the uncertainty of how auditors should respond. "In auditing, the profession remains largely reactive to emerging technologies," she said. "Firms and standard setters are making progress however the pace of innovation means they are still catching up—there are no definitive standards yet for auditing AI systems."

Although major regulatory and professional bodies have begun to issue guidance, most of it remains incremental. The Public Company Accounting Oversight Board (PCAOB) has recently amended two standards to address the use of advanced technologies in audits, specifically evidence and risk-assessment standards. "These updates tell auditors that when they use technology-assisted analyses, they must evaluate general controls, automated application controls, and the integrity and completeness of the data," Brown-Liburd explained. "It's an amendment to existing guidance rather than a new framework. There is still no explicit guidance on providing assurance over AI systems—how auditors should gather sufficient, appropriate evidence for evaluating the integrity of processes and outputs that underpin the AI system."

Auditors, she said, remain ultimately responsible for the systems they rely on. "If you're relying on an ERP system, you need to understand its IT controls, and if you can't evaluate them yourself, you need experts who can. The same logic applies when

(continues)

EXPERT ADVICE: AUDITING AI (CONTINUED)

a client uses AI. Auditors must understand how the AI is being used, how algorithms are evaluated, and how clients ensure their systems comply with regulatory and privacy requirements." For many financial auditors, this may mean working closely with IT specialists or expanding training programs to develop AI literacy across audit teams.

Public accounting firms themselves are experimenting with AI, mostly in research and documentation. "They're using it to help auditors research standards, to create prompt libraries that bring consistency to how they query large databases of professional guidance," she said. "But firms are not transparent about how extensively they use AI or whether these tools are applied to engagements for publicly held companies, where the standards are stricter and the risk is higher."

When asked about what external auditors should be looking for as companies begin to use AI in financial processes, Brown-Liburd emphasized that their responsibility remains tied to financial reporting. "To the extent that AI is part of a company's financial reporting framework it shares a conceptual similarity with auditing complex ERP systems in that auditors must demonstrate the existence and effectiveness of controls that mitigate risks of material misstatement." However, Brown-Liburd stressed, this analogy has limits. "Unlike ERP systems, which are rule-based and relatively static, AI systems introduce fundamentally different risks—such as model opacity, continuous learning, and susceptibility to data bias—that challenge traditional audit approaches. These characteristics make it harder for auditors to understand how outputs are generated, assess reliability, and determine whether governance and monitoring controls are sufficient," she noted. As a result, auditing AI requires not only evaluating the integrity of processes and outputs that underpin the system but also adapting audit methodologies to address dynamic behavior and algorithmic complexity.

Brown-Liburd added that auditing AI systems will require continuous oversight, not periodic checks. "AI models change," she said. "You need to understand the design, but you also have to evaluate them on an ongoing basis. Were there any modifications to how the system was originally developed? How are companies monitoring the information coming out of it?" She sees parallels with auditing complex accounting estimates, where auditors must understand the inputs and assumptions that produce a number. "You have to ask, why did you use these inputs and not others? The same logic applies to auditing AI."

Internal auditors, she noted, are in a different position. "They have access and can be involved much earlier in the process," she said. "They might participate in the development of systems or advise management on the controls that will be needed. They can't design and audit the same system, of course, but they can play an advisory role early on." For that reason, she believes companies would benefit from including internal audit functions in AI development and governance discussions from the outset.

(continues)

EXPERT ADVICE: AUDITING AI (CONTINUED)

Brown-Liburd also pointed to the importance of explainable AI as a mechanism to support auditing. "Continuous monitoring and explainability go hand in hand," she said. "If I can see how you got from step A to step C, I can verify it. That's what auditors need, the ability to trace the reasoning and understand why an output looks the way it does."

While technology is moving faster than regulation, she remains pragmatic. "Auditors are not trying to slow innovation," she said. "They just need to know enough to verify, to ensure integrity, completeness, and compliance. That's the essence of the profession."

For leaders, then, AI governance is more than a safeguard; it is the system for responsible innovation. It requires sustained attention, clear accountability, and collaboration across technical, legal, and business teams. Organizations that treat governance as an enabler rather than an obstacle will move faster with less risk, gain the trust of regulators, partners, and customers. In the coming years, the most successful enterprises will be those that make governance a core competency in terms of how they design, deploy, and manage AI.

Executive sponsorship is key to embedding AI governance enterprise-wide. Boards and senior leaders should regularly review AI risk reports, allocate funding for governance infrastructure, and ensure that accountability for AI outcomes is explicit in leadership performance objectives. Without sustained top-level commitment, governance remains policies written on paper rather than practice in action.

What's Next?

At this point, we've gone through all of the considerations for AI readiness: the organizational and cultural/change considerations, the solid data foundation and platforms, the AI tools and frameworks, the skills, and roles needed for AI, and both data and AI governance. In the next (and last) chapter, we tie it all together.

Part

VI

Putting It All Together

CHAPTER

21

Tying It All Together

In this chapter, we will examine the following:

- The minimum viable checklist
- A comprehensive checklist
- Conclusion

We've covered a lot of ground in this book, but none of it stands alone. AI succeeds when all the pieces: strategy, data, skills, operations, and governance come together in the right way. In this chapter I'll tie the major themes together to give you an overall picture of what it takes to build an AI-ready enterprise.

As you look back across the chapters in this book, I hope a clear narrative emerges. It starts with the fact that AI does not succeed over the long haul because organizations install new tools or implement isolated projects. Yes, that can work for a little while as organizations gain efficiency benefits. For long-term success, AI succeeds when the enterprise is prepared for it. That preparation shows up in the strategy, the data, the architecture, the processes, the skills, the governance, and the culture. Every chapter, in one way or another, has shown that AI magnifies whatever operating conditions already exist inside an organization. Strong foundations accelerate momentum. Weak or inconsistent foundations can slow progress, create friction, and can even amplify problems. AI can expose the gaps quickly and, as we have seen, sometimes unforgivingly.

From the beginning of the book, I emphasized that strategy comes first. Not a vague "let's use AI," but a clear articulation of the problems worth solving and the outcomes worth improving. Where does your organization need AI to help? When is AI not a good idea? This strategy needs to be concrete.

Throughout the book, experts echoed this point, explaining that successful AI efforts begin with strong leadership sponsorship and a well-defined vision and need rather than experimentation for its own sake.

My goal for the book was to both explain what is involved in building out AI, how it works, as well as what it takes to do it right. The AI Readiness Model in Chapter 3 set the stage for the path forward and the five dimensions of AI readiness. Organizational readiness determines whether leaders align, sponsor, and sustain the work. Data readiness determines whether AI has anything reliable to operate on. Skills and tools readiness determines whether teams can build and manage the systems responsibly. Operational readiness determines whether those systems can function in real environments, under real constraints. Governance readiness determines whether AI can be deployed safely and ethically. These are not theoretical constructs. They are the conditions that consistently show up as predictors of AI success or failure.

A solid data foundation is critical for AI success. The foundation can grow in phases, but it needs to be there. AI will add pressure to the data foundation. Modern workloads require data that is connected, interpretable, high quality, and governed. Modern workloads utilize diverse data, and this is especially true with the popularity of GenAI. Trustworthy AI depends on knowing where the data came from and how it has changed. Modern architectures including the lakehouse and the data fabric exist because the traditional ways of managing data no longer fit the scale or complexity of what organizations need to do today. In other words, there is no path toward enterprise AI without a strong data foundation. This was echoed in the expert advices. Experts spoke about the dangers of “vanity projects,” particularly with GenAI, when foundational data issues were ignored. Others described how their companies moved faster specifically because they had already invested in governance, semantic layers, lineage, and observability.

Another important component for AI success is skills and expectations. This is not just about hiring a few specialists. It is about raising the overall literacy of the enterprise so people understand how AI works, when it works, and when it should not be trusted. Technical roles evolve as organizations move from traditional analytics to machine learning to generative and agentic AI. At the same time, the business must mature. Business users need to validate outputs, interpret insight, and collaborate with AI systems. Leadership must understand risk, oversight, and the practical implications of using AI in decision-making. AI does not remove the need for human judgment. In some ways, it actually raises the stakes around exercising human judgment.

Operational readiness is another key for success. Building a model is often not the hard part. Running it is. Deploying, integrating, monitoring, and updating AI systems require discipline and maturity. A model in a lab environment is one thing. A model running in production, under real workloads, repeatedly delivering value while being monitored for drift, bias, hallucinations, and performance degradation is another. Many organizations underestimate this part of the work. They celebrate the prototype and ignore the operational burden. But the organizations that scale AI treat operationalization rigorously, with pipelines, observability, life cycle management, and well-defined human oversight. This applies to predictive models, generative systems, and to agentic AI.

Governance sits underneath and surrounds everything discussed in the book. Governance cannot be an afterthought or a compliance obligation; it is how organizations create trust. It is an enabler; in fact, some organizations are moving away from the governance term to call it enablement. Data governance and AI governance are different because they solve different problems, although they complement each other. And AI governance introduces risks that organizations have never confronted before. Generative and agentic AI raise the bar for transparency, provenance, documentation, oversight, and risk management. AI can produce incorrect content, biased outputs, hallucinations, or overly confident answers. Agentic systems can take actions. These capabilities require governance structures that keep the organization safe and accountable. Without governance, AI simply creates risk at scale.

In the final chapters on generative and agentic AI, I tried to reinforce rather than contradict the earlier parts of the book. GenAI offers powerful new capabilities, but it is still dependent on data quality and governance. Without grounding, it is unreliable. Without oversight, it is risky. Retrieval augmented generation can be helpful because it can anchor models in enterprise context (although it is not the complete solution). However, agentic AI expands the space further by enabling systems that reason, plan, act, and learn, and this introduces new demands around permissions, autonomy boundaries, behavioral testing, and human accountability. These chapters reflect the next stage of enterprise AI that will ultimately require greater readiness, not less.

A thread runs consistently throughout the book. AI is as much about people as it is about technology. Change management, organizational psychology, communication, and incentives play a central role. People experience AI as change, disruption, and sometimes as loss. If an organization does not address the emotional, structural, and social aspects of adoption, resistance will undermine even the best AI systems. The organizations that succeed are the ones that take this seriously and invest the time to help people adapt.

Stepping back from the details, the overall message is that AI is not a project. It is not a tool. It is an enterprise capability that touches strategy, data, architecture, workflows, governance, skills, and culture. It reveals the strengths and weaknesses of the organization that adopts it. If those foundations are solid, AI can help a company become innovative and competitive and achieve a great deal of success. If they are not, AI can cost a lot and provide some efficiency gains but can hit that “value ceiling” and even accelerate negative consequences.

What comes next are a few practical checklists and an operating model that translate these insights into concrete guidance. Specifically, first I provide the minimum viable items your organization needs to address to succeed with AI. Then I'll provide a more comprehensive checklist.

The Minimum Viable AI Checklist

Before moving into the full operating model and comprehensive checklists, it is helpful to step back and look at the minimum areas an organization needs to succeed with AI. Not every company will start with mature governance, perfect data, or fully automated pipelines. But there is a baseline set of capabilities that every organization must have, regardless of whether they are deploying consumerized, off-the-shelf AI tools or building more advanced, enterprise-scale systems. These minimum elements determine whether AI will create value or create problems. They represent the baseline of responsible AI adoption.

READINESS AREA	MINIMUM REQUIREMENTS	WHY IT MATTERS
Strategy	A short list of clearly defined high-value AI use cases tied to business needs and outcomes	AI must be anchored in measurable goals. Ambiguity leads to potential wasted investment.
Leadership Alignment	Executive sponsor, agreement on priorities, funding commitment	Without alignment, AI can become fragmented or political.
Data Foundation	Reliable access to core data, basic quality controls, lineage awareness, metadata, and secure access	Poor data produces unreliable AI. Minimum quality is essential even for off-the-shelf tools.
Architecture and Platforms	Scalable environment, cloud or equivalent compute, pipelines, ability to work with unstructured data, vector store for RAG	AI workloads require modern infrastructure. Legacy systems can break under load.
MLOps and Monitoring	Versioning, deployment path, monitoring for performance and drift, and rollback capability	AI will degrade. Even small tools need oversight.
Governance	Basic policies for data access and AI usage, approval for sensitive use cases, guardrails for GenAI, human oversight	Prevents legal, ethical, and reputational risk.

(continues)

READINESS AREA	MINIMUM REQUIREMENTS	WHY IT MATTERS
AI Literacy	Core training so business users understand limitations, validation, and escalation procedures	Prevents blindly trusting and potential misuse of AI outputs.
Core Skills	Small team covering data engineering, ML/LLM basics, deployment, and privacy/risk	AI requires cross-functional expertise even when using vendor solutions. This can be a three- to five-person team for smaller organizations.
Change Management	Communication about goals, impact on roles, training for workflow changes	People resist when they do not understand what AI means for them. You may not need an entire change management program, but it is important to consider the changes AI will bring.
Safe Experimentation Environment	A secure sandbox with curated data and approved models or APIs	Prevents shadow AI and accidental exposure of sensitive data.

Following is a more comprehensive checklist to consider as your organization matures in its AI capabilities.

A Comprehensive Checklist

Organizations don't need to build every capability at once to succeed with AI, but they do need a clear understanding of the full landscape. This comprehensive checklist lays out the end-to-end elements that matter across strategy, data, governance, technology, and culture. Think of it as a journey roadmap: you start with the minimum viable foundations, then expand and mature each area as your ambitions, use cases, and risks grow. The checklist is intentionally comprehensive, so leaders can see what "good/mature" ultimately looks like, without implying that every box must be checked on day one.

Organizational Readiness Checklist

Strategy and Leadership

- Executive leadership understands AI's possibilities and limitations.
- AI strategy aligns with enterprise strategy, and it can be measured.

- Leadership accepts AI requires ongoing investment.
- A senior sponsor removes roadblocks and champions the work.
- Leaders understand why foundations matter before scaling generative or agentic AI.

Culture

- Culture supports cross-functional collaboration across business, data, analytics, and IT.
- The organization acknowledges that AI introduces uncertainty, loss, and new ways of working.
- People are encouraged to raise risks, challenge assumptions, and ask questions.
- Curiosity and experimentation coexist with accountability.

Change Management and Organizational Psychology

- Structured change frameworks are considered/used (ADKAR, Kotter).
- Leaders understand common resistance patterns ("don't get it," "don't like it").
- WIIFM ("What's in it for me?") is addressed for employees and teams.
- Stakeholders are engaged based on their position along the adoption curve.
- Success stories are used to build momentum and reduce fear.
- Leaders understand timing differences: the change leader has emotionally moved on before the rest of the organization.

Data and AI Literacy

- Organizational literacy programs exist beyond one-off training.
- Business users understand when they can trust AI outputs and when they cannot.
- Executives understand model limitations, failure modes, bias, drift, and hallucinations.

Data Readiness Checklist

Architecture and Infrastructure

- Architecture supports structured, semi-structured, and unstructured data.
- Architecture is designed with business and capabilities in mind.

- Lakehouse, warehouse, and lake architectures are coherent and aligned.
- Distributed and hybrid environments are intentional, not accidental.
- The data environment has been modernized to support AI workloads.

Pipelines and Integration

- Pipelines reliably move data from source to target.
- Teams address data quality at the source, not just downstream.
- Automation exists to handle increasing data volume and new sources.
- APIs and middleware support integrating AI into existing business systems.
- Orchestration and workflow automation tools are in place.

Data Quality

- Data quality dimensions are defined (accuracy, completeness, consistency, timeliness, validity).
- Checks occur both at certain control points (e.g., ingestion and before model training).
- Checks occur on diverse data types.
- Data quality issues are tracked with clear remediation paths.
- A data quality function or equivalent role exists.
- Teams recognize that "good decisions made on bad data are still bad decisions."

Metadata, Lineage, and Findability

- Metadata is captured consistently across systems.
- Lineage is documented from source → transformations → consumption.
- Data provenance is tracked, including for unstructured sources.
- Metadata quality is sufficient to support GenAI grounding.
- Data is findable, accessible, and not buried in silos.
- Data catalogs are in place.

Unstructured and Multimodal Data

- Unstructured data (documents, text, images) is governed and accessible.
- RAG and embedding pipelines are supported.
- Unstructured data are screened for toxicity, bias, and copyright concerns.

Data Products

- Data products have clear owners, SLAs, and life cycle plans.
- Data products solve real business problems, not just technical ones.
- Data products are documented, discoverable, and monitored.

Skills and Tools Readiness Checklist

Technical Skills

- Data engineers can build and maintain reliable pipelines.
- Data scientists can design, train, and validate robust ML models.
- ML engineers or developers can deploy models responsibly.
- Skills extend to open source.
- Teams understand key evaluation metrics, drift detection, and robustness testing.
- Teams understand foundational GenAI concepts (embeddings, RAG, grounding).
- Teams understand agentic AI workflows (perceive, reason, plan, act, learn).

Enterprise Literacy

- Business users can validate AI outputs and know when to escalate concerns.
- Executives understand limitations, failure modes, and risks associated with AI.
- Product owners understand how AI changes workflows and responsibilities.

Tools Readiness

- Versioning exists for models, datasets, prompts, and configurations.
- Model monitoring tools are available and used.
- Interpretability tools (e.g., SHAP, LIME, etc.) are available for governed environments.
- Data/feature catalog and lineage tools are used consistently.
- Pipelines, orchestration tools, and deployment tools are reliable and documented.

Generative and Agentic Tools

- A RAG stack or equivalent system is available, if needed.

- Guardrails exist for input filtering, output filtering, toxicity detection, and safe execution.
- Tools exist for testing, evaluating, and debugging agent behavior.
- Teams understand when to use RAG vs. fine-tuning and the implications of each.
- Teams take the time to understand agentic AI and the risks involved.

Training and Enablement

- Training is ongoing and tied directly to readiness gaps.
- Training includes governance expectations and responsible AI requirements.
- Training is role-based, not generic.

Operational Readiness Checklist

Model Life Cycle and Deployment

- A full life cycle is defined: preparation → build → validate → deploy → monitor → retrain/retire.
- Life cycle includes steps specific to generative and agentic AI.
- Pipelines for training, validation, and deployment are automated.
- Deployment is reproducible, version-controlled, and auditable.
- Monitoring covers model performance and data integrity.

Workflow Integration

- AI is integrated into business workflows, not isolated pilot systems.
- APIs enable AI to interact with existing tools and processes.
- Business processes can adapt as AI behavior changes.
- Human-in-the-loop or human-on-the-loop oversight is clearly defined.

Scalability

- Infrastructure can handle increases in data, interactions, and complexity.
- Scalability testing is performed for ML, GenAI, and agent workloads.
- Teams understand how operational load changes as AI adoption grows.

Monitoring and Incident Response

- Monitoring includes drift, accuracy changes, hallucinations, toxicity, and anomalous behavior.

- Alerts route to the right people with defined SLAs.
- An incident response plan exists for AI failure, misuse, or harm.
- Rollback and remediation processes are documented and tested.

Agentic Operational Controls

- Agent permissions are in place.
- All agent actions are logged and auditable.
- Autonomy boundaries are defined and enforced.
- Multiagent interactions are tested for failure modes and unsafe behaviors.

Governance Readiness Checklist

Data Governance

Structure and Roles

- Data owners, stewards, and custodians have clearly defined responsibilities.
- There is accountability.
- The governance function has authority, sponsorship, and resources.
- Governance is positioned as enablement as opposed to policing.

Policies and Standards

- Policies exist for data quality, lineage, retention, classification, and access.
- Policies are updated to address GenAI and unstructured data.
- Data governance integrates with privacy, compliance, and enterprise risk.

Data Intelligence and Transparency

- Metadata is documented and findable.
- Data quality metrics are transparent and monitored.
- Lineage is detailed enough to build trust and support auditability.

Regulatory Awareness

- Governance supports GDPR/CPRA/HIPAA and emerging regulations.
- Data subject rights (access, delete, correct) are operationalized.
- External data sources (web data, crawled data, third-party data) have documented provenance.

Enablement

- Governance does not create bottlenecks or unnecessary barriers.
- Governance informs teams what they can do safely.
- Business stakeholders participate in governance forums.

AI Governance

Risk Classification

- AI systems are classified by risk.
- High-risk systems have enhanced validation, documentation, and controls.

Model Documentation and Transparency

- Model cards or system cards exist for all deployed models.
- Training data sources and assumptions are documented.
- Known limitations and failure modes are clearly stated.

Testing and Validation

- Models are tested for fairness, robustness, and reliability.
- Generative models undergo grounding and hallucination testing.
- Agentic systems undergo behavioral testing and red-teaming.

Oversight and Accountability

- An AI governance council or equivalent body exists.
- There is accountability.
- Builders, validators, and approvers are separated to prevent conflicts.
- Human oversight policies (HITL/HOTL) are documented.

Operational Governance

- Monitoring includes drift, hallucinations, bias, and security issues.
- Incident response plans are defined for AI misuse or harm.
- Audit trails exist for all high-risk systems.

Regulatory Compliance

- Teams understand implications of the EU AI Act and similar frameworks.
- Copyright and IP risks are reviewed for training data and outputs.
- External data/model sources undergo governance review.

Agent Governance

- Agent permissions have clear boundaries.
- Autonomy levels are documented and enforced.
- All agent interactions are logged.
- "Red lines" for unacceptable agent behaviors are defined.

Conclusion

AI is a capability that organizations will build, refine, and govern over time. The principles in this book point to the fact that long-term success depends on getting the fundamentals right: the strategy, data, architecture, skills, operations, and governance. Those elements determine whether AI delivers real value or becomes another short-lived initiative. The organizations that move forward effectively will be the ones that approach AI deliberately, invest in the foundations, and remain disciplined as the technology evolves. AI will continue to change and evolve, but the responsibility for using it wisely and safely remains with us.

Index

Note: Page numbers in *italics* and **bold** refers to figures and tables respectively.